From Surface to Core: Opening New Windows into the Chemistry of Rocky Planets through Novel Methods in Mass Spectrometry and Machine Learning

Thesis by
Ben J. Thyer

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Ben J. Thyer
ORCID: 0000-0002-5341-0316

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ABSTRACT

[TODO: Write abstract.]

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[TODO: Write acknowledgements.]

I am deeply grateful...

The fieldwork at Draaikraal would not have been possible...

Finally, I thank family/friends for their unwavering support...

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INTRODUCTION

[TODO: Write short dissertation introduction, longer than abstract, detail each chapter.]

Chapter 1

A SIMPLE METHOD FOR HELIUM STEP HEAT EXPERIMENTS AT HIGH WATER OR SULFUR FUGACITY

Helium isotope studies and U-Th/He geochronology of sulfur-bearing and hydrous minerals offer direct windows into the timing and context of hydrothermal mineralization, provided the conditions governing radiogenic helium retention and loss in nature are known. However, assessing helium retentivity at elevated temperatures is currently hampered by these minerals' instability under the high-temperature, high-vacuum conditions of conventional helium step-heating experiments. This paper introduces a novel method for conducting helium step-heating experiments under hydrothermal and sulfur-saturated conditions, designed to extend our knowledge of helium diffusion kinetics in hydrous and sulfur-bearing mineral phases. In this approach, silica glass ampoules encapsulate the mineral of interest, maintaining internally sulfur- or water vapor-saturated conditions against the ultra-high vacuum of the surrounding extraction line, while the high helium permeability of fused silica permits quantification of evolved helium using standard temperature control and mass spectrometry procedures. This work presents the experimental framework and key equations necessary to deconvolve the processes of helium permeation through the silica ampoule from the thermally activated degassing of helium from the encapsulated mineral. Permeation lag is quantified with a first-order permeation model and removed by one of two routes: the permeation step method*, which measures the ampoule's residual helium directly with low-temperature permeation steps, or, where those yields fall to blank level, the* bounding method*, which constrains that residual from above by assuming steady-state permeation during long steps. Both rest on the ampoule's permeation rate constant, which is measured during the experiment itself rather than taken from the literature. Validation experiments measuring helium permeation through empty ampoules and ampoules containing only water, sulfur,*

or Durango apatite demonstrate that permeation lag is readily correctable in well-designed experiments and characterize the extent to which water and sulfur influence the helium permeability of fused silica. While sulfur-saturated experiments are straightforward to carry out with this method, several tractable complications are identified for experiments when significant water is present. This method expands standard experimental capabilities to encompass mineral phases that decompose under high vacuum, including sulfides and iron oxyhydroxides, opening new avenues for thermochronometric investigation of hydrothermal systems.

1.1 Introduction

Noble gas geochronology and thermochronometry ($^{40}Ar/^{39}Ar$ and U-Th/He) on hydrous and sulfur-bearing mineral phases such as hornblende, clay minerals, biotite, goethite, and pyrite have already been leveraged as a powerful tool for understanding regional thermal histories as well as the timing of iron or rare metal mineralization (McInnes et al., 2005; Heim et al., 2006; P. M. Vasconcelos et al., 2013; Nomade et al., 2004; O. Yakubovich, Podolskaya, et al., 2020; O. Yakubovich, Vikentyev, et al., 2021). The interpretation of these data, however, is often complicated by the fact that helium (or argon) within hydrous or sulfur-bearing phases often degasses differently in the laboratory and in nature (Hanson et al., 1975; Lee, 1993; Lo et al., 2000; Farley, H. B. Monteiro, et al., 2024).

Experiments aiming to characterize the migration mechanisms and rates of noble gasses through hydrous phases yield different results when conducted under hydrothermal conditions as opposed to traditional experimental methods in vacuo. Hydrothermal step heat experiments for argon in biotite and hornblende showed higher retentivity and comparatively favored diffusive loss mechanisms relative to experiments conducted in vacuo, and these results are more likely to accurately reflect loss mechanisms in nature (Harrison, 1982; Harrison et al., 1985); however, such experiments are labor-intensive and comparatively inflexible relative to modern helium step heat methods (Farley, Reiners, et al., 1999).

Recently, some workers have taken advantage of the high helium permeability and selectivity of pure silica glass when designing novel experimental methods

for U-Th/He geochronology (O. Yakubovich, Podolskaya, et al., 2020; O. Yakubovich, Vikentyev, et al., 2021), but these ideas have only been applied in a limited way to characterize helium migration in sulfur-bearing or hydrous phases (e.g., monotonic ramped heating experiments: O. V. Yakubovich, Gedz, et al., 2019). This work aims to apply decades of research on helium permeability, solubility, and diffusion in fused silica glass (e.g., Jones, 1953; Norton, 1953; Rogers et al., 1954; Altemose, 1961; Swets et al., 1961; Doremus, 1966; Masaryk, 1968; Shelby, 1971; Shelby, 1972; Shelby, 1976; Shackelford et al., 1972) to adapt existing agile step-heat methods (Farley, Reiners, et al., 1999) to investigate helium loss rates and mechanisms in sulfur-bearing and hydrous mineral phases.

Encapsulating minerals in silica glass ampoules allows experimentalists to maintain hydrothermal or sulfur-saturated conditions within the ampoule without otherwise altering the experimental design. Depending on the thermodynamics of the phase of interest, this method may stabilize the phase at elevated experimental temperatures, allowing for detailed characterization of helium migration in sulfide and hydrous minerals that hitherto has remained poorly understood (Fig. 1.1).

1.2 Experimental Methods

Five helium step heat experiments of sealed silica glass ampoules were undertaken at Caltech to demonstrate the applicability of equations derived here to real helium step heat experiments of geologic materials using a projector-lamp heating apparatus, described in detail in Farley, Reiners, et al. (1999). Four of these experiments involved introducing helium into empty, water-filled, and sulfur-filled ampoules, then measuring helium permeation to assess the applicability of theory outlined in this chapter to helium step heat experiments at Caltech, and to reproduce measurements of permeability rates from the literature (Norton, 1953; Jones, 1953; Swets et al., 1961; Altemose, 1961). The fifth experiment was a standard helium step heat experiment on encapsulated Durango apatite, a widely used standard for helium diffusion experiments and U-Th/He thermochronometry (Zeitler et al., 1987; Wolf, Farley, and Silver, 1996; Farley, 2000; Shuster and Farley, 2005; Shuster, Flowers, et al., 2006; Flowers, Shuster, et al., 2007). The aim of that experiment was to be a practical test in deconvolving the process of helium permeation through silica glass ampoules from thermal helium loss of a well-studied geologic material.

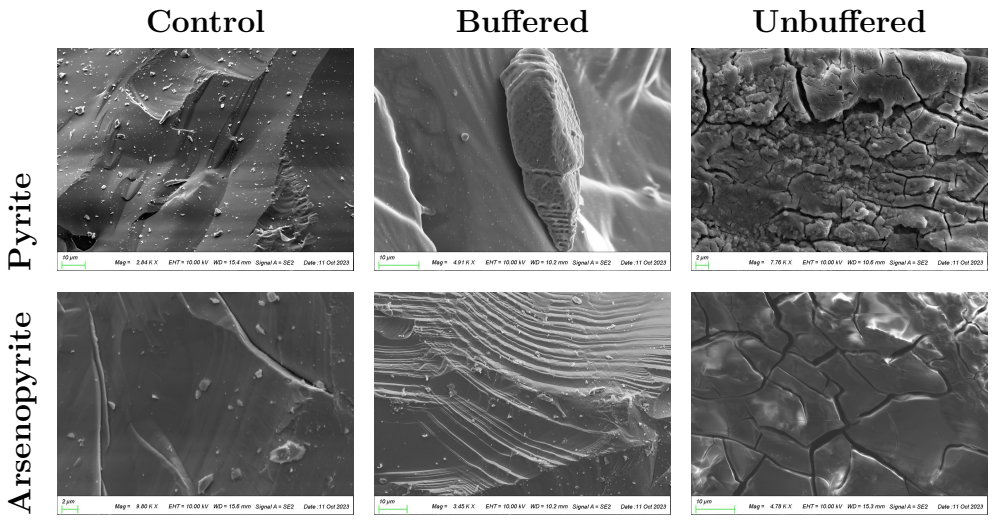

Figure 1.1: Secondary electron SEM images of pyrite (top row) and arsenopyrite (bottom row) surface textures before heat treatment (control), and after heat treatment at 400°C for at least 4 hours, both unbuffered and buffered in a silica glass ampoule with fused sulfur or mineral realgar (AsS) for pyrite and arsenopyrite respectively. The buffered samples maintain the smooth surfaces and clear cleavage faces that the controls have, showing no evidence of decomposition. The unbuffered samples show clear evidence of decomposition by devolatilization, indicated by the formation of contraction cracks and surface roughening. Using silica glass ampoules to buffer samples within controlled microenvironments can prevent phase decomposition at the elevated experimental temperatures typical of helium step heat experiments. See the supplement §1.6s for details on the experimental methods and sample preparation.

Table 1.1 summarizes the dimensions, geometric factors, and encapsulated materials of this work.

To prepare the silica glass ampoules, ~10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thin pipettes with a jewelry welding torch and their thin ends were fused closed.

After the glass had cooled, degassed lab-grade sulfur, distilled water, or Durango apatite were added to some of the tubes. Two tubes were left empty. To seal, the top of each pipette was interfaced with a vacuum line and was pumped to a pressure of <0.1 torr. The encapsulated water tube was flash-frozen in liquid nitrogen and then pumped and sealed in a saltwater ice bath to selectively

Figure 1.2: A typical silica glass ampoule used for helium step heat experiments. The ampoule shown here contains Durango Apatite crushate sieved to 300 µm (radii ≈ 150 µm). The ampoule is roughly one cm long and one mm wide. The bars visible in the background are each separated by 1 mm.

Table 1.1: Silica Glass tubes that were the subject of helium step heat experiments in this work. Ampoules labeled "Quad" had evolved 4He quantified by isotope dilution using a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer, whereas ampoule labeled "MC" had both evolved 3He and 4He quantified using a Helix-MC-Plus^{10K} multi-collector mass spectrometer. Where 3He was measured, the ampoule was soaked in tank helium with an isotope ratio of ~0.15 Ra

Ampoule Contents	Mass Spec.	Inner Diameter µm	Outer Diameter µm	Geometric factor G cm^{-2}	Volume Ratio $\left(\frac{V_{glass}}{V_{gas}}\right)$	Content Mass	Helium Source
Empty	Quad	630	960	4785.3	1.32	—	Atmosphere
Empty	MC	747	1159	3263.9	1.41	—	Tank He
Sulfur	MC	529–550	835–872	5992.1	1.5	~0.2 mg	Tank He
Water	MC	487–515	734–776	7789	1.27	~0.5 mg	Tank He
Durango Apatite	Quad	850	1350	2393.5	1.52	3.18 mg	Radiogenic

pump away atmospheric gasses and not water. Sublimation of sulfur and water ice during pumping was minor. The primary motivation for pumping the tube to rough vacuum was to aid the sealing process. After rough vacuum was achieved, a fine-tipped welding torch was used to seal the top of each ampoule. To determine the geometric factors for permeation of the ampoules, the inner and outer diameters of each ampoule were determined using the measurement function of the Amscope stereo microscope software, which was calibrated and tested during early development of this method.

Except for the encapsulated Durango apatite ampoule, helium gas needed to be introduced into the sealed ampoules before they were loaded into the projector lamp heating cells. About 15 nccSTP of atmospheric helium was introduced into the empty ampoule used for the quadrupole mass spectrometer experiment by heating the ampoule in a muffle furnace to >400°C in atmosphere overnight, which was sufficient to support a short permeation experiment of 4He. However, to conduct step heat permeation experiments on 3He, much larger quantities of helium were necessary. The encapsulated sulfur and water ampoules and the other empty ampoule were soaked in 2-20 torr of high purity tank helium with an isotope ratio of about ~0.15 Ra at 300°C for about 24 hours. This procedure introduced 0.6-10 pccSTP 3He and up to 52,000 nccSTP 4He into each tube, allowing for measurements of both 3He and 4He permeation simultaneously.

The sealed ampoules were placed in copper foil packets and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in more detail by Farley, Reiners, et al. (1999). Before each experiment began, each ampoule was washed with ethanol, air-dried, loaded in the lamp cell, pumped to high vacuum, and was held at 100°C for 30 minutes under dynamic vacuum to desorb ethanol, water, and other atmospheric species on the ampoule, copper packet, and lamp cell. Less than 0.5 ‰ of the radiogenic helium in 150µm radius Durango apatite would be expected to degas in 30 minutes at 100°C, which would amount to ~0.01 nccSTP for this experiment, using the diffusion parameters of Farley (2000) and the forward equations of Fechtig and Kalbitzer (1966). The atmospheric helium component within the encapsulated Durango Apatite ampoule constrained by permeation steps at the beginning of the experiment was much larger: 1.47 ± 0.18 nccSTP 4He.

We found that the fused silica ampoules were capable of maintaining surprisingly high internal pressures, but ampoules filled with water would often explode at temperatures above 315°C. It is recommended that experimentalists limit the temperature of water-filled ampoules to below 300°C. We have not observed any explosions of sulfur-filled ampoules at any temperature.

Heating steps and isotopic measurements for all samples were automated using custom LabView routines with the exception of the tank helium-soaked empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer after the 7th measurement. For all other ampoules, a Watlow temperature controller held the ampoule at the temperature set point for a pre-programmed

duration, after which the ampoule was allowed to cool while the evolved gas was prepared for measurement for about 25 minutes. After the first 7 measurements, the empty tube experiment on the Helix-MC-Plus^{10K} was instead set to an elevated temperature manually while the evolved gas was allowed to accumulate continuously in an attempt to mitigate uncertainties related to frequently changing temperatures (supplementary Eq. (1.48); Fig. 1.19) and continued permeation at room temperature during gas cleanup and measurement. The accumulated gas was measured every 40-120 minutes. For this experiment, the temperature was changed either during the 1 minute pump-out of the lamp cell after inletting gas into the mass spectrometer, or a period of helium accumulation for measurement. When the temperature was changed during an accumulation period, no permeability was calculated for that step, and the measured abundances of those steps were used only to track the helium contents of the tube.

Helium isotope measurements of evolved gasses from steps heats of one empty silica glass ampoule and glass-encapsulated sulfur and water ampoules equilibrated with 2-20 torr of high purity tank helium ($R \approx 0.15\ R_a$) were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from November 19th, 2023 to Feburary 15th, 2024. Helium was purified by exposing evolved gasses to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. The majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech have shown this approach to be reliable, reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray. 4He abundances were very large during these experiments, resulting in a small but measurable 4He overspray of up to 0.68 cps at mass 3.5. These overspray signals at mass 3.5 were always less than 1% of the 3He signal at mass 3, suggesting that 4He overspray did not significantly affect 3He signals. Measurements of high 3He MM gas standard were analyzed every 8 hours (totaling 116 measurements) to track instrumental

drift of 3He and 4He sensitivities and to quantify internal reproducibility. These standards show 3He was internally reproducible to within 5% (2σ), whereas 4He was internally reproducible to within 2% (2σ), where reproducibility is estimated as the standard deviation divided by the mean.

Helium-4 evolved during step heats of the empty silica glass ampoule equilibrated with atmosphere and the silica glass-encapsulated Durango apatite was quantified using isotope dilution with a high purity 3He spike on a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer from October 30$^{\text{th}}$ to November 10$^{\text{th}}$, 2023. Evolved gas from the ampoule in the lamp heating cell was allowed to mix with the 3He spike and the mixture was frozen on charcoal held at 14°K within a cryostat, then helium was selectively released to the mass spectrometer by raising the temperature to 36°K.

A standard gas with a known amount of 4He is routinely measured, at least one time per experiment, to monitor drift in the relative sensitivity of 3He and 4He on the Pfeiffer Prisma QMS 200 quadrupole mass spectrometer. Relative sensitivities are determined for each standard measurement and are linearly interpolated in time for sample measurements. Blanks are approximately 5 pccSTP ^{4}He, which is very small compared to all interpreted ^{4}He measurements.

1.3 Theoretical Framework of Permeation Lag

The process of helium permeation through thin walls of an ampoule is rapid but not instantaneous on experimental timescales. Helium evolved from encapsulated minerals is only available for measurement following permeation through the glass ampoule (Fig. 1.3), producing a predictable "permeation lag" effect in helium measurements that commonly manifests with undermeasurement of helium during prograde heating sequences and corresponding overmeasurement of helium produced during retrograde sequences (Fig. 1.5). This is not merely a theoretical concern: Fig. 1.6 shows the same artifact in real, uncorrected step heat data for the encapsulated Durango apatite experiment described above (§1.2); the correction developed below is applied to this dataset in full in §1.6 (Fig. 1.6).

This effect can be modeled (and corrected for) for a system obeying the simple equation:

$$n_t = n_0 + \omega - m \tag{1.1}$$

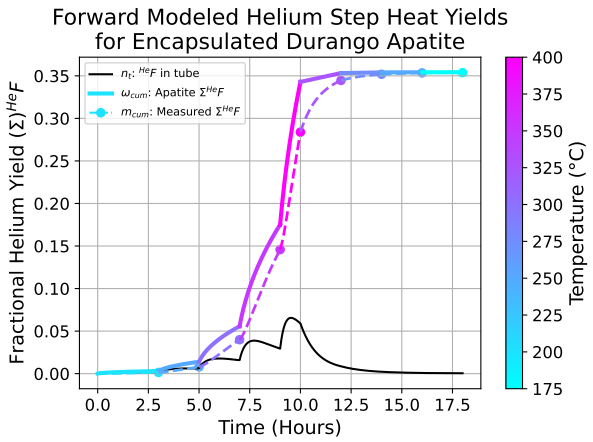

Figure 1.3: An illustrative simulation of the effects of permeation lag on step heat measurements of 150 µm radii spheres of Durango apatite encapsulated in a "standard ampoule" with an outer diameter of 1mm and inner diameter of 0.6mm, corresponding to a geometric factor G of 4350.26 cm^{-2}. Some amount of helium degassed from the apatite during an experimental step remains in the tube during measurement and permeates out in later steps. The heating schedule of this model system is given in Table 1.2. This simple model assumes new temperatures and their corresponding steady-state diffusive concentration profile within the glass ampoule are reached instantaneously. Effects of diffusive disequilibrium, if considered, would change the permeated amount for each step by less than 5%.

Where n_t is the amount of helium within the ampoule after time t, n_0 is the initial amount of helium within the ampoule (which is inherited from previous heating steps), ω is the amount of helium evolved by the encapsulated sample during time t, and m is the amount of measurable helium that has permeated out of the ampoule into the high-vacuum chamber during time t.

Heating steps are consecutive, so the helium inventory quantified between two steps applies to both the previous and the subsequent step:

$$n_t^i = n_0^{i+1}, \qquad n_0^i = n_t^{i-1} \tag{1.2}$$

Recovering ω for a step therefore requires knowledge of the ampoule's helium content immediately before and immediately after it. Sections 1.4 and 1.4 give the two routes used in this work to obtain those quantities.

Helium permeation through fused silica in diffusive steady state is a first order process that can be described with the simple differential equation:

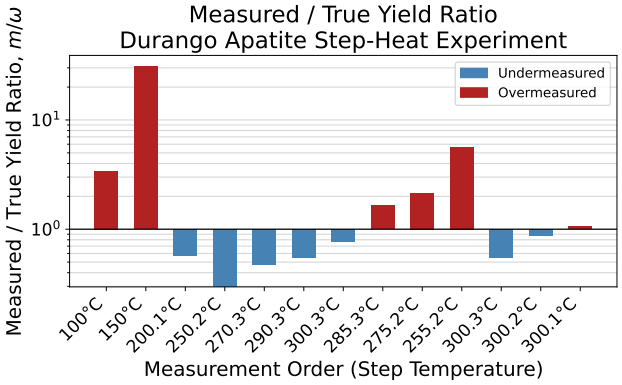

Figure 1.4: Measured-to-true yield ratio, m/ω, for the first 13 experimental steps (through the isothermal 300°C triplicate) of the real encapsulated Durango apatite step heat experiment described above (§1.2) and tabulated in full in Table 1.3, plotted in measurement order and labeled by nominal step temperature. Here m is the raw measured helium yield for a step and ω is its permeation lag-corrected yield; bars are anchored at a ratio of 1 (no anomaly) on a logarithmic axis, so a doubling and a halving of the ratio are given equal visual weight. Steps taken while ramping to higher temperatures are generally undermeasured (blue), as some of the sample's evolved helium has not yet fully permeated out of the tube at the time of measurement, while steps taken while ramping to lower temperatures are generally overmeasured (red), as helium retained in the tube from earlier, hotter steps continues to permeate out. The first two steps (100-150°C) are strongly overmeasured despite being on the heating ramp because little helium had yet evolved from the apatite, so measured yield there is dominated by pre-existing atmospheric helium draining from the tube rather than fresh sample degassing.

$$\frac{dn}{dt} = -GKnRT \tag{1.3}$$

Where n is the number of helium atoms, t is the time, G is the geometric parameter in units of cm^{-2}, which is a function only of the geometry of the glass membrane, K is the permeation constant for helium in fused silica, which follows the Arrhenius law: $\left(K(T) = K_0 e^{-\frac{E_a}{RT}}\right)$ and has been repeatedly measured for fused silica (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), R is the gas constant, and T is the temperature in Kelvin. Equation (1.3) is identical to the permeation equation used in early studies (Norton, 1953; Altemose, 1961; Swets et al., 1961; Studt, 1962; Masaryk, 1968), except that it is formulated in terms of absolute atomic abundances instead of helium gas pressure by the ideal gas law. It is easily integrated by separation of variables to yield $n_t = n_0 e^{-GKRTt}$ (see supplement, §1.1s, for details). Relating that

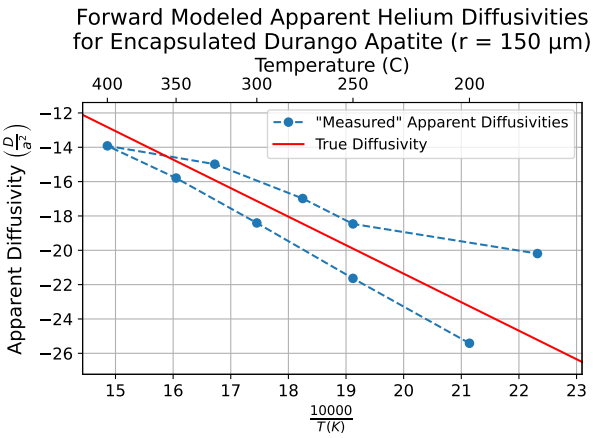

Figure 1.5: An Arrhenius plot of apparent diffusivities inverted from modeled hypothetical measurements of an encapsulated sample of Durango apatite. Permeation lag produces a characteristic effect where measurements of prograde sequences of increasing temperature produce erroneously low apparent diffusivities when inverted using the equations of Fechtig and Kalbitzer (1966), whereas retrograde (decreasing) temperature sequences exhibit the opposite effect. Diffusion parameters are taken from Farley (2000).

Table 1.2: Heating schedule for numerically modeled step heat experiment of silica glass encapsulated Durango apatite.

Temperature (°C)	Time (min)
200	180
250	120
300	120
350	120
400	60
325	120
275	120
250	120
175	120

result to the measured quantity through Eq. (1.1) in the absence of any release from the sample ($m(\omega = 0) = n_0 - n_t$) gives an expression most intuitively stated in terms of the fraction of the ampoule's initial helium reservoir that permeates during time t:

$$^{He}F = \frac{m}{n_0} = 1 - e^{-\lambda t}, \qquad \lambda = KGRT \tag{1.4}$$

Where ^{He}F is the fractional permeation yield: the fraction of the ampoule's

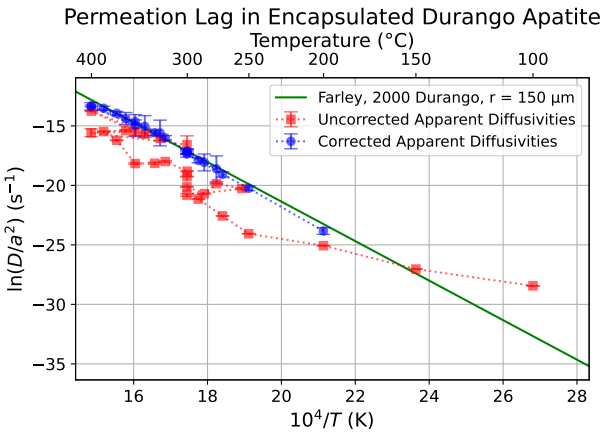

Figure 1.6: An Arrhenius plot of a helium step heat experiment of silica-glass encapsulated Durango Apatite with 2σ error bars. Without permeation lag corrections, the apparent diffusivities display highly irregular behavior that would be unrecognizable to this well-behaved sample – exactly the artifact anticipated by the simulation of Fig. 1.3 above. With permeation lag corrections, the sample agrees well with the reported diffusive parameters of Farley (2000) for the appropriate sieve size ($r = 150\mu$m). As the experiment proceeds beyond temperatures of 300°C, the activation energy of diffusion decreases to reveal a slight downward curl, a phenomenon well-documented for Durango apatite likely due to annealing of crystal defects and radiation damage (Wolf, Farley, and Silver, 1996; Farley, 2000). Apparent diffusivities are calculated with the equations of Fechtig and Kalbitzer (1966) with and without permeation lag corrections. Four measurements of the permeation lag corrected apparent diffusivities are within error of 0 at the 2σ level and thus are excluded from the plot.

initial helium reservoir (n_0) that permeates as m (Eq. (1.1)) during a time t of steady state permeation, and λ is the ampoule's permeation rate constant. Collecting K, G, R, and T into λ is useful because λ is the quantity that can be measured directly during an experiment (§1.4), whereas K and G cannot be separated without independent knowledge of the ampoule geometry.

The geometric factor G for a cylindrical membrane with negligible permeation through the top or bottom is determined using the effective area of permeation given by Masaryk, 1968:

$$G = \frac{2}{a^2 \ln\left(\frac{b}{a}\right)} \tag{1.5}$$

where a is the cylinder's internal radius, and b is the external radius. There

is no dependence on the cylinder length when permeation through the top and bottom are negligible. This geometric constant will vary according to the experimental design and therefore must be measured by the experimentalist.

Equation (1.4) is only strictly valid with two assumptions:

1. The helium partial pressure in the "measurable" reservoir outside of the ampoule is negligible compared to the partial pressure within the ampoule. Because the ratio of typical ampoule and the diffusion cell volumes is ~1/10,000 (Farley, Reiners, et al., 1999), this assumption is essentially always reasonable.

2. The helium concentration profile within the ampoule's glass walls is in diffusive steady state between the inner and outer walls. The real steady state helium concentration profile is well-approximated by the ideal steady state profile for ampoules with a glass/gas volume ratio less than 2 (Figs. 1.11, 1.15). For these ampoules, (1.4) is accurate to within 1% after the establishment of a diffusive steady state. Additionally, the diffusive steady state assumption is ephemerally violated when the temperature is changed due to the temperature dependence of the glass/gas partitioning of helium. The ampoule should therefore be held at the permeation temperature for at least one diffusive timescale $\tau = L^2/D$ after every temperature change before measuring, where L is the wall thickness and D the diffusivity of helium in silica glass (Swets et al., 1961). In this work, low-temperature heating steps are repeated to allow for the new diffusive steady state concentration profile to become established. The ampoules of this work reach diffusive equilibrium in 30-70 minutes at 80° (Fig. 1.7, supplementary figures 10 and 11). Comparisons between the abundance of helium within glass as a function of temperature limits the error introduced by thermal perturbation to less than 5% (supplementary figure 9).

1.4 Experimental Approaches Addressing Permeation Lag

A combination of analytical and experimental methods may be used to correct for permeation lag and properly account for all helium evolved from the sample. Some common practices for noble gas measurements may introduce error when using this method. For example, it is common practice to purify

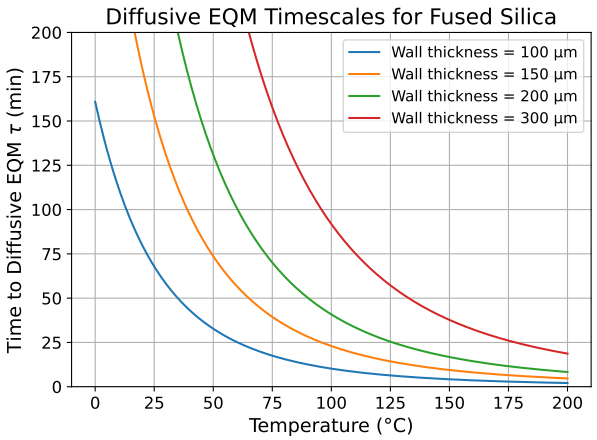

Figure 1.7: Timescales as a function of temperature to reestablish a steady state diffusive concentration profile for different glass thicknesses. These curves correspond to the timescales at which the Fourier number equals 1. 4He diffusivities are taken from Swets et al. (1961).

and collect emitted helium with the diffusion cell open to the vacuum line at room temperature for extended times, often 15-30 minutes, and it is assumed that no additional helium evolution occurs. This assumption is reasonable for commonly measured geological materials, but it can be problematic when unexamined during permeation steps of silica glass encapsulated materials because helium permeation in silica glass is significant even at room temperature. In this work, permeation during these waiting times is often mathematically corrected for using Eq. (1.4), assuming permeation continues at room temperature. Alternatively, there are experimental approaches to address ongoing permeation during these waiting periods: they may be shortened to minimize their impact on the measurement, or the ampoule may be held at the warm permeation temperature during the gas purification and cryofocusing processes, so that permeation continues at the same temperature during this time without introducing complicated thermal histories.

From Eq. (1.1), it is immediately clear that when the evolved helium ω is large with respect to the amount of helium within the ampoule, n_t or n_0, then $\omega \approx m$ and permeation lag can be neglected. Unfortunately, this is not the case for most phases commonly subjected to helium step heat experiments with routine heating schedules (e.g. hour-timescale steps at 100-400°C for apatite or zircon), but it is sometimes possible to design an experiment with long time-steps where the total evolved helium is large compared to the amount of helium in-residence

within the tube. Nonetheless, permeation lag can be quantified and corrected for in *any* experiment where the encapsulated material is effectively closed to helium loss at room temperature. Two approaches are described here to recover the helium evolved from an encapsulated sample (ω) from measured amounts (m): the *permeation step method* (§1.4), which measures the ampoule's helium inventory directly between experimental steps, and the *bounding method* (§1.4), which constrains that inventory from above when it cannot be measured. Both require the ampoule's permeation rate constant, which should be measured during the experiment itself rather than assumed from the literature (§1.4).

The Permeation Step Method

In typical experimental heating steps, there are two fluxes of helium controlling the ampoule's helium abundance: an influx of helium evolved from the sample and an outflux of helium due to permeation. For lower temperature steps, for which helium release from the sample is negligible ($\omega = 0$), the helium abundance within the ampoule is controlled only by the process of helium permeation, simplifying Eq. (1.1) and allowing for Eqs. (1.1) and (1.4) to form a system of equations that can be solved for n_t and n_0 given a measurement m for an isothermal heating step of known duration.

$$n_0 = \frac{m}{{}^{He}F} \tag{1.6}$$

Where ${}^{He}F$ is the fractional permeation yield defined in Eq. (1.4). Such a low temperature heating step is hereby referred to as a *permeation step* in this work. A permeation step does double duty: through the step-to-step continuity of the ampoule's inventory, it constrains both the helium remaining after the preceding high temperature step and the helium inherited by the following one. Placing a permeation step after every high temperature *experimental step* ($\omega > 0$) therefore supplies n_t and n_0 throughout the experiment, and Eq. (1.1) returns the helium evolved from the natural sample, ω.

Equation (1.6) requires the rate constant of permeation for the ampoule in use, $\lambda = KGRT$. This may be assembled from the ampoule's measured dimensions through the geometric factor G (Eq. (1.5)) together with a literature permeability coefficient K for helium in fused silica, which has been measured repeatedly as a function of temperature (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968). Published values should not be relied upon, however,

especially for hydrothermal experiments: water entering the glass changes its permeability (Shelby, 1972; Tsuru et al., 2011), and water-bearing ampoules show anomalously high rate constants (§1.7). The rate constant should instead be measured on the ampoule actually in use, during the experiment itself (§1.4). That measurement is best understood as a prerequisite and extension of the permeation step method rather than as an alternative to it: the consecutive isothermal permeation steps it requires are already part of the method, so the only additional cost is the discipline of running enough of them in sequence.

Measuring the Ampoule's Rate Constant In Situ

The permeability coefficient K of helium through nominally anhydrous silica glass is well known at relevant experimental temperatures for water-free glass (Norton, 1953; Altemose, 1961; Swets et al., 1961; Rogers et al., 1954; Masaryk, 1968). Literature values need not describe the ampoule actually in use. Water diffuses into glass and changes its helium permeability (Shelby, 1972; Tsuru et al., 2011), so K drifts during hydrothermal experiments; and the volume occupied by a sample or buffering phase raises the effective rate constant above the value implied by the ampoule's nominal internal geometry (§1.7). Neither effect is captured by G and a tabulated K.

Fortunately, the rate constant can be measured without knowing either quantity separately. Regardless of the values of K and G, helium permeation through glass remains a first order process obeying Eq. (1.4), which is the recognizable decay equation written with the rate constant $\lambda = KGRT$ defined above:

$$n_t = n_0 e^{-\lambda t} \tag{1.7}$$

λ can be experimentally determined alongside n_0 by using a non-linear least squares optimization to fit a consecutive series of at least three isothermal permeation step measurements to the following equation, expressed in terms of cumulative measured helium over cumulative time:

$$m_{cum} = n_0 \left(1 - e^{-\lambda t_{cum}}\right) \tag{1.8}$$

We fit individual step yields rather than cumulative yields, because cumulative fitting propagates the errors of early measurements into later measurements, unduly biasing the solver to fit the earlier measurements.

$$m_{i+1} = n_0 \left(1 - e^{-\lambda t_{i+1}}\right) - n_0 \left(1 - e^{-\lambda t_i}\right) \tag{1.9}$$

We used the Levenberg-Marquardt implementation in SciPy (Virtanen et al., 2020), weighted by each measurement's own uncertainty, with initial guesses $\lambda \approx 6 \times 10^{-6}\,\mathrm{s}^{-1}$ and $n_0 \approx m_i / \left(1 - e^{-\lambda t_i}\right)$ evaluated at the first measurement of each series. Uncertainties come from the fit covariance matrices. The same optimization can be accomplished with the "Solver" function in Microsoft Excel.

This fit requires that enough cumulative time has passed to reveal the curvature of cumulative helium yield against time, or else the problem will be poorly conditioned. This can be assured in practice by waiting until the rate of helium permeation has slowed markedly after three or more isothermal permeation measurements before advancing to the next high-temperature experimental step. The exercise can technically be done with as few as two consecutive measurements, although the solver would then be unable to estimate uncertainties on the fit parameters.

In addition to the initial helium contents of the ampoule n_0, this procedure simultaneously determines the product of the permeability and geometric factor for the silica glass ampoule. Once $\lambda = KGRT$ has been satisfactorily constrained over a range of temperatures, the activation energy E_a of permeation and the product of the geometric factor and permeability at infinite temperature, $K_0 G$, can be fit using linear regression. Determining these values allows for extrapolation of the permeability to the high temperatures of the experimental steps, which can be useful to an experimentalist who wishes to design a heating schedule wherein the permeation lag is negligible for some experimental steps but does not know the permeability coefficient K and/or the geometric factor G.

The rate constant costs little to measure. A series of consecutive isothermal permeation steps is already required by the permeation step method (§1.4), so the additional expense is the discipline of running enough of them in sequence to resolve curvature rather than a single step in isolation. We therefore treat in-experiment measurement of λ as a routine part of the permeation step method rather than as an optional refinement, and as indispensable for hydrothermal experiments, where the glass takes up water during the run and no tabulated

value can be expected to hold. Where λ is measured repeatedly through an experiment, its drift can be tracked and interpolated; where it is measured once, it at least applies to the ampoule at hand rather than to an idealized dry one.

The Bounding Method

Permeation steps require a measurable signal at low temperature. When the encapsulated sample is retentive and its yields are small, the low-temperature signal can fall to blank level and Eq. (1.6) cannot be applied. An alternative is available when steps are long compared with the permeation timescale of the ampoule, which is common for retentive phases: rather than measuring the ampoule's residual helium, it can be bounded from above. Permeation lag is a significant effect for typical heating schedules operating on timescales of a few hours (e.g., Farley, 2000; Shuster, Farley, et al., 2004; Figs. 1.3 and 1.5), but its relative significance decreases with step duration and increased temperature (Fig. 1.8). Consider an arbitrarily long experimental step: after some time, a steady state $\left(\frac{dn}{dt} = 0\right)$ will be reached where the helium permeation rate out of the ampoule is balanced by the production rate of helium within the ampoule from the degassing of the natural sample. The upper bound for this steady state concentration as a function of measured helium abundance (Eq. (1.10)) is derived in the supplement, §1.3s (Eq. (1.58)).

$$n^{ss} \leq \frac{m}{\lambda t - 1} \tag{1.10}$$

Equation (1.10) only applies when step heats are long enough (roughly 5-24 hours depending on temperature and ampoule geometry, Fig. 1.22) to assume steady state permeation ($\lambda t >\sim 5$), which requires that G and K are known or can be conservatively estimated. It also assumes that the production rate of helium within the ampoule due to degassing of the encapsulated natural sample is constant over time at a given temperature. Typically, the helium production rate Ω decreases over time in most experiments, which would produce a smaller steady state He concentration than what is predicted by Eq. (1.10). Ω decreases over time for helium loss governed by volume diffusion (Wolf, Farley, and Silver, 1996; Reiners and Farley, 1999; Farley and McKeon, 2015; Farley, 2018; Farley, H. B. Monteiro, et al., 2024), or first order processes (O. V. Yakubovich, Y. A. Shukolyukov, et al., 2010; Wilson and Bisson, 1973), except for the case of

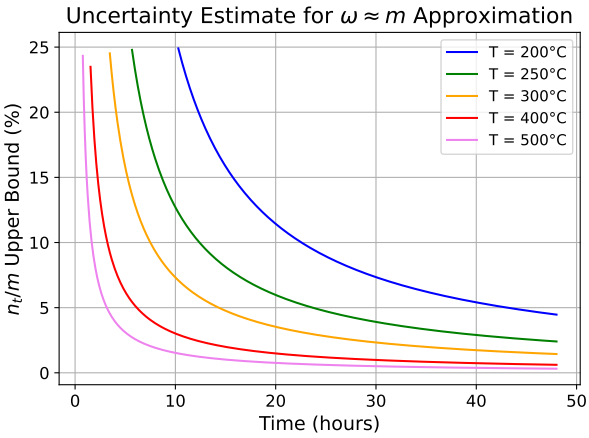

Figure 1.8: For long steps and high temperatures, the abundance of helium within the ampoule will reach a steady state and will be constant as the measurable helium outside the ampoule in the lamp cell increases. As the step gets longer, this amount becomes small relative to the amount of measurable helium, and the measurable helium can be used to approximate the total amount of degassed helium with only minor inaccuracy. Eq. (1.10) is plotted here to show the upper bound relative amount of helium trapped within the ampoule normalized to the measurable amount. The geometric factor for these calculations is 4350 cm^{-2}, corresponding to an ampoule with an inner and outer diameter of 0.6 and 1 mm respectively. Permeability values (K) are for 4He in silica glass and are taken from Swets et al. (1961).

initially inhomogeneous distributions in at low helium loss fractions (Goldsmith et al., 2020; Ketcham et al., 2011; Shuster, Farley, et al., 2004; Farley, Wolf, et al., 1996) or radiation damage annealing (Cherniak, 2019; Guenthner et al., 2013; Flowers, Shuster, et al., 2007; Flowers, Ketcham, et al., 2009; Shuster, Flowers, et al., 2006). For experiments where these factors are of interest, the permeation step method (§1.4), which handles permeation lag explicitly, is superior. Additionally, Eq. (1.10) is conservative whenever the value of λ used does not exceed the true one, because a lower bound on λ yields an upper bound on the residual. Any λ that can be independently shown to underestimate the ampoule's true rate constant is therefore admissible. The bounding method is consequently an excellent fallback when the small permeation step yields of retentive phases cannot be well resolved.

Using the steady state concentrations calculated with Eq. (1.10) from a long duration step (1) and the previous step (0) as upper bound estimates in Eq. (1.1) yields the inequality that can be used to bound the amount of helium evolved

from the sample during step 1 without permeation steps:

$$m_1 - \frac{m_0}{(\lambda t)_0 - 1} \leq \omega_1 \leq m_1 \left(1 + \frac{1}{(\lambda t)_1 - 1}\right) \tag{1.11}$$

The relative uncertainty $\left(\frac{m}{n^{ss}} = \frac{1}{\lambda t - 1}\right)$ is plotted in Fig. 1.8 as a function of time and temperature for a standard ampoule with geometric factor G of 4350.26 cm^{-2}. To achieve relative uncertainties below ~ 5%, steps longer than 24 hours are typically required at temperatures below 300°C.

1.5 Analytical Methods

Permeabilities were calculated for the empty ampoule experiments by solving Eq. (1.4) for permeability K.

Besides the encapsulated Durango experiment, which contained a helium source, the helium contents (n_0, n_t) of the other ampoules changed only by decreasing due to helium permeation through silica glass throughout the experiments. These experiments were concluded when measured helium permeate of a high temperature step was indistinguishable from blank. The values of n_0 and n_t for each step were directly quantified after the experiments concluded using the total yield. Assuming that the helium permeated during the short pump-out time of the lamp cell is negligible, the total yield is simply the sum of all the measured abundances of helium for each step. The atmosphere-equilibrated empty tube experiment on the Pfeiffer Prisma QMS 200 quadrupole mass spectrometer was pumped for 15 minutes between steps, the encapsulated apatite experiment was pumped for 5 minutes between steps, and the Helix-MC-Plus^{10K} experiments on silica glass ampoules equilibrated in 2-20 torr tank helium were pumped for only 1 minute between steps. This pump-out time was short enough to neglect for all but the atmosphere-equilibrated empty tube, whose longer 15-minute pump-out and short (45-minute) heated steps required small explicit pump-out and gas-preparation corrections using Eq. (1.4); see the supplement, §1.5s, for the full derivation.

One high temperature measurement – the last step of the empty ampoule experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer – showed unusual, non-linear signal evolution and required a bespoke recalculation to recover its 4He and 3He yields; see the supplement, §1.5s, for details.

The encapsulated apatite experiment included high temperature steps ranging

from 100-400°C, each preceded and followed by a sequence of two or three permeation steps. Equation (1.6) was applied to solve for the helium contents of the ampoule (n_0) between steps for permeability corrections. Only the last of each sequence of room temperature measurements (shown in **bold** in Table 1.3) was used to solve for n_0 using Eq. (1.6), because the temperature records showed that the encapsulated apatite had not yet cooled to room temperature by the time the first permeation step at room temperature began, although each pair of permeation steps always yielded values of n_0 within uncertainty (2σ) of one another. Measured helium abundances of the other permeation steps taken before reestablishing an equilibrium steady state concentration profile (shown in *italics* in Table 1.3) were added or subtracted to the n_0 value to propagate the determined helium contents of the ampoule back or forward in time to the high temperature experimental steps for permeation lag corrections using Eq. (1.1). Measurement uncertainties are estimated using standard deviations of sensitivities and blanks determined by standards. These measurement uncertainties are propagated through Eqs. (1.4) and (1.1) in quadrature, in addition to a general 10% uncertainty for the use of Eq. (1.4) following the recommendations of this work (see discussion). 4 of the 25 high temperature experimental steps of the encapsulated Durango apatite produced corrected yields within error (2σ) of 0 and were excluded from Fig. 1.6 (see figure caption). Uncertainty for a step's fractional cumulative yield was estimated as the sum of the uncertainty of its fractional yield plus the uncertainties of the measured fractional yields of all previous measurements, added in quadrature. The total yield was estimated based on the precisely-measured mass of the encapsulated Durango apatite (3.18 mg) and its helium concentration of 194.2 nccSTP/mg (Farley, 2000), a value independently verified in-lab (supplement, §1.5s). No uncertainty in the total yield estimation is propagated through to the diffusivities reported here.

Numerical Diffusive Permeation Model for Arbitrary Thermal History and Diffusive Disequilibrium

Most of the permeation measurements performed on the Helix-MC-Plus^{10K} mass spectrometer were complicated by additional permeation occurring at variable temperature during long cooling times following high temperature experimental steps. Such a complication can be effectively avoided by limiting the duration of time for which the diffusion cell is opened to the rest of the

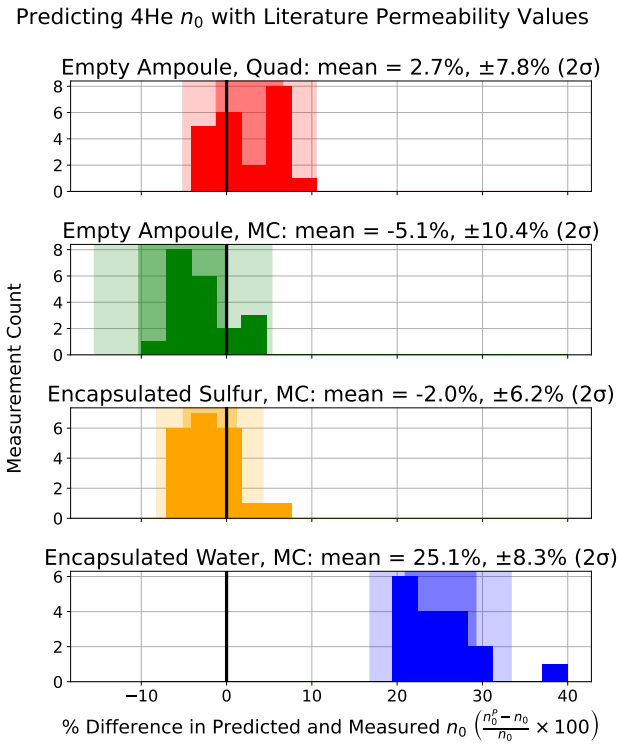

Figure 1.9: Histograms showing the accuracy of 4He n_0 predictions using permeabilities of Swets et al. (1961) from either Eq. (1.22) or a numerical diffusion model from the measured value, with shaded regions behind the histograms corresponding to 1 and 2 sigma (σ) from the average relative difference between measured and predicted values. The measured n_0 values of the empty and sulfur-filled ampoules are within 1σ of the predicted values, suggesting that the permeation model and literature values apply well to those systems, and such models can be used to reliably recover helium contents of permeating ampoules for use in permeation lag corrections. On the other hand, measured and predicted 4He n_0 for water-filled ampoules are well resolved beyond 2σ, suggesting that their helium permeability differs strongly from the literature values. Internal reproducibilities (1σ) of a gas standard repeatedly analyzed during these experiments were better than 1% for 4He, requiring that most of the scatter in these data be related to factors outside of analytical precision, such as long/inconsistent ramp times during heating or uncertainties or rounding of heating times.

vacuum line, or by keeping the sample at-temperature during this equilibration time. To address this, a numerical diffusive permeation model is used to calculate the ampoule's expected permeation fraction (n_t/n_0) for each step with a complex thermal history (see supplement, §1.2s, for details).

1.6 Results

Predicted and measured fractional yields for all permeation steps agree well (within 1σ; Fig. 1.9, supplementary Figs. 1.23–1.24) with the predictive equations presented in this work and with the permeabilities reported by Swets et al. (1961) and Jones (1953) for the empty and sulfur-filled ampoules across all studied temperatures (24-300°C) and step fractional yields (up to ~80%), but not for the water-filled ampoule.

Conversely, helium permeation for the ampoule containing water was inconsistent with the permeabilities reported by Swets et al. (1961) and Jones (1953). For these hydrothermal step heat experiments, measured helium partial yields are systematically high by about 25% relative to the expected values (supplementary Figs. 1.9, 1.24), meaning that permeation was systematically faster than expected for the ampoule filled with water. An analysis of this discrepancy and a hypothesis for its origin is presented in the discussion.

Permeation ratios for 3He and 4He measured in the empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer are plotted in supplementary Fig. 1.25. These data present significant scatter, but generally agree with the reported results of Jones (1953), except for measurements at room temperature, which cluster tightly at a permeation ratio of 1:1 for the two isotopes. Additionally, fields corresponding to the ratios of apparent permeabilities for the water-filled and sulfur-filled experiments are plotted for the temperature ranges of those experiments. Results for those experiments are plotted as fields because each step integrated permeation behavior over a complex (non-isothermal) cooling history. These experiments also agree with the results of Jones (1953) and the empty ampoule experiment. The qualitative result of Jones (1953) that silica glass is only slightly more permeable for 3He than 4He is supported by these data, which are in marked contrast to the reported extreme differences in 3He and 4He diffusivity in silica glass of McMurtry et al. (2019), who report instantaneous isotopic fractionations up to a factor of 1000.

The permeation lag-corrected helium diffusivities for the encapsulated Durango apatite experiment are within uncertainty of the Arrhenius relationship for 150µm radii Durango apatite reported by Farley (2000), except for the first two plotted points and the 400°C step (Fig. 1.6). Farley (2000) also observed some low apparent diffusivities for helium in Durango apatite in early steps,

although rounding of the concentration profile by diffusive loss or alpha ejection is not expected for interior fragments of large gemmy apatites. The high temperature downward curvature of the apparent diffusivities in Fig. 1.6, however, is well-understood. Wolf, Farley, and Silver (1996) and Farley (2000) discuss similar downward curvature of the diffusion array of Durango apatite at high temperatures. The annealing of radiation damage 'traps' and crystalline defects is the favored explanation for this irreversible change in diffusivity at high temperatures for Durango apatite (Farley, 2000; Shuster, Flowers, et al., 2006). Overall, the permeation lag-corrected apparent diffusivities for encapsulated Durango Apatite reproduce the same results of unencapsulated Durango apatite (Fig. 1.6), suggesting that permeation lag can be reasonably deconvolved from thermal helium emission from minerals or other solid materials of interest when performing step heat experiments to characterize helium retentivity and loss mechanisms in solids.

Table 1.3: Step heat data for the silica glass encapsulated Durango apatite experiment. ${}^{He}F_{Tube}$ is the fractional permeation yield (Eq. (1.4)) of the adjacent permeation step; ${}^{He}F_{Apatite}$ is the permeation lag-corrected cumulative fractional yield of helium released from the apatite. n_0 values (initial ^{4}He content of the ampoule, Eq. (1.6)) are shown in **bold** for permeation steps used directly to solve for n_0, and in *italics* for all other steps, where n_0 is instead propagated forward or backward in time from the nearest bolded permeation step using Eq. (1.1). Blank "Type" entries denote room-temperature measurements preceding the permeation step used to solve for n_0, taken before the diffusive steady state concentration profile was reestablished (see text). Uncertainties are 2σ. Corrected $\ln(D/a^2)$ values shown as "–" correspond to steps where the corrected apatite yield (ω) was within 2σ of 0 and were excluded from the Arrhenius plot (Fig. 1.6).

Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	${}^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	${}^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	24.3	0.00704 ± 0.00046	4.76‰	**1.48 ± 0.18**				
Experimental	240	100.0	0.1700 ± 0.0011		*1.47 ± 0.18*	0.05 ± 0.24	0.00803%	–	-28.39
	15	28.0	0.00721 ± 0.00039		*1.35 ± 0.16*				
	15	24.0	0.00639 ± 0.00039		*1.34 ± 0.16*				
Permeation	15	24.0	0.00630 ± 0.00039	4.71‰	**1.34 ± 0.16**				
Experimental	240	150.0	0.3090 ± 0.0023		*1.33 ± 0.16*	0.01 ± 0.21	0.00983%	–	-26.98
	15	28.1	0.00542 ± 0.00039		*1.03 ± 0.13*				
	15	24.3	0.00484 ± 0.00039		*1.03 ± 0.13*				
Permeation	15	24.2	0.00485 ± 0.00039	4.73‰	**1.02 ± 0.13**				
Experimental	240	200.1	0.9416 ± 0.0085		*1.02 ± 0.13*	1.64 ± 0.23	0.276%	-23.80 ± 0.28	-25.02
	15	29.9	0.00947 ± 0.00041		*1.72 ± 0.19*				
	15	23.8	0.00786 ± 0.00040		*1.71 ± 0.19*				
Permeation	15	24.0	0.00804 ± 0.00040	4.72‰	**1.70 ± 0.19**				
Experimental	60	250.2	1.097 ± 0.011		*1.69 ± 0.19*	3.70 ± 0.48	0.874%	-20.21 ± 0.20	-24.01
	15	33.0	0.02653 ± 0.00050		*4.30 ± 0.44*				

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
	15	23.5	0.02016 ± 0.00046		*4.27 ± 0.44*				
Permeation	15	24.0	0.02003 ± 0.00047	4.72‰	**4.25 ± 0.44**				
Experimental	60	270.3	2.508 ± 0.027		*4.23 ± 0.44*	5.26 ± 0.83	1.73%	-19.04 ± 0.22	-22.51
	15	33.9	0.04414 ± 0.00067		*6.98 ± 0.70*				
Permeation	15	23.6	0.03246 ± 0.00057	4.68‰	**6.93 ± 0.70**				
Experimental	60	290.3	4.979 ± 0.058		*6.90 ± 0.70*	9.0 ± 1.3	3.18%	-17.87 ± 0.20	-21.10
	15	33.6	0.06816 ± 0.00095		*10.9 ± 1.1*				
Permeation	15	23.6	0.05050 ± 0.00078	4.66‰	**10.8 ± 1.1**				
Experimental	60	300.3	7.243 ± 0.091		*10.8 ± 1.1*	9.5 ± 1.7	4.72%	-17.34 ± 0.25	-20.07
	15	34.2	0.0843 ± 0.0012		*13.1 ± 1.3*				
	15	23.5	0.06160 ± 0.00094		*13.0 ± 1.3*				
Permeation	15	24.1	0.06097 ± 0.00095	4.72‰	**12.9 ± 1.3**				
Experimental	60	285.3	5.919 ± 0.080		*12.9 ± 1.3*	3.6 ± 1.7	5.30%	-18.04 ± 0.71	-20.65
	15	36.8	0.0729 ± 0.0011		*10.5 ± 1.1*				
Permeation	15	23.6	0.04846 ± 0.00084	4.64‰	**10.4 ± 1.1**				
Experimental	80	275.2	5.382 ± 0.077		*10.4 ± 1.1*	2.5 ± 1.3	5.71%	-18.6 ± 1.1	-19.80
	15	37.0	0.05243 ± 0.00092		*7.53 ± 0.76*				
Permeation	15	23.6	0.03487 ± 0.00070	4.66‰	**7.48 ± 0.76**				
Experimental	100	255.2	3.382 ± 0.052		*7.44 ± 0.76*	0.60 ± 0.90	5.81%	–	-20.22
	15	37.1	0.03206 ± 0.00069		*4.66 ± 0.48*				
Permeation	15	23.6	0.02147 ± 0.00056	4.64‰	**4.63 ± 0.48**				
Experimental	60	300.3	3.684 ± 0.059		*4.61 ± 0.48*	6.68 ± 0.91	6.89%	-17.17 ± 0.20	-20.76
	15	36.3	0.0518 ± 0.0010		*7.60 ± 0.77*				
Permeation	15	23.6	0.03513 ± 0.00077	4.65‰	**7.55 ± 0.77**				
Experimental	120	300.2	9.78 ± 0.17		*7.51 ± 0.77*	11.1 ± 1.2	8.69%	-17.13 ± 0.22	-19.21
	15	35.7	0.0595 ± 0.0012		*8.85 ± 0.90*				

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	23.6	0.04087 ± 0.00089	4.65‰	**8.79 ± 0.90**				
Experimental	180	300.1	14.58 ± 0.27		*8.75 ± 0.90*	13.8 ± 1.2	10.9%	-17.08 ± 0.18	-18.75
	15	35.9	0.0547 ± 0.0012		*7.96 ± 0.81*				
Permeation	15	23.6	0.03782 ± 0.00089	4.78‰	**7.91 ± 0.81**				
Experimental	60	320.3	7.49 ± 0.14		*7.87 ± 0.81*	11.4 ± 1.5	12.8%	-15.96 ± 0.18	-17.94
	15	36.9	0.0813 ± 0.0017		*11.7 ± 1.2*				
Permeation	15	23.5	0.0541 ± 0.0012	4.64‰	**11.7 ± 1.2**				
Experimental	60	330.4	11.18 ± 0.22		*11.6 ± 1.2*	14.5 ± 1.9	15.1%	-15.54 ± 0.21	-18.08
	15	37.4	0.1076 ± 0.0023		*14.9 ± 1.5*				
Permeation	15	23.7	0.0691 ± 0.0016	4.67‰	**14.8 ± 1.5**				
Experimental	60	350.4	19.78 ± 0.41		*14.7 ± 1.5*	28.3 ± 2.8	19.7%	-14.61 ± 0.15	-18.11
	15	37.0	0.1615 ± 0.0035		*23.3 ± 2.4*				
Permeation	15	23.5	0.1071 ± 0.0024	4.64‰	**23.1 ± 2.4**				
Experimental	60	370.4	32.97 ± 0.71		*23.0 ± 2.4*	41.3 ± 4.0	26.4%	-13.90 ± 0.15	-16.15
	15	37.3	0.2207 ± 0.0049		*31.4 ± 3.2*				
Permeation	15	23.5	0.1447 ± 0.0033	4.65‰	**31.1 ± 3.2**				
Experimental	60	385.4	42.32 ± 0.94		*31.0 ± 3.2*	44.8 ± 4.8	33.6%	-13.48 ± 0.17	-15.41
	15	37.8	0.2393 ± 0.0055		*33.5 ± 3.4*				
Permeation	15	23.5	0.1548 ± 0.0037	4.66‰	**33.2 ± 3.4**				
Experimental	60	360.4	27.53 ± 0.64		*33.1 ± 3.4*	15.3 ± 4.1	36.1%	-14.35 ± 0.51	-15.33
	15	39.6	0.1620 ± 0.0039		*20.9 ± 2.1*				
Permeation	15	23.6	0.0963 ± 0.0024	4.65‰	**20.7 ± 2.1**				
Experimental	60	340.4	14.78 ± 0.35		*20.6 ± 2.1*	7.4 ± 2.6	37.3%	-15.00 ± 0.90	-15.63
	15	40.0	0.1029 ± 0.0026		*13.2 ± 1.4*				
Permeation	15	23.5	0.0610 ± 0.0017	4.64‰	**13.1 ± 1.4**				
Experimental	100	325.2	12.37 ± 0.31		*13.1 ± 1.4*	6.8 ± 1.6	38.4%	-15.6 ± 1.1	-13.69

continued on next page

(continued)

Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
	15	40.1	0.0589 ± 0.0017		*7.50 ± 0.78*				
Permeation	15	24.0	0.0351 ± 0.0011	4.72‰	**7.44 ± 0.78**				
Experimental	240	400.1	101.1 ± 2.7		*7.40 ± 0.78*	110.0 ± 3.2	56.2%	-13.281 ± 0.076	-13.70
	15	34.9	0.1096 ± 0.0031		*16.3 ± 1.7*				
Permeation	15	23.8	0.0757 ± 0.0022	4.67‰	**16.2 ± 1.7**				
Experimental	90	350.3	14.88 ± 0.41		*16.1 ± 1.7*	5.8 ± 1.9	57.2%	-14.92 ± 0.92	-15.42
	15	39.5	0.0567 ± 0.0017		*7.06 ± 0.74*				
Permeation	15	24.0	0.0330 ± 0.0011	4.71‰	**7.00 ± 0.74**				
Experimental	60	400.4	14.21 ± 0.40		*6.97 ± 0.74*	18.1 ± 1.4	60.1%	-13.31 ± 0.31	-15.50
	15	40.4	0.0866 ± 0.0026		*10.9 ± 1.1*				
Permeation	15	24.0	0.0508 ± 0.0016	4.71‰	**10.8 ± 1.1**				
Experimental	120	300.2	7.65 ± 0.22		*10.7 ± 1.1*	0.7 ± 1.2	60.2%	–	-16.50
	15	39.4	0.0291 ± 0.0011		*3.79 ± 0.41*				
Permeation	15	23.7	0.01758 ± 0.00077	4.68‰	**3.76 ± 0.41**				

1.7 Discussion

Except for ampoules with encapsulated water, the experimental results of this work are predicted within 10% when assuming the literature values for permeabilities for 4He from Swets et al. (1961) and the 3He/4He permeability ratio of Jones (1953). This suggests that the complication of permeation lag is a tractable complexity, enabling the use of fused silica ampoules to control the sulfur and water fugacity of helium step heat experiment, but that literature values for helium permeability do not apply to hydrothermal experiments, which instead require the ampoule's rate constant to be measured in situ (§1.4).

Resolving the Paradoxically High Apparent Helium Permeability in Water Filled Tubes

The high measured permeability of helium for water-filled silica glass ampoules (supplementary Figs. 1.9, 1.24) is initially surprising and warrants further explanation, as the dissolved water and/or hydroxyl ion content of fused silica glass is only known to decrease helium permeability (Tsuru et al., 2011; Shelby, 1972; Masaryk, 1968). It is hypothesized here to result from an increase in the partial pressure of helium within the gas phase of the ampoule due to volume displacement of the gas phase by liquid water. This results from the low solubility and high diffusivity of helium in water, which allow for helium to be concentrated within the gas phase while remaining in equilibrium with the glass throughout the entire ampoule, even where obstructed by water.

The diffusivity of helium in water is about 7×10^{-5} cm^2/s at room temperature (Verhallen et al., 1984), whereas the diffusivity of helium in glass is three orders of magnitude slower: about 2×10^{-8} cm^2/s (Swets et al., 1961). Helium diffusion in water can be considered instantaneous relative to the timescales of helium diffusion through glass, so the helium concentration in the gas phase is expected to be in equilibrium with the glass throughout the entire tube even where mediated by liquid water. Therefore, adding water to an ampoule changes the volume available for the gas phase while not affecting the effective area of permeation.

Helium partial pressures in the gas phase can be increased by volume displacement assuming that helium strongly partitions into the gas phase over glass and water. Helium solubilities in water are similar to but slightly lower than glass. At room temperature, the solubility of helium in water is ~

$4 \times 10^{-6} \frac{moles}{m^3 * Pa}$ (Sander, 2015) or $\sim 2 \times 10^{17} \frac{atoms}{cm^3 * atm}$. The solubility of helium in silica glass is $\sim 6 \times 10^{17} \frac{atoms}{cm^3 * atm}$ (Swets et al., 1961), a factor of 3 higher. At room temperature, the concentration of helium atoms in glass and in water is about 3% and 1% respectively of the concentration of helium atoms in the gas phase (supplementary Eq. (1.41)). The net result of this differential partitioning is that the effective permeability of helium roughly scales with the inverse volume fraction of the gas phase, following the equation:

$$K_{eff} = K_{dry} \frac{V_{water} + V_{gas}}{V_{gas}} \tag{1.12}$$

Where K_{eff} is the effective helium permeability, K_{dry} is the nominal helium permeability at anhydrous conditions, and V_{water} and V_{gas} are the relative volumes of water and gas respectively. This effect is potentially larger than the decrease in helium permeability of fused silica glass due to elevated water contents; therefore an ampoule that is filled with significant water may paradoxically permeate helium faster than the same ampoule without water, despite the lower helium permeability of the glass itself.

Evolving Helium Permeability in Water Filled Tubes

Helium permeability measurements of water-filled ampoules show decreasing fractional yields for identical steps (Fig. 1.10). This observation is consistent with previous studies that show decreased helium permeability in silica glass following hydrothermal treatment (Tsuru et al., 2011; Masaryk, 1968). Other studies have shown that diffusion of water in silica glass is complex, but it is always significantly slower than diffusion of helium in silica glass. At 250°C, the effective diffusivity of both hydroxyl groups and molecular water in silica glass is initially ~10^{-11} cm^2/s and further slows by a factor of 4 over the first 500 hours of observation (Oehler and Tomozawa, 2004). Helium, on the other hand, has a diffusivity of ~10^{-6} cm^2/s at 250°C (Swets et al., 1961). For a 200 µm thick silica ampoule, the diffusive timescale ($\tau = \frac{L^2}{D}$) of water is about 76 years, whereas the diffusive timescale of helium in the same ampoule is less than 7 minutes. Even so, we observe a gradual decrease in helium permeability of about 7% over the course of a few days at 100°C in this work (Fig. 1.10). The implication of this observation is that the helium permeability in water-filled tubes must be continuously monitored by measuring the rate constant in situ (§1.4; Eqs. (1.8), (1.9)) during a hydrothermal step heat experiment to track

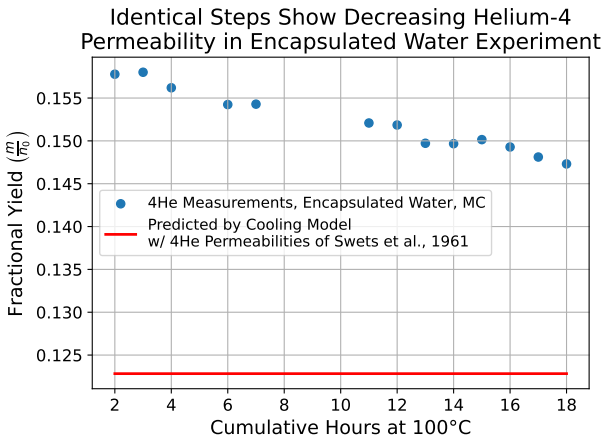

Figure 1.10: Identical permeation steps at 100°C with identical cooling times of the encapsulated water tube show higher 4He fractional yields than predicted with a numerical diffusion model and literature values of permeability from Swets et al. (1961). These identical steps also show a striking systematic decrease in fractional yields as the ampoule is held at 100°C, suggesting that the permeability of the ampoule is slowing during the experiment, possibly due to diffusion of water and hydroxyl groups into the silica glass (Tsuru et al., 2011; Oehler and Tomozawa, 2004). This trend is not visible in the 3He data due to higher scatter in the data.

(or exclude) significant temporal evolution of helium permeability throughout the experiment.

Previous studies have shown that the diffusion of water-derived species (OH^-, H_2O, H^+) into silica glass slows significantly as the water concentration increases within the glass (Oehler and Tomozawa, 2004; Davis and Tomozawa, 1995). Thus, it may be possible that a hydrothermal pretreatment (or early high-temperature heating steps) could sufficiently increase the trace water content of the silica glass for it to enter a regime of slower solubilization of water, thereby stabilizing the trace water content and helium permeability of silica glass for hydrothermal experiments. This hypothesis would be a fruitful direction for future work related to hydrothermal helium step heat experiments in silica glass ampoules.

1.8 Conclusions

Encapsulating minerals in pure silica glass allows helium step heat experiments to be conducted within controlled microenvironments of high water or sulfur partial pressure. Helium permeation through fused silica glass is a rapid but

non-instantaneous first-order process that complicates the interpretation of step heat data; however, its effects can be reliably deconvolved from the sample's true helium release using the equations and procedures described above. A general workflow combining the two approaches described above is suggested below:

1. Pull a fused silica tube (e.g., 3 mm inner diameter, 4 mm outer diameter stock) into a pipette with the desired inner diameter and wall thickness for the experiment.

2. Load the mineral sample and the desired buffering phase (e.g., water or sulfur) into the ampoule and seal it, taking care not to heat the sample itself during sealing.

3. If possible, measure the ampoule's inner and outer diameters with a calibrated microscope. This is not strictly necessary, but is helpful for planning the experiment: these measurements give the geometric factor G (Eq. (1.5)), which, combined with literature permeability values, allows the rate constant of permeation λ to be estimated in advance.

4. Program the step heat schedule so that, except during the elevated-temperature experimental steps themselves, the ampoule is held at all times at a low-but-warm permeation temperature at which helium is expected to be immobile in the sample (e.g., ~40–100°C). Minimize pump-out times between steps: 1-5 minutes is ideal.

5. After every heating step, program an automated resting period, before measurement, in which the ampoule is held at the permeation temperature for at least the diffusive timescale $\tau = \frac{L^2}{D}$, where L is the ampoule's wall thickness and D is the diffusivity of helium in fused silica glass (e.g., from Swets et al., 1961). Doubling this estimated timescale is a reasonable safety margin that should be conservative enough even for hydrothermal experiments. This resting period allows a steady state helium concentration profile to establish within the glass immediately after each heating step, which is required for accurate permeation corrections (Eqs. (1.1), (1.6)). Because the resting period alone is already long enough to permeate a significant portion of the ampoule's residual helium, a

permeation step can simply be programmed as a step with no additional heating time.

6. At several points throughout the experiment, run a series of at least three consecutive permeation steps to determine the permeation rate constant λ in situ (§1.4; Eq. (1.9)). Do this in every experiment rather than relying on literature permeabilities, and repeat it through the run for hydrothermal experiments, where λ drifts as the glass takes up water.

7. Follow every high-temperature experimental step with at least one permeation step, using either the interpolated rate constant for experiments where this value resolvably drifts or the average measured rate constant for experiments where there is no resolvable drift, to calculate the ampoule's helium contents between steps (Eqs. (1.1), (1.6)).

For high temperature experiments of retentive phases, where low temperature permeation steps may yield signals indistinguishable from blank, experimentalists may instead use the bounding method (§1.4) to constrain the ampoule's residual helium from above rather than measuring it. This requires only a lower bound on the rate constant of permeation at the experimental temperature, and can only be done for steps where $\lambda t > 5$. No permeation steps are necessary for this approach, although a series of them remains the most reliable way to establish the rate constant it depends on.

Supplementary Information

for Chapter 1: *A Simple Method for Helium Step Heat Experiments at High Water or Sulfur Fugacity*

1.1s A Fick's Law Formulation Describing Helium Migration Across Semi-Permeable Membranes

We derive the relevant equations describing helium permeation through glass by beginning with Fick's Law, similarly to previous works (Rogers et al., 1954; Altemose, 1961).

$$J = -D\frac{d\varphi}{dx} \tag{1.13}$$

Where J is the area-normalized Fickian diffusive flux in units of $\frac{atoms}{s \times cm^2}$, D is the diffusivity (of helium in glass) in units of $\frac{cm^2}{s}$, and $\frac{d\varphi}{dx}$ is the concentration gradient, in units of $\frac{\left(\frac{atoms}{cm^3}\right)}{cm}$ or $\frac{atoms}{cm^4}$.

For steady state permeation across a planar membrane, the concentration gradient is constant. If we take l to be the glass thickness and we rewrite the Fickian flux J as $\frac{F}{A}$, where F is the absolute atom flux in units of $\frac{atoms}{s}$, and A is simply the membrane area, we can rewrite Fick's law as:

$$F = -DA\frac{\Delta\varphi}{l} \tag{1.14}$$

Finally, we assume that the concentration of helium within the glass membrane is Henrian, i.e. proportional to the gas pressure in contact with the membrane. Experimental work has shown that helium permeation through silica glass is Henrian at pressures of several hundred bars and lower (Shelby, 1976), well above the experimental helium partial pressures for this work. We define this proportionality factor as the solubility, S, in units of $\frac{atoms}{atm \times cm^3}$. Now we may write the expression in terms of the gas pressure gradient, or the difference of helium partial pressures on each side of the membrane.

$$F = -DA\frac{S\Delta P}{l} \tag{1.15}$$

Diffusivity and solubility are material properties that are functions of temperature following the Arrhenius equation, and therefore their mathematical product (permeability, K) also follows an Arrhenius relation with temperature (Rogers et al., 1954; Swets et al., 1961). Now, recognizing that K = DS, we may express the equation in terms of permeability $\frac{atoms}{s \times cm \times atm}$.

$$F = -\frac{KA\Delta P}{l} \tag{1.16}$$

Studt (1962) and Masaryk (1968) also use this equation to describe helium permeation through silica glass, taken from Darcy's Law for gas flow through semi-permeable membranes (Carman, 1956). This ordinary, first-order differential equation is well suited for describing permeation through a silica membrane when the permeate pressures on each side of the membrane are constant (Norton, 1953; Altemose, 1961; Swets et al., 1961). Cases where permeate partial pressure changes during the experiment (such as this work) require solving the first order ODE above to relate the abundance of an analyzed permeate to the abundance of unanalyzed permeate remaining in the depleting reservoir.

Recalling that F is an atomic flux in units of $\frac{atoms}{s}$ that could be notated as $\frac{dn}{dt}$:

$$\frac{dn}{dt} = -\frac{KA\Delta P}{l} \tag{1.17}$$

We assume that the helium partial pressure in the ultra-high vacuum (UHV) heating cell surrounding the ampoule is 0, so that the pressure gradient equals the partial pressure within the ampoule ($\Delta P = P$). This is a valid assumption for this work because the heating cell is 4-5 orders of magnitude more voluminous than the glass ampoule. The assumption therefore is only appreciably violated for the trivial case in which effectively all the helium present within the tube is permeated out and the helium partial pressures within and outside of the ampoule are equal. In this case, net helium flux goes to zero even though the ampoule is not entirely evacuated. Relative to the amount of helium within the ampoule at time 0 (n_0), the amount of helium remaining in the ampoule after infinite time *(n_{eqm})* will be set by the volume ratio of the ampoule and the lamp cell.

$$\frac{n_{eqm}}{n_0 - n_{eqm}} \approx \frac{n_{eqm}}{n_0} = \frac{V_{ampoule}}{V_{lamp\ cell}} \approx 10^{-4}$$

For this experimental setup, only 1 part in 10000 helium atoms will remain in the ampoule after infinite time in the closed heating cell.

Before integrating Eq. (1.17), it is useful to experimentalists working with mass spectrometry to express the pressure term P in terms of atoms n, the atomic abundance of helium within the ampoule. Helium is well-approximated as an ideal gas at experimental conditions, so the ideal gas law (PV = nRT) relates the pressure, P, to the atomic abundance, n, yielding Eq. (1.18):

$$\frac{dn}{dt} = -\frac{AKnRT}{lV} \tag{1.18}$$

Where R is the gas constant $\left(1.36 \times 10^{-22} \frac{cm^3 atm}{K\ atoms}\right)$, T is the temperature in Kelvin, and V is the volume in cm^3. It is now obvious that the helium atoms in the gas phase (n) decrease over time. Because Eq. (1.18) assumes a steady state helium concentration profile in the glass walls of the ampoule in equilibrium with the helium partial pressure inside the ampoule, one must make another essential assumption to continue: that the permeation timescale be small with respect to the diffusive timescale so that the helium concentration profile of the glass effectively remains in an equilibrium with the helium atoms in the gas phase (n). If the inner gas volume is small relative to the volume of the ampoule glass, then $\frac{dn}{dt}$ will be relatively large and the helium atoms in the gas phase (n) will decrease faster than diffusion's ability to maintain an equilibrium concentration profile in the glass, violating the assumption of a steady state concentration profile in equilibrium with the inner gas phase. Then, Eq. (1.18) and all equations derived from it will be invalid. In other words, the timescale of diffusion must be much smaller than the timescale of permeation ($\tau_D \ll \tau_P$) for Eq. (1.18) and all derived equations to apply. A more thorough analysis of this assumption is presented in *Diffusive Steady State Assumption for Permeation.*

Eq. (1.18) can be simplified by defining a geometric factor (G) in units of cm^{-2} that incorporates the spatial parameters for area, volume, and length of the experimental glass ampoule:

$$\frac{dn}{dt} = -GKnRT \tag{1.19}$$

Eq. (1.19) can be integrated using a straightforward separation of variables to get a useful general equation describing helium reservoirs depleting via permeation.

$$n_t = n_0 e^{-KGRTt} \tag{1.20}$$

Where n_t is the amount of helium atoms within the reservoir after time t in seconds, and n_0 is the amount of helium atoms within the reservoir at time t = 0.

Eq. (1.20) can equivalently be expressed in terms of the fraction of the initial helium reservoir that has permeated during time t, which is often a more intuitive quantity when relating this equation to a measured amount of helium. Defining the fractional permeation yield ${}^{He}F \equiv \frac{m}{n_0}$, where $m \equiv n_0 - n_t$ is the amount of helium permeated and available to be measured during time t, Eq. (1.20) can be recast as:

$$ {}^{He}F = \frac{m}{n_0} = 1 - e^{-KGRTt} \tag{1.21}$$

Eq. (1.20) can be solved for permeability (K) for use in experimentally determining permeability for experiments in which the permeating helium source is significantly depleted during the experiment, yielding Eq. (1.22):

$$ K = -\frac{\ln(n_t) - \ln(n_0)}{GRTt} \tag{1.22}$$

In typical experimental measurements of helium permeability through glass, permeation rates are measured in real time, and the pressure gradient is constant throughout the measurement, so an equation like Eq. (1.16) is used to solve for permeability instead (Norton, 1953; Swets et al., 1961; Altemose, 1961).

Permeation Through Cylindrical Membranes

For non-planar membranes, the effective area of permeation, A, is not straightforward to evaluate because the surface area on each side of the membrane differs. Masaryk (1968) gives the effective area of permeation for a cylinder (such as for this experiment) as:

$$ A = \frac{2\pi hl}{\ln\left(\frac{b}{a}\right)} \tag{1.23}$$

Where a is the cylinder's internal radius, b is the cylinder's external radius, and h is its length. This effective area of permeation (Eq. (1.23)) can be used with the equation for the internal volume of the cylinder: $V = h\pi a^2$ to evaluate the geometric factor G (Eq. (1.19)) for a cylinder in terms of the ampoule geometry. Reducing yields Eq. (1.24):

$$G = \frac{2}{a^2 \ln\left(\frac{b}{a}\right)} \tag{1.24}$$

Interestingly, h cancels so that the permeation rate has no dependence on the ampoule length when permeation through the top and bottom are negligible. Eq. (1.24) displays the intuitive behavior that thinner walls and narrower tubes lead to more efficient permeation. Where G is used elsewhere in this work, it refers to the cylindrical geometric factor given in Eq. (1.24). Most silica ampoules sealed at Caltech for helium step heat experiments were 0.6-1.2 mm wide with 150-300 µm thick glass, translating to values of G between 2000-7000 cm^{-2}. In this work, if no value of G is specified, then a typical 200 µm thick ampoule with an outer diameter of 1 mm is assumed, corresponding to $a = 0.03$ cm, $b = 0.05$ cm and $G = 4350.26$ cm^{-2}.

1.2s Diffusive Steady State Concentration Profiles in Cylindrical Membranes

To establish a steady state concentration profile, the diffusive timescale must be much faster than the permeation timescale, otherwise the helium concentration profile within the glass will respond too slowly to maintain equilibrium with the rapidly depleting internal gas phase (Fig. 1.11). The ratio of the permeation timescale and diffusive timescale is a strong function of the ratio of the ampoule's glass and internal gas volumes with a very minor dependence on temperature (Fig. 1.12 below). An ampoule with a glass/gas volume ratio of 2 has a diffusive timescale that is ~5.6% of the permeation timescale (Eq. (1.28) below). Numerical diffusion-permeation simulations show that Eq. (1.22) accrues less than 1% error due to deviation from the idealized helium concentration profile (Figs. 1.13 and 1.14 below). Smaller glass/gas volume ratios are better: the ampoules in the main text have glass/gas volume ratios ~1.25-1.5. Numerical simulations show that this error is determined nearly entirely by the volume ratio of the ampoule glass and gas and is otherwise insensitive to ampoule geometry (Figs. 1.15 and 1.16 below).

The derivation for Eq. (1.22) always assumes an equilibrium steady state helium concentration profile across the ampoule. The most important corollary of this assumption is that the timescale of permeation must be much larger than the diffusive timescale ($\tau_D \ll \tau_P$) for Eq. (1.22) and all derived equations to apply.

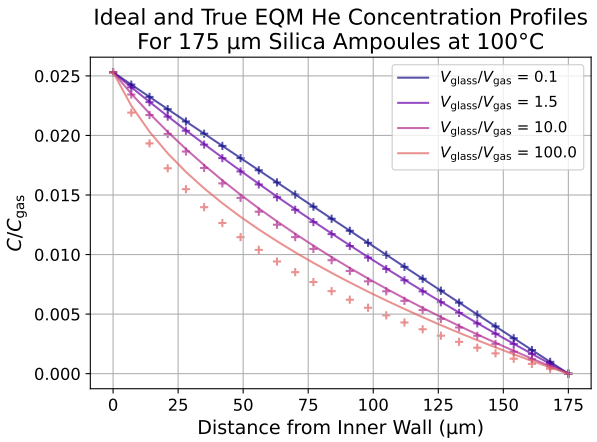

Figure 1.11: A visualization of ideal diffusive steady state concentration profiles (demarcated by the '+' symbols) are contrasted with the numerically calculated "effective" steady state concentration profile, represented by the solid lines. These effective concentration profiles are not strictly in steady state, but they are effectively in equilibrium with a rapidly decreasing internal helium reservoir. These profiles are retrieved by numerically simulating a model ampoule permeating helium for at least one diffusive timescale. At low glass/gas volume ratios, the idealized and true concentration profiles are indistinguishable. Ideal steady state concentration profiles for cylindrical membranes are strongly curved at high glass/gas volume ratios and become linear as the glass/gas volume ratio approaches 0. For closed ampoules where internal pressure is not constant, but instead decreases due to permeation, the internal pressure decreases faster than the glass concentration profile can respond to the new internal helium pressure, so the concentrations are higher (and less curved) than the ideal profile, because of memory of higher internal helium pressures. 4He diffusivities and solubilities are taken from Swets et al. (1961).

An analysis is presented here exploring the ampoule geometries that permit the use of the analytical solutions for helium permeation through silica glass.

The diffusive timescale is the timescale at which diffusion is generally expected to establish a steady state concentration profile between two (effectively) constant reservoirs. For a diffusive domain size L, the timescale is given as:

$$\tau_D = \frac{L^2}{D} \tag{1.25}$$

The permeation timescale is analogous to an "e-folding time" or "mean life" in radioactive decay, the time after which point the original abundance of something decaying by a first order process following the form $\frac{dn}{dt} = -\lambda n$ would have decreased by a factor of Euler's number $(1/e)$. After one permeation

timescale has passed, more than half of the original helium has permeated through the ampoule walls, leading to the intuition that the dimensionless timescale ratio $\frac{\tau_D}{\tau_P} \ll 1$ in order for Eq. (1.22) and Eq. (1.20) to apply, because only vanishingly small changes in the total helium are allowed within one diffusive timescale in order for an equilibrium helium concentration profile to be stabilized and maintained.

$$\tau_P = \frac{1}{\lambda} = \frac{1}{GRTK} \tag{1.26}$$

The ratio of these two timescales is a weak function of temperature and a strong function of the ratio of ampoule glass and internal gas volumes $\left(\frac{V_{glass}}{V_{gas}}\right)$ (Fig. 1.12). By the equation for the volume of cylinders:

$$V_{\frac{glass}{gas}} = \frac{h\pi\left(b^2 - a^2\right)}{h\pi a^2} = \frac{\left(b^2 - a^2\right)}{a^2} \tag{1.27}$$

$\frac{\tau_D}{\tau_P}$ can be expanded and recast to be a function of only temperature and $V_{\frac{glass}{gas}}$. Recalling that L = b-a and K = SD, and the identity of G for cylinders from Eq. (1.24).

$$\frac{\tau_D}{\tau_P} = \frac{L^2}{D}GRTK = \frac{(b-a)^2}{D}\frac{2}{a^2 \ln\left(\frac{b}{a}\right)}RTSD$$

Factoring $(b^2 - a^2) = (b - a)(b + a)$, substituting, and simplifying:

$$\frac{\tau_D}{\tau_P} = V_{\frac{glass}{gas}} \bullet \frac{(b-a)}{(b+a)}\frac{2}{\ln\left(\frac{b}{a}\right)}RTS$$

Now using Eq. (1.27) to relate the terms containing a and b to $V_{glass/gas}$, all spatial terms can be expressed in terms of this ratio.

$$\left(\frac{b}{a}\right) = \sqrt{V_{\frac{glass}{gas}} + 1}$$

For example:

$$b = a\sqrt{V_{\frac{glass}{gas}} + 1}$$

$$(b-a) = a\left(-1+\sqrt{V_{\frac{glass}{gas}}+1}\right)$$

$$(b+a) = a\left(1+\sqrt{V_{\frac{glass}{gas}}+1}\right)$$

$$\frac{\tau_D}{\tau_P} = S(T)\bullet\, RT\, \bullet V_{\frac{glass}{gas}} \bullet \frac{-1+\sqrt{V_{\frac{glass}{gas}}+1}}{1+\sqrt{V_{\frac{glass}{gas}}+1}} \bullet \frac{2}{\ln\left(\sqrt{V_{\frac{glass}{gas}}+1}\right)} \quad (1.28)$$

Eq. (1.28) is plotted for three temperatures as a function of glass/gas volume ratio in Fig. 1.12. $\frac{\tau_D}{\tau_P}$ increases monotonically and nearly linearly in log-log space with the glass/gas volume ratio of ampoules, whereas temperature plays only a minor role. Ampoules with high glass/gas volume ratios and $\frac{\tau_D}{\tau_P}$ ratios can be considered "glass dominated". For these glass-dominated ampoules, the relative difference in surface area between the inner and outer surfaces is large; resulting in a more profoundly curved ideal concentration profile (Fig. 1.11 above, Eq. (1.33)). For glass dominated ampoules, this ideal curved concentration profile is only realized if the internal pressure is held constant, otherwise, this internal helium partial pressure decays rapidly due to permeation relative to the rate of diffusion within the glass, thereby stabilizing a helium concentration profile in the glass that will be less curved and higher than the idealized steady state of Eq. (1.33) (Fig. 1.11 above). When normalized to the concentration, quantity, or pressure of helium within the gas phase of the ampoule, this quasi-steady state concentration profile is reached within one diffusive timescale and remains constant as diffusion and permeation proceeds, regardless of the ratio of glass and gas volumes (this property was tested by varying the permeation time between one and ten diffusive timescales, which had no effect of the concentration profile normalized to the concentration, abundance, or partial pressure of helium in the gas phase). Indeed, all model ampoule calculations from Figs. 1.13–1.15 were initialized to have this effective steady state concentration profile, which is done by allowing the model ampoule to permeate and diffuse for one diffusive timescale, after which the concentration profile is multiplied by the simulated $\frac{n_0}{n_t}$ to renormalize it to the original helium contents of the ampoule. The helium content of the gas is then recalculated using the relation $n_{gas} = n_0 - n_{glass}$, and the permeated helium outside of the

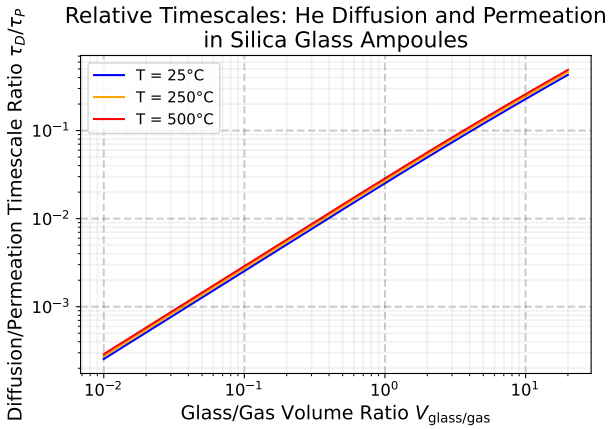

Figure 1.12: The accuracy of Eq. (1.20) and equations derived from it is dependent on a steady state concentration profile in equilibrium with the helium gas concentration inside the ampoule. If the timescale of permeation is comparable to or lower than the timescale of diffusion for the ampoule, then the gas concentration inside the ampoule will change faster than the response of the concentration profile within the glass, and these equations will become inaccurate. In other words, these equations are only accurate when $\tau_D/\tau_P \ll 1$. For helium permeation in silica glass ampoules, the ratio of the diffusive and permeation timescales is a strong function of the ratio of glass and gas volume, and a weak function of temperature. τ_D/τ_P is nearly linear with V_{glass}/V_{gas} in log-log space, forming subtly-offset parallel arrays at different temperatures. 4He solubilities are taken from Swets et al. (1961).

ampoule is reset to 0. Numerical simulations of model ampoules with glass/gas volume ratios above ~ 0.1 that were initialized to the ideal steady state of Eq. (1.33) recorded an early spike in relative error with Eq. (1.22) that was maximized before one diffusive timescale because of an incorrect initial steady state assumption. This artifact vanished completely when the true steady state profile was used.

Numerical simulations of model ampoules of varying dimensions, thicknesses, and geometric factors show that the only geometric parameter that directly affects the accuracy of the analytical solutions, e.g. Eq. (1.22), is the glass/gas volume ratio of the ampoules (Figs. 1.13–1.15). This geometric dependence is much more important than subtle effects of temperature (Figs. 1.12; 1.15). Ampoules from this work have glass/gas volume ratios between 1.2 and 1.6; for which Eq. (1.22) is accurate to within 1% for temperatures less than 500°C (Fig. 1.14). For more precise handling, odd thermal schedules, or for ampoules with higher glass/gas volume ratios, permeation fractions would need

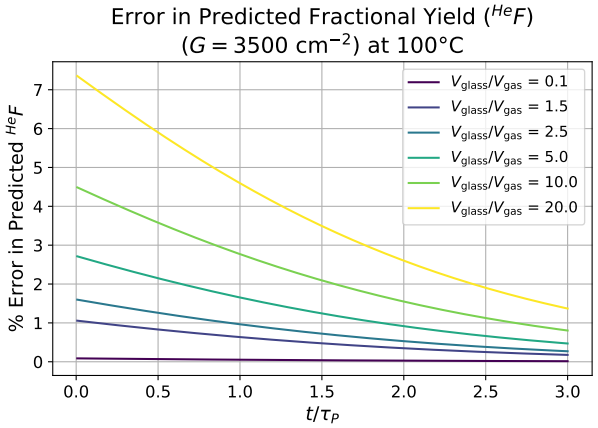

Figure 1.13: Numerical simulations confirm that Eq. (1.22) is accurate for ampoules with low glass/gas volume ratios. For ampoules with higher glass/gas volume ratios, the relative error is not constant but depends on the duration of the step. The relative error reaches a maximum at a step shorter than the permeation timescale (τ_P) for ampoules with glass/gas volume ratios higher than 1, then decreases as steps continue and most of the original gas within the ampoule has been permeated. 4He diffusivities, solubilities, and permeabilities are taken from Swets et al. (1961).

to be numerically solved to achieve sufficient accuracy to use a permeation measurement to constrain the helium contents of an ampoule.

To understand the steady state helium concentration profile in the glass, it is necessary to model the radial diffusion through a symmetrical, hollow cylinder. Carslaw and Jaeger (1959) give the equation for the change in concentration by diffusion in a cylinder (eqn. 5, page 17, second edition), expressed here in terms of concentration and diffusivity.

$$\frac{dC}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r} + \frac{1}{r^2}\frac{\partial^2 C}{\partial \theta^2} + \frac{\partial^2 C}{\partial z^2}\right] \tag{1.29}$$

For typical ampoules used in this work, the surface area of the top and bottom together is about 5-10% of the surface area of the walls, but they are neglected here for simplicity. Assuming that the concentration profiles are radially symmetrical and uniform with height (z), Eq. (1.29) is reduced to:

$$\frac{dC}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r}\right] \tag{1.30}$$

At steady state, $\frac{dC}{dt} = 0$, so

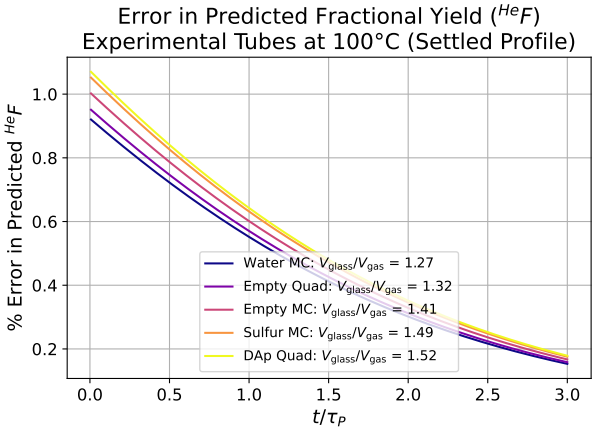

Figure 1.14: Numerical experiments of this work show that these equations will be accurate to within 1% for ampoule glass/gas volume ratios less than 2 for temperatures less than 500°C, which corresponds to $\tau_D/\tau_P < 0.056$. Experimental ampoules at Caltech have glass/gas volume ratios generally between 1.25 and 1.5, and thus Eq. (1.22) is accurate to within 1% for ampoules with concentration profiles in steady state. 4He diffusivities, solubilities, and permeabilities are taken from Swets et al. (1961).

$$\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r} = 0$$

Which, by the product rule, can be expressed as:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C}{\partial r}\right) = 0$$

Now, the $\frac{1}{r}$ term can be dropped, and both sides can be integrated with respect to r to yield:

$$r\frac{\partial C}{\partial r} = A$$

Where A is a constant of integration. Now both sides are integrated again with respect to r, leaving the equation describing the geometry of the diffusive steady state concentration profile of a cylinder (Eq. (1.31)):

$$\frac{\partial C}{\partial r} = \frac{A}{r}$$

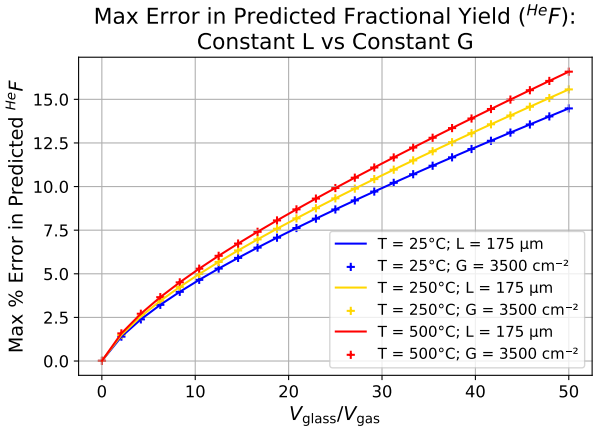

Figure 1.15: Numerical simulations of two suites of model ampoules with varying glass/gas volume ratios and a constant thickness (L) or permeation geometric factor (G) show that the only factors that determine the applicability of Eq. (1.22) for helium permeation of closed ampoules are glass/gas volume ratio (primarily) and temperature (secondarily). These two suites have different geometries and permeation rates, yet the deviation of the numerical model from Eq. (1.22) is the same for different ampoules with the same glass/gas volume ratio at the same temperature. This is clear evidence that the ratio of diffusive and permeation timescales is the primary control on the accuracy of Eq. (1.22). 4He diffusivities, solubilities, and permeabilities are taken from Swets et al. (1961).

$$C(r) = \int \frac{A}{r} dr$$

$$C(r) = Aln(r) + B \tag{1.31}$$

Where B is another constant of integration. Constants A and B can be solved for once the boundary conditions are defined, that is, the helium concentrations at the inner and outer walls of the glass ampoule. The concentration of helium at the outer wall is effectively 0 because the high vacuum heating cell is essentially an infinite volume compared to the glass ampoule. At the inner wall, the concentration is governed by the solubility and helium partial pressure of the encapsulated volume. Letting a be the radius of the ampoule inner wall, b be the radius of the outer wall, and P_{He}^{gas} be the partial pressure of helium in the encapsulated gas of the ampoule, the following boundary conditions are enforced:

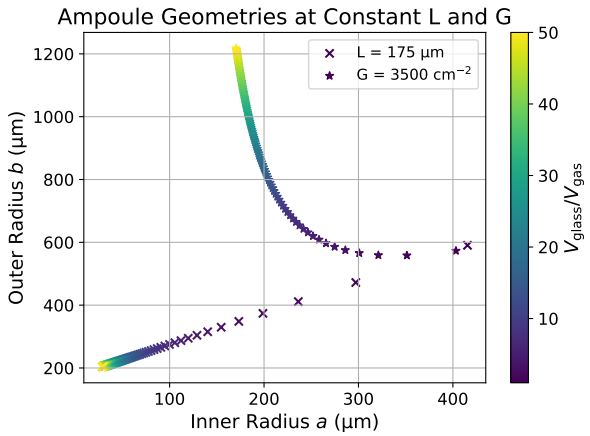

Figure 1.16: Model ampoules with uniformly distributed glass/gas volume ratios between 0.01 and 50 and one other constraint (either constant thickness or constant geometric permeation factor G, Eq. (1.24)) have defined inner and outer radii. These are the same suites of ampoules for which helium permeation was numerically simulated in Fig. 1.15.

$$C(r = b) = 0$$

$$C(r = a) = S(T) \bullet P_{He}^{gas}$$

Now a system of equations can be formulated so that A can be solved first, then B.

$$0 = Aln(b) + B$$

$$S(T) \bullet P_{He}^{gas} = Aln(a) + B$$

$$S(T) \bullet P_{He}^{gas} = Aln(a) - Aln(b)$$

$$S(T) \bullet P_{He}^{gas} = A\ ln\left(\frac{a}{b}\right)$$

$$A = \frac{S(T) \bullet P_{He}^{gas}}{ln\left(\frac{a}{b}\right)}$$

$$B = -\frac{S(T) \bullet P_{He}^{gas}}{\ln\left(\frac{a}{b}\right)} ln(b)$$

Substituting A and B back into Eq. (1.31):

$$C(r) = \frac{S(T) \bullet P_{He}^{gas}}{\ln\left(\frac{a}{b}\right)} \ln(r) - \frac{S(T) \bullet P_{He}^{gas}}{\ln\left(\frac{a}{b}\right)} \ln(b)$$

$$C(r) = S(T) \bullet P_{He}^{gas} \bullet \frac{\ln\left(\frac{r}{b}\right)}{\ln\left(\frac{a}{b}\right)} \tag{1.32}$$

Using the ideal gas law and the volume for the cylinder, the partial pressure of helium within the ampoule can be related to the geometry of the ampoule and the abundance of helium atoms within the ampoule:

$$P_{He}^{gas} = \frac{n_{He}^{gas} RT}{V_{gas}}$$

$$V_{gas} = \pi a^2 h$$

$$P_{He}^{gas} = \frac{n_{He}^{gas} RT}{\pi a^2 h}$$

$$C(r) = S(T) \bullet \frac{n_{He}^{gas} RT}{\pi a^2 h} \bullet \frac{\ln\left(\frac{r}{b}\right)}{\ln\left(\frac{a}{b}\right)} \tag{1.33}$$

Where h is the length of the ampoule.

Because the concentration profile of the ampoule is radially symmetrical, Eq. (1.33) can be expressed as a radial density of helium atoms instead to eliminate the dependence on h. The concentration ($C(r)\frac{atoms}{cm^3}$) can be converted into a radial density ($\epsilon(r)\frac{atoms}{cm}$) by multiplying by the circumference of a circle with radius r and the cylinder height h.

$$\epsilon(r) = C(r) * 2\pi r h \tag{1.34}$$

Substituting in $C(r)$ from Eq. (1.33) and the geometric constant of permeation G from Eq. (1.24) $\left(G = \frac{2}{a^2 \ln\left(\frac{b}{a}\right)} = -\frac{2}{a^2 \ln\left(\frac{a}{b}\right)}\right)$:

$$\epsilon(r) = \ S(T) \bullet \frac{2n_{He}^{gas} rRT}{a^2} \bullet \frac{\ln\left(\frac{r}{b}\right)}{\ln\left(\frac{a}{b}\right)}$$

$$\epsilon(r) = \ -S(T) \bullet n_{He}^{gas} GRTr \ln\left(\frac{r}{b}\right) \tag{1.35}$$

To find the proportion of helium within the glass at steady state, the radial density is integrated from $r = a$ to $r = b$.

$$n_{He}^{glass} = \int_a^b \epsilon(r)dr = \int_a^b -S(T) \bullet n_{He}^{gas} GRTr \ln\left(\frac{r}{b}\right)dr$$

$$n_{He}^{glass} = -S(T) \bullet n_{He}^{gas} GRT \bullet \int_a^b \left[rln(r) - rln(b)\right] dr$$

Applying the sum rule, then integration by parts to one term yields the indefinite integral:

$$\int \left[rln(r) - rln(b)\right] dr = \frac{1}{2}r^2 \ln(r) - \frac{r^2}{4} - \frac{1}{2}r^2 \ln(b) + C$$

Evaluating the definite integral from the bounds of $r = a$ to $r = b$ yields:

$$\int_a^b \left[rln(r) - rln(b)\right] dr = \frac{1}{2}b^2 \ln(b) - \frac{b^2}{4} - \frac{1}{2}b^2 \ln(b) - \frac{1}{2}a^2 \ln(a) + \frac{a^2}{4} + \frac{1}{2}a^2 \ln(b)$$

Reducing and simplifying:

$$\int_a^b \left[rln(r) - rln(b)\right] dr = \frac{1}{2}a^2 \left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}$$

$$n_{He}^{glass} = -S(T) \bullet n_{He}^{gas} GRT \bullet \left(\frac{1}{2}a^2 \left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right) \tag{1.36}$$

Eq. (1.36) can be easily expressed in terms of total helium in the ampoule by substituting for the relationship $n_{He}^{gas} = n_0 - n_{He}^{glass}$

$$n_{He}^{glass} = -S(T) \bullet \left(n_0 - n_{He}^{glass}\right) GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)$$

$$\frac{\left(n_0 - n_{He}^{glass}\right)}{n_{He}^{glass}} = \left[-S(T) \bullet GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)\right]^{-1}$$

$$\frac{n_0}{n_{He}^{glass}} = 1 + \left[-S(T) \bullet GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)\right]^{-1} \tag{1.37}$$

A Numerical Model for Diffusive Permeation of Helium Through Silica Glass Ampoules

A numerical model to estimate diffusive permeation is invaluable, as it provides a flexible way to estimate the permeation fraction $\left(\frac{m}{n_0}\right)$ even when the required assumptions for the analytical solutions are not met. For example, a numerical approach can calculate the expected permeation fraction for a non-isothermal time-temperature history, and it needn't obey the assumption that the helium concentration profile in the glass is in diffusive equilibrium with the partial pressure of helium in the encapsulated gas phase. As will be investigated below, the equilibrium concentration profile assumption is ephemerally broken when temperatures are changed, and for ampoules with high glass/gas ratios it is never true.

We numerically solve the radial diffusion equation for cylinders (Eq. (1.30)) using the Forward-Time Central-Space (FTCS) Finite Difference Method (FDM) along a radial transect of a model ampoule of arbitrary geometry for small, discrete time intervals. The system is spatially discretized into 100 cells along a radial transect, and numerical stability was insured by solving for the time interval of a single step, Δt, such that $\frac{D\Delta t}{\Delta x^2} = 0.5$, satisfying the Courant-Friedrichs-Lewy (CFL) condition.

Helium concentrations along the transect (except for the boundaries, where r = a and r = b) are updated according to the following equation, an FTCS discretization of the radially symmetric diffusion equation for a cylinder (Eq. (1.30)):

$$\frac{dC_r}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r}\right]$$

$$\Delta C_r \approx \Delta t \ \bullet D\left[\frac{C_{r-\Delta r} - 2C_r + C_{r+\Delta r}}{\Delta r^2} + \frac{1}{r}\frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r}\right] \tag{1.38}$$

Dirichlet boundary conditions were enforced at the inner and outer boundaries. The concentration at the model ampoule's outer boundary (where r = b) was held at 0, while the concentration at the inner boundary (where r = a) was calculated at each discrete step by multiplying the solubility (in units of $\frac{atoms}{atm \times cm^3}$) by the helium partial pressure within the ampoule, according to Eq. (1.39) or (1.40).

$$C_{He}^{glass} = S(T) \bullet P_{He}^{gas} \tag{1.39}$$

Eq. (1.39) can be related to the atomic abundance of helium in the gas phase using the ideal gas law and the formula for the volume of a cylinder:

$$P_{He}^{gas} = \frac{n_{gas} RT}{V_{gas}}$$

$$V_{gas} = \pi a^2 h$$

$$C_{He}^{glass} = S(T) \bullet \frac{n_{gas} RT}{\pi a^2 h} \tag{1.40}$$

For simulations where the internal pressure is not constant, n_{gas} changes after every step and Eq. (1.39) is reapplied to the inner boundary cell where r = a. While the helium concentrations in the glass and in the encapsulated gas phase are both dependent on the length of the ampoule (h), they compensate for one another such that the absolute atom flux into and out of the ampoule is not a function of h. Within the numerical model, h was varied by orders of magnitude and the results of the simulations did not change, confirming this property.

It is sometimes useful to normalize the helium concentration of the glass to the concentration of helium in the gas. To do this, the internal helium partial pressure is related to an atomic or molar concentration using the ideal gas law:

$$C_{He}^{gas} = \frac{n}{V} = \frac{P_{He}^{gas}}{RT}$$

The pressure term vanishes in the ratio of the two concentrations and depends only on temperature:

$$C_{He}^{glass/gas} = \frac{C_{He}^{glass}}{C_{He}^{gas}} = S(T) \bullet RT \tag{1.41}$$

Eq. (1.41) is plotted in Fig. 1.17. It shows that this partitioning ratio strongly favors the gas and is only a weak function of temperature, varying between 2.5 and 2.8% between 0 and 300°C. This implies that departures from the equilibrium steady state assumption due to temperature changes will be small and perhaps negligible, which is confirmed in Fig. 1.19.

In order to simulate permeation, the atomic abundances of helium within the gas phases both outside the ampoule (at high vacuum) and inside the ampoule must be tracked, as do the fluxes between the glass and these two gas phases. These fluxes were calculated simply as the diffusive flux in $\frac{atoms}{s \bullet cm^2}$ as is given by Fick's law (Eq. (1.13)). For spatially discrete cells throughout the ampoule at any radius except for the boundaries, the diffusive flux can be approximated with the finite difference:

$$J_r = -D\frac{\partial C_r}{\partial r} \approx -D\frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r} \tag{1.42}$$

However, Eq. (1.42) cannot be applied at the ampoule's inner wall when r = a or outer wall when r = b because there is no remaining glass on one side of the cell. In that case, second order terms from Taylor expansions centered at $a + \Delta r$ and $b - \Delta r$ can be used to better approximate the $\frac{\partial C_a}{\partial r}$ and $\frac{\partial C_b}{\partial r}$ terms. Letting $f(r) = \frac{\partial C_r}{\partial r}$ and $f'(r) = \frac{\partial^2 C_r}{\partial r^2}$:

$$J_a = -D\frac{\partial C_a}{\partial r} \approx -D\left(\frac{\partial C_{a+\Delta r}}{\partial r} + \frac{\partial^2 C_{a+\Delta r}}{\partial r^2}\Delta r\right) \approx -D\left(\frac{C_{a+2\Delta r} - C_a}{2\Delta r} + \frac{C_a - 2C_{a+\Delta r} + C_{a+2\Delta r}}{2\Delta r^2}\Delta r\right) \tag{1.43}$$

$$J_b = -D\frac{\partial C_b}{\partial r} \approx -D\left(\frac{\partial C_{b-\Delta r}}{\partial r} - \frac{\partial^2 C_{b-\Delta r}}{\partial r^2}\Delta r\right) \approx -D\left(\frac{C_b - C_{b-2\Delta r}}{2\Delta r} - \frac{C_b - 2C_{b-\Delta r} + C_{b-2\Delta r}}{2\Delta r^2}\Delta r\right) \tag{1.44}$$

The helium contents in the measurable (external, high vacuum) reservoir and the inner encapsulated gas phase are updated every step using fluxes determined from Eq. (1.44) and (1.43) respectively. To convert these area-normalized fluxes into absolute atomic abundances, these fluxes are multiplied

by the time increment (Δt) and the surface area of the cylinder at the outer and inner walls respectively.

Modeling permeation for ampoules that experienced complex thermal histories

In order to compare the experimental results for ampoules that experienced complex cooling histories with the expectations of the first order permeation model developed in this work, the numerical model developed above was used. The model predicts the relative helium permeation for the three experiments on the Helix-MC-Plus^{10K} multi-collector mass spectrometer for both 4He and 3He by simulating helium permeation of a model ampoule with the same dimensions and thermal history of the real experiment. For ampoules with multiple measurements of the geometric dimensions, the smaller set of measurements is used for the numerical model.

To simulate permeation during cooling of an ampoule, the temperature is decreased at every step following a parabolic rate law determined using the Watlow temperature records of the cooling ampoule. A parabolic function is fit to cooling rates evaluated as a function of temperature. Cooling rates were algorithmically calculated for discrete time intervals using the Watlow temperature files of the relevant ampoule in Python. Temperatures were taken to be the mid-point between the maximum and minimum temperatures of the interval over which the temperature change was evaluated. The quadratic functions fit to these data were forced to have a single root at T = 24°C so that the simulated ampoules would cool to a final temperature of 24°C, a common room temperature of the Noble Gas Laboratory at Caltech.

The model ampoules were initially assigned an arbitrary and unitless amount of helium, and the permeation predictions were output as step fractional yields $\left(\frac{n_t}{n_0}\right)$, where n_0 was the amount of helium in the ampoule at the beginning of the step. Before the first heating step was simulated, the ampoule was assumed to be in diffusive steady state at 24°C. Afterwards, the ampoule was not assumed to be in diffusive equilibrium except for isothermal steps at times longer than the diffusive timescale at that temperature, so that the concentration profile of the model ampoule was tracked throughout the entire heating history. For steps above room temperature, the ampoule was simulated to immediately transition to the target temperature, then the helium concentration profile,

influx, and outflux of the glass was tracked for the nominal (programmed) heating times for times smaller than the diffusive timescale at that temperature. After a heated step, an equilibrium concentration profile was reinitialized at that temperature only if the equilibrium equation (Eq. (1.22)) was used for permeation timescales longer than the diffusive timescale, then the temperature of the ampoule was cooled continuously according to the quadratic rate law determined from the experimental ampoule's thermal history while helium diffusion and permeation was continuously simulated.

Errors related to violation of the steady state concentration profile assumption due to temperature change

The equilibrium concentration profile is always a function of temperature because the partitioning of helium between the gas phase and the glass at the ampoule's inner boundary is temperature dependent (Shackelford et al., 1972). A new steady-state helium concentration profile must be established every time the temperature is changed. Because of this behavior, it is advantageous to characterize the timescales necessary to achieve a steady state diffusive profile at different temperatures, as well as the magnitude of the uncertainties that result from applying Eq. (1.22) immediately after a temperature change, when the diffusive steady state concentration profile is violated.

A new steady state concentration profile must first be established when the temperature is changed for Eq. (1.22) to strictly apply. It is possible to design experiments where permeation measurements are taken only after a diffusive steady state concentration profile is established within the ampoule, which is further discussed below.

It may unduly inconvenience the experimentalist to experimentally enforce the diffusive steady state condition when the uncertainties introduced by violating the assumption are small. In Fig. 1.17, Eq. (1.41) is plotted for relevant experimental temperatures using the solubilities reported by Swets et al. (1961). Fig. 1.17 shows that the helium volume density of glass at the ampoule's inner boundary will be between 2.5 and 3% of the volume density of equilibrated gaseous helium. The overall variation in helium partitioning is small ($\leq$12%) across temperatures relevant to step heat experiments on encapsulated goethite (0-300°C). There will be excess helium within the glass when changing from a higher temperature experimental step ($\geq$175°C) to a lower temperature

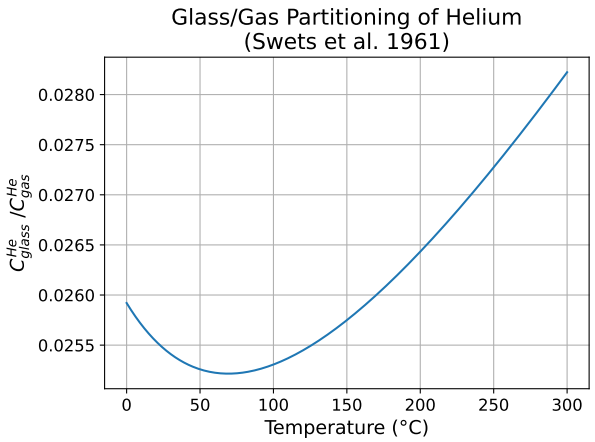

Figure 1.17: Helium preferentially partitions into the gas phase over silica glass. Eq. (1.41) is plotted here: $\frac{C^{glass}_{He}}{C^{gas}_{He}} = S \cdot RT$. It is the ratio of concentrations (in units of atoms/mL) for helium in silica glass over gas at equilibrium using solubility values of 4He from Swets et al. (1961). This partitioning is only a weak function of temperature, so the concentration ratio is between 2.5% and 2.9% for temperatures between 0 and 300°C.

permeation step (≤120°C) that would result in a slightly higher measured value than what would be predicted by Eq. (1.20). Other temperature shifts could also result in a deficit of dissolved helium instead of an excess (e.g. when shifting temperatures from 70°C to 25°C: Figs. 1.18, 1.19).

$$Disequilibrium\ Excess = \ n^{glass}_{He}\ (T_1) - n^{glass}_{He}\,(T_2) \tag{1.45}$$

The disequilibrium excess of Eq. (1.45) can be made into a relative error by dividing by the ideal amount of permeated helium measured in a permeation step from Eq. (1.20).

$$Relative\ Disequilibrium\ Error\ = \frac{n^{glass}_{He}\ (T_1) - n^{glass}_{He}\,(T_2)}{n_0(1 - e^{-\lambda(T_2)t})} \tag{1.46}$$

To evaluate Eq. (1.45) and (1.46), n^{glass}_{He} must be calculated. An upper bound can be done by assuming a steady state diffusion profile and an ideal open tube geometry (with small tops and bottoms neglected for simplicity). Assuming that the helium partial pressure outside of the ampoule is 0 and defining the abundance of helium within the gas phase as n^{gas}_{He} . The radial concentration profile at diffusive steady state as a function of radius and temperature is

given in Eq. (1.35). Radially integrated, the total amount of helium dissolved within the glass is given in Eq. (1.37), which can be solved for n_{He}^{glass} , yielding Eq. (1.47).

$$n_{He}^{glass} = \frac{n_0}{1 + \left[-S(T) \bullet GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)\right]^{-1}} \tag{1.47}$$

Where a is the radius of the ampoule's inner wall and b is the radius of the ampoule's outer wall. For legibility, the terms that are not a function of temperature are grouped as the parameter H.

$$H = -GR \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)$$

Now Eq. (1.47) can be expressed as:

$$n_{He}^{glass} = \frac{n_0}{\left(\frac{1}{HTS} + 1\right)}$$

$\frac{n_{He}^{glass}}{n_0}$, the fraction of helium atoms within the ampoule dissolved within glass, is a function only of temperature for a given ampoule. This quantity is evaluated using Eq. (1.47) for several tubes of varying glass/gas volume ratios that share a geometric factor for permeation of $G = 4350$ cm^{-2} and plotted in Fig. 1.18. Although these model tubes would all permeate helium at the same rate in diffusive steady state, they have different sizes, thicknesses, and relative internal (gas) and glass volumes, which affect how long they would need to achieve a steady state diffusive concentration profile after thermal perturbation and how severely diffusive disequilibrium would cause the amount of permeated helium to vary from the predicted amount from Eq. (1.19). Fig. 1.18 gives the intuitive result that ampoules containing more glass relative to gas will produce larger helium excesses/deficits when permeating immediately after thermal perturbation.

Now that the amount of helium atoms dissolved in the glass is defined as a function of temperature, Eq. (1.47) evaluated at the old and new temperatures (T_1 and T_2 respectively) can be substituted into Eq. (1.46) to estimate the relative measured helium excess caused by thermal perturbation as a function of time. Considering that the predicted value of n_0 is proportional to the

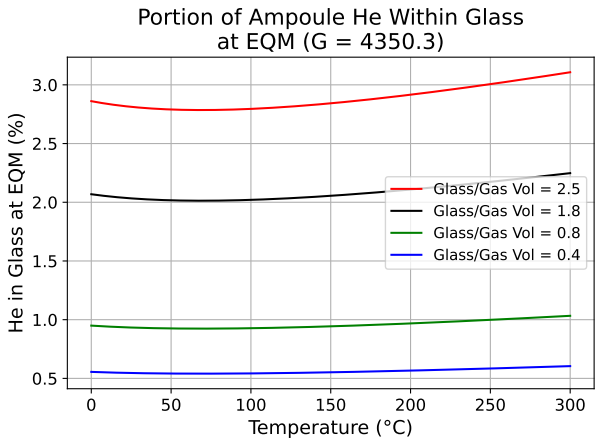

Figure 1.18: Most helium trapped within the ampoules of this work resides in the gas phase. Permeation from ampoules with more glass relative to gas will deviate more strongly from the expected permeation at diffusive steady state (Eq. (1.20)) after thermal perturbation. At diffusive equilibrium, the concentration of helium at the ampoule's inner wall is defined by the equilibrium concentration ratio with the gas phase plotted in Fig. 1.17. The concentration at the outer wall is effectively 0. The concentration profile at any point along a transect through the tube is defined by Eq. (1.35). Eq. (1.36) integrates this concentration profile along the radius to calculate the total amount of helium within the glass, and this value is divided by the total amount of helium within the gas to show the relative distribution of helium between glass and gas for ampoules as a function of inner and outer radii and temperature. It is solved here using solubilities of 4He from Swets et al. (1961) for several tubes with varying glass/gas volume ratios that share the same geometric constant of permeation G. Red, black, green, and blue lines correspond to inner radii a of 270, 300, 400, and 500 microns and outer radii of 507, 500, 533, and 601 microns respectively. The black line follows the "standard ampoule" of this work.

measured amount of helium in a permeation step according to Eq. (1.19), the *Relative Disequilibrium Excess* defined below applies to both the measured value of helium atoms m and to the calculated value of helium within the ampoule at the beginning of the permeation step n_0.

$$Relative\ Disequilibrium\ Excess = \frac{\left(\frac{n_0}{\left(\frac{1}{HT_1S_1}+1\right)} - \frac{n_0}{\left(\frac{1}{HT_2S_2}+1\right)}\right)}{n_0(1-e^{-KGRT_2t})}$$

Upon factoring and simplifying, n_0 cancels out, removing the dependence of this error estimation on knowledge of the true initial helium content of the

ampoule.

$$\textit{Relative Disequilibrium Excess} = \frac{\left(\frac{1}{\left(\frac{1}{HT_1S_1}+1\right)} - \frac{1}{\left(\frac{1}{HT_2S_2}+1\right)}\right)}{1-e^{-KGRT_2t}} \tag{1.48}$$

The relative excess helium that would be measured in a permeation step following a high-temperature experimental step is plotted in percent in Fig. 1.19 as a function of permeation time according to Eq. (1.48) for several initial experimental temperatures and permeation temperatures. This relative error decreases as the new equilibrium concentration profile is established, approaching the permeation behavior expected from Eq. (1.19). These excesses assume an instantaneous transition from experimental step to permeation step and therefore are upper bounds. These excesses are dependent also on the ampoule dimension; in Fig. 1.19, a standard ampoule ($a = 0.03$ cm, $b = 0.05$ cm) is assumed. Thicker ampoules will produce higher magnitudes of excesses (or deficits) for the same heating schedules. Within the temperature range 0-300°C the maximum excess helium that could be introduced into a permeation step measurement due to diffusive disequilibrium would be for an 80°C permeation step following an experimental step at 300°C. Even for those circumstances, the maximum relative error would be $\leq$2% for a 3-hour permeation step (Fig. 1.19). It is important to recognize, however, that this error estimation does not account for non-instantaneous cooling, although fast thermal response for similar experiments and the exponential temperature dependence of diffusion and permeation typically permit experimentalists to neglect fast but non-instantaneous heating and cooling (Farley, Reiners, et al., 1999). These calculations show that, at least for ampoules of similar dimensions to those sealed at Caltech, the error in Eq. (1.20) resulting from violating the diffusive steady state assumption of Eq. (1.22) is small.

For cases where more precise determinations of n_0 and n_t from the application of Eq. (1.20) on a permeation step are desired, it is possible to experimentally ensure compliance with the diffusive steady state concentration assumption. For example, the ampoule could be programmed to be held at the target permeation temperature (for example, 80°C) immediately after a high temperature experimental step instead of completely cooling (to ~25°C) while evolved gasses are processed and helium is purified from the high temperature

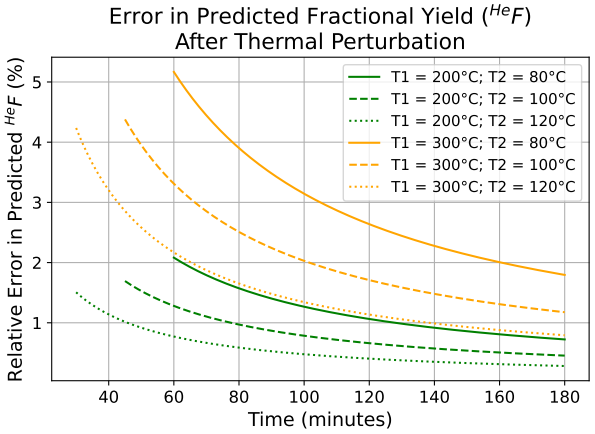

Figure 1.19: When changing temperatures, the inner boundary condition and diffusive steady state concentration profile of helium within a silica glass ampoule change, so the amount permeated will be different from the expected permeate amount of Eq. (1.20) that assumes a diffusive steady state profile has already been established at the new temperature. This difference is calculated here as the difference of helium dissolved within the tube at the starting and final temperature (neglecting the change in helium abundance in the gas phase) and is normalized to the expected permeation amount at the final temperature as a function of time for several temperature jumps, using Eq. (1.36) and solubilities of 4He from Swets et al. (1961) for a standard ampoule (with inner and outer radii of $a = 300$ microns and $b = 500$ microns respectively). Generally, larger temperature jumps produce larger relative errors. For a standard ampoule, the error of Eq. (1.20) will be less than 2% for any permeation step 3 hours or longer following a 300°C experimental step.

experimental step. To do this, one must be aware of the timescale necessary to reach diffusive equilibrium at a given temperature. At Caltech, helium purification after heating steps typically takes 30 minutes, which is only enough time to establish steady state concentration profiles for ampoules <175 µm thick at 100°C (main text Fig. 1.7). For larger ampoules or cooler permeation step temperatures, an additional delay may need to be programmed in order to experimentally enforce the diffusive steady state assumption. Alternatively, expected permeation fractions for any arbitrary thermal schedule can be numerically solved, provided that the solubility and diffusivity are each known for the experimental temperature range.

The Fourier number can be used to determine the time necessary to establish diffusive equilibrium of the helium concentration profile within the ampoule. The Fourier number is a dimensionless number that describes the extent of

diffusive equilibration or homogenization of a system by heat or mass transfer given a characteristic length scale and time (Ostrogorsky and Glicksman, 2015). It is the time divided by the diffusive timescale:

$$Fo = \frac{Dt}{L^2} = \frac{t}{\tau} \tag{1.49}$$

Where t is time and L is the diffusive length scale. For the purposes of helium permeation through silica glass ampoules, L is the glass thickness of the ampoule. When $Fo > 1$, a system is generally said to be effectively homogenized or at equilibrium with respect to diffusion, but it is somewhat arbitrary to define a specific Fourier number at which one can confidently say that diffusion has effectively re-established an equilibrium concentration profile (Ostrogorsky and Glicksman, 2015). To test the value of $Fo = 1$ as the relevant minimum value, we use the numerical diffusive permeation model described above for a characteristic ampoule where a = 300 µm and b = 500 µm. Dirichlet boundary conditions are imposed at the ampoule's inner and outer boundaries, where r = a and r = b, holding $C^{Glass}_{r=b} = 0$ and $C^{Glass}_{r=a} = C^{Gas}_{He} \bullet \left(\frac{C^{Glass}_{He}}{C^{Gas}_{He}}\right)_{EQM}$ as calculated in Eq. (1.41) and plotted in Fig. 1.17. For the analysis of the Fourier number during the approach to the diffusive equilibrium concentration profile in Figs. 1.20 and 1.21, C^{Gas}_{He} is arbitrary and held constant, so plotted concentrations are normalized to C^{Gas}_{He}. A steady state concentration profile (Eq. (1.33)) is taken as the initial condition. The absolute concentration is technically dependent on the ampoule height, but this dependence on h vanishes when normalizing the glass concentration to the gas concentration. The results of these calculations are plotted in Figs. 1.20 and 1.21, where Fig. 1.20 shows the transition from an equilibrium concentration profile at 300°C to 100°C, and Fig. 1.21 shows the more dramatic establishment of an equilibrium concentration profile at 100°C from an initial helium concentration of 0 throughout, such as could be expected after suddenly introducing helium into an ampoule that previously contained no helium. Both figures show that an equilibrium concentration profile is well-established for Fourier number $Fo = 1$, and Fig. 1.20 in particular shows that the difference in concentration profiles is small and nearly reestablished even when the Fourier number $Fo = 0.1$. We conclude that the timescale associated with $Fo = 1$ is the relevant time needed to hold the ampoule to ensure that Eq. (1.20) strictly applies without errors due to thermally-forced disequilibrium in the helium concentration profile.

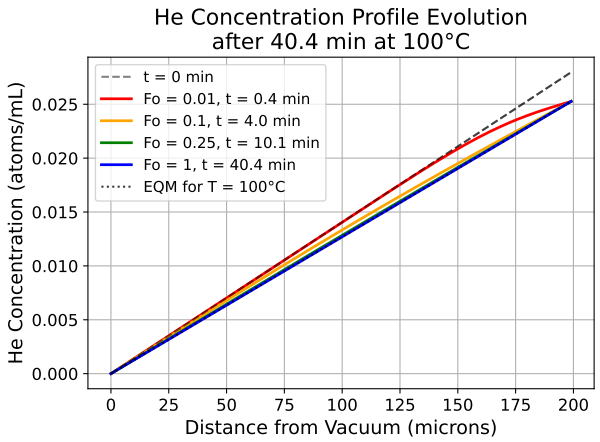

Figure 1.20: A numerical simulation of the evolution of 4He concentration profiles in silica glass ampoules from steady state at 300°C to steady state at 100°C. The steady state concentration profile is only subtly different for different temperatures. The diffusive timescale $\tau = L^2/D$ corresponding to a Fourier number of 1 is effectively sufficient to reestablish a steady state concentration profile at the new temperature. The concentration is arbitrary and is normalized to the concentration of helium in the gas phase of the ampoule, which is held constant for this simulation (therefore the ideal steady state of Eq. (1.35) applies regardless of the glass/gas volume ratio). The inner boundary condition is determined as a function of temperature using the values of Eq. (1.41), plotted in Fig. 1.17. The ampoule simulated here is a standard ampoule with an outer radius of 500 microns and inner radius of 300 microns.

The diffusive timescales associated with $Fo = 1$ for permeability values taken from Swets et al. (1961) are plotted in main text Fig. 1.7 for a few different thickness values (L) as a function of temperature for typical permeation step temperatures used in this study.

1.3s Estimating permeation lag uncertainty for uncorrected, long heating steps

The uncertainty of the helium evolution of the encapsulated natural material during a heating step (ω) due to permeation lag correction can be estimated by calculating the maximum possible helium residing in the tube between steps, i.e. for the values for $n_{0,}$ and n_t: n_0^{UB} and n_t^{UB}. From the permeation lag equation (main text Eq. (1.1)), upper bound estimates for $n_{0,}$ and n_t bound the encapsulated material contribution ω within:

$$m - n_0^{UB} \leq \omega \leq m + n_t^{UB} \tag{1.50}$$

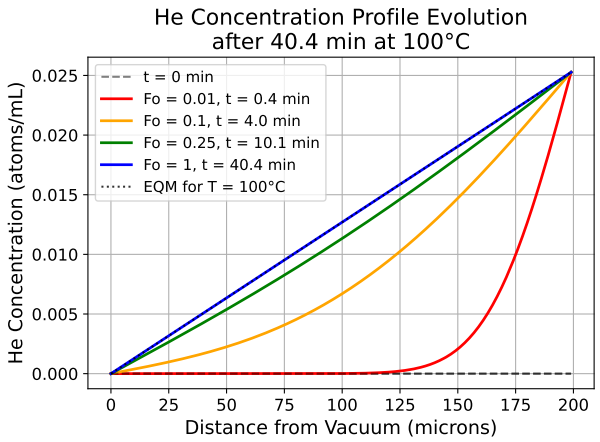

Figure 1.21: Similar to Fig. 1.20, but starting from a helium concentration of 0 throughout the ampoule. This is most useful to visualize the incomplete establishment of a steady state concentration profile for timescales corresponding to Fourier numbers up to 0.25. When the Fourier number is 1, the concentration profile is in diffusive steady state.

Upper bounds for n_t for a given high temperature experimental step can be calculated if step heats are long enough to assume steady state permeation, assuming G and K are known or can be conservatively estimated. To proceed, one must make the generally conservative assumption that the production rate of helium within the ampoule due to degassing of the encapsulated natural sample ($\Omega = \frac{\omega}{t}$) is constant over time at a given temperature. In reality, Ω will decrease over time for helium loss governed by first order processes (O. V. Yakubovich, Y. A. Shukolyukov, et al., 2010; Wilson and Bisson, 1973) or diffusion (except for the case of initially inhomogeneous distributions, the memory of which is often erased early in any diffusion experiment: Shuster, Farley, et al., 2004). Eq. (1.52), given below, shows that the amount of helium trapped in the tube (n_t) is proportional to the rate of helium outgassing from the natural encapsulated sample (Ω), so the approximation that this rate is constant will overestimate the amount of unanalyzed helium remaining in the ampoule after a high temperature experimental step. The rate of helium degassing over time will decrease for degassing controlled by second-order volume diffusion *except* for the minor case of the alpha-ejection effect at low He-release fractions (Farley, Wolf, et al., 1996; Ketcham et al., 2011) or for the more serious effects of radiation damage annealing (Goldsmith et al., 2020) and mineral decomposition (Farley, H. B. Monteiro, et al., 2024). In any case, sequential isothermal steps immediately reveal when $\frac{\omega}{t}$ increases over time.

A new first order differential equation like Eq. (1.19) is formulated, now incorporating a constant source term (Ω) to conservatively approximate the helium influx into the ampoule from the degassing natural sample.

$$\frac{dn}{dt} = \Omega - \lambda n \tag{1.51}$$

Where $\Omega = \frac{\omega}{t}$ and $\lambda = KGRT$. Immediately, the steady state quantity of helium atoms (n_{ss}) present in the ampoule can be found by solving Eq. (1.51) for the steady state condition where $\frac{dn}{dt} = 0$.

$$n_{ss} = \frac{\Omega}{\lambda} = \frac{\omega}{\lambda t} \tag{1.52}$$

Additionally, the helium concentration within the ampoule during the approach to the steady state concentration can be solved for by integrating Eq. (1.51) with respect to t. First, the integrating factor for Eq. (1.51) is identified: $I = e^{\lambda t}$, which is multiplied to both sides to yield Eq. (1.53).

$$e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n = e^{\lambda t}\Omega \tag{1.53}$$

Then both sides are integrated with respect to *t:*

$$\int \left(e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n \right) dt = \int e^{\lambda t}\Omega dt$$

Because $\frac{d}{dt}(ne^{\lambda t}) = e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n$ by the product rule Eq. (1.53) reduces to:

$$n_t e^{\lambda t} = \frac{\Omega}{\lambda}e^{\lambda t} + C \tag{1.54}$$

Where C is an unknown constant introduced by indefinite integration. This expression becomes more intuitive when both sides are divided by the integration factor and by substituting the steady state concentration from Eq. (1.52).

$$n_t = n_{ss} + Ce^{-\lambda t} \tag{1.55}$$

The integration constant C can be solved by enforcing the condition that $n_t(t = 0) = n_0$, leading to:

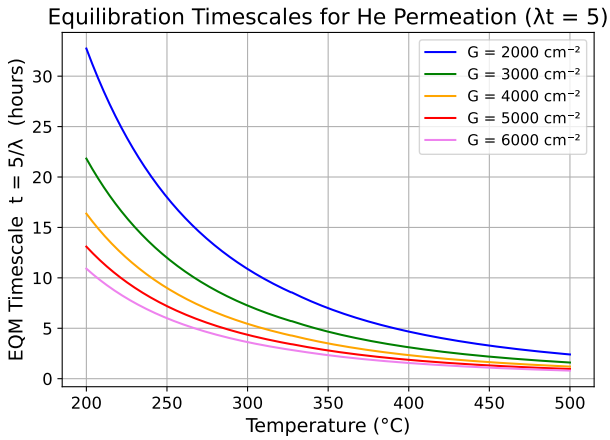

Figure 1.22: The upper limit of helium contents within the ampoule between high temperature experimental steps can be reasonably estimated without further measurement, provided permeative equilibrium has been reached. For typical ampoules with geometric factors between 3000 and 4000 cm^{-2}, this criterion is generally met for steps longer than 24 hours, or within 5 hours at 350°C or higher.

$$n_0 = n_{ss} + C \tag{1.56}$$

Solving Eq. (1.56) for C and substituting into Eq. (1.55) yields Eq. (1.57). Now an inequality is assigned to the equation to reflect the initial assumption that the isothermal helium outgassing rate (Ω) is constant *or decreasing*.

$$n_t \leq \ n_{ss} + (n_0 - n_{ss})\, e^{-\lambda t} \tag{1.57}$$

For long timescales (i.e. when $e^{-\lambda t} \approx 0$), Eq. (1.57) reduces to $n_t \leq \ n_{ss}$. This is effectively true for steps long enough such that $t \geq \frac{5}{\lambda}$, for which $e^{-\lambda t} \leq 0.007$. It is advisable to only consider the Approximation with Uncertainty method for steps of sufficiently long duration such that $t \geq \frac{5}{\lambda}$ so that the estimation for n_t can be reduced to simply the steady state value and not be dependent upon the estimation for n_0 or its uncertainties. The timescales required to reach permeation equilibrium are plotted as a function of temperature in Fig. 1.22 below for some select values of G, the geometric factor.

To estimate the upper bound on the permeation lag (n_t^{UB}), one must consider the circumstances that would produce the highest possible value for n_t. In addition, the constraint that $t \geq \frac{5}{\lambda}$ is enforced so that previous heating steps

are not significant, simplifying the problem significantly. Because n_t scales with ω by Eq. (1.52), the upper bound estimate n_t^{UB} must assume the upper bound estimate for ω, which is given in Eq. (1.50) as $m + n_t^{UB}$. Substituting $\omega^{UB} = m + n_t^{UB}$ into Eq. (1.52) to express n_t^{UB} in terms of the measured value m:

$$n_t^{UB} = n_{ss}(\omega^{UB} = m + n_t^{UB}) = \frac{m + n_t^{UB}}{\lambda t}$$

Solving for n_t^{UB} yields Eq. (1.58):

$$n_t^{UB} = \frac{m}{\lambda t - 1} \tag{1.58}$$

The upper bound amount of helium-4 within the ampoule normalized to the measured value in percent at steady state permeation is plotted in main text Fig. 1.8 using Eq. (1.58) $\left(\frac{n_t^{UB}}{m} = \frac{1}{\lambda t - 1}\right)$, assuming a standard tube where G = 4350.26 cm^{-2} and the permeability values given in Swets et al. (1961). Main text Fig. 1.8 shows that, in order to push this uncertainty to be less than 5%, even very hot steps require long step durations. For a 5% uncertainty using this method, a 10-hour step duration is required even at 400°C. Much longer durations are required for the same precision at lower temperatures: at 250°C, a step duration of 35 hours is required to push the uncertainty beneath 5%.

1.4s Comparison to Literature Permeability Values

Predicted and measured fractional yields for all permeation steps in ampoules besides the encapsulated water experiment agree well (within 1σ: Figs. 1.9, 1.24) with the predictive equations presented in this work and with the permeabilities reported by Swets et al. (1961) and Jones (1953) for all studied temperatures (24-300°C; Fig. 1.23) and step fractional yields (up to ~80%). One outlier measurement on the empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer was excluded from Figs. 1.9 and 1.24.

Permeation ratios for 3He and 4He measured in the empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer are plotted in Fig. 1.25. These data present significant scatter, but generally agree with the reported results of Jones (1953), except for measurements at room temperature, which cluster tightly at a permeation ratio of 1:1 for the two isotopes. Additionally, fields corresponding to the ratios of apparent permeabilities for the water-filled

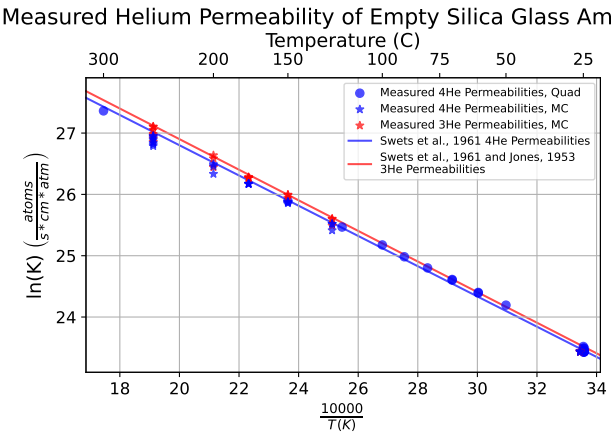

Figure 1.23: Helium permeabilities measured in silica glass ampoules at Caltech on a quadrupole mass spectrometer (4He only) and on a Helix-MC-Plus^{10K} (both 3He and 4He) multi-collector mass spectrometer are compared to literature values. Permeabilities of 3He and 4He are very similar, but 3He permeates slightly faster. The final measurement on the Helix-MC-Plus^{10K} mass spectrometer is an outlier and is excluded from the plot, as it is most sensitive to the value of the total yield remaining in the ampoule. These experiments on empty tubes at Caltech show excellent agreement with the literature values with allowance for some scatter. This result shows that existing understanding of helium permeability in glass can be reasonably leveraged for use in helium step heats of silica glass encapsulated geological samples. Literature 4He permeabilities are taken from Swets et al. (1961); whereas 3He permeabilities from the literature are calculated by multiplying the 3He/4He permeability ratios of Jones (1953) with the 4He permeabilities of Swets et al. (1961).

and sulfur-filled experiments are plotted for the temperature ranges of those experiments. Results for those experiments are plotted as fields because each step integrated permeation behavior over a complex (non-isothermal) cooling history. These experiments also agree with the results of Jones (1953) and the empty ampoule experiment. The qualitative result of Jones (1953) that silica glass is only slightly more permeable for 3He than 4He is supported by these data, which are in marked contrast to the reported extreme differences in 3He and 4He diffusivity in silica glass of McMurtry et al. (2019), who report instantaneous isotopic fractionations up to a factor of 1000.

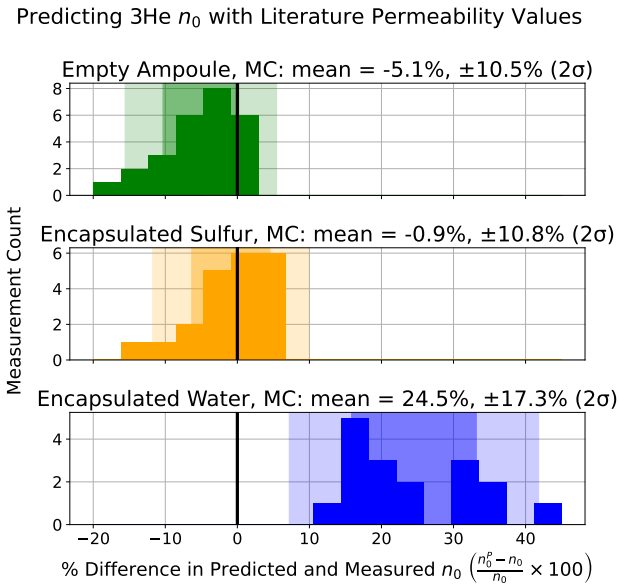

Figure 1.24: Similar to Fig. 1.9, but with 3He permeabilities taken from Jones (1953) and Swets et al. (1961). Shaded regions behind the histograms correspond to 1 and 2 sigma (σ) from the average relative difference between the predicted and measured values. The 3He permeation model and literature permeability values apply well to empty and sulfur-filled ampoules, and such models can be used to reliably recover 3He contents of permeating ampoules for use in permeation lag corrections. On the other hand, measured and predicted values of 3He n_0 for water-filled ampoules are well resolved beyond 2σ, suggesting that their permeability differs strongly from the literature values. Internal reproducibilities (1σ) of a gas standard repeatedly analyzed during these experiments were better than 3% for 3He, requiring that much of the scatter in these data be related to factors outside of analytical precision such as long/inconsistent ramp times during heating or uncertainties or rounding of heating times.

1.5s Additional Analytical Corrections for the Empty Ampoule Calibration Experiments

Pump-Out and Gas-Preparation Corrections for the Atmosphere-Equilibrated Empty Ampoule

Using the helium permeability values of Swets et al. (1961) for silica glass, an ampoule with a geometric factor (G) of 6000 cm^{-2} would permeate 0.2‰ of its helium in 1 minute at 26°C (Eq. (1.4)), which is negligible for the scope of this work. The atmosphere-equilibrated empty ampoule with a geometric factor (G) of ~4785 cm^{-2} would permeate 2.8‰ of its helium in its 15-minute pumping time between steps. For that experiment only, helium loss during pumping is accounted for by applying the permeation relationship of Eq. (1.4)

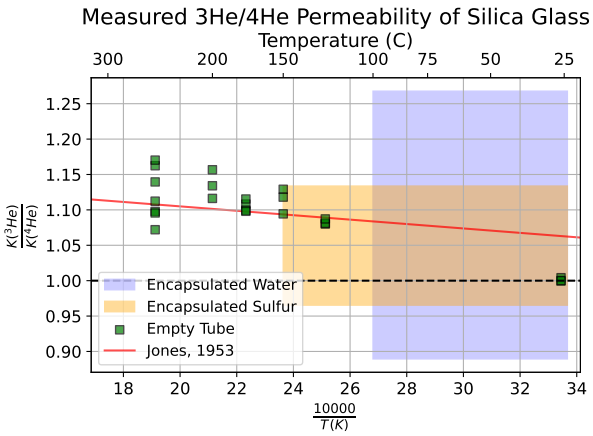

Figure 1.25: Ratios of calculated permeabilities from step heat experiments on silica glass ampoules on the Helix-MC-Plus^{10K} mass spectrometer at Caltech. The last measurement on the empty tube experiment is an outlier and was excluded from the plot. Experiments on encapsulated water and encapsulated sulfur were allowed to cool before isotopic measurements. Because of slow cooling rates, simple corrections could not be applied to these two experiments, which effectively queried permeabilities at a range of temperatures within a single step. Therefore, the ratios of their apparent permeabilities are plotted as a field with 2σ standard deviation across the experimental temperature range. These three experiments show general, albeit imprecise, agreement to the results of Jones (1953), except for the measurements at room temperature. Jones (1953) was likely able to measure the relative permeabilities of 4He and 3He with much higher precision because his experimental vessel was charged with 3He at a partial pressure of around 0.5 atmospheres, over one billion times higher than the 3He partial pressure within the fused silica ampoule used in this work. Interestingly, three room-temperature measurements of helium permeation of an empty tube at Caltech yielded essentially identical permeability of 3He and 4He.

($n_0 = n_t/(1 - {}^{He}F)$) to solve for n_0, the amount of helium in the ampoule before pump-out, from n_t, the amount remaining after pumping with T = 25°C and t = 15 minutes. A permeability coefficient (K) for helium in silica glass had to be assumed in order to apply this correction, so the value of 1.54×10^{10} $\frac{atoms}{s \times cm \times atm}$ at 25°C given by the Arrhenius law reported by Swets et al. (1961) was used. The validity of this choice is supported by the data; the average calculated permeability at room temperature for this experiment after correcting for helium lost during the 15-minute pump out is also 1.54×10^{10} $\frac{atoms}{s \times cm \times atm}$.

For that experiment only, a correction was also applied to the high temperature

steps because they were heated for short durations (45 minutes for all but one heated step), so that the permeation of the ampoule at room temperature during gas cryofocusing represented a significant portion of the step's yield. This was similarly corrected by calculating the expected fractional yield at room temperature during the gas preparation time, which is the step's total time minus the pump out time and the heated time, (29-31 minutes for all heated steps). For that correction, the permeation relationship of Eq. (1.4) ($n_0 = n_t/(1 - {}^{He}F)$) is applied again to solve for the unknown n_0, the amount of helium within the ampoule immediately after the completion of the heated step, from n_t, the amount of helium within the ampoule when the gas is inlet into the mass spectrometer about 30 minutes later after spiking and helium purification. This cooling and gas preparation correction is not applied to the room temperature measurements for which the temperature remained constant during gas preparation.

Anomalous Signal Recalculation for the Tank Helium-Soaked Empty Ampoule

The last high temperature step of the empty ampoule experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer had an unusual and non-linear signal evolution at the beginning of the measurement for both 3He and 4He. For this measurement, 4He was recalculated using a linear regression on the last three of six measurement cycles with an R^2 of 0.985 to regress to the 4He content at time 0. Typically, the signal evolution for a measurement like this would have a negative slope because the helium in the mass spectrometer is consumed as it is ionized and measured, but the slope for this measurement was positive. 3He from this measurement was calculated based on the recalculated 4He and an assumed isotope ratio for the measurement of 0.07 Ra. Three more measurements were taken over the next few days to ensure complete degassing of the ampoule. Two of these measurements yielded helium above blank, but were small compared to the final high temperature step. The yields of these steps were all considered in the calculation of the experiment's total yield.

Independent Verification of the Encapsulated Durango Apatite Mass

The total yield for the encapsulated Durango apatite experiment (main text, §1.5) is estimated from the precisely-measured mass of the encapsulated apatite (3.18 mg) and a helium concentration of 194.2 nccSTP/mg (Farley, 2000). This

concentration is independently supported by the measurement of 8 different >0.5 mg aliquots of the same Durango apatite sample in-lab, which yield a concentration of 194 ± 9 nccSTP/mg.

1.6s Sample Preparation and SEM Imaging of Buffered and Decomposed Sulfides

Samples of pyrite, arsenopyrite, and marcasite were crushed under ethanol, and grains larger than 100 µm were dried for each mineral and subdivided into four populations: one control, and three for heat treatment in atmosphere, ultra-high vacuum, or encapsulated within a glass ampoule along with an excess of fused sulfur (or realgar: As3S3 for arsenopyrite). The encapsulated populations were loaded into 3 mm inner diameter, 4 mm outer diameter silica glass tubes with fused bottoms and open tops, along with sufficient fused sulfur or crystalline realgar to guarantee vapor saturation at 400°C. These tubes were interfaced with a vacuum line and were pumped to 60 millitorr before the tops of the tubes were sealed.

The open-air populations for each mineral were carefully placed on glass microscope slides within borosilicate petri dishes and the encapsulated populations were submerged in pure quartz sand within capped, deep petri dishes. These petri dishes containing the encapsulated and open-air populations of sulfides were placed on a temperature controlling hot plate under a fume hood and were held at 400°C for 8-11 hours, at which point the hot plate was turned off and the samples were left to cool.

The ultra-high vacuum population for pyrite and marcasite were loaded into copper packets and mounted to a type K thermocouple within a high vacuum line. A silver wool based sulfur trap was installed between the samples and the turbo pump. The high-vacuum line was closed and pumped to <10-5 torr. A Watlow temperature controller held the samples at 400°C for 4.5 hours with a projector lamp illuminating the sample through a sapphire window. This procedure was repeated for each mineral sample of pyrite and marcasite. The arsenopyrite sample was not heated in vacuum to avoid contaminating the line with arsenic. To simulate thermal decomposition in an anoxic environment, a population of arsenopyrites was thermally treated in a large glass capsule similarly to the encapsulated populations, but without a realgar "buffer".

The control and thermally treated samples were mounted to carbon tape and

carbon coated, then were loaded in the Zeiss 1550 Scanning Electron Microscope (SEM) at Caltech. Images of the surface topography of each population were taken using the secondary electron detector and an Electron High Tension (EHT) of 10 kV at a working distance of 10-15 mm.

1.7s Supplementary Data Tables

The following tables present the reduced step-by-step helium permeation data underlying the empty-tube, encapsulated-water, and encapsulated-sulfur calibration experiments summarized in the ampoule dimensions table of the main text, Table 1.1 and the permeability/isotope-ratio comparison figures above. m is the measured helium abundance of the step; n_0 and n_t are the helium contents of the ampoule at the beginning and end of the step (Eq. (1.20)); n_t/n_0 predicted and measured are the fractional yields expected from the permeability of Swets et al. (1961) (and, for 3He, the isotope ratios of Jones (1953)) and as measured, respectively.

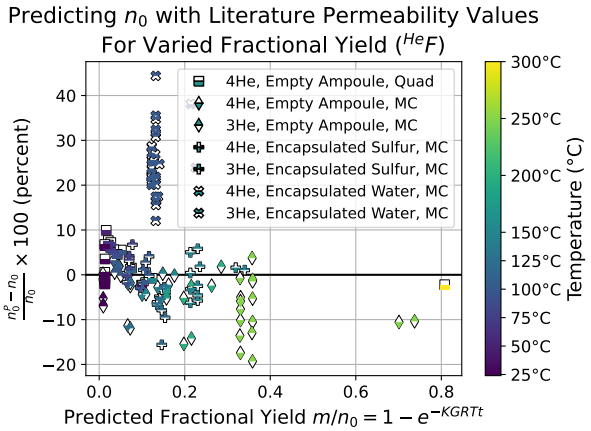

Figure 1.26: Helium measurements (m) of a permeating ampoule are related to the initial helium contents of the ampoule (n_0) by Eq. (1.22), using permeability values taken from Swets et al. (1961) and Jones (1953) (for 3He only). The n_0^P variable is the predicted value of n_0, the true value. For the empty ampoules, except for 3 measurements on the Helix-MC-Plus^{10K}, predicted values (n_0^P) are calculated using Eq. (1.20) (with some corrections applied to the quadrupole experiment: see main text for details). For the encapsulated sulfur and water ampoules, as well as 3 measurements of the blank ampoule on the Helix-MC-Plus^{10K}, predicted values (n_0^P) were determined by forward modeling diffusion and permeation on ampoules as they were held at high temperature and subsequently allowed to cool (see main text for details). This plot shows that the predicted values are correct to within ~10% for empty and sulfur-filled ampoules regardless of temperature or extent of permeation, whereas n_0^P values for water-filled ampoules are systematically high. The final Helix-MC-Plus^{10K} measurement on the empty ampoule yields an outlier n_0^P that was ~80% high — this measurement is not plotted here.

Table 1.4: Distilled helium permeation data for the empty silica glass ampoule equilibrated with atmosphere, measured by isotope dilution on the Pfeiffer Prisma QMS 200 quadrupole mass spectrometer ("Empty Tube, Pfeiffer Quad" in the ampoule dimensions table of the main text, Table 1.1). n_0 and n_t are the initial and final helium contents of the ampoule for each step (Eq. (1.20)); n_t "before cooling" is only populated for high-temperature experimental steps, for which the room-temperature pump-out/cool-down correction discussed in the main text is applied to obtain the "after cooling" value used to calculate permeability. $\ln(K)$ values are the natural log of the permeability coefficient in $\frac{atoms}{s \times cm \times atm}$.

Temp (°C)	Time (min)	m 4He (nccSTP)	n_0 4He (nccSTP)	n_t 4He (before cooling) (nccSTP)	n_t 4He (after cooling) (nccSTP)	n_t/n_0 measured	$10000/T$ (K^{-1})	$\ln(K)$ measured
25	77	0.1877	12.87		12.68	0.9854	33.56	23.52
25	77	0.1793	12.64		12.47	0.9858	33.56	23.49
50	45	0.2902	12.43	12.21	12.14	0.9819	30.96	24.19
24.8	74	0.1593	12.11		11.95	0.9868	33.58	23.46
100	45	0.7129	11.92	11.26	11.2	0.9452	26.81	25.18
24.7	74	0.145	11.17		11.03	0.9870	33.59	23.44
60	45	0.3094	11	10.75	10.69	0.9771	30.03	24.40
70	45	0.3619	10.66	10.35	10.3	0.9712	29.15	24.60
80	45	0.4225	10.27	9.902	9.849	0.9640	28.33	24.80
24.7	74	0.1272	9.823		9.696	0.9870	33.59	23.44
24.8	74	0.1252	9.67		9.545	0.9871	33.58	23.44
24.8	75	0.1238	9.519		9.395	0.9870	33.58	23.43
70	45	0.3209	9.37	9.098	9.049	0.9709	29.15	24.61
60	45	0.2534	9.025	8.82	8.771	0.9773	30.03	24.39
24.8	74	0.113	8.748		8.635	0.9871	33.58	23.44
90	45	0.4231	8.611	8.231	8.188	0.9558	27.55	24.98
119.8	45	0.6625	8.166	7.543	7.504	0.9236	25.46	25.47
24.8	74	0.09578	7.484		7.388	0.9872	33.58	23.43
150	45	0.9371	7.368	6.465	6.431	0.8775	23.64	25.89
25	581	0.6179	6.414		5.796	0.9037	33.56	23.43
300	90	4.549	5.78	1.238	1.231	0.2141	17.45	27.36
24.7	74	0.01673	1.228		1.211	0.9864	33.59	23.49
400	90	1.178	1.208		0.03073			
24.7	74	0.00117	0.03065					

Table 1.5: Distilled 3He permeation data for the empty silica glass ampoule equilibrated with high-purity tank helium, measured on the Helix-MC-Plus^{10K} multi-collector mass spectrometer ("Empty Tube, Helix MC" in the ampoule dimensions table of the main text, Table 1.1). Rows marked '–' in the predicted and measured n_t/n_0 and permeability columns are cooling-only intervals for which no isothermal permeability was calculated (see main text).

Temp (°C)	Time (min)	m 3He (pccSTP)	$\pm 2\sigma$ (pccSTP)	n_0 3He (pccSTP)	n_t 3He (pccSTP)	n_t/n_0 predicted	n_t/n_0 measured	3He K measured ($\frac{atoms}{s\ cm\ atm}$)
26	73.6	0.0516	6.16×10^{-4}	5.787	5.736	0.9902	0.9911	1.53×10^{10}
–	113.3	0.269	3.89×10^{-3}	5.736	5.467			
26	73.7	0.048	8.51×10^{-4}	5.467	5.419	0.9902	0.9912	1.50×10^{10}
26	75.3	0.0485	9.92×10^{-4}	5.419	5.37	0.9900	0.9910	1.50×10^{10}
–	112.9	0.271	5.67×10^{-3}	5.37	5.099			
–	126.6	0.418	8.80×10^{-3}	5.099	4.682			
125	52.9	0.299	6.31×10^{-3}	4.682	4.382	0.9275	0.9361	1.18×10^{11}
125	53.1	0.311	6.57×10^{-3}	4.382	4.071	0.9273	0.9290	1.31×10^{11}
125	53.1	0.291	6.15×10^{-3}	4.071	3.78	0.9273	0.9285	1.32×10^{11}
–	73.6	0.422	8.67×10^{-3}	3.78	3.358			
150	73.6	0.502	8.56×10^{-3}	3.358	2.856	0.8512	0.8506	1.95×10^{11}
150	52.9	0.299	4.47×10^{-3}	2.856	2.557	0.8907	0.8952	1.86×10^{11}
150	88.4	0.45	6.28×10^{-3}	2.557	2.107	0.8240	0.8240	1.94×10^{11}
175	53.1	0.318	4.55×10^{-3}	2.107	1.788	0.8427	0.8489	2.59×10^{11}
175	53.1	0.272	4.00×10^{-3}	1.788	1.516	0.8427	0.8478	2.61×10^{11}
175	53.1	0.228	3.54×10^{-3}	1.516	1.288	0.8427	0.8498	2.57×10^{11}
175	53.1	0.193	3.15×10^{-3}	1.288	1.095	0.8427	0.8504	2.56×10^{11}
–	53.1	0.164	2.82×10^{-3}	1.095	0.9317			
200	53.2	0.174	3.23×10^{-3}	0.9317	0.7582	0.7843	0.8139	3.07×10^{11}
200	53.1	0.16	3.16×10^{-3}	0.7582	0.5978	0.7847	0.7887	3.55×10^{11}
200	73.6	0.175	3.80×10^{-3}	0.5978	0.4224	0.7146	0.7070	3.75×10^{11}
250	53.2	0.152	3.93×10^{-3}	0.4224	0.2704	0.6410	0.6406	6.02×10^{11}
250	53.1	0.0977	2.95×10^{-3}	0.2704	0.1727	0.6413	0.6396	6.06×10^{11}
250	53.1	0.0607	2.12×10^{-3}	0.1727	0.1119	0.6413	0.6499	5.85×10^{11}
250	53.1	0.0379	1.56×10^{-3}	0.1119	0.07407	0.6414	0.6639	5.58×10^{11}
250	52.9	0.0249	1.13×10^{-3}	0.07407	0.04914	0.6427	0.6634	5.57×10^{11}
250	53.1	0.016	7.70×10^{-4}	0.04914	0.03315	0.6415	0.6746	5.32×10^{11}
250	159.9	0.026	1.31×10^{-3}	0.03315	0.007191	0.2625	0.2169	6.86×10^{11}
250	52.9	0.00494	2.52×10^{-4}	0.007191	0.00225	0.6424	0.3129	1.58×10^{12}
High T	52.2	0.00225		0.00225	0			

Table 1.6: Distilled 4He permeation data for the same empty silica glass ampoule experiment as Table 1.5. $K(3/4)$ is the ratio of the measured 4He to 3He permeability coefficients for each step; 3/4 He measured (Ra) is the measured 3He/4He ratio of the step, normalized to the atmospheric ratio.

Temp (°C)	Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	4He K measured ($\frac{atoms}{s\,cm\,atm}$)	K_3/K_4	3He/4He measured (Ra)
26	73.6	263.7	1.33	29670	29410	0.9908	0.9911	1.52×10^{10}	1.0031	0.1408
–	113.3	1282	7.99	29410	28130					0.1509
26	73.7	247.4	1.95	28130	27880	0.9908	0.9912	1.51×10^{10}	0.9984	0.1396
26	75.3	250	2.34	27880	27630	0.9905	0.9910	1.50×10^{10}	0.9990	0.1397
–	112.9	1322	13.1	27630	26310					0.1475
–	126.6	1971	20	26310	24340					0.1524
125	52.9	1446	14.9	24340	22890	0.9332	0.9406	1.09×10^{11}	1.0786	0.1489
125	53.1	1509	15.6	22890	21380	0.9330	0.9341	1.21×10^{11}	1.0795	0.1483
125	53.1	1413	14.6	21380	19970	0.9330	0.9339	1.21×10^{11}	1.0856	0.1483
–	73.6	2032	20.8	19970	17940					0.1495
150	73.6	2399	22.2	17940	15540	0.8630	0.8663	1.73×10^{11}	1.1271	0.1505
150	52.9	1498	12.7	15540	14040	0.8996	0.9036	1.70×10^{11}	1.0922	0.1438
150	88.4	2238	16.7	14040	11800	0.8377	0.8406	1.74×10^{11}	1.1155	0.1447
175	53.1	1626	10.8	11800	10170	0.8556	0.8622	2.34×10^{11}	1.1053	0.1409
175	53.1	1404	8.46	10170	8771	0.8556	0.8620	2.34×10^{11}	1.1121	0.1395
175	53.1	1211	6.28	8771	7560	0.8556	0.8619	2.34×10^{11}	1.0959	0.1353
175	53.1	1042	4.8	7560	6518	0.8556	0.8622	2.34×10^{11}	1.0939	0.1332
–	53.1	895.4	3.67	6518	5623					0.1314
200	53.2	952.2	3.41	5623	4670	0.8020	0.8306	2.77×10^{11}	1.1107	0.1311
200	53.1	887.5	2.99	4670	3783	0.8023	0.8100	3.15×10^{11}	1.1275	0.1300
200	73.6	987.1	3.27	3783	2796	0.7370	0.7391	3.26×10^{11}	1.1485	0.1278

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(continued)

Temp (°C)	Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	4He K measured $(\frac{atoms}{s\,cm\,atm})$	K_3/K_4	3He/4He measured (Ra)
250	53.2	898	3.01	2796	1898	0.6693	0.6788	5.23×10^{11}	1.1520	0.1218
250	53.1	609.8	2.07	1898	1288	0.6696	0.6787	5.24×10^{11}	1.1569	0.1153
250	53.1	412.2	1.42	1288	875.7	0.6696	0.6799	5.21×10^{11}	1.1231	0.1059
250	53.1	278.8	0.995	875.7	596.9	0.6697	0.6817	5.18×10^{11}	1.0778	0.0978
250	52.9	187.7	0.689	596.9	409.2	0.6709	0.6855	5.13×10^{11}	1.0868	0.0955
250	53.1	128.8	0.486	409.2	280.4	0.6697	0.6852	5.11×10^{11}	1.0411	0.0893
250	159.9	215.4	0.953	280.4	64.98	0.2990	0.2318	6.56×10^{11}	1.0452	0.0867
250	52.9	41.86	0.205	64.98	23.12	0.6706	0.3558	1.40×10^{12}	1.1246	0.0849
High T	52.2	23.12		23.12	0					0.0700

Table 1.7: Distilled 3He isothermal permeation data for the silica glass ampoule containing distilled water, equilibrated with high-purity tank helium, measured on the Helix-MC-Plus^{10K} multi-collector mass spectrometer ("Encapsulated Water, Helix MC" in the ampoule dimensions table of the main text, Table 1.1). Predicted n_t/n_0 values marked '–' correspond to steps for which the ampoule had not yet reached a diffusive steady state concentration profile (see main text).

Temp (°C)	Hot Time (min)	Cold Time (min)	m 3He (pccSTP)	$\pm 2\sigma$ (pccSTP)	n_0 3He (pccSTP)	n_t 3He (pccSTP)	n_t/n_0 predicted	n_t/n_0 measured
100	60	26	0.116	1.22×10^{-3}	0.6447	0.5289	–	0.8204
100	60	25.1	0.0942	1.00×10^{-3}	0.5289	0.4347	0.8677	0.8219
100	60	25	0.0757	8.06×10^{-4}	0.4347	0.359	0.8678	0.8259
100	60	25.1	0.0578	6.18×10^{-4}	0.359	0.3012	0.8678	0.8389
100	60	53.5	0.053	5.72×10^{-4}	0.3012	0.2482	0.8589	0.8241
100	60	25.2	0.0382	4.34×10^{-4}	0.2482	0.21	0.8678	0.8459
100	60	25.1	0.0337	4.06×10^{-4}	0.21	0.1763	0.8678	0.8396
100	60	795.6	0.0493	2.52×10^{-4}	0.1763	0.127	0.7741	0.7201
100	60	28.1	0.0197	9.42×10^{-5}	0.127	0.1072	0.8666	0.8446
100	60	43.7	0.0173	8.39×10^{-5}	0.1072	0.08988	0.8622	0.8382
100	60	24.9	0.0138	6.99×10^{-5}	0.08988	0.0761	0.8673	0.8467
100	60	24.9	0.0132	7.60×10^{-5}	0.0761	0.06295	0.8678	0.8272
100	60	25	0.00977	6.32×10^{-5}	0.06295	0.05318	0.8678	0.8448
100	60	25.1	0.00952	6.36×10^{-5}	0.05318	0.04366	0.8679	0.8210
100	60	25.1	0.00764	5.16×10^{-5}	0.04366	0.03602	0.8678	0.8250
100	60	25	0.00572	3.86×10^{-5}	0.03602	0.0303	0.8678	0.8413
100	60	25	0.00578	3.90×10^{-5}	0.0303	0.02451	0.8678	0.8091
100	60	25.1	0.00363	2.45×10^{-5}	0.02451	0.02089	0.8679	0.8520
REST			0.0209	1.39×10^{-4}	0.02089	0		

Table 1.8: Distilled 4He isothermal permeation data for the same encapsulated-water ampoule experiment as Table 1.7.

Temp (°C)	Hot Time (min)	Cold Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	3He/4He measured (Ra)
100	60	26	604	1.77	3871	3267	–	0.8440	0.1379
100	60	25.1	515.5	1.52	3267	2751	0.8772	0.8422	0.1315
100	60	25	434.8	1.29	2751	2317	0.8772	0.8420	0.1252
100	60	25.1	361.9	1.07	2317	1955	0.8772	0.8438	0.1150
100	60	53.5	325.4	0.965	1955	1629	0.8688	0.8335	0.1171
100	60	25.2	251.3	0.745	1629	1378	0.8772	0.8458	0.1095
100	60	25.1	212.6	0.63	1378	1165	0.8772	0.8457	0.1140
100	60	795.6	344.8	0.802	1165	820.7	0.7858	0.7042	0.1029
100	60	28.1	127.2	0.276	820.7	693.5	0.8761	0.8450	0.1116
100	60	43.7	111.3	0.222	693.5	582.2	0.8719	0.8395	0.1122
100	60	24.9	88.54	0.166	582.2	493.7	0.8768	0.8479	0.1120
100	60	24.9	74.96	0.121	493.7	418.7	0.8773	0.8481	0.1262
100	60	25	62.69	8.84×10^{-2}	418.7	356	0.8772	0.8503	0.1121
100	60	25.1	53.28	7.18×10^{-2}	356	302.7	0.8773	0.8503	0.1285
100	60	25.1	45.45	6.11×10^{-2}	302.7	257.3	0.8772	0.8499	0.1209
100	60	25	38.41	5.30×10^{-2}	257.3	218.9	0.8773	0.8507	0.1071
100	60	25	32.42	4.81×10^{-2}	218.9	186.4	0.8772	0.8519	0.1284
100	60	25.1	27.46	4.35×10^{-2}	186.4	159	0.8773	0.8527	0.0950
REST			159	0.258	159	0			0.0945

Table 1.9: Distilled 3He isothermal permeation data for the silica glass ampoule containing fused sulfur, equilibrated with high-purity tank helium, measured on the Helix-MC-Plus^{10K} multi-collector mass spectrometer ("Encapsulated Sulfur, Pfeiffer Quad" in the ampoule dimensions table of the main text, Table 1.1; note this isothermal calibration series was run on the Helix-MC-Plus^{10K}).

Temp (°C)	Hot Time (min)	Cold Time (min)	m 3He (pccSTP)	$\pm 2\sigma$ (pccSTP)	n_0 3He (pccSTP)	n_t 3He (pccSTP)	n_t/n_0 predicted	n_t/n_0 measured
80	60	25.6	0.699	2.78×10^{-2}	9.569	8.87	0.9299	0.9270
80	60	61.3	0.736	2.89×10^{-2}	8.87	8.134	0.9222	0.9170
80	60	25.1	0.591	2.17×10^{-2}	8.134	7.543	0.9300	0.9274
80	60	25	0.544	1.83×10^{-2}	7.543	6.999	0.9301	0.9279
80	60	25	0.494	1.60×10^{-2}	6.999	6.505	0.9301	0.9294
100	60	91.8	0.759	2.42×10^{-2}	6.505	5.746	0.8831	0.8834
100	60	25.9	0.601	2.01×10^{-2}	5.746	5.145	0.8966	0.8955
100	60	52.0	0.575	2.06×10^{-2}	5.145	4.57	0.8912	0.8882
100	60	26	0.463	1.72×10^{-2}	4.57	4.108	0.8966	0.8988
100	60	89.9	0.498	2.04×10^{-2}	4.108	3.609	0.8834	0.8787
120	60	24.8	0.51	2.27×10^{-2}	3.609	3.099	0.8537	0.8586
120	60	52.8	0.457	2.18×10^{-2}	3.099	2.641	0.8481	0.8524
120	60	24.9	0.356	1.76×10^{-2}	2.641	2.286	0.8537	0.8653
120	60	61.3	0.318	1.56×10^{-2}	2.286	1.967	0.8460	0.8607
120	60	25.1	0.243	1.17×10^{-2}	1.967	1.724	0.8537	0.8766
150	60	25.0	0.42	2.00×10^{-2}	1.724	1.304	0.7695	0.7562
150	60	25.0	0.283	1.32×10^{-2}	1.304	1.021	0.7696	0.7828
150	60	690.8	0.351	1.03×10^{-2}	1.021	0.6699	0.6598	0.6563
150	60	25	0.147	3.98×10^{-3}	0.6699	0.5226	0.7696	0.7802
150	60	53.3	0.125	3.33×10^{-3}	0.5226	0.3972	0.7643	0.7601
150	60	25.0	0.0893	2.40×10^{-3}	0.3972	0.308	0.7695	0.7752
remainder			0.308	8.48×10^{-3}				

Table 1.10: Distilled 4He isothermal permeation data for the same encapsulated-sulfur ampoule experiment as Table 1.9.

Temp (°C)	Hot Time (min)	Cold Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	3He/4He measured (Ra)
80	60	25.6	3354	10.8	52330	48980	0.9348	0.9359	0.1499
80	60	61.3	3617	12	48980	45360	0.9276	0.9262	0.1464
80	60	25.1	2949	10	45360	42410	0.9349	0.9350	0.1441
80	60	25	2760	9.6	42410	39650	0.9350	0.9349	0.1419
80	60	25	2587	9.13	39650	37060	0.9350	0.9348	0.1373
100	60	91.8	3980	15.3	37060	33080	0.8913	0.8926	0.1371
100	60	25.9	3088	13.5	33080	30000	0.9041	0.9066	0.1399
100	60	52.0	2978	14.7	30000	27020	0.8990	0.9007	0.1389
100	60	26	2517	13.1	27020	24500	0.9041	0.9068	0.1322
100	60	89.9	2687	14.4	24500	21810	0.8916	0.8903	0.1334
120	60	24.8	2769	14.6	21810	19040	0.8646	0.8731	0.1327
120	60	52.8	2551	13	19040	16490	0.8592	0.8661	0.1290
120	60	24.9	2097	10.4	16490	14400	0.8646	0.8728	0.1221
120	60	61.3	1948	9.38	14400	12450	0.8572	0.8647	0.1176
120	60	25.1	1573	7.36	12450	10870	0.8645	0.8736	0.1110
150	60	25.0	2435	11.3	10870	8439	0.7868	0.7761	0.1242
150	60	25.0	1806	8.2	8439	6633	0.7868	0.7859	0.1128
150	60	690.8	2151	8.5	6633	4481	0.6807	0.6756	0.1173
150	60	25	924.6	3.42	4481	3557	0.7868	0.7937	0.1146
150	60	53.3	753.9	2.75	3557	2803	0.7818	0.7880	0.1197
150	60	25.0	560.4	2.08	2803	2242	0.7868	0.8001	0.1146
remainder			2242	8.72					0.0988

Chapter 2

HYDROTHERMAL HELIUM STEP HEATS ON GOETHITE REVEAL HIGH RETENTIVITY AND MULTIPLE DIFFUSION DOMAIN SYSTEMATICS

U-Th/He geochronology of Goethite (FeOOH) is a powerful method to quantitatively date weathering and erosion processes on Earth's surface; however, precise interpretation of goethite U-Th/He dates is challenging because uncertainty persists regarding the mineralogical and environmental conditions under which radiogenic helium is retained and how it may be lost over time. Standard vacuum step heating experiments have been successfully used on a variety of mineral phases to characterize helium migration, but the results of such experiments on goethite have been dominated by dehydration-induced helium loss, raising questions about their relevance to helium retention and loss in nature (Farley, H. B. Monteiro, et al., 2024). Observations that thermal decomposition of goethite is greatly slowed in the presence of high water pressure led us to develop a new method to conduct helium step heat experiments under hydrothermal conditions (up to 86 bars and 300°C) within standard experimental setups. Using this method, we report results from the first helium step heat experiment of proton-irradiated goethite under hydrothermal conditions. Irradiated goethite sample Win-06-01b of the Flinders Ranges, Southern Australia and distilled water were encapsulated in a pure silica glass ampoule and were subjected to multiple prograde and retrograde thermal cycles over a temperature range of 170°C to 250°C under water vapor saturated conditions. Step durations varied from 2 hours to over a week, with low-temperature (<120°C) permeation steps bracketing higher-temperature steps to account for residual helium within the ampoule. Because 4He yields from those permeation steps were at blank level, permeation lag was removed by the bounding method, which constrains the ampoule's residual helium from above using the rate constant measured during the experiment itself; the permeation step method, applied to 3He alone,

reproduces the same result. The results show apparent helium diffusivities approximately seven log units lower than those observed in vacuum, with apparent diffusivities following a consistent linear array on an Arrhenius plot for cumulative fractional 3He loss exceeding 7%. Delta plots generated using this reference array with a calculated activation energy of 157.1± 5.3kJ/mol are consistent with a volume-diffusion helium loss mechanism with MDD (Multiple Diffusion Domain) systematics. These results show high helium retentivity in natural goethite under hydrothermal conditions that extrapolate to robust helium retention at surface conditions even over billion-year timescales. The method presented in this work can be systematically applied to experimentally investigate effects of crystallite size, chemical composition, and radiation damage to refine these results and to assess the applicability of existing vacuum step heat data to helium retention and loss in goethite in nature.

2.1 Introduction

Goethite (FeOOH) crystallization is datable only by U-Th/He geochronology (Shuster, P. M. Vasconcelos, et al., 2005; Farley, H. B. Monteiro, et al., 2024). Goethite is a powerful chronometer for dating surface processes such as weathering and ferruginization directly: laterite and duricrust formation and recrystallization (H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Allard et al., 2018; Ansart et al., 2022), and the ferruginization of organic matter and paleosols (Heim et al., 2006; Yapp and Shuster, 2011; P. M. Vasconcelos et al., 2013; Danišík et al., 2013; Hofmann, Reichenbacher, et al., 2017).

Interpreting a goethite U-Th/He date requires knowing the conditions under which the mineral retains radiogenic helium. As with any radiochronometer, meaningful geological interpretation of a measured age depends on knowing when the system is open to loss or gain of parent or daughter nuclides. For goethite, step heats of proton-irradiated, crystalline samples are used to reveal the degree of partial helium loss by the 4He/3He method (Shuster, P. M. Vasconcelos, et al., 2005), and suggest that helium loss fractions in nature are always non-zero but modest, ranging from 2 to 30% (Shuster, P. M. Vasconcelos, et al., 2005; Heim et al., 2006; Hofmann, Reichenbacher, et al., 2017; Deng et al., 2017). Although one study attempts to explain the variable loss principally as a result of variable radiation damage (Bassal et al., 2022), the

effects of temperature, crystallite size, and chemical composition have not been determined experimentally and remain unresolved, because goethite decomposes to hematite (Fe_2O_3) in the step heat experiments used to characterize it (Farley, H. B. Monteiro, et al., 2024).

Laboratory experiments of helium migration under controlled conditions have proven extremely useful in understanding the mechanisms and rates of helium migration in geological materials and have produced sensible results that are predominantly explainable by thermally activated volume diffusion (Reiners and Farley, 1997; Farley, 2000; Farley, 2007; Cherniak and Watson, 2011; Farley, 2018), albeit variably complicated by the effects of radiation damage and annealing (Shuster, Flowers, et al., 2006; Guenthner et al., 2013). This approach has also been leveraged for goethite; however, some complications arise when applying this approach to goethite. Above surface temperatures, goethite decomposes to hematite and water, often along crystallite elongation directions (Goss, 1987). Farley, H. B. Monteiro, et al. (2024) measured evolved water by capacitance manometry during vacuum step heats and found helium and water release to be tightly correlated even below 200°C, indicating that decomposition, rather than helium migration through an unchanging crystal, governs the release. Vacuum step-heat results for goethite therefore need not describe helium behavior in nature.

Still, existing helium step heat measurements of goethite in vacuum likely remain informative to an extent, as goethite dehydration kinetics are highly influenced by some of the same factors that could be responsible for variable helium retention in natural goethite, like variable grain size (Navrotsky et al., 2008; Guo and Barnard, 2011; Goss, 1987), crystallinity (Schwertmann et al., 1985) or substitution of aluminum or other cations within the goethite structure (Majzlan et al., 2003; Bassal et al., 2022). For example, Shuster et al. (2005) initially conducted vacuum step-heating experiments on two vitreous, polycrystalline goethite samples, in which uniformly distributed 3He was introduced by proton irradiation for comparison with radiogenic 4He that was partially lost in nature. The results of Shuster, P. M. Vasconcelos, et al. (2005) could be well-fit with a model of two distinct diffusion domains or migration mechanisms with significant differences in helium retention under surface conditions, e.g. crystallite interiors and intergranular space. Although these results may have been impacted by dehydration in vacuum, certain aspects

may remain applicable to helium loss in goethite in nature if helium loss in nature and the laboratory advance through the samples with similar geometries. One important point is that if laboratory experiments degas domains in order of their helium retentivity in nature, even if the rates of loss are affected by goethite dehydration, then a $^4He/^3He$ experiment in vacuum can still be used to measure helium loss (Shuster, P. M. Vasconcelos, et al., 2005; P. M. Vasconcelos et al., 2013).

An additional complexity is that natural goethite almost always occurs as aggregates of minute crystallites with cross-sections smaller than 1 µm in nature. Neglecting to account for multiple diffusion domains can lead to inaccurate determinations of activation energy (E_a) and frequency factor (D_o/a^2) for helium diffusion (Farley, 2018). Detailed step-heating with temperature cycling, as has been done for hematite (Farley and Flowers, 2012; Farley, 2018), is needed to characterize domain size and activation energy for polycrystalline samples.

Molecular simulations indicate that helium diffusion in perfectly crystalline goethite should be highly anisotropic, essentially reducing to 1-dimensional diffusion along the b-axis (typically the long axis of the polycrystalline aggregate) that would amount to near-instantaneous helium escape from goethite (Bassal et al., 2022), which is clearly disputed by observed (U–Th)/He dates as old as ~275 Ma (Waltenberg, 2012) and dates that are concordant with $^{40}Ar/^{39}Ar$ ages of other phases (P. M. Vasconcelos et al., 2013). However, the models presented by Bassal et al. (2022) suggest that crystallographic defects caused by radiation damage could block these channels, thereby reducing He loss. While these simulations are informative, they alone are insufficient for drawing conclusions about the mineralogical and environmental conditions needed for accurately dating the crystallization of goethite with the (U–Th)/He method and require experimental verification.

Goethite dehydration is suppressed at high water pressure, so a step heat conducted under hydrothermal conditions may interrogate helium loss rather than decomposition kinetics. Until now, no studies have reported helium step-heating results for goethite under hydrothermal (high pH_2O) conditions, which could help stabilize goethite or slow its decomposition to illuminate the processes relevant to helium loss in natural settings. If experimental conditions can be controlled to slow the decomposition of goethite such that the rate of natural helium loss in goethite greatly exceeds the rate of goethite

decomposition into hematite and water, it would open up an experimental approach to systematically assess how factors like crystallite size, temperature, radiation damage, and chemical composition could affect helium mobility and retentivity in nature, enabling estimates of helium retention in previously untested natural samples.

2.2 Experimental Methods

This study builds upon previous helium step heat studies of the dense colloform goethite sample Win-06-01b (occasionally referred to as Win0601b) collected within float at the Flinders Ranges in Southern Australia (Waltenberg, 2012; H. Monteiro, 2017). It yields one of the oldest known goethite U-Th/He ages: 180-135 Ma (Waltenberg, 2012). To produce a homogeneous distribution of 3He, Win-06-01b was irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following standard procedures (Shuster and Farley, 2005) and subsequently was subjected to helium step heat measurements in vacuum by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024). In their step heat experiment of Win-06-01b, Farley, H. B. Monteiro, et al. (2024) additionally measured evolved water using capacitance manometry to show that helium and water release were tightly correlated for goethite step heats in vacuum. Although they used subtly different heating schedules, the results of Farley, H. B. Monteiro, et al. (2024) clearly agree well with H. Monteiro (2017).

In this study, an aliquot of the same irradiated Win-06-01b goethite used by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) was encapsulated with MilliQ water within a silica glass tube. Silica glass was chosen over other laboratory glasses such as borosilicate because it is the most permeable glass type to helium (Norton, 1953; Altemose, 1961). To prepare the silica glass tube, 10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thinner pipettes using a natural gas and oxygen glass-welding torch and the pipette ends were fused closed. After the glass had cooled, 1.18 mg of irradiated Win-06-01b goethite blades less than 2mm in length were carefully transferred to the bottom of the tube in the presence of an ionizing fan. Less than a microliter of distilled water was transferred to the sealed pipette using a thin glass capillary, submerging

the goethite, after which the tube was gently tapped to liberate the bubbles and ensure that the water and goethite were efficiently packed at the bottom of the pipette. To seal the pipette, the contents of the ampoule were flash-frozen in liquid nitrogen and transferred to a saltwater-ice bath at a temperature of -10°C. While the pipette end of the tube remained frozen in the saltwater-ice bath, the top was interfaced with a vacuum line and the tube was pumped to a pressure of <0.1 torr. The advantage of this approach was that water would remain solid within the tube, only slowly sublimating to vacuum while atmospheric gasses would be readily pumped away. While it was desirable to remove atmospheric gasses to reduce experimental complexity by removing potentially reactive gasses like oxygen, the primary motivation for pumping the tube to rough vacuum was to aid the sealing process. When the tube contents were under vacuum, fused glass would deform inwards and self-seal, greatly improving the ease and quality of the seal while requiring far less heat input from the torch. After an acceptably low pressure was attained, the tube was closed to the vacuum line and the saltwater-ice bath was quickly replaced with a liquid nitrogen bath, ensuring that no ice was melted. The liquid nitrogen bath was preferable during sealing as it more effectively protected the goethite and ice from the high temperatures of the glass-welding torch. A fine-flamed welding torch was used to seal the top of the ampoule, which dropped into the liquid nitrogen bath and was retrieved by passing the liquid nitrogen through a sieve. Fig. 2.1 shows the sealed ampoule after the completion of the step heats with all of the original water remaining, a testament to the quality of the ampoule's seal, which had sustained a massive pressure gradient of over 11 orders of magnitude across the ampoule walls, with an inner vapor pressure of about $3x10^4$ torr (~40 bars: Haar et al. (1984)) against the ~10^{-7} torr ultra-high vacuum of the lamp cell while the ampoule was held at 250°C for over a week.

Figure 2.1: A view of the entire experimental ampoule after the completion of the step heat experiment before it was broken for the final step in the resistance furnace. At room temperature, the tube is approximately half-filled with gas, one-third filled with liquid water, and one-sixth filled with <2 mm long shards of irradiated Win-06-01b goethite. A gel-like substance was observed lining the inner walls of the tube, which is likely to be amorphous silica dissolved from the tube and redeposited by the thermally-cycled, high pressure liquid water phase.

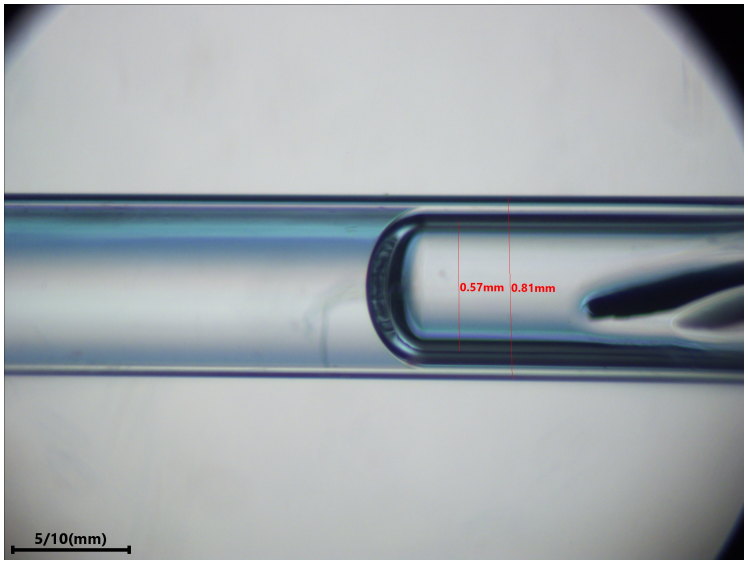

Figure 2.2: Measurements of the inner and outer diameter of the ampoule used in the hydrothermal step heats of Win-06-01b. The inner walls of the ampoule are only clearly visible within air bubbles.

The sealed ampoule, now referred to as HT-Win-06-01b, was placed in a copper foil packet and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in more detail by Farley, Reiners, et al. (1999). Before the experiment began, the ampoule was loaded in the lamp cell, pumped to high vacuum, and was held at 70°C for 4 days to degas atmospheric helium entrained within the ampoule.

HT-Win-06-01b was subjected to multiple prograde-retrograde heating cycles to investigate possible MDD behavior. Step durations varied from 2 hours to about a week; the total experiment was conducted over about 40 days and included experimental steps ranging from 170-250°C. We elected not to heat the sample hotter because high temperatures generated very high vapor pressures. In early trials, we found that similar ampoules most often withstood the ~86 bar vapor pressures associated with liquid water saturation at 300°C (Haar et al., 1984), but they would explode when further heated to temperatures beyond 315°C.

Experimental temperature records show ~1°C amplitude oscillations around the target temperature. While we cannot exclude the possibility that sustained temperature gradients were present throughout the sample due to the use of the comparatively large ~1 mm thick ampoule, we are confident that the conclusions of this work are robust. The ampoule was in direct contact with the thermocouple junction and was in front of thermocouple junction with respect to the projector lamp heating source, suggesting that if there were a temperature discrepancy, the sample would've been hotter than the recorded temperature, which wouldn't affect the conclusions of this work. Additionally, during early trials of this method for helium step heats, silica-glass encapsulated Durango apatite had been subjected to step heats at Caltech and gave indistinguishable corrected 4He yields to unencapsulated Durango apatite, suggesting that the temperature recorded by the thermocouple for our experimental setup is sufficiently accurate.

Helium isotope measurements of evolved gasses from hydrothermal steps heats of glass-encapsulated irradiated HT-Win-06-01b goethite were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from January 31st, 2024 to March 13th, 2024. Helium was purified by exposing evolved gasses to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding

into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. The majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech have shown this approach to be reliable, reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray, which was never significant during the experiment. Measurements of high 3He MM gas standard were analyzed every 8 hours (totaling 116 measurements) to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility.

A cold blank was measured at the beginning of the experiment, showing a negligible 3He blank essentially at 0 cps (technically this measurement yielded a slightly negative intercept to time 0) and a 4He blank signal corresponding to about 1.25 pccSTP, which was similar to 4He values attained during low temperature permeation step measurements. Permeation steps were thus assumed to measure shifting 4He blanks during the experiment, and the entire blank correction was added to all 4He uncertainties.

For the final step, the HT-Win-06-01b ampoule was broken and dried overnight in a 50°C oven, then completely degassed by heating to ~1100 °C in a resistance furnace. The final step measurement was conducted on a Thermo Fisher Helix SFT mass spectrometer. A hot furnace blank subjected to the same volume splits as the final HT-Win-06-01b measurement gave a 3He blank of 6.8 fccSTP and a 4He blank of 0.31 nccSTP. This blank correction was less than 1% of the measured 4He and negligible for 3He. A re-extract of HT-Win-06-01b immediately following the measurement yielded above-blank 3He and 4He signals, which were added to the total measurement, although both constituted less than 1% of the final step yield. Sensitivity was calculated by measuring a 3He-enriched air standard with known 3He and 4He contents immediately before the measurement.

Permeation Lag Correction Because HT-Win-06-01b was encapsulated in an imperfectly helium-transmissive silica glass ampoule, permeation lag was an

expected complication in these measurements. To address this, experimental steps at temperatures at or above 170°C were bracketed by low temperature (<120°C) steps where the helium in goethite at high pH2O was effectively immobile at experimental timescales, following the permeation step method (Chapter 1, §1.4). For 3He, measured signals at these low temperature permeation steps were entirely controlled by the process of permeation through silica glass, and therefore could be used to estimate the amount of helium remaining within the ampoule between high-temperature experimental steps. However, for 4He, the measured signals at low temperature permeation steps were most often indistinguishable from the blank, so Eq. (1.6) (Chapter 1) could not be applied to 4He and the permeation step method could not be carried through for that isotope.

As a result, permeation lag is instead addressed using the bounding method described in Chapter 1 (§1.4), which constrains the ampoule's residual helium from above rather than measuring it. In this approach, an upper bound constraint for the helium content of the ampoule at the end of each high-temperature step was calculated using Eq. (1.10) (Chapter 1). 3He was reduced this way as well, for consistency with 4He; because 3He can also be reduced by the permeation step method, the two reductions can be compared directly, and that comparison is used below (§2.5) as an internal test of the bounding method.

Both routes require the ampoule's permeation rate constant λ, which was measured during the experiment itself rather than assumed from the literature (Chapter 1, §1.4). This in-experiment measurement is not an alternative method of handling permeation lag but a prerequisite of the permeation step method and, here, the empirical basis for choosing a conservative λ for the bounding method. It is indispensable in this experiment: the permeability of ampoules used in hydrothermal step heats cannot reasonably be assumed to match literature values for anhydrous silica glass.

10 series of 3-4 consecutive permeation steps held at a single temperature between 90-110°C for 120-180 minutes each with 26 minutes of cooling time before being inlet into the Helix-MC-Plus10k were identified as suitable for this exercise. The 3He measurements in these series were fit to Eq. (1.9) using a non-linear least squares optimization (Fig. 2.3). This is accomplished with the Levenberg-Marquardt algorithm implemented in the curve_fit function

in the SciPy package of python (Virtanen et al., 2020), weighted by each measurement's own uncertainty. One additional series that otherwise qualified was not well fit by this model and was not used further for clarity, though its fit value of λ was within uncertainty of the other series' fit values for λ.

An initial guess of $\lambda \approx 6\text{x}10^{-6}$ sec^{-1} and $n_0 \approx \frac{m_i}{1-e^{-\lambda t_i}}$ (evaluated at the first measurement of a series) pccSTP was passed to the optimizer for each series. Uncertainties from the model fits were calculated using the covariance matrices generated for each of the model fits. Although the initial helium amount n_0 was a free parameter in the optimizations, the rate constant of permeation (λ) was the only parameter of interest for this exercise. The fitted rate constants span 2.7×10^{-5} to 6.3×10^{-5} s^{-1}, with 2σ uncertainties of 0.08 to 0.46 log units and no trend resolvable against step duration:

All measured apparent 3He permeabilities were equal to or in excess of the literature values for anhydrous silica glass (Fig. 2.4), similar to observations discussed in Chapter 1 (§1.7). 4He permeability itself could not be independently measured in this experiment due to blank-level signals during the low temperature permeation steps and is assumed to similarly exceed the literature values for anhydrous silica glass. To evaluate the uncertainty due to the permeation lag, we use the literature values as a conservative (lower bound) estimate of the true permeability; a λ that underestimates the ampoule's true rate constant yields a conservative (higher) upper bound on the residual helium (Chapter 1, §1.4). λt (where $\lambda = GRTK$) is estimated here using the 4He permeability K reported by Swets et al. (1961), while the 3He permeability is taken to be the product of the 4He permeability and the relative 3He/4He permeability reported by Jones (1953). The geometric constant G is calculated using the measured inner and outer diameter of the ampoule (Fig. 2.2).

The first three experimental steps had values of λt lower than 4, and therefore the upper bound of the helium content of these steps could not be determined. These measurements are omitted from further analysis. Later experimental steps with values of λt lower than 4 could be determined using Eq. (1.57) from the supplement to Chapter 1 because upper bounds on previous steps' residual helium content could be evaluated.

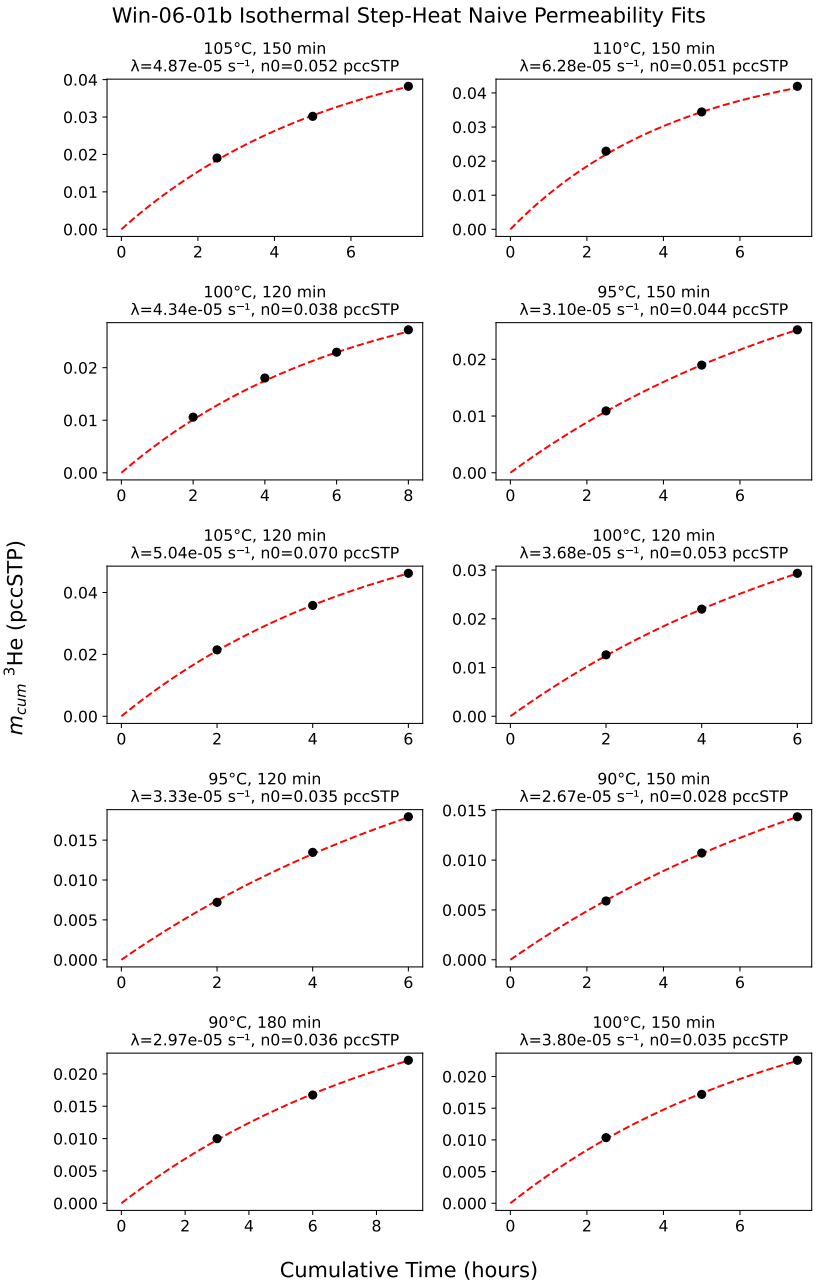

Figure 2.3: Cumulative yield against cumulative time for helium-3 yields for isothermal permeation steps. A non-linear least squares optimization fits λ and n_0 of Eq. (1.9) (Chapter 1) to data from a series of permeation steps like these. The permeability K is calculated according to the relation $K = \lambda/GRT$ and is plotted as a function of temperature in Fig. 2.4. The curvature of each series is what conditions the fit: a series in which the permeation rate has not yet slowed appreciably constrains λ and n_0 only weakly.

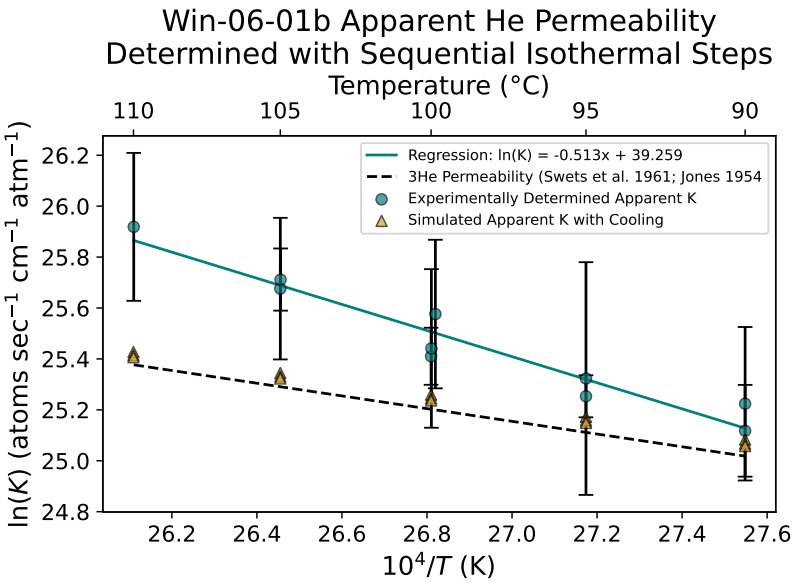

Figure 2.4: Arrhenius plot of apparent permeation coefficients as a function of inverse temperature for HT-Win-06-01b, converted from the rate constants fitted in Fig. 2.3 via $K = \lambda/GRT$. The activation energy is higher than anhydrous silica glass, as expected (Masaryk, 1968; Shelby, 1972). It is unexpected that the glass ampoule of HT-Win-06-01b has higher apparent 3He permeability than the literature values of Swets et al. (1961) and Jones (1953), but this can be explained by an increase in the partial pressure of the gas phase within the tube due to displacement by high He diffusivity, low He solubility water. Yellow points labeled "Diffusion Model with cooling, Anhydrous" were produced to evaluate the expected excess apparent permeability due to the 25 minute cooldown period before measurement. This is accomplished by simulating the measured apparent permeability for a dry silica glass ampoule that continued to permeate helium while cooling for an additional 26 minutes before measurement after being held at 90–110°C for 120–180 minutes, similarly to the measured values. Variability in these points is due to differences in the length of time the ampoule was heated but is very small. They show a near constant offset from the expected values of Swets et al. (1961) and Jones (1953) that is much smaller than the uncertainty in the regression of the fit values determined experimentally in teal.

2.3 Analytical Methods

Finite Difference Diffusion Model of Cooling Tube One potential concern about the results of the internally measured permeabilities is that some additional helium was permeated during the cooling period after each permeation step. In order to evaluate the significance of this effect, a finite difference model was built in python to simulate diffusion, influx, and outflux of helium across a hollow glass cylinder containing an arbitrary amount of helium gas within a helium-free chamber using the procedure described in Chapter 1 (§1.5). The glass cylinder was given the same dimensions of the ampoule used in the Win-06-01b hydrothermal experiment, and was subjected to the same heating and cooling schedule as the hydrothermal experiment's permeation steps. The tube was charged with 100 arbitrary units of helium atoms at the end of each experimental step and a steady state concentration profile was assumed for the new arbitrary charge at T = 24°. Permeability values for 3He were assumed as the product of the 4He permeability values given by Swets et al. (1961) and the 3He/4He relative permeabilities given by Jones (1953).

To simulate cooling, a cooling rate had to be assumed. To most accurately simulate the cooling of the experimental ampoule, Watlow temperature records of the experimental ampoule during cooling steps were used to calculate the cooling rate dT/dt for discrete time intervals. A quadratic function was fit to these data points to model the cooling rate as a function of temperature with a single root at T = 24°C such that the temperature would stabilize at T = 24°C by design.

$$\frac{dT}{dt} = -\left(9.23081 \times 10^{-6}\right)(T-24)^2 - \left(7.74304 \times 10^{-4}\right)(T-24)$$

(dT/dt in °C/s, T in °C; $R^2 = 0.9633$).

Helium-in-Goethite Diffusivity Inversions Apparent values of D/a^2 for 3He and 4He were calculated from times and cumulative measured yields using the equations of Fechtig and Kalbitzer (1966). Lower and upper bounds for diffusivity were calculated using the upper bound for the helium contents of the ampoules before and after each step, evaluated using (1.10). Diffusivity for step i is a function of the cumulative fractional yield for step i $\Sigma^i F$ and the cumulative fractional yield of the previous step $\Sigma^{i-1} F$. The diffusivity is maximized when the difference between the fractional yields is largest, and minimized when the difference is smallest. This difference is maximized for

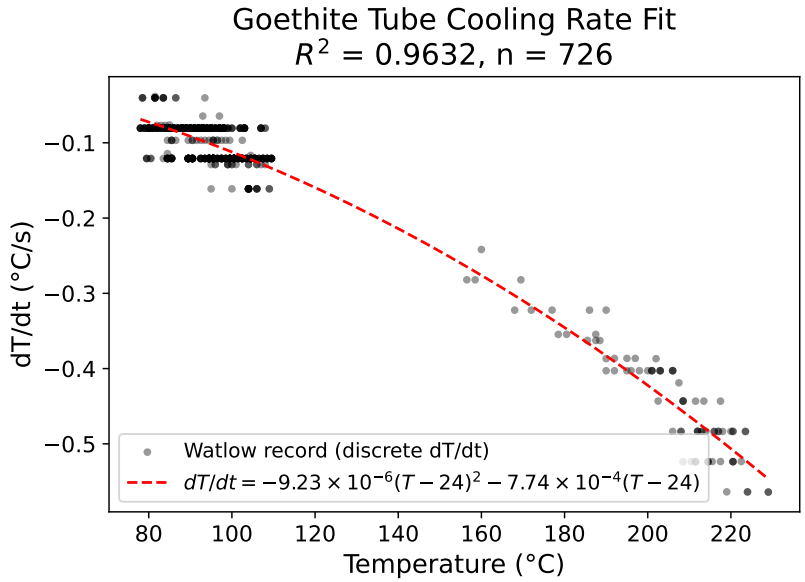

Figure 2.5: Discrete cooling rate (dT/dt) values calculated from the Watlow temperature record for the Winsor Goethite experiment, plotted against temperature, with the quadratic fit used to simulate cooling in the finite difference diffusion model overlain.

the extreme case where the previous step ended with no permeation lag, and the current step ended with the maximum amount of permeation lag (${}^{i}_{SS}n_t$), whereas the opposite is true for the minimum diffusivity.

$$\begin{aligned} {}^{UB}D_\omega &= D\Big(\Sigma^{i-1}F_m,\ \Sigma^{i}F_m + {}^{i}_{SS}n_t\Big) \\ {}^{LB}D_\omega &= D\Big(\Sigma^{i-1}F_m + {}^{i-1}_{SS}n_t,\ \Sigma^{i}F_m\Big) \end{aligned} \tag{2.1}$$

For the 3He diffusivities, a linear regression was fit to the steps with cumulative 3He fractional yield ≥7% (12 steps, Fig. 2.7), yielding an R^2 statistic of 0.989. The fit's slope and intercept are used as a reference line to calculate "Delta" values for the diffusivities of 3He and 4He (Fig. 2.8). No uncertainty on the total amounts of 3He and 4He is propagated into the diffusivities because the experiment operated at low cumulative yields where this error affects all diffusivities approximately homogeneously, so it is only relevant for the y-intercept of the regression and the corresponding calculated closure temperature. The uncertainty from the intercept and slope parameters of the linear regression is significantly larger than the uncertainty related to the total yield, so it is neglected.

Helium Isotope Ratio Uncertainty for Step Heats The upper and lower bounds of the $\frac{^4He}{^3He}$ isotope ratios for each heating step are taken as the maximum and minimum $\frac{^4He}{^3He}$ isotope ratio calculated out of the 4 hypothetical end-member permeation scenarios, which would have been experienced by both helium isotopes systems:

1. $n_t = 0,\ n_0 = 0$
2. $n_t = n_t^{\mathrm{UB}},\ n_0 = 0$
3. $n_t = n_t^{\mathrm{UB}},\ n_0 = n_0^{\mathrm{UB}}$
4. $n_t = 0,\ n_0 = n_0^{\mathrm{UB}}$

Intuitively, higher differential permeability between the two helium isotopes ($\delta = \frac{K_{3He}}{K_{4He}}$ here) will result in greater isotope fractionation between the measured $\left(\frac{m_4}{m_3}\right)$ and true $\left(\frac{\omega_4}{\omega_3}\right)$. Furthermore, lower absolute permeabilities will result in stronger permeation lag, additionally producing higher isotope fractionation in the measured value. The 4He permeability is thus conservatively taken to be identical to Swets et al. (1961), an anhydrous literature value which this experiment's own isothermal 3He permeation measurements show to underestimate the ampoule's true permeability (Fig. 2.4); 4He permeability itself could not be independently measured in this experiment. For the purposes of evaluating the measured isotope ratios, the relative 3He/4He permeability (δ) is taken to be 1.29, the 2σ upper bound of the value measured independently for a water-filled ampoule in Chapter 1 (§1.6): 1.10 ± 0.19 (2σ). 1.29 is in great excess of realistic values of the relative 3He/4He permeability, which is expected to be between 1.06-1.11 for the temperature range relevant to this experiment (Jones, 1953). These conservative assumptions taken together guarantee that the calculated upper and lower bounds for the isotope ratios accurately constrain the true value. The upper bounds of the ampoule contents are calculated using the steady-state approximation of Eq. (1.10) (Chapter 1): $n_{ss} = \frac{m}{\lambda t - 1}$. For the four extreme cases listed above, the following equations apply:

1. $\dfrac{\omega_4}{\omega_3} = \dfrac{m_4}{m_3}$

2. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1}}$

3. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1} - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1} - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$

4. $\frac{\omega_4}{\omega_3} = \frac{m_4 - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$

Where m_x^{i-1} and $(\lambda t)^{i-1}$ refer to the measured helium content and λt of the previous step, respectively. λ here applies explicitly to 4He, whereas $\delta\lambda$ applies to 3He. The upper and lower bounds of the isotope ratios are then taken to be the maximum and minimum of these four extreme cases.

SEM Study of Thermally Treated Polycrystalline Goethite In early trials of the "glass tube method" for helium step heat experiments with controlled microenvironments presented here, sub-samples of goethite morphologically similar to Win-06-01b from Igarapé Bahia (Shuster, P. M. Vasconcelos, et al., 2005; H. Monteiro, 2017) were heated in vacuum using the projector lamp heating apparatus of Farley, Reiners, et al. (1999) and in sealed glass ampoules with gypsum or water additionally added to provide moisture to slow/prevent dehydration of goethite. All samples were held at 300°C for 4-11 hours, and the glass ampoules were broken and the goethites were allowed to air dry before all samples were placed on carbon tape and carbon coated. These samples were imaged at Caltech on Secondary Electron mode by the ZEISS 1550VP Field Emission SEM.

2.4 Results

The complete corrected step-heat dataset underlying the diffusivities, delta plots, and Arrhenius regressions discussed below is distilled in Tables 2.1 and 2.2, reporting the lower-bound, central, and upper-bound values from the bounding method's permeation-lag correction for ^{3}He and ^{4}He respectively.

Table 2.1: ^{3}He step-heat dataset for HT-Win-06-01b under the bounding method: measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), the lower- and upper-bound true step production (ω_{LB}, ω_{UB}), and the corresponding lower-bound, central, and upper-bound apparent diffusivities. "–" denotes early steps with $\lambda t < 4$, for which bounds could not be determined (see 2.2). Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots (Fig. 2.16 and others derived from this dataset).

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	ω_{LB} (pccSTP)	ω_{UB} (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
170	300	0.9630	0.0000	1.1660	–	–	–	–	–
180	463	0.5442	0.9630	0.2413	–	–	–	–	–
190	550	0.5292	1.6241	0.1538	–	–	–	–	–
200	500	0.5190	2.2112	0.1430	0.3651	0.6619	-22.158	-21.775	-21.503
210	475	0.5770	2.7646	0.1440	0.4341	0.7210	-21.715	-21.406	-21.160
220	450	0.6333	3.3799	0.1447	0.4894	0.7781	-21.345	-21.068	-20.843
230	425	0.7899	4.0551	0.1685	0.6452	0.9584	-20.823	-20.604	-20.392
225	908	0.7513	4.8885	0.0716	0.5827	0.8229	-21.516	-21.246	-21.148
215	450	0.1415	5.6842	0.0349	0.0699	0.1764	-22.836	-22.124	-21.900
205	520	0.0763	5.8529	0.0184*	0.0413*	0.0947*	-23.479*	-22.863*	-22.646*
195	1106	0.0670	5.9491	0.0080*	0.0487*	0.0750*	-24.053*	-23.731*	-23.618*
185	740	0.0227	6.0246	0.0049*	0.0147*	0.0276*	-24.839*	-24.404*	-24.207*
175	920	0.0180	6.0610	0.0036*	0.0131*	0.0216*	-25.168*	-24.847*	-24.665*
200	260	0.0224	6.0814	0.0163*	0.0189*	0.0388*	-23.534*	-23.360*	-22.812*
210	520	0.0812	6.1081	0.0181*	0.0649*	0.0993*	-22.984*	-22.758*	-22.554*
220	1167	0.5766	6.1914	0.0447	0.5585	0.6213	-21.587	-21.553	-21.475
230	370	0.3577	6.7796	0.0903	0.3130	0.4480	-20.948	-20.811	-20.579
240	320	0.6767	7.1563	0.1776	0.5864	0.8543	-20.062	-19.913	-19.667
245	250	0.5549	7.8582	0.1888	0.3773	0.7437	-20.176	-19.778	-19.474
250	210	0.6934	8.4743	0.2786	0.5047	0.9721	-19.626	-19.296	-18.942

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Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	ω_{LB} (pccSTP)	ω_{UB} (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
245	270	0.4659	9.2086	0.1413	0.1872	0.6071	-20.801	-19.874	-19.601
240	320	0.2857	9.7206	0.0749	0.1445	0.3606	-21.176	-20.487	-20.250
235	370	0.2199	10.0357	0.0517	0.1450	0.2716	-21.284	-20.864	-20.650
230	420	0.1400	10.2736	0.0303	0.0883	0.1703	-21.885	-21.421	-21.224
225	652	0.1494	10.4279	0.0206	0.1191	0.1700	-22.009	-21.781	-21.650
230	500	0.1550	10.5908	0.0271	0.1344	0.1820	-21.605	-21.462	-21.299
235	450	0.1951	10.7582	0.0361	0.1681	0.2313	-21.258	-21.107	-20.936
240	400	0.2425	10.9638	0.0483	0.2063	0.2908	-20.913	-20.750	-20.566
245	350	0.3150	11.2328	0.0690	0.2667	0.3840	-20.495	-20.326	-20.125
250	300	0.3716	11.5689	0.0939	0.3026	0.4656	-20.181	-19.973	-19.743
250	300	0.3640	11.9627	0.0899	0.2700	0.4539	-20.261	-19.959	-19.734
250	300	0.3452	12.3490	0.0852	0.2553	0.4304	-20.285	-19.979	-19.755
250	10250	8.4829	12.7167	0.0494	8.3977	8.5324	-19.989	-19.976	-19.969
240	785	0.2261	21.2187	0.0209	0.1767	0.2470	-21.026	-20.778	-20.689
Total			146.1483						

Table 2.2: ^{4}He step-heat dataset and absolute ^{4}He/^{3}He ratios for HT-Win-06-01b under the bounding method, structured as in Table 2.1. "–" denotes steps for which bounds could not be determined. The Total row reports the bulk (total-yield) ^{4}He/^{3}He ratio, i.e. the normalizing value used to bulk-normalize these ratios for the plot in Fig. 2.16. Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots, as in Table 2.1.

Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	$n_{t,\mathrm{UB}}$ (nccSTP)	ω_{LB} (nccSTP)	ω_{UB} (nccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{LB}}$	$(^4\mathrm{He}/^3\mathrm{He})_m$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{UB}}$
170	300	0.0192	0.0000	0.0280	–	–	–	–	–	–	–	–
180	463	0.0305	0.0192	0.0152	–	–	–	–	–	–	–	–
190	550	0.0558	0.0660	0.0179	–	–	–	–	–	–	–	–
200	500	0.1024	0.1375	0.0313	0.0845	0.1337	-26.207	-25.965	-25.619	–	197.3	–
210	475	0.1460	0.2500	0.0404	0.1147	0.1864	-25.314	-25.022	-24.717	248.2	253.0	268.8
220	450	0.2091	0.4055	0.0530	0.1686	0.2620	-24.407	-24.152	-23.876	326.3	330.1	349.3
230	425	0.3243	0.6319	0.0760	0.2714	0.4004	-23.425	-23.213	-22.956	403.7	410.6	432.3
225	908	0.3483	0.9799	0.0363	0.2722	0.3845	-23.807	-23.527	-23.413	442.6	463.6	474.4
215	450	0.0716	1.3487	0.0196	0.0353	0.0912	-24.945	-24.225	-23.976	429.1	505.9	537.1
205	520	0.0385	1.4359	0.0103*	0.0189*	0.0487*	-25.661*	-24.942*	-24.701*	392.4*	504.2*	534.6*
195	1106	0.0336	1.4912	0.0043*	0.0233*	0.0379*	-26.165*	-25.796*	-25.675*	446.7*	500.9*	515.3*
185	740	0.0079	1.5467	0.0019*	0.0037*	0.0098*	-27.586*	-26.808*	-26.592*	224.7*	350.1*	369.1*
175	920	0.0152	1.6020	0.0033*	0.0133*	0.0185*	-26.475*	-26.340*	-26.141*	843.2*	843.2*	982.0*
200	260	0.0107	1.6351	0.0091*	0.0073*	0.0198*	-25.787*	-25.411*	-24.791*	375.6*	475.5*	554.7*
210	520	0.2166	1.6547	0.0535*	0.2075*	0.2701*	-23.067*	-23.021*	-22.786*	2667.1*	2667.1*	3103.8*
220	1167	0.3187	1.8714	0.0269	0.2652	0.3456	-23.499	-23.302	-23.214	471.7	552.7	563.2
230	370	0.2113	2.6766	0.0590	0.1844	0.2703	-22.391	-22.249	-21.993	575.1	590.8	628.0
240	320	0.4286	3.1616	0.1243	0.3697	0.5529	-21.360	-21.203	-20.931	612.1	633.4	674.8
245	250	0.3988	3.6430	0.1521	0.2746	0.5510	-21.289	-20.899	-20.556	665.9	718.7	778.4
250	210	0.4440	4.2359	0.2020	0.2918	0.6460	-20.906	-20.469	-20.072	541.7	640.2	702.8
245	270	0.3123	4.7263	0.1053	0.1103	0.4176	-22.043	-20.981	-20.680	469.1	670.4	721.4

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Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	$n_{t,\mathrm{UB}}$ (nccSTP)	ω_{LB} (nccSTP)	ω_{UB} (nccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{LB}}$	$(^4\mathrm{He}/^3\mathrm{He})_m$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{UB}}$
240	320	0.1927	5.0846	0.0559	0.0874	0.2485	-22.376	-21.575	-21.315	505.5	674.3	718.4
235	370	0.1517	5.3031	0.0393	0.0958	0.1910	-22.386	-21.922	-21.688	599.8	689.6	729.9
230	420	0.0991	5.4714	0.0236	0.0598	0.1227	-22.956	-22.448	-22.232	608.3	707.9	745.9
225	652	0.0995	5.5835	0.0151	0.0760	0.1146	-23.136	-22.863	-22.721	608.2	666.5	689.2
230	500	0.1051	5.6969	0.0202	0.0900	0.1253	-22.679	-22.523	-22.345	652.1	678.1	707.5
235	450	0.1369	5.8251	0.0279	0.1167	0.1648	-22.290	-22.128	-21.940	674.6	701.7	734.0
240	400	0.1781	5.9762	0.0390	0.1502	0.2171	-21.892	-21.719	-21.518	705.1	734.6	770.9
245	350	0.2305	6.1807	0.0556	0.1915	0.2861	-21.479	-21.290	-21.070	694.4	731.9	771.7
250	300	0.2777	6.4385	0.0758	0.2221	0.3535	-21.134	-20.906	-20.659	703.3	747.3	793.3
250	300	0.2654	6.7374	0.0725	0.1896	0.3379	-21.250	-20.908	-20.661	653.7	729.2	774.1
250	300	0.2489	7.0231	0.0680	0.1764	0.3169	-21.282	-20.932	-20.647	646.8	721.0	765.4
250	10250	6.8042	7.2901	0.0430	6.7362	6.8472	-20.704	-20.691	-20.682	800.6	802.1	803.3
240	785	0.1853	14.1156	0.0187	0.1423	0.2040	-21.688	-21.422	-21.325	770.5	819.4	838.0
Total			145.8145								997.7	

The diffusivities determined from the corrected step heat 3He and 4He yields for HT-Win-06-01b are always significantly lower than the diffusivities determined from step heat experiments on the same sample in vacuum, and evolutions that are distinct from those determined from the vacuum experiments. The Arrhenius plot of the 3He diffusivities initially follow a declining concave-up trend until ~3.5% cumulative 3He yield, then slowly decreasing in sub-parallel arrays during prograde and retrograde heating cycles from 3.5-7% cumulative 3He yields (Fig. 2.7, Fig. 2.8). The Arrhenius plot of the 4He diffusivities differs in shape: it is concave-down to about 5% cumulative 3He yield, after which it follows a roughly linear array about one log unit below the 3He array (Fig. 2.7). The 3He diffusivities settled into a linear array which was suitable for linear regression after ~7% of the cumulative 3He yield was lost (Fig. 2.6). All remaining diffusivity measurements of the experiment were included in the linear regression, so that diffusivities from 3He loss fractions of about 7-15% were included. The linear regression yields an R^2 of 0.989, an activation energy of 157.1± 5.3kJ/mol, and a closure temperature of 98.1± 8.6°C assuming a cooling rate of 10°C/Ma and considering only the uncertainty in the regression intercept (D_0/a^2).

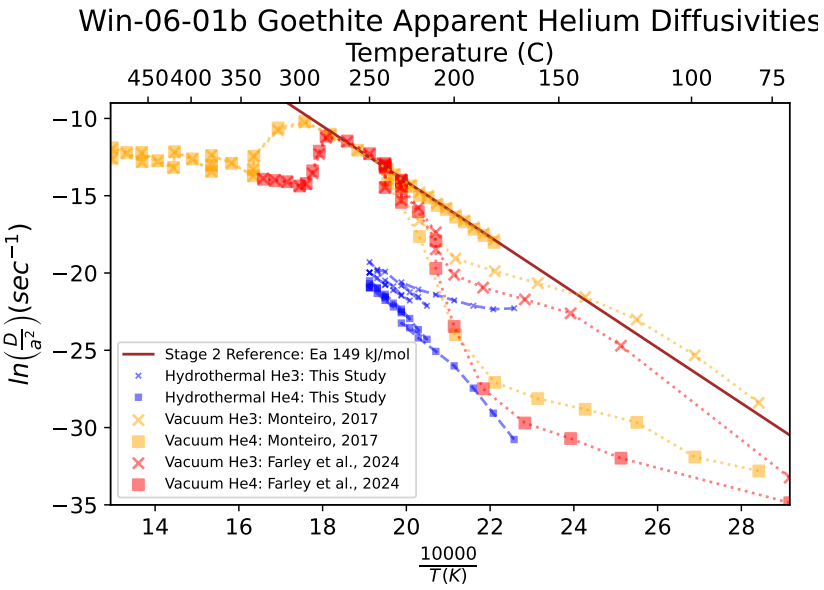

Figure 2.6: Arrhenius plot of 3He and 4He diffusivities from HT-Win-06-01b with other step heat experiments of Win-06-01b from H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024). The "Stage 2 reference line" is fit to helium step heats from H. Monteiro (2017). Error bars on the data from this study are not plotted for clarity. A diffusive domain corresponding to the Stage 2 reference line would have a closure temperature of about 45°C assuming a cooling rate of 10°C/Ma.

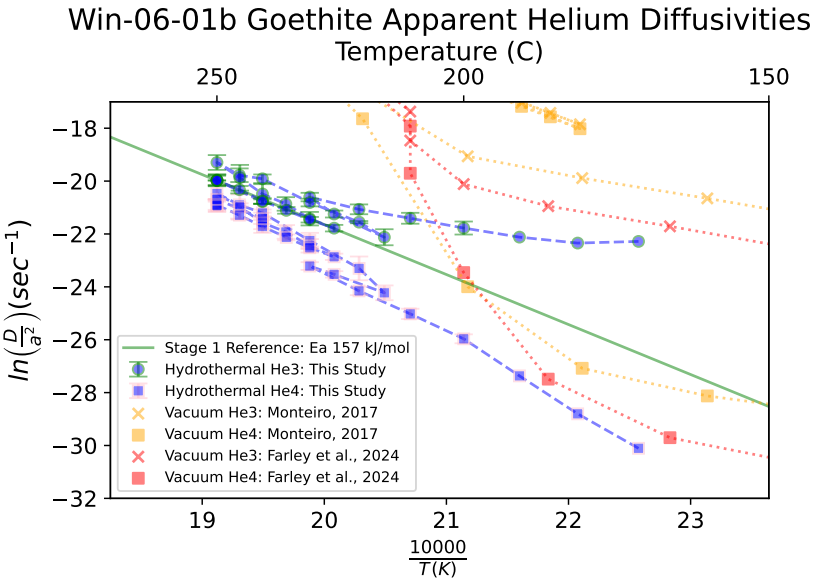

Figure 2.7: Linear regression of the last 11 measurements of 3He diffusivities in HT-Win-06-01b defining the "Stage 1" reference line. Apparent diffusivities under hydrothermal conditions are approximately seven natural log units lower than in vacuum. A diffusion domain corresponding to the Stage 1 reference line has a closure temperature of 98.1±8.6°C assuming a cooling rate of 10°C/Ma. Apparent diffusivities calculated from 4He step heats are sometimes within error of those calculated for 3He, however they are occasionally resolved at the 2σ level and are about a single log unit lower than apparent diffusivities calculated for 3He after a cumulative fractional yield of 14% 3He.

A delta plot was generated using the intercept and slope parameters from the 3He regression as a reference line, similarly to previous work (Farley, 2018). The 3He apparent diffusivity delta values monotonically decrease until a cumulative 3He loss of 7%, after which the delta values are within error of Delta = 0: the reference line (Fig. 2.8). A series of low-yield measurements on the first retrograde cycle produce high-error diffusivities which appear as scattered, uncertain delta values just after a cumulative 3He loss of 4% for both 3He and 4He.

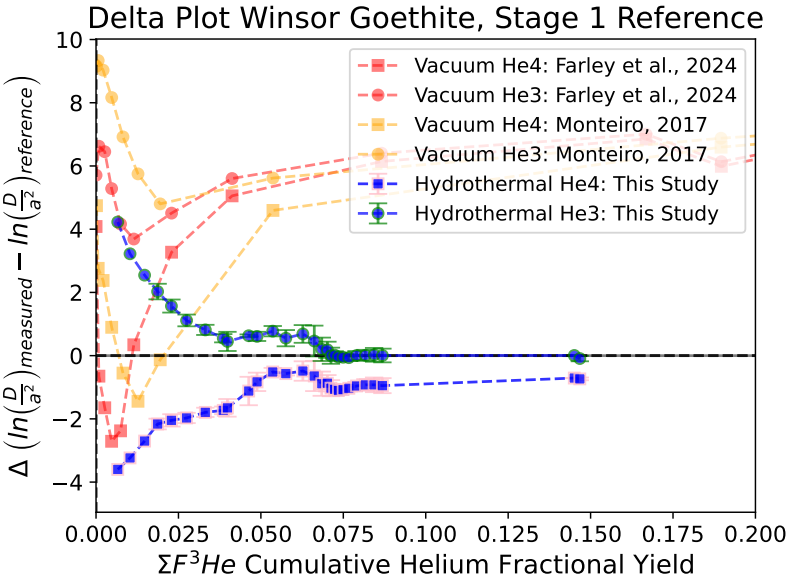

Figure 2.8: Delta plot similar to those presented in (Farley, 2018), or r/r0 plots in Lovera et al. (1991) and Farley and Flowers (2012). Delta plots are the difference between apparent log diffusivities and a reference line; they are designed to remove the temperature dependence of the dominant process of helium migration to reveal how apparent diffusivities evolve as an experiment proceeds. 3He monotonically decreases until settling on a temporary plateau from 4–6% 3He fractional yield, then settling on the reference line from 7–15% 3He fractional yield, which is consistent with MDD systematics. 4He diffusivities and delta values are generally highly uncertain but initially show an increase in delta values that settles to a slow increase or plateau about one log unit below the 3He reference line by a 3He fractional yield of 5%, which is consistent with diffusive loss of radiogenic 4He in the least retentive domains interrogated in HT-Win-06-01b. Apparent diffusivities for helium in Win-06-01b in hydrothermal conditions are up to 7 log units lower than the same sample in vacuum.

SEM imagery taken of thermally treated natural goethites in vacuum and

open air showed the development of contraction cracks on the outside and throughout the interior of goethite grains heat treated in vacuum and in open air (Figs. 2.10, 2.12), while goethite heat-treated when encapsulated with water or gypsum did not form cracks (Fig. 2.11). Goethite heat treated in open air turned red, whereas goethite heat treated in vacuum remained black.

2.5 Discussion

Measurement and Numerical Simulation of Apparent Permeability of Experimental Ampoule The apparent permeability values used in the numerical simulation (§2.3) are taken from the literature (Swets et al., 1961; Jones, 1953) and explicitly apply to an ampoule without water. However, the purpose of these simulations was simply to quantify the excess apparent permeabilities that would be observed for ampoules with a cooling history similar to that of the hydrothermal experiments. Because the predicted offset is universally small and a weak function of temperature or permeability, these simulations demonstrate that the much larger experimentally observed permeability is not an artifact of the cooling history of the hydrothermal experiments, reaffirming the conclusion that the hydrothermal experiments are indeed more permeable than the vacuum experiments.

The difference in log permeability between the actual and apparent permeabilities of the anhydrous simulation were significantly smaller than nearly all the 2σ uncertainties of the apparent permeabilities determined above for the hydrothermal Win-06-01b experiment, which ranged from 0.08 to 0.46 log units. Thus, no trends with step duration could be identified in the effective permeability values.

Therefore, the literature values underestimate the helium permeabilities of the ampoule used in this experiment, which is permissible because these values are used to calculate an upper bound constraint of the helium content of the ampoule, and lower permeabilities produce more conservative (higher) upper bound constraints.

A linear regression of $\ln(K)$ against inverse temperature (Fig. 2.4) fit to these isothermal 3He permeation measurements over 90-110°C yields an R^2 of 0.936 and an activation energy of 42.6±7.9 kJ/mol (2 s.e.), significantly higher than expected for pure silica glass with or without water (Swets et al., 1961; Jones, 1953; Masaryk, 1968; Shelby, 1972), but similar to glasses with a high

abundance of network-modifying cations such as Na, K, Pb, or Li (Norton, 1953; Altemose, 1961). Because 4He signals during these isothermal steps were blank-level, only the ampoule's 3He permeability could be measured directly this way, using the fitting procedure described above.

Every 3He permeability measurement for HT-Win-06-01b equals or exceeds the anhydrous literature value, which runs counter to expectations for a hydrothermal experiment: water and hydroxyl groups dissolved in silica glass are reported to *decrease* helium permeation at all temperatures (Tsuru et al., 2011; Shelby, 1972; Masaryk, 1968), under the common paradigm that species such as $(OH)^-$ occupy interstitial space within the amorphous glassy network (Norton, 1953; Shackelford, 1999). The offset is nonetheless real: it cannot be accounted for by the ampoule's extended permeation during cooling before measurement (§2.3), and a similarly elevated permeability was independently observed at 100°C in a separate silica glass ampoule at Caltech partially filled with MilliQ water and equilibrated with ~0.1 torr of tank helium, as discussed in Chapter 1 (§1.7).

The likely explanation is geometric. Helium is only sparingly soluble in water and diffuses through it rapidly, roughly 7×10^{-5} cm^2/s at room temperature (Verhallen et al., 1984) against about 2×10^{-8} cm^2/s in silica glass (Swets et al., 1961). On the timescale of permeation through the glass wall, transport through the water is effectively instantaneous, so the gas phase stays in equilibrium with the glass over the entire inner surface of the ampoule, including where liquid water contacts it. Helium partitions strongly into the gas phase over both water and glass: helium concentrations in the glass and water are about 3% and 1%, respectively, of the gas-phase concentration, using solubilities from Sander (2015) and Swets et al. (1961). Displacing gas-phase volume with water therefore raises the helium partial pressure without changing the surface area available for permeation, and because the rate of permeation is proportional to the gas-phase partial pressure (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), the ampoule's rate constant scales with the inverse volume fraction of the gas phase (Chapter 1, Eq. (1.12)), generalized here to any condensed payload: $\lambda_{ampoule} = \lambda_{empty} \left(V_{condensed} + V_{gas}\right) / V_{gas}$. This observed complexity emphasizes the need to measure the ampoule's rate constant in-experiment as described in Chapter 1 (§1.4).

Consistency of the Bounding Method with Direct Permeation-Step Quantification: The Complete Degas Test All Win-06-01b step heat data presented in this chapter are reduced using the bounding method (Eq. (1.10)), because 4He permeation-step yields were near blank level and could not support the permeation step method directly. As an internal check on this choice, the ampoule's 3He inventory was also reconstructed directly from the permeation steps themselves, using the in-experiment rate constants of Fig. 2.4 (§2.1s). Reducing 3He by the bounding method gives Stage 1 activation energies between 157.1 and 160.6 kJ/mol and closure temperatures between 97.5 and 101.2°C across its lower-bound, central, and upper-bound reductions, a spread comparable to the regression uncertainty itself. Fig. 2.9 compares the two: n_t, the ampoule's residual helium quantified directly from a permeation step, against $n_{t,SS}$ (equivalent to $n_{t,\mathrm{UB}}$ used elsewhere in this chapter), the upper bound on that same residual calculated independently from the preceding high-temperature step. For every ordinary permeation step in the experiment, $n_t \leq n_{t,SS}$, exactly as expected if the steady-state approximation genuinely bounds the ampoule's residual helium from above. This agreement is the justification for reducing the 4He dataset by the bounding method alone, since no analogous direct check of 4He is possible.

Beyond the ampoule's residual helium content itself, the two methods' derived apparent diffusivities agree just as closely (Fig. 2.17, §2.1s): $\ln(D/a^2)$ from the permeation step method tracks the bounding method's central value along the 1:1 line within uncertainty across the full range of the experiment, excepting the two steps whose permeation step n_0 depended on an atypically long, ~13-hour room-temperature permeation rather than the ~26-minute window used to calibrate every other step (§2.1s). Fitting the permeation step method's own diffusivities with the same Stage 1 procedure used for the bounding method above (Fig. 2.18, §2.1s) yields an activation energy of 157.0±4.8 kJ/mol (R^2=0.991, n=12) and a closure temperature of 97.5±7.8°C, indistinguishable from the bounding method's own 157.1±5.3 kJ/mol and 98.1±8.6°C (Fig. 2.7).

The two "attempted complete degas" steps depart from this pattern. For the longer of the two (7200 minutes; predicted $n_t/n_0 \approx 1 \times 10^{-8}$ under simple first-order decay), the directly measured 3He inventory exceeds $n_{t,SS}$ by a factor of 1.5. For the shorter (1580 minutes; predicted $n_t/n_0 \approx 0.02$), the excess is smaller, 25% over $n_{t,SS}$. Because $n_{t,SS}$ is a conservative upper bound

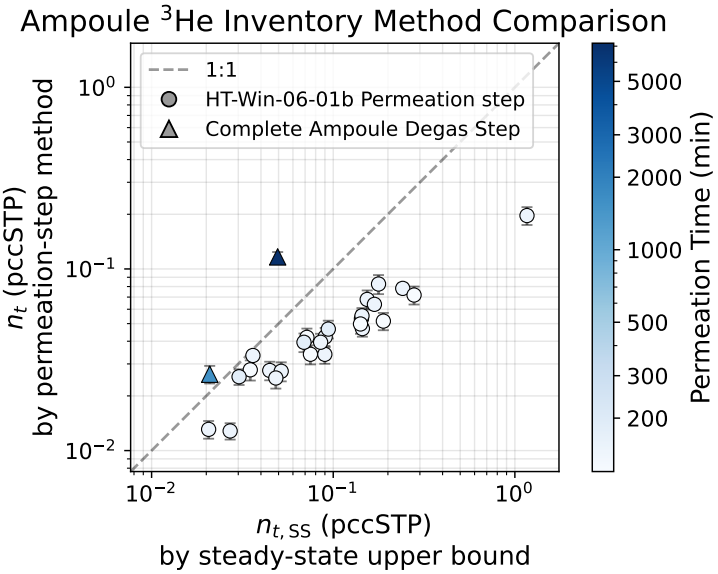

Figure 2.9: Two independent estimates of the ampoule's residual 3He inventory, n_t, following each permeation step of HT-Win-06-01b: quantified directly from the permeation step itself (*permeation step method*, y-axis, §2.1s), against the upper bound calculated independently from the preceding high-temperature step (*bounding method*, x-axis, Eq. (1.10)). Color indicates the duration of the permeation step. Circles are ordinary steps; triangles are the two "attempted complete degas" steps, held at 100°C for far longer than any other permeation step (1580 and 7200 minutes) in an attempt to drive the ampoule's 3He inventory toward zero. The dashed line is 1:1. Excepting the two complete-degas steps, n_t never exceeds $n_{t,SS}$, as required if permeation lag is genuinely bounded above by the steady-state approximation. Both complete-degas steps instead record more helium than their own $n_{t,SS}$ permits.

that assumes the sample's helium production rate is constant or decreasing during the step, a measured inventory in excess of $n_{t,SS}$ is not simply unlikely, it is inconsistent with that assumption, and requires the source term to have increased during the hold rather than decreased.

The size of the two excesses tracks the magnitude of the preceding, rather than the concurrent, thermal history. The 7200-minute step, with the larger (1.5×) excess, immediately followed a high-temperature step with a comparatively large 8.5 pccSTP 3He yield; the 1580-minute step, with the smaller (25%) excess, immediately followed a high-temperature step yielding only 0.2 pccSTP. An artifact intrinsic to the ampoule itself, such as an underestimated rate constant, a slow leak, or a memory effect in the extraction line, has no reason

to scale with how much helium the preceding high-temperature step released from the goethite. That the excess does scale this way points to the sample rather than the ampoule: whatever reservoir is being interrogated by these extended low-temperature holds was charged in proportion to the immediately preceding high-temperature step, and released its contents slowly and with only weak temperature dependence, remaining measurable at 100°C over days to weeks.

Dehydration is not precluded by this pattern, but it is a harder case to make for these two steps specifically. Goethite dehydration in vacuum step heats is associated with anomalously elevated apparent diffusivity (§2.5; (Farley, H. B. Monteiro, et al., 2024)), yet the high-temperature steps immediately following each complete-degas hold, both at 250°C, return delta values indistinguishable from zero relative to the Stage 1 regression (Fig. 2.8) rather than the elevated delta expected if a fraction of the aggregate had newly dehydrated. The preferred explanation is instead diffusion of residual helium held within interstitial space of the goethite crystallite aggregate (§2.5; (Naono et al., 1987)), recharged during the preceding high-temperature step and released slowly, and with weak temperature dependence, during the following extended low-temperature hold. This requires communication between a lattice or grain-boundary source and an interstitial reservoir, a departure from the independent, non-communicating domains of a strict MDD interpretation, but it is consistent with the interstitial framework already invoked to explain the early, low-retention "Stage 1" fraction of 3He release (§2.5), and unlike dehydration, it requires no change in the intrinsic retentivity of the crystalline lattice recorded by the bulk of the experiment.

Consistency with MDD Systematics Previous work on helium step heats of goethite in vacuum have demonstrated degassing behavior in the laboratory that is inconsistent with thermally activated volume diffusion through goethite. For example, apparent diffusivities calculated from helium step heat experiments of goethites with uniform distributions of 3He produced by proton irradiation following the methods of Shuster, Farley, et al. (2004) will increase in an isothermal experiment (Farley, H. B. Monteiro, et al., 2024), which is inconsistent with volume diffusion through an unchanging crystal.

Farley, H. B. Monteiro, et al. (2024) reported that helium and water evolution

in goethite vacuum step heating experiments are highly correlated and concluded that the crystal was decomposing into water and hematite during the experiment.

The goal of this work was to attempt to slow this decomposition process below the rate of thermal helium loss in unchanging goethite. SEM imagery shows that goethite that is thermally treated when encapsulated with water (or water for gypsum) does not extensively dehydrate like goethite heat-treated in vacuum or open air (Figs. 2.10–2.12). While it is difficult to prove that decomposition or recrystallization is not playing a role in the results presented in this work, the apparent diffusivities observed for the Win-06-01b goethite under hydrothermal conditions are much lower than those observed in previous step heats of Win-06-01b in vacuum (Fig. 2.6, Fig. 2.7, Fig. 2.8, Fig. 2.13), implying that processes other than dehydration may be responsible for the observed apparent diffusivities.

All helium step heat data on goethites are from naturally occurring polycrystalline aggregates with unclear and potentially widely variable grain sizes (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024), which are expected to follow MDD systematics. If volume diffusion is responsible for the data reported here, they too would be expected to follow MDD systematics. In MDD systems, apparent bulk diffusivities of a homogeneously distributed diffusant at a single temperature will decrease as an experiment proceeds, as would delta or r/r0 values at any temperature (where $\Delta = ln\left(\frac{D}{a^2}\right)_{measured} - ln\left(\frac{D}{a^2}\right)_{reference}$) (Lovera et al., 1991; Farley, 2018), assuming that the activation energy of the reference line used in such plots matches that of the process controlling diffusion. For the activation energy of 160 kJ/mol retrieved by regression in this work, the homogeneously distributed 3He step heat data follow the expected MDD systematics and monotonically decrease within a delta plot (Fig. 2.8), which is consistent with volume diffusion in a MDD system.

The strong plateau in the delta plot and statistically robust regression over a wide range of fractional cumulative 3He yield (from 7-15% ΣF_{3He} with an R^2 of 0.989) suggests that the frequency factor obtained by the regression may be dominated by a particular diffusion domain, rather than a mix of diffusion domains as is most often observed in the apparent diffusivities for polycrystalline hematite (Farley, 2018). The 4He apparent diffusivities are generally lower than the 3He apparent diffusivities and increased at the beginning of the experiment,

which is consistent with rounding of 4He concentration profiles due to loss of radiogenic helium in the least retentive domains (Shuster, Farley, et al., 2004; Shuster, P. M. Vasconcelos, et al., 2005).

Many vacuum experiments in goethite also record declining delta values for early fractional yields of 3He (Shuster, P. M. Vasconcelos, et al., 2005; Farley, H. B. Monteiro, et al., 2024) that are generally not represented in the 4He data. This effect may be from helium migrating out of fast pathways along grain boundaries that would retain little helium in nature (Shuster, P. M. Vasconcelos, et al., 2005). Goethite aggregates are known to have interstices that are filled by chemisorbed non-stoichiometric water that represents the earliest water to be lost in vacuum (Naono et al., 1987). Non-stoichiometric hydrogen is known to persist in goethite even after heating to 140°C (Yapp and Poths, 1995). It is therefore reasonable to posit that artificial 3He observed in less-retentive domains may be trapped by non-stoichiometric water or hydroxyl groups within interstices of goethite aggregates. Dehydration of non-stoichiometric interstitial water may be associated with early fractions of degassing water and 3He from goethite samples during step heats, and could potential represent the LRD (low retention domain) of Shuster, P. M. Vasconcelos, et al. (2005), and the 3He offgassed during "Stage 1" in vacuum step heats (Farley, H. B. Monteiro, et al., 2024). The high variability of the portion of 3He hosted in early-degassing "Stage 1" fractions within vacuum step heats is sensible considering observations that goethite formed at different temperatures is known to have highly variable crystallinity, crystallite ordering, surface area, and non-stoichiometric water contents (Schwertmann et al., 1985). This population of 3He would likely also be the first to be lost under hydrothermal conditions.

While the results presented here are strong experimental evidence for high helium retentivity in goethite (Fig. 2.13, Fig. 2.14), they do not yet independently provide any evidence to support or refute hypotheses that helium diffusion in crystalline goethite is anisotropic or affected by radiation damage (Bassal et al., 2022).

Goethite Dehydration and/or Recrystallization During Helium Step Heat Experiments The aim of helium step heat experiments is to characterize the mechanisms and rates helium loss in nature to aid in the interpretation of geochronologic and helium geochemical data. While the results of this

experiment are encouraging in that they differ remarkably from experiments on the same and similar samples in vacuum that are primarily controlled by dehydration processes (Farley, H. B. Monteiro, et al., 2024), this experiment may also be affected to some degree by processes aside from volume diffusion that may potentially be unrelated to helium loss in goethite in nature.

Aside from nanoparticles (Guo and Barnard, 2011; Navrotsky et al., 2008), bulk goethite is unstable with respect to hematite and water at the pressures and temperatures of this experiment (Majzlan et al., 2003). The SEM study of goethite heat-treated in vacuum and open air shows that clearly visible volume contraction cracks form throughout the interiors and surfaces of highly-dehydrated goethite grains, while goethite heat-treated in water-saturated conditions does not (Figs. 2.10–2.12). Clearly, dehydration is occurring at a much smaller scale, if at all, in the hydrothermal experiments. While this evidence excludes the possibility of pervasive dehydration in the hydrothermal experiments, it does not exclude the possibility of some dehydration occurring at a scale too small to be observed in the SEM images. Future work on textural evolution of natural goethite aggregates during low fraction dehydration would be useful to resolve the extent of "hidden" dehydration that may occur before the formation of SEM-visible cracks.

Dehydration processes may be responsible for some or all of the helium loss of Win-06-01b-HT, although if so they are markedly slowed relative to step heat experiments in vacuum (Fig. 2.14–Fig. 2.15). Even if dehydration completely controls helium loss in the step heat experiment of Win-06-01b-HT, these new data presented here would represent upper bounds for the apparent diffusivities of helium at the temperatures and fractional cumulative yields (ΣF_{He}).

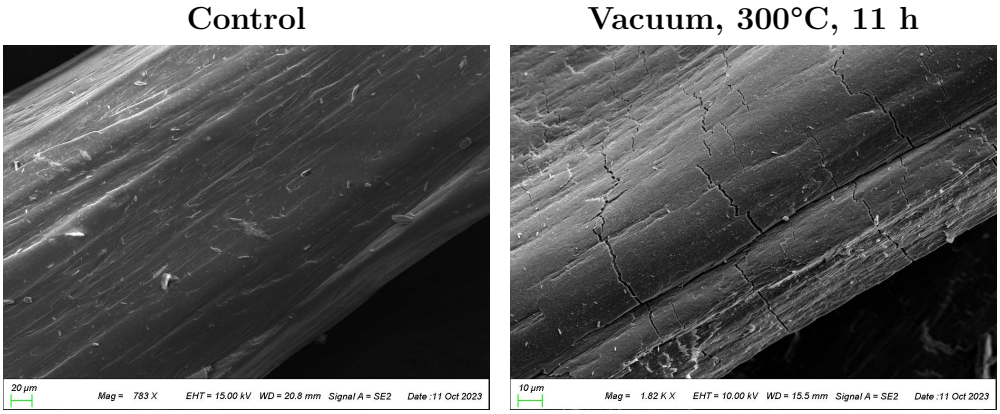

Figure 2.10: Secondary electron images of Bahia goethite aggregate before heat treatment (control) and after heat treatment at 300°C for 11 hours in vacuum. Cracks developed in the vacuum-heated sample both parallel and perpendicular to the long axis of the polycrystalline aggregate, representing volume contraction due to dehydration/decomposition of goethite into hematite. The buffered treatments at the same temperature are shown in Fig. 2.11.

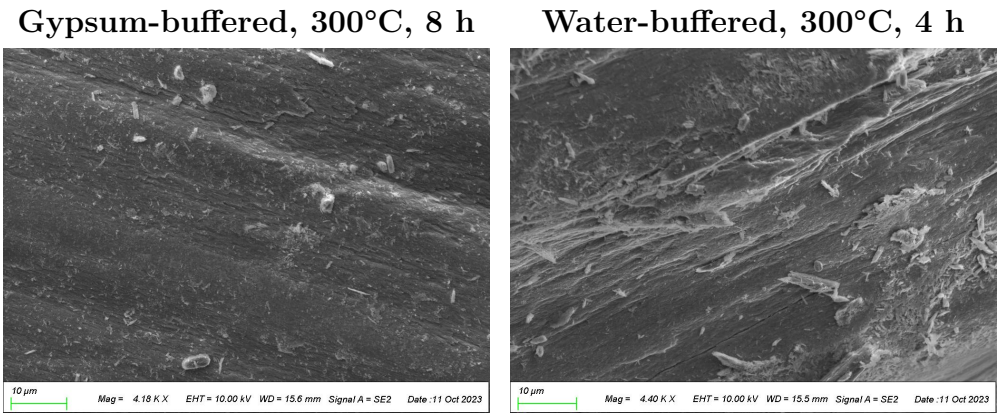

Figure 2.11: Secondary electron images of Bahia goethite aggregate heated at 300°C in a sealed glass tube with gypsum (8 hours) and with water (4 hours). Despite treatment at high heat, no cracks indicative of dehydration were visible on either goethite surface. In the gypsum experiment, the gypsum dehydrated first and produced enough water pressure to stop or slow dehydration of the goethite and prevent the development of microscopic cracks; at higher temperatures goethite and gypsum reacted, forming a surface layer of other minerals. For the water-buffered sample, no cracks were observed when investigating the goethite at scales similar to Fig. 2.10, and most of the aggregate's surface resembled the gypsum-buffered panel; the area pictured had the roughest texture on the grain. A very small crack parallel to the long axis may indicate some amount of dehydration had occurred, or it could have been inherited from before hydrothermal treatment, since even the untreated control grain (Fig. 2.10) had some rough surfaces and rare long-axis-parallel cracks.

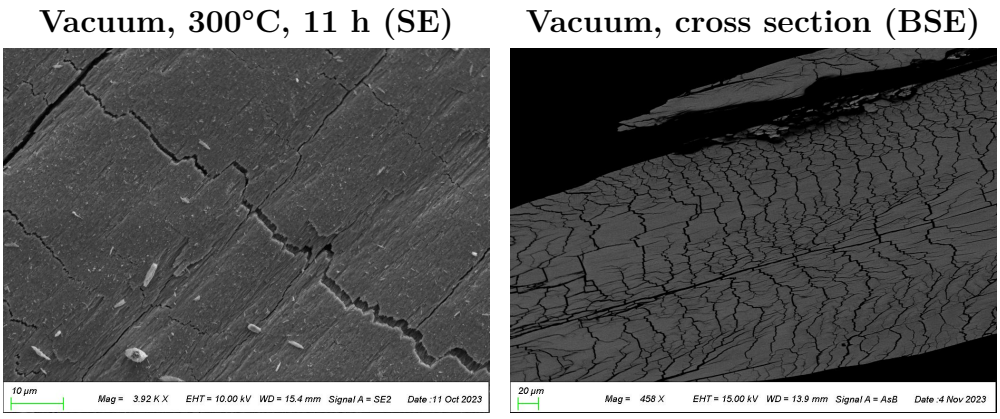

Figure 2.12: Vacuum-heated Bahia goethite at higher magnification and in cross section. Left: secondary electron image of goethite heated at 300°C for 11 hours in vacuum, at the same scale as the buffered samples of Fig. 2.11, showing rough cracks at various scales both parallel and orthogonal to the long axis of the goethite aggregate. These are the typical decomposition features that would be expected in Fig. 2.11 if significant decomposition had occurred there. Right: backscatter electron image of a polished cross section of epoxy-mounted, vacuum-heated Bahia goethite, from the same subsample as the vacuum-heated grain of Fig. 2.10, exposing the grain's interior. This image clearly shows that the cracks visible on the exterior of dehydrated goethite are pervasive throughout the interior as well.

The process of recrystallization is another potential complication in interpreting helium step heats of goethite in hydrothermal conditions. Poorly crystalline synthetic goethite held in hydrothermal conditions is known to recrystallize into more orderly, crystalline goethite aggregates, even at temperatures as low as 125°C in as little as 16 hours (Schwertmann et al., 1985). In this study, natural goethite was held in contact with liquid water and water vapor at hydrothermal conditions as harsh as ~40 bars and 250°C for as long as a week. These conditions appear to have visibly dissolved and reprecipitated silica from the ampoule's inner walls (Fig. 2.1), qualitatively testifying to the ability of the hydrothermal fluid to dissolve and recrystallize rock-forming oxides. It is unquestionable that recrystallization of some goethite could have occurred under these conditions; however, it is unclear if conditions were favorable to recrystallize this specimen of goethite, Win-06-01b, which is amongst the most helium-retentive and crystalline dated goethite samples (Waltenberg, 2012; Farley, H. B. Monteiro, et al., 2024). Less crystalline goethites in nature are commonly recrystallized, which appears to result in helium loss, resetting the U-Th/He age (H. S. Monteiro, P. M. P. Vasconcelos, et al., 2018; H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Heim et al., 2006). Intuitively, it is difficult to imagine that dissolution-precipitation mediated recrystallization would not result in complete loss of helium within recrystallized regions. Therefore, assuming an initially uniform distribution of 3He throughout the crystal from proton irradiation, the upper bound extent of recrystallization would be equivalent to the 3He cumulative fractional yield, which is only 14.7% by the end of the hydrothermal experiment. Remaining helium is unlikely to pass through these now helium-free recrystallized areas if grain boundaries in goethite serve as fast pathways like they do in hematite (Farley, 2018). Recrystallization could be the dominant process controlling helium loss in this experiment, but only if it is faster than diffusive loss, in which case the data presented here would represent new upper bounds for apparent diffusivities for helium in Win-06-01b goethite. In that case, it stands to reason that processes such as dehydration and recrystallization may be faster than volume diffusion of helium in crystalline goethite in nature and may be the primary drivers of helium loss.

High Helium Retentivity of Win-06-01b-HT Apparent diffusivities calculated from helium step heats on the hydrothermal experiment on Win-06-01b

goethite are clearly resolved and much lower than the apparent diffusivities calculated for the same sample in vacuum by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) for all steps after the first few (Fig. 2.6, Fig. 2.7, Fig. 2.8). Diffusivities calculated from the hydrothermal experiment are approximately seven natural log units lower than the earlier experiments in vacuum on the same sample. This high helium retentivity is explored through three lenses: closure temperature of cooling samples (Dodson, 1973), isothermal diffusion (Wolf, Farley, and Kass, 1998), and through comparison with step heats of other polycrystalline samples. Strictly speaking, both closure temperatures and retention estimations under isothermal diffusion only apply to helium loss occurring exclusively via thermally activated volume diffusion with known domain size. While the results presented here on a polycrystalline sample are consistent with a thermally activated volume diffusive process for helium loss in goethite under hydrothermal conditions, they do not exclude other processes (such as goethite dehydration/decomposition/recrystallization) or potential complexities related to radiation damage and crystal defects on bulk diffusion (Bassal et al., 2022), nor do they permit estimation of the diffusivity at infinite temperature (D0).

If the slope and intercept on the regression of 3He apparent diffusivities from this experiment are taken to represent a real diffusion domain for goethite, then a closure temperature can be calculated for this domain. Considering uncertainty only in the intercept (D_0/a^2) term and a cooling rate of 10°C/Ma, the closure temperature for that domain would be 98.1± 8.6°C. A diffusion domain corresponding to the "Stage 2" reference line (Fig. 2.6) would instead have a closure temperature of about 45°C at the same cooling rate. That is the closure temperature of a ~0.2µm radius hematite crystal, or a ~650µm radius apatite crystal.

Goethite is formed near Earth's surface and thus U-Th/He geochronology is most typically applied to date surface processes (Heim et al., 2006; Yapp and Shuster, 2011; P. M. Vasconcelos et al., 2013; Danišík et al., 2013; H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Hofmann, Reichenbacher, et al., 2017; Allard et al., 2018). For crystals of goethite grown near the surface that are found near the surface, the concept of a closure temperature is poorly applicable, as it necessarily assumes a cooling rate (Dodson, 1973). In such cases, a better measure of retentivity is an analysis of isothermal retentivity

over time (Wolf, Farley, and Kass, 1998), which is commonly expressed in terms of t'/t, where t' is the helium age and t is the isothermal holding time (most often the 'real age'). For an isothermal holding time of 100 Ma at 40°C, a diffusive domain with the parameters obtained from the regression of 3He apparent diffusivities of Win-06-01b-HT has a t'/t = 0.968, where t'/t = 1 corresponds to perfect retentivity. Fig. 2.13 shows isothermal holding time against helium age for this same domain for 4 different temperatures. At 25°C, this domain is nearly quantitatively retentive of helium even at billion-year time scales. The results presented here qualitatively support that goethite is highly retentive of helium at surface conditions, even at 100-million-year timescales.

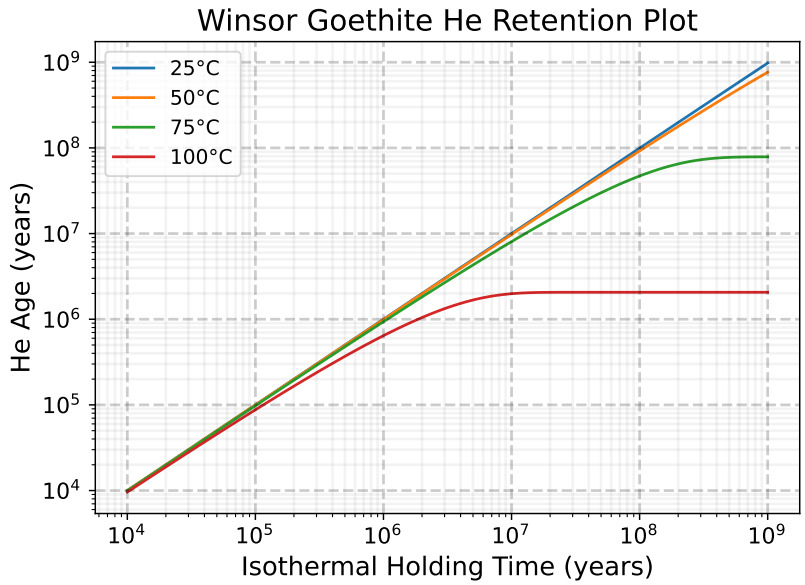

Figure 2.13: Isothermal retention plot (Wolf, Farley, and Kass, 1998) for helium in goethite corresponding to the "Stage 1" reference line regression on the hydrothermal experiment. Helium hosted in a diffusion domain with these diffusion parameters would be nearly perfectly retentive at billion-year timescales at 25°C. At 40°C, the equilibrium age for this domain (the helium age at which radiogenic helium production and diffusive loss are equal) is 33.2 billion years, attained after 50.4 billion years.

It is difficult to use Arrhenius plots to compare many samples' helium retentivities. Farley, H. B. Monteiro, et al. (2024) report that linear regressions on apparent diffusivities calculated from helium step heat data on goethite in vacuum can be fit to yield widely variable diffusive parameters even for a single sample (e.g. activation energies ranging from 107 to 176 kJ/mol within a single experiment on YAN-0201 AC). Similarly, regressions on helium step heat data of polycrystalline materials can yield erroneous activation energies when not applied carefully (Farley, 2018). Because of the complexity of comparing Arrhenius plots of calculated apparent diffusivities for polycrystalline samples with differing heating schedules, it is advantageous to convert heating schedules to some parameter that can be compared easily between experiments. One effective way to reduce the complexity of step heat experiments to visualize helium retentivity is to "flatten" a thermal history to a dimensionless diffusive timescale: the Fourier number ($Fo = Dt/a^2$). This approach offers similar benefits to Delta plots (Farley, 2018) or r/r0 plots (Lovera et al., 1991), but is optimized to qualitatively compare the helium retentivities of polycrystalline samples for a given thermal history (expressed as diffusive time).

For a given diffusive domain, a Fourier number of $Fo \approx 1$ corresponds to the complete degassing of that domain. Complex heating schedules can be converted to Fourier numbers by assuming a reference line similarly to a Delta or r/r0 plot (Lovera et al., 1991; Farley, 2018). This procedure is especially useful for MDD (Multiple Diffusion Domain) systems, where the apparent diffusivities decrease over the course of a step heat experiment as the least retentive domains are degassed. For a single spherical diffusion domain, the fraction of helium released is a function of the Fourier number alone (Crank, 1979; Fechtig and Kalbitzer, 1966). If the activation energy of the reference Fourier number is the same as the activation energy of diffusion, then the amount of helium released in that domain then is an exact function of the reference Fourier number multiplied by a constant. Thus, a plot with reference Fourier numbers against cumulative fractional helium yields for different samples will yield the same geometries and interrelationships regardless of the intercept used to calculate the reference Fourier numbers; the values of the reference Fourier numbers will be arbitrary and only meaningful as relative diffusive time. Another implication is that different aliquots of the same sample should appear identical in this space regardless of heating schedule (see Win-06-01b in Fig. 2.15, or overlapping hematite samples in Fig. 2.14). Many such plots were generated with a variety

of activation energies between 149 kJ/mol to 171 kJ/mol, but the relative positions of the degassing curves for all samples are preserved, illustrating the utility of this approach even for comparing helium retentivity between different minerals.

Fig. 2.14 compares the step heat results of Win-06-01b-HT with 11 step heat experiments of 3 samples of polycrystalline hematite and 9 step heat experiments of 2 samples of goethite in Vacuum, including Win-06-01b. In a thermal-history normalized space, it is obvious that goethite in vacuum and hematite define well-separated populations and goethite in vacuum is far less retentive of helium than hematite. Most interestingly, the most retentive sample on these plots is clearly the Win-06-01b-HT goethite under hydrothermal conditions, at least for the 14.7% of the total 3He hosted in the least retentive domains. This implies that the helium retentivity of stable goethite may even be higher than for hematite of similar size, although step heat experiments on irradiated single-crystal goethite of controlled size under hydrothermal conditions would be needed to fully assess this possibility. This plot also makes it clear that step heat results for different, unrelated goethite samples in vacuum are more similar to one another than step heat results of the same sample at different water activities.

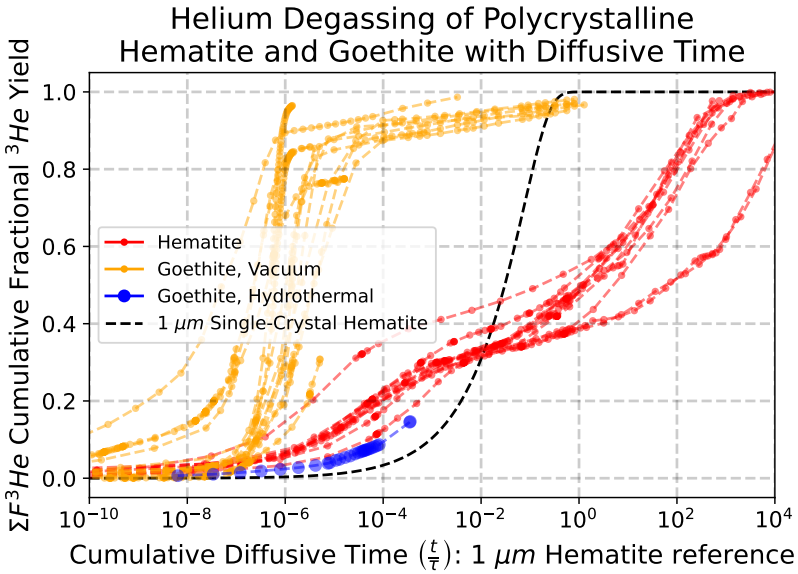

Figure 2.14: Heating schedules for polycrystalline goethite and hematite samples are flattened to reference Fourier numbers, dimensionless numbers that can be thought of as integrated thermal history, allowing for easy comparison of entire helium step heat experiments. The 3He cumulative fractional yield is plotted on the y axis as a function of Fourier number in log scale on the x axis. Curves on the left of the plot degas earlier within a given thermal history than the curves on the right. Hematite and goethite step heats in vacuum each cluster closely together, forming distinct populations. Most strikingly, the hydrothermal Win-06-01b-HT experiment is the most helium-retentive polycrystalline sample on the plot, at least until the experiment's end at a cumulative fractional yield ∼15% F3He. In vacuum, all hematite samples are significantly more retentive of helium than all goethite samples, with some limited exceptions for the least retentive domains holding <5% of the total helium. Win-06-01b step heat experiments in vacuum are indistinguishable from the Capao samples in vacuum (the two leftmost goethites are Capao samples; the rest of the Capao samples closely cluster with Win-06-01b in vacuum). The differences in the curves here reflect differences in grain size distributions for hematite, and some combination of grain size and dehydration rate (due to variable water pressure in the experimental vessel and crystal aggregate size) for goethite. Vacuum step heats of 11 aliquots of 4 samples of polycrystalline hematite from the Grand Canyon, Arizona (Farley and Flowers, 2012) and the Gogebic Range, Michigan (Farley and McKeon, 2015), and 9 aliquots of Win-06-01b and Capao goethite from Minas Gerais, Brazil (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024) are plotted here. The reference D/a^2 values are calculated for 1µm domains of hematite ($\ln(D_0) = 0.66$; $E_a = 171$ kJ/mol: (Farley, 2018)), and the reference curve for the degassing of 1µm spherical domains of hematite with uniformly distributed 3He is calculated using the forward equations of Fechtig and Kalbitzer (1966).

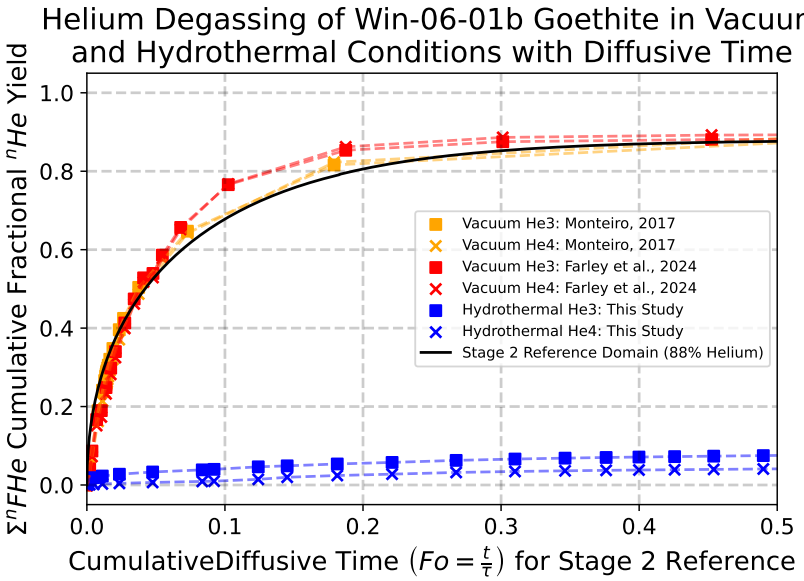

Figure 2.15: Win-06-01b step heats flattened to the Fourier number calculated from the "Stage 2" reference domain from the Win-06-01b step heat of H. Monteiro (2017). In this space, the helium retentivity contrast between the same goethite sample in vacuum and under hydrothermal conditions is striking. This plot shows that nearly 88% of helium loss for Win-06-01b in vacuum can be well-explained by a single thermally activated process following an Arrhenius law, corresponding to the "Stage 2" reference line from the experiment of H. Monteiro (2017), unlike polycrystalline hematite samples, which show a complex helium release pattern against a reference Fourier number, as expected for polycrystalline samples with complex grain size distributions. Whether or not the frequency factor D_0/a^2 for the "Stage 2" process Arrhenius law corresponds to grain size, aggregate size, or another parameter relevant to helium loss in goethite in nature is unclear. This process seems to be slightly faster for the experiment of Farley, H. B. Monteiro, et al. (2024), potentially due to lower water pressures from a liquid nitrogen trap. Both experiments of H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) show that helium diffusivity (and the 4/3 ratio) suddenly decreases after a cumulative fractional yield of 3He around 88%, signifying the beginning of "Stage 3" (Farley, H. B. Monteiro, et al., 2024), despite different heating schedules.

Profiling Radiogenic Helium Loss with $\frac{^4He}{^3He}$ Ratios of Win-06-01b-HT Bulk-normalized 4He/3He ratios in step heats of irradiated samples measure rounding concentration profiles by natural helium loss, if the loss mechanism in nature and in the laboratory share the same geometry (Shuster, Farley, et al., 2004; Farley, H. B. Monteiro, et al., 2024). Aside from the first experimental step, bulk-normalized 4/3 ratios for all experiments begin at vanishingly small values and steeply increase at first, with the rate of increase slowing as the experiments proceed (Fig. 2.16). This behavior is consistent with complete then partial loss of radiogenic helium in nature in the least retentive domains and is observed in all step heat experiments on irradiated goethite (Farley, H. B. Monteiro, et al., 2024). The hydrothermal experiment, however, records lower bulk-normalized 4/3 ratios than the vacuum experiments at any given $\Sigma F_{3\mathrm{He}}$ for the duration of the experiment. Similarly, the approach to the 4/3 plateau varies between the three experiments: a high bulk normalized 4/3 He plateau is most decisively represented in the Farley, H. B. Monteiro, et al. (2024) experiment, whereas the 4/3 ratios in the step heats of H. Monteiro (2017) transition more gradually before reaching a plateau later in the experiment, while the increase in 4/3 ratios is slowest for the hydrothermal experiment (Fig. 2.16). This suggests that the geometry of helium loss differs between the hydrothermal and vacuum experiments, and could be explained by the relative dominance of dehydration processes or their mode of spatial propagation as a function of water pressure, where the smoothest 4/3 spectra are generated by higher water pressures. We interpret this experimental control on the 4He/3He spectra to be the relative importance of goethite dehydration and helium diffusion through crystalline goethite in governing the geometry of helium loss from the sample. If the geometry of helium loss due to dehydration and due to diffusion differ, then the 4He/3He spectra of goethite measured in vacuum may not straightforwardly encode the geometry or history of helium loss in the way that such data faithfully encode this history in phases such as hematite or apatite (Farley and McKeon, 2015; Shuster, Farley, et al., 2004). This claim rests entirely upon the correctness of the 4He data reported here, for which each step yield is somewhat small, most often between 0.15 and 0.5 nccSTP. More 4He/3He spectra of goethite under hydrothermal conditions, with stronger 4He signals, are needed to investigate this hypothesis further. The experiment in Farley, H. B. Monteiro, et al. (2024) was open to a liquid nitrogen trap during heating, which would maintain lower water pressures than H. Monteiro

(2017), who in turn held Win-06-01b at much lower water pressures than the hydrothermal experiment presented in this work. While the hydrothermal experiment did not reach a plateau, it may have if the experiment was allowed to proceed further.

Interestingly, both vacuum experiments show a maximum bulk normalized 4/3 He ratio of ~ 1.1 at or before $\Sigma F_{^3He} = 0.88$, after which the bulk normalized 4/3 He decreases to ~0.9, suggesting that these values are independent of heating schedule and may reflect a material property of Win-06-01b.

Both vacuum experiments record a somewhat high 4/3 ratio for the first measurement that is not observed in the hydrothermal experiment (Fig. 2.16). This feature represents a very small amount of the total helium and could be related to dehydration of poorly retentive domains in the vacuum, or it may have been lost in the hydrothermal experiment while the ampoule was held at 70°C to degas entrained atmosphere before the initiation of the experiment. This small abundance of poorly retained 4He observed in the vacuum experiments could be related to physisorption of atmospheric helium (Protin et al., 2016). In any case, it represents a negligible fraction of the 4He contained in Win-06-01b.

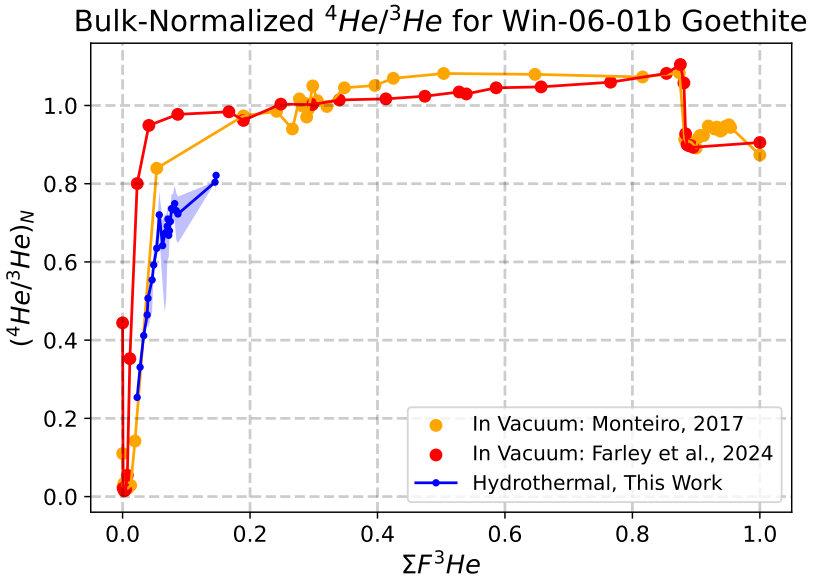

Figure 2.16: Bulk-normalized 4/3 ratios for Win-06-01b goethite step heats against cumulative fractional 3He yield. The hydrothermal experiment shows lower bulk-normalized 4/3He ratios at a given ΣF_{3He} for the entire duration of the experiment, besides the very first step. All experiments' 4/3He ratios begin at very small values and increase steeply at first, which is consistent with complete then partial loss of radiogenic helium in nature for the least retentive helium domains. The rate of increase slows between ΣF_{3He} of 0.05 to 0.2 for all experiments. The hydrothermal experiment is plotted as a shaded envelope spanning the bounding method's lower- and upper-bound 4/3He ratios; 10 of 34 analyses are excluded, either because the upper and lower bounds of ω cannot be computed for the early steps before λt was large enough to assume a steady-state permeation condition, or because the measured 3He signal for that step is below 0.1 pccSTP and too low-signal to give a reliable ratio.

2.6 Conclusions

Predicting the degree of helium retention and loss to guide interpretation of U-Th/He dates of natural goethite based on factors such as grain size, crystallinity, radiation damage, defect density, impurities, and aluminum substitution remains challenging. Previous efforts to understand Helium loss in natural goethite with step heating experiments in vacuum have instead queried the kinetics of dehydration (Farley, H. B. Monteiro, et al., 2024). It remains unclear how informative existing vacuum helium step heat data are with respect to helium retention and loss in natural samples. Because goethite dehydration is slowed or stopped at high water pressure, helium step heats under hydrothermal conditions are more likely to interrogate the processes controlling radiogenic helium loss in natural goethite. The method presented in this study leverages the high helium permeability of silica glass to easily conduct helium step heat experiments under hydrothermal conditions, achieving internal water pressures up to 86 bars at 300°C using standard ultra-high vacuum experimental setups (Farley, Reiners, et al., 1999).

The experiment conducted on irradiated Win-06-01b represents the first hydrothermal helium step heat experiment for goethite. It gives apparent diffusivities approximately seven natural log units below those obtained from the same sample in vacuum. The corrected 3He array is linear between 7–15% cumulative yield, with an activation energy of 157.1± 5.3kJ/mol, and its delta plot follows multiple diffusion domain systematics. Extrapolated to surface conditions, a domain with these parameters is essentially quantitatively retentive of helium at billion-year timescales at 25°C. The 4He data require that this domain experienced helium loss in nature.

The 4He/3He spectrum observed for this sample is distinct from that of the same sample in vacuum, suggesting that the geometry of helium loss induced by goethite dehydration is distinct from that of diffusion. More hydrothermal step heat experiments of goethite are necessary to confirm this hypothesis.

Although hydrothermally treated goethite does not develop the visible dehydration-induced contraction cracks that goethite heated in vacuum does, we cannot exclude the possibility that subtle dehydration or recrystallization occurred during the hydrothermal step heat experiment. If either contributed, the diffusivities reported here would be upper bounds and the retentivity a lower bound. Distinguishing these possibilities, and separating the effects of crystallite size,

chemical composition, and radiation damage, will require further hydrothermal experiments on irradiated single-crystal goethite of controlled size. The method described here makes such experiments straightforward.

Supplementary Information

for Chapter 2: *Hydrothermal Helium Step Heats on Goethite Reveal High Retentivity and Multiple Diffusion Domain Systematics*

2.1s Goethite Permeation Rate Constant Dataset

The in-experiment permeation rate constants used by the permeation step method below – and the isothermal-permeation-fit methodology, cooling-tube numerical model, and Arrhenius regression used to determine them – are described in full in the main text (§2.2). The complete fitted dataset underlying that regression is reported here: Table 2.3 gives the raw per-step data for all 10 isothermal sequences, and Table 2.4 gives the lambda/n_0 fit to each sequence (Fig. 2.3) that together define the Arrhenius regression of Fig. 2.4.

Table 2.3: Raw step-heat data ('Fit Lambdas' sheet, cross-checked against 'Raw Data and Corrections') for the 10 isothermal 3He permeation sequences used to determine the ampoule's apparent permeability (Fig. 2.3, Fig. 2.4). Series are separated by horizontal rules; within a series, steps are listed in the order measured.

Series	Temp (°C)	Nominal Time (min)	Cooling Time (min)	^{3}He measured (pccSTP)	$\pm 2\sigma$
1	105.00	150	26	0.0190	1.1×10^{-4}
1	105.00	150	26	0.0111	7.4×10^{-5}
1	105.00	150	26	0.0080	5.3×10^{-5}
2	109.99	150	26	0.0229	1.4×10^{-4}
2	110.00	150	26	0.0115	7.3×10^{-5}
2	110.00	150	26	0.0075	5.2×10^{-5}
3	99.86	120	26	0.0106	6.1×10^{-5}
3	100.00	120	26	0.0074	4.0×10^{-5}
3	100.00	120	26	0.0049	2.5×10^{-5}
3	99.99	120	26	0.0043	2.1×10^{-5}
4	95.00	150	26	0.0109	2.8×10^{-4}
4	95.00	150	26	0.0081	2.0×10^{-4}
4	95.00	150	26	0.0062	1.4×10^{-4}
5	104.99	120	26	0.0215	8.9×10^{-5}
5	104.99	120	26	0.0144	6.0×10^{-5}
5	104.99	120	26	0.0104	4.3×10^{-5}
6	100.00	120	26	0.0126	5.6×10^{-5}
6	100.00	120	26	0.0094	3.5×10^{-5}
6	100.00	120	26	0.0074	2.0×10^{-5}
7	95.00	120	26	0.0072	3.0×10^{-5}
7	95.00	120	26	0.0063	2.6×10^{-5}
7	95.00	120	26	0.0045	1.9×10^{-5}
8	90.00	150	26	0.0059	2.7×10^{-5}
8	89.90	150	26	0.0048	2.4×10^{-5}

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Series	Temp (°C)	Nominal Time (min)	Cooling Time (min)	^{3}He measured (pccSTP)	$\pm 2\sigma$
8	90.00	150	26	0.0036	1.9×10^{-5}
9	90.00	180	26	0.0100	6.2×10^{-5}
9	89.92	180	26	0.0067	5.1×10^{-5}
9	90.00	180	26	0.0054	4.8×10^{-5}
10	99.99	150	26	0.0104	4.3×10^{-5}
10	99.98	150	26	0.0068	3.5×10^{-5}
10	100.00	150	26	0.0054	3.4×10^{-5}

Table 2.4: Fitted λ (permeation rate constant) and n_0 (residual helium at the start of the series) for each of the 10 isothermal sequences in Table 2.3, from the weighted non-linear least-squares fit of Eq. (1.9) (Chapter 1) described in the main text (§2.2). n is the number of steps in the sequence. These are the same per-sequence fits plotted in Fig. 2.3 and used to build the Arrhenius regression in Fig. 2.4.

Series	Temp (°C)	Duration (min)	n	λ (s^{-1})	$\pm 2\sigma$	n_0 (pccSTP)	$\pm 2\sigma$
1	105.00	150	3	4.9×10^{-5}	1.4×10^{-5}	0.0521	7.8×10^{-3}
2	109.99	150	3	6.3×10^{-5}	1.8×10^{-5}	0.0508	7.3×10^{-3}
3	99.86	120	4	4.3×10^{-5}	1.3×10^{-5}	0.0377	5.2×10^{-3}
4	95.00	150	3	3.1×10^{-5}	2.6×10^{-6}	0.0444	2.2×10^{-3}
5	104.99	120	3	5.0×10^{-5}	6.2×10^{-6}	0.0696	4.7×10^{-3}
6	100.00	120	3	3.7×10^{-5}	4.1×10^{-6}	0.0534	3.4×10^{-3}
7	95.00	120	3	3.3×10^{-5}	1.5×10^{-5}	0.0348	1.1×10^{-2}
8	90.00	150	3	2.7×10^{-5}	4.8×10^{-6}	0.0279	3.5×10^{-3}
9	90.00	180	3	3.0×10^{-5}	9.0×10^{-6}	0.0357	7.0×10^{-3}
10	99.99	150	3	3.8×10^{-5}	1.2×10^{-5}	0.0351	7.0×10^{-3}

Permeation Step Method Step-Heat Dataset

The complete corrected step-heat dataset produced by the permeation step method, using the in-experiment rate constants above (Fig. 2.4), is reported in Table 2.5, giving the central value and 2σ uncertainty for each quantity, analogous to Table 2.1 in the main text for the bounding method. 4He is not treated by the permeation step method here: 4He permeation-step signals were blank level, so no direct, step-by-step quantification of the ampoule's 4He inventory was possible, and 4He throughout this chapter is instead reduced exclusively by the bounding method of the main text (§2.3). The permeation step dataset reported here is retained only for 3He, as the internal check on the bounding method discussed in §2.5.

Table 2.5: ^{3}He step-heat dataset for HT-Win-06-01b under the permeation step method: measured and cumulative ^{3}He, the initial and residual ampoule content (n_0, n_t), the true step production (ω), and the apparent diffusivity $\ln(D/a^2)$, each with its 2σ uncertainty. Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal, as in Table 2.1.

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
170	300	0.9630	0.0000	0.00×10^{0}	0.00×10^{0}	0.1968	2.22×10^{-2}	1.1597	2.63×10^{-2}	-21.910	0.045
180	463	0.5442	0.9630	7.26×10^{-2}	1.29×10^{-2}	0.0782	4.93×10^{-3}	0.5497	1.60×10^{-2}	-22.184	0.089
190	550	0.5292	1.6241	1.95×10^{-2}	2.54×10^{-3}	0.0679	8.20×10^{-3}	0.5777	1.30×10^{-2}	-21.976	0.085
200	500	0.5190	2.2112	3.28×10^{-2}	5.42×10^{-3}	0.0532	5.77×10^{-3}	0.5394	1.15×10^{-2}	-21.702	0.088
210	475	0.5770	2.7646	1.41×10^{-2}	2.11×10^{-3}	0.0552	5.71×10^{-3}	0.6182	1.17×10^{-2}	-21.311	0.082
220	450	0.6333	3.3799	1.10×10^{-2}	1.80×10^{-3}	0.0469	4.58×10^{-3}	0.6692	1.24×10^{-2}	-20.991	0.080
230	425	0.7899	4.0551	5.87×10^{-3}	1.86×10^{-3}	0.0639	3.48×10^{-3}	0.8479	1.25×10^{-2}	-20.514	0.066
225	908	0.7513	4.8885	1.95×10^{-2}	2.45×10^{-3}	0.0420	4.94×10^{-3}	0.7738	1.42×10^{-2}	-21.200	0.121
215	450	0.1415	5.6842	1.31×10^{-2}	1.85×10^{-3}	0.0278	3.54×10^{-3}	0.1563	6.48×10^{-3}	-22.015	0.316
205	520	0.0763	5.8529	$7.18\times10^{-3*}$	$1.23\times10^{-3*}$	0.0125*	$1.63\times10^{-3*}$	0.0816*	$8.10\times10^{-3*}$	-22.788*	0.829*
195	1106	0.0670	5.9491	$2.12\times10^{-3*}$	$4.02\times10^{-4*}$	0.0307*	$2.86\times10^{-3*}$	0.0956*	$3.23\times10^{-3*}$	-23.370*	1.094*
185	740	0.0227	6.0246	$1.25\times10^{-2*}$	$1.64\times10^{-3*}$	0.0090*	$1.06\times10^{-3*}$	0.0192*	$2.00\times10^{-3*}$	-24.567*	2.822*
175	920	0.0180	6.0610	$3.68\times10^{-3*}$	$5.02\times10^{-4*}$	0.0045*	$4.42\times10^{-4*}$	0.0189*	$7.42\times10^{-4*}$	-24.796*	3.797*
200	260	0.0224	6.0814	$6.53\times10^{-4*}$	$1.17\times10^{-4*}$	0.0095*	$1.23\times10^{-3*}$	0.0313*	$1.65\times10^{-3*}$	-23.022*	1.495*
210	520	0.0812	6.1081	$7.48\times10^{-3*}$	$9.98\times10^{-4*}$	0.0145*	$1.32\times10^{-3*}$	0.0882*	$3.85\times10^{-3*}$	-22.669*	1.135*
220	1167	0.5766	6.1914	2.82×10^{-3}	1.44×10^{-3}	0.0276	3.24×10^{-3}	0.6014	2.45×10^{-2}	-21.504	0.184
230	370	0.3577	6.7796	8.68×10^{-3}	3.96×10^{-3}	0.0422	5.13×10^{-3}	0.3912	1.48×10^{-2}	-20.712	0.146
240	320	0.6767	7.1563	1.72×10^{-2}	2.42×10^{-3}	0.0826	9.94×10^{-3}	0.7420	1.57×10^{-2}	-19.807	0.099
245	250	0.5549	7.8582	2.13×10^{-2}	8.87×10^{-3}	0.0516	5.65×10^{-3}	0.5852	1.46×10^{-2}	-19.710	0.125
250	210	0.6934	8.4743	1.08×10^{-2}	4.52×10^{-3}	0.0718	8.15×10^{-3}	0.7545	1.41×10^{-2}	-19.200	0.102
245	270	0.4659	9.2086	2.48×10^{-2}	3.48×10^{-3}	0.0498	5.84×10^{-3}	0.4909	1.01×10^{-2}	-19.810	0.197

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Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
240	320	0.2857	9.7206	2.07×10^{-2}	2.81×10^{-3}	0.0339	4.08×10^{-3}	0.2989	6.58×10^{-3}	-20.433	0.312
235	370	0.2199	10.0357	1.66×10^{-2}	2.20×10^{-3}	0.0273	3.29×10^{-3}	0.2307	5.23×10^{-3}	-20.810	0.401
230	420	0.1400	10.2736	1.32×10^{-2}	1.76×10^{-3}	0.0255	2.53×10^{-3}	0.1523	4.00×10^{-3}	-21.332	0.603
225	652	0.1494	10.4279	1.26×10^{-2}	1.62×10^{-3}	0.0131	1.47×10^{-3}	0.1499	3.42×10^{-3}	-21.772	0.747
230	500	0.1550	10.5908	7.34×10^{-4}	1.67×10^{-3}	0.0128	1.32×10^{-3}	0.1671	3.37×10^{-3}	-21.383	0.461
235	450	0.1951	10.7582	2.57×10^{-3}	4.18×10^{-4}	0.0334	1.85×10^{-3}	0.2259	3.29×10^{-3}	-20.956	0.364
240	400	0.2425	10.9638	6.90×10^{-3}	1.45×10^{-3}	0.0251	3.16×10^{-3}	0.2607	1.23×10^{-2}	-20.672	0.318
245	350	0.3150	11.2328	5.79×10^{-3}	1.05×10^{-3}	0.0395	4.64×10^{-3}	0.3487	9.39×10^{-3}	-20.219	0.240
250	300	0.3716	11.5689	1.65×10^{-2}	2.24×10^{-3}	0.0467	5.24×10^{-3}	0.4018	9.62×10^{-3}	-19.888	0.209
250	300	0.3640	11.9627	2.43×10^{-2}	4.37×10^{-3}	0.0339	3.82×10^{-3}	0.3735	7.99×10^{-3}	-19.927	0.242
250	300	0.3452	12.3490	1.13×10^{-2}	1.60×10^{-3}	0.0395	4.48×10^{-3}	0.3733	7.91×10^{-3}	-19.895	0.244
250	10250	8.4829	12.7167	1.91×10^{-2}	2.53×10^{-3}	0.1165	7.38×10^{-3}	8.5804	1.28×10^{-1}	-19.958	0.189
240	785	0.2261	21.2187	2.90×10^{-9}	6.85×10^{-9}	0.0263	2.92×10^{-3}	0.2525	2.21×10^{-2}	-20.660	0.580
Total			146.1483								

Permeation Step Approach and Concordance with Bounding Approach

Beyond the ampoule's residual helium content itself (§2.5), the two methods' derived apparent diffusivities agree just as closely. Fig. 2.17 compares $\ln(D/a^2)$ from the permeation-step method against the bounding method's central value for every step in the experiment, with the bounding method's own lower- and upper-bound spread on one axis and the permeation step method's propagated uncertainty on the other; the two steps marked "†" in Table 2.5 above are excluded for the reason given there. The two diffusivity datasets track the 1:1 line within uncertainty across the full range of the experiment. Fitting the permeation-step method's own diffusivities with the same Stage 1 procedure used for the bounding method in the main text (Fig. 2.18) yields an activation energy of 157.0±4.8 kJ/mol (R^2=0.991, n=12) and a closure temperature of 97.5±7.8°C, indistinguishable from the bounding method's own 157.1±5.3 kJ/mol and 98.1±8.6°C (Fig. 2.7, main text).

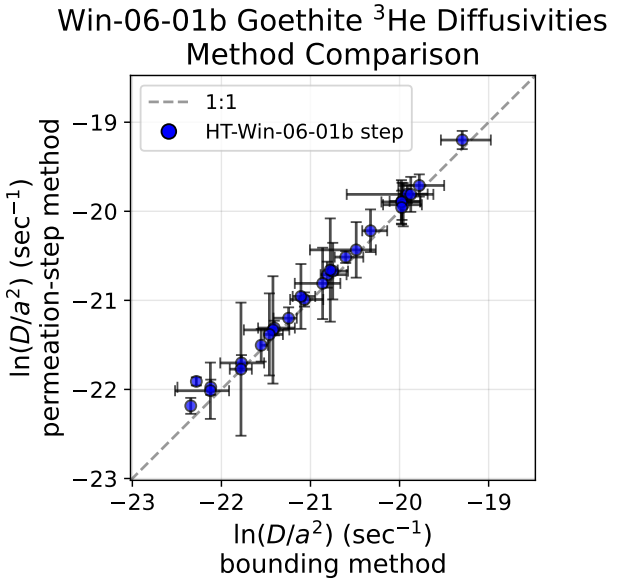

Figure 2.17: Apparent ^{3}He diffusivities for HT-Win-06-01b from the permeation-step method (y-axis) against the bounding method's central value (x-axis, Eq. (1.10)), with the bounding method's lower/upper-bound spread and the permeation step method's own propagated uncertainty plotted on their respective axes. The dashed line is 1:1. Two steps (195°C/1106 min and 185°C/740 min) are excluded for the reason given in Table 2.5's caption.

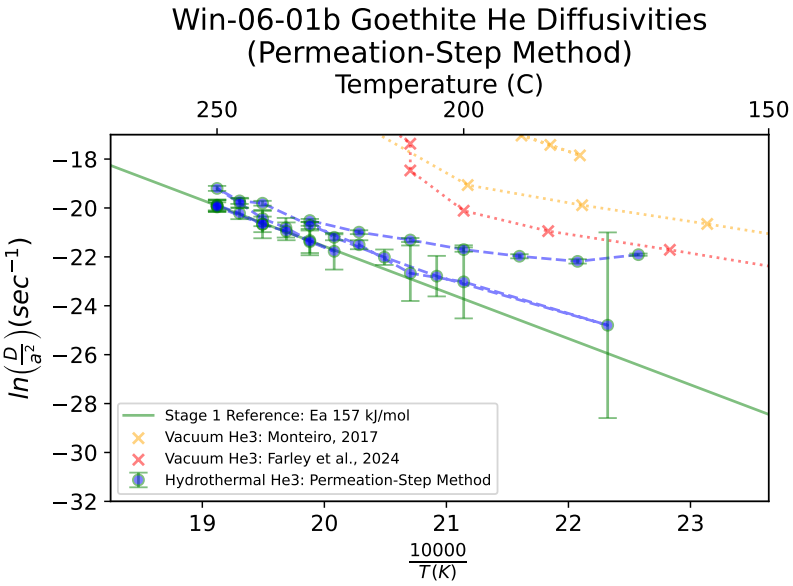

Figure 2.18: Analogue of Fig. 2.7 (main text) using the permeation-step method's own ^{3}He diffusivities and its own Stage 1 linear regression, rather than the bounding method's. The resulting activation energy and closure temperature (157.0±4.8 kJ/mol; 97.5±7.8°C) agree closely with the bounding method's own (157.1±5.3 kJ/mol; 98.1±8.6°C).

Chapter 3

(U-TH-SM)/HE SULFIDE GEOCHRONOLOGY

Sulfide mineralization is a near-ubiquitous process in the formation of ore deposits of valuable chalcophile and siderophile elements. Helium isotope studies of sulfides have been applied to infer the origins of mineralizing hydrothermal fluids, and some pioneering studies have applied the U-Th/He method to date sulfides, especially pyrite (O. Yakubovich, Podolskaya, et al., 2020; O. Yakubovich, Vikentyev, et al., 2021). Yet, the interpretation of helium isotope data or U-Th/He ages requires an understanding of the conditions under which helium in sulfides behaves as an open or closed system, as well as the mechanisms of helium loss, a process which is poorly understood in sulfides. Furthermore, sulfides systematically reject U-Th and only produce appreciable radiogenic helium when they contain U-Th bearing inclusions, further complicating the interpretation of U-Th/He data in sulfides. We explore the systematics of U-Th/He in sulfides through two lenses: 1) an example application of the sulfide U-Th/He geochronometer to two test cases, and 2) helium step heat experiments of sulfide phases. We report pyrite/arsenopyrite U-Th/He ages for Au-ores in the Permian-Triassic Gympie formation of Queensland, Australia, and the Archean Kidd Creek VMS deposit of Ontario, Canada. The Gympie arsenopyrite yields a mean age of 258 ± 35 Ma and spans the range between reported Ar/Ar and K/Ar ages of the country rock and the mineralized quartz veins (Holcombe et al., 1997; Little et al., 1995), suggesting the sulfides predate and were variably reset by the primary mineralization event. Pyrite/arsenopyrite from the 2.7 Ga Kidd Creek VMS deposit of Ontario contain large apatite inclusions ($r \leq 30$ μm) and span a range of ages from 465-1600 Ma. Although one aliquot can be explained as a perfectly retentive 2.7 Ga sulfide host of apatite inclusions that record a ∼370 Ma cooling age, the remaining aliquots are too young to be explained by a closed sulfide host and instead require that the sulfide host lost helium over geologic time.

Helium step heat experiments in sulfides are challenging because sulfides readily react/decompose in the high vacuum, high temperature conditions of the experiments, producing results that reflect phase decomposition/reaction kinetics that are not geologically relevant. We address this challenge by performing helium step heat experiments at temperatures up to 600°Con natural single-crystal pyrite, arsenopyrite, and galena implanted with uniform 3He through proton bombardment sealed in helium-permeable silica glass tubes stabilized under high sulfur (± arsenic) fugacity to stabilize the phase against decomposition during analysis. We were unsuccessful in extending this method to stabilize chalcopyrite, chalcocite, or marcasite during step heat experiments: the helium retentivity of these phases in nature remains unclear. Retrograde-prograde thermal cycles on sulfur-stabilized galena support a Multiple Diffusion Domain (MDD) helium loss mechanism with an activation energy around 174.8 kJ/mol, but with realtively low helium retentivity that makes it unlikely to be useful as a target for helium isotope studies orU-Th/He geochronology. Comparisons of helium step heat data of sulfur-stabilized pristine and deformed pyrite point towards a multipath diffusion model for deformed pyrites, in which dislocations and fine fractures serve as high activation energy (300.6 ± 7.3 kJ/mol) fast pathways, with bulk volume diffusion following a lower activation energy that is constrained by step heats on pristine pyrite samples: 199.1 ± 3.5 kJ/mol. Helium step heat data of arsenopyrite are complex, potentially complicated by exchange of sulfur and arsenic with the vapor phase, but qualitatively demonstrate very high helium retentivity. These experiments demonstrate that pyrite is similarly helium-retentive to monzonite-structure phosphates and is more retentive of helium than natural zircon, apatite, hematite, or goethite. Arsenopyrite is less retentive, more similar to the most helium-retentive zircon.

3.1 Introduction

Sulfide minerals are nearly ubiquitous in the formation histories of economic ore deposits, and indeed are themselves the primary repositories for many base and precious metals in the crust. The question of when and how they formed

is central to understanding ore-forming processes, crustal fluid circulation, and the tectonic contexts in which economic deposits develop. Despite decades of study, the timing of sulfide mineralization remains contested or poorly resolved across a wide range of deposit types, from shallow epithermal systems to deep, high temperature magmatic deposits in mafic intrusions. This uncertainty reflects both the genuine complexity of ore-forming systems (which are frequently multistage, hydrothermal, and overprinted) and the limitations of the geochronological tools available for dating sulfide phases directly.

An Alternative and Compliment to Re-Os Dates in Sulfide Geochronology

The rhenium-osmium isotope system is the most widely applied method for direct radiometric dating of sulfide minerals, exploiting the relatively high Re concentrations that pyrite, molybdenite, arsenopyrite, and pyrrhotite can accommodate. Re-Os geochronology has produced geologically meaningful ages across a wide range of sulfide-bearing deposit types, and remains indispensable for directly dating ore-forming events (M. Chen et al., 2015; Hnatyshin et al., 2026; Schoenberg et al., 1999; Ostendorf et al., 2019; Selby et al., 2002). Nevertheless, the Re-Os system is vulnerable to a set of well-documented failure modes, and a growing number of published ages exhibit scatter, implausible geologic interpretations, or internal inconsistencies that motivate the development of independent dating methods (Stein et al., 2003; Suzuki et al., 2001; Li et al., 2025; Lyu et al., 2020).

More practically, the highly specialized and costly n-TIMS method for Re-Os dating restricts its application to a select few labs, limiting the scale of its application (Ackerman, 2025). Furthermore, Re-Os dating of sulfides (other than the rare mineral molybdenite) requires gram-quantities (Qi et al., 2010) and is therefore poorly suited to resolve distinct mineralization episodes in multistage systems.

For multistage systems in particular, the co-application of independent radiochronometers is often useful to distinguish the timings for hydrothermal alteration from uplift, cooling, exhumation, and potentially earlier emplacement of host rocks. For example, Selby et al. (2002) demonstrate that Re-Os molybdenite is more likely to accurately record the age of sulfide mineralization in the Tintina Gold Belt, Alaska, relative to the comparatively thermally-sensitive

Ar-Ar mica ages. However, molybdenite is far from a ubiquitous phase in sulfide-bearing ores, unlike pyrite. A pyrite U-Th/He geochronometer would expand such opportunities to co-apply radiochronometers that interrogate processes operating at different times and temperatures to a much more diverse and expansive array of locales that experienced complex histories. Similar geochronometers developed for other sulfide phases such as arsenopyrite, galena, and sphalerite may offer the opportunity to disentangle multiple episodes of sulfide mineralization and alteration within an ore body, allowing geologists to better understand the sequence of events that lead to economic ore formation.

One such example of an opportunity for applying U-Th/He sulfide geochronology is in the world's largest known platinum-group element deposit, the Bushveld Complex of South Africa, which hosts the sulfide-bearing Merensky Reef and Platreef PGE deposits in the sulfide-bearing upper critical zone, which for which geochronometers yield a surprising diversity of ages: zircon U/Ph ages vary from ≈2057-2060 Ma, while rutile ages are significantly younger at ≈2053 Ma (Scoates et al., 2021), yet Re-Os systematics show evidence for multiple sulfide mineralization events within the same horizon (Schoenberg et al., 1999). The Re/Os and Rb/Sr isochron ages they report are unable to resolve these distinct mineralization events from the zircon U/Pb emplacement age. Unlike Re/Os dating, the U-Th/He system produces distinct ages for every dated aliquot, potentially offering the opportunity to resolve the timings of prolonged volatile activity or late hydrothermal Re-Os disturbance.

Similarly, at the Norilsk-Talnakh deposits of the Siberian Traps, the ore-bearing intrusions can be correlated with a specific transition in flood basalt source and eruptive style (Yao and Mungall, 2021), yet only three of many differentiated intrusions in the region are economic, and the precise timing and mechanism of sulfide saturation relative to trap emplacement is not fully understood. In the 1.1 Ga Sonju Lake Intrusion of Minnesota, USA, stable isotopic evidence suggests that meso/epithermal fluids reacted with primary magmatic sulfides raising questions about when redistribution occurred and how much metal was moved (Park et al., 2004). These are deposits where independent thermochronometric constraints capable of distinguishing primary crystallization from later hydrothermal reworking are needed.

The Single-Jump Migration Model and its Application to Pyrite

The Russian school of noble gas geochronology spearheaded by Shukolyukov and Yakubovich, approaches the problem of helium migration in minerals through a theoretical framework that differs fundamentally from the volume diffusion model used for to describe noble gas migration dominant in the Western thermochronological literature (Dodson, 1973; Harrison et al., 1985; Fortier and Giletti, 1989; Lovera et al., 1991; Wolf, Farley, and Silver, 1996; Farley, 2000; Lee, Williams, et al., 1997). In the Shukolyukov "single-jump model," helium atoms occupy discrete energy minima within the crystal structure, and migration proceeds through thermally activated jumps between these sites (Y. A. Shukolyukov et al., 2009; O. V. Yakubovich, Y. A. Shukolyukov, et al., 2010).

Within this framework, Y. Shukolyukov et al. (2012) developed the ^{190}Pt-^{4}He system as a new geochronometer for native platinum minerals, demonstrating through thermal desorption analysis that radiogenic ^{4}He in isoferroplatinum is stable to temperatures approaching the melting point of platinum, and obtaining ^{190}Pt-^{4}He ages from the Galmoenan and Kondyor massifs consistent with independent Sm-Nd and Rb-Sr estimates. He storage in native platinum was attributed to accumulation in vesicles and clusters at dislocation cores and grain boundaries. Western scientists similarly model helium migration in metals with single-jump mechanistic single-jumps, although these single jumps are not typically modeled to instantaneously liberate helium, instead the modeled single-jump allows the atom to spontaneously travel a distance of some angstroms with some frequency, introducing a spatial scale from which a diffusivity can be evaluated (Rätzke et al., 1992; Cost and Hickman, 1975; Wilson and Bisson, 1973).

O. V. Yakubovich, Mochalov, et al. (2015) extends this method to sperrylite ($PtAs_2$) a rare mineral found in magmatic PGE deposits that is isostructural with pyrite. ^{190}Pt-^{4}He ages from sperrylite at the Kondyor and Norilsk-Oktyabrsky deposits were geologically consistent, and the high He retentivity demonstrated in this arsenide prophesies similar behavior in pyrite.

Without providing any sulfur buffer, O. V. Yakubovich, Gedz, et al. (2019) interpreted ramped-heating experiments for a variety of silica glass encapsulated sulfides under the single-jump step-heating method. Pyrite, in particular, was found to be anomalously high retentivity of helium. The first application

of U-Th/He pyrite geochronometry to a natural ore deposit was reported by O. V. Yakubovich, Gedz, et al. (2019) and O. Yakubovich, Podolskaya, et al. (2020) for the Uzelga Cu-Zn VMS deposit of the South Urals, yielding a consistent weighted mean age of 382 ± 12 Ma that is older than regional prehnite-pumpellyite metamorphism, demonstrating that pyrite can survive a metamorphic overprint without complete isotopic resetting. While these results establish the viability of pyrite as a U-Th/He geochronometer, their analytical approach employs a simple, low-resolution monotonic heating schedule without grain-size scaling experiments, sulfur buffers, and proton-implanted ^{3}He domain mapping, leaves the diffusion/migration physics of sulfides undercharacterized. Unsurprisingly, given the rapid heating rates used in their experiments, their results for pyrite generally show abrupt helium loss at temperatures corresponding to the upper stability field of pyrite around 700°C Toulmin and Barton (1964), and therefore are unlikely to contain substantial information about helium loss in natural pyrite over geologic timescales. Extending these initial observations into a robust, predictive thermochronometric framework requires the more rigorous experimental and theoretical treatment that is one of the aims of this work.

Geologic Setting

Pyrite and Arsenopyrite for the geochronologic study was collected from two deposits that strongly differ in the timing and conditions of their formation: the Archean Kidd Creek volcanogenic massive sulfide (VMS) deposit, Ontario, Canada, and the Permo-Triassic orogenic gold system at Gympie, Queensland, Australia. Pairing a ∼2.7 Ga Cu-Zn-Ag VMS sulfide with a ∼235 Ma Au-bearing arsenopyrite provides two independently well-constrained age brackets against which to test He retentivity and closure behavior in pyrite across very different formation temperatures, host lithologies, and post-depositional thermal histories.

Kidd Creek, Ontario, Canada

Kidd Creek is among the largest and most profitable VMS deposits in the world, with past production and reserves exceeding 138.5 Mt at average grades of 2.4 percent Cu, 6.5 percent Zn, 0.23 percent Pb, and 90 g/t Ag (Hannington et al., 1999). The deposit occurs within the Late Archean Kidd-Munro assemblage of the Abitibi subprovince, Superior Province, hosted at the top of a thickened pile

of massive rhyolite flows, felsic volcaniclastics, and coarse epiclastic rocks that attain a maximum thickness of ∼300 m beneath the mine (Hannington et al., 1999). The three main stratiform orebodies (North, Central, and South) formed largely by subseafloor infilling and replacement of permeable volcaniclastic rocks within a subsiding, rift-related graben, rather than by high-temperature venting at the seafloor (Hannington et al., 1999). U-Pb zircon geochronology brackets the ore-forming interval tightly: the oldest footwall rhyolite is dated at 2716.1 ± 0.6 Ma, the onset of bimodal komatiite-rhyolite volcanism at 2717 ± 1.1 Ma, and the youngest, uppermost quartz porphyry unit at 2711.5 ± 1.2 Ma, indicating that mineralization took place during at least one and possibly several million years of intermittent volcanic activity (Hannington et al., 1999). Oxygen isotope studies of the host rhyolites show that, while primary igneous zircon retains magmatic δ^{18}O values (5.4‰ ± 0.3‰), coexisting quartz phenocrysts are pervasively shifted to much higher and more heterogeneous values (14.2‰ to 16.1‰), recording extensive subsolidus recrystallization during the same long-lived hydrothermal convection that deposited the sulfides (King et al., 1997).

The interior of the Canadian shield has been a stable platform for hundreds of millions of years, as documented by U-Th/He and fission track dating of apatite (Sturrock et al., 2024). Kidd Creek in particular has been of particularly interest for its famously ancient groundwaters that sustain a low-biomass but metabolically active community of sulfate-reducing and alkane-oxidizing microorganisms (Lollar et al., 2019). The Kidd Creek deposit is thus an ideal natural laboratory for testing the long-term He retentivity of pyrite and arsenopyrite.

Gympie, Queensland, Australia

The Gympie arsenopyrite comes from a very different setting: an accreted, Early Permian bimodal (basalt-dacite) submarine volcanic sequence, the Gympie Group, interpreted as a fragment of an intra-oceanic island arc that rifted from the Gondwana margin before docking against the New England Orogen of eastern Australia (Sivell and Waterhouse, 1988). Drillcore near the Gympie Eldorado Gold Mine resolves at least nine sequentially emplaced magmatic suites and a >2 km thick volcano-sedimentary succession. This arc terrane is inferred to have accreted to the Gondwana margin by about 220 Ma (Early to

Middle Triassic), based on cross-cutting granitoid relationships with adjacent New England Orogen terranes (Sivell and McCulloch, 2001).

Gold-bearing pyrite-quartz mineralization at Gympie occurs in fibrous extensional veins developed within sheared, carbonaceous black slates. Alteration associated with this vein set has yielded a K/Ar age of about 235 Ma (Holcombe et al., 1997). Refractory dolerite dykes and later microdiorite-shoshonite dykes cross-cut the mineralized structures and bracket the timing of the auriferous quartz veining from below and above, respectively, giving an independent field constraint on the age of mineralization that is broadly consistent with the K/Ar alteration ages (Sivell and McCulloch, 2001).

The nominal "Early Permian" age assigned to the host rocks to the Gympie deposit, the Highbury Volcanics and Rammutt Formation, however, rests mainly on sparse invertebrate and plant fossil biostratigraphy higher in the sequence and on lithologic correlation, rather than on direct radiometric dates on the volcanic pile itself. Holcombe et al. (Holcombe et al., 1997) note that the lower Gympie Block volcanics, like the correlative Berserker beds, "may contain older volcanic units," and fold the entire Gympie succession into a long-lived (∼320-280 Ma, peaking at ∼305 Ma) regional magmatic episode. These volcanics may very well be related to related to nearby ∼300-310 Ma granodiorite magmatism and rapid uplift of the regional epidote-blueschist-facies basement, which record $^{40}Ar/^{39}Ar$ hornblende cooling ages around 297 Ma Little et al. (1995) and Holcombe et al. (1997).

3.2 Experimental Methods

U-Th/He Geochronology

Sample Preparation

Intact crystals of arsenopyrite (and pyrite from the Kidd Creek deposit) were extracted from samples from the Kidd Creek and Gympie deposits with a hammer and chisel at Caltech and SELFRAG high-voltage pulse fragmentation at the University of Queensland, respectively. Pyrite is present but small from the gympie sample and no pyrite dates are reported for it. The majority of the Kidd Creek samples were silver arsenopyrite, but brassy cubes (presumed pyrite) were also analyzed. A subsample of the gympie arsenopyrites were leached in 1M HCl for 4 hours to remove any surface contamination and then rinsed in MilliQ water and dried. Individual crystals were washed in a sequence

of acetone, ethanol, and MilliQ water baths, then promptly dried using clean, lint-free wipes. These samples, along with the hematite MI-43 standard from Gogebic, Michigan (Farley and McKeon, 2015) and the Apatite standard from Durango, Mexico (Farley, 2000) were weighed, then encapsulated in silica glass ampoules similarly to the sulfides in section 3.2, but without any additionaly buffer materials. An empty tube was also encapsulated to serve as a procedural blank. Photos and masses of the prepared ampoules were recorded so that the helium and trace metal analyses could be correctly matched.

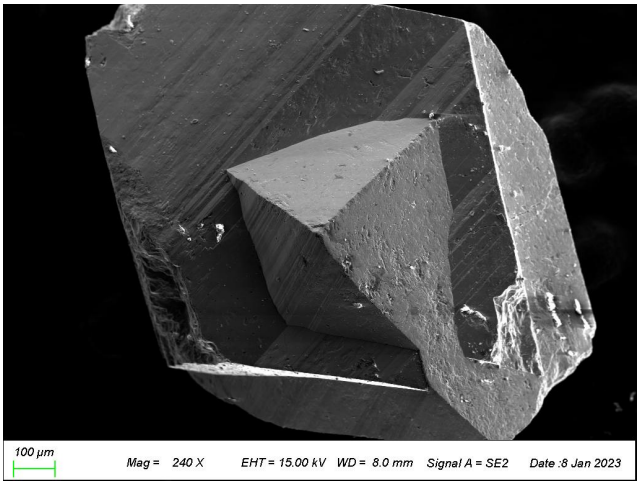

Figure 3.1: Secondary-electron SEM image of crystallographic arsenopyrite twins exceeding 1 cm across, recovered intact from Kidd Creek ore (sample ST05) with a hammer and chisel. Fine-scaled growth striations are preserved on the crystal faces. Euhedral crystals of this size were extracted routinely, showing that simple mechanical excavation preserved crystal faces well.

Helium Analysis

Encapsulated samples were wrapped in Ni-foil and loaded into a custom-built autosampler with the procedural blank loaded first. The autosampler was wrapped in heating tape, which was made to warm the samples to 80-100°C for at least 12 hours in ultra-high vacuum before the first step heat to allow for atmospheric helium to escape. Initial efforts in method development showed that atmospheric helium contamination was significant (Fig. 3.2). This atmospheric helium payload of an ampoule equilibrated with air is on the order of 10s of nccSTP helium per ampoule (Chapter 1, §1.2). It is difficult to mathmatically correct for even when 3He can be accurately measured, because helium diffusion and permeation through silica glass fractionates 3He/4He ratios

(Shelby, 1971; Jones, 1953), and the ampoules were each geometrically unique and therefore had different permeation rates. Still, we found the best results when we applied a simple correction to the helium data after substantially mitigating this contamination with the warm (80-100°C) degassing step. To correct for the much smaller remaining atmospheric contamination, we subtract 4He proportionally to measured 3He by the atmospheric 3He/4He isotope ratio: 1.39×10^{-6}. The procedural blanks showed that atmospheric helium contamination was reduced by a factor of 5-10, which was sufficient for these helium-rich samples with 100s of nccSTP helium (Fig. 3.2). Future work may develop a more rigorous approach to experimentally remove atmospheric helium contamination or to mathematically correct for it using the first order framework presented in Chapter 1 (§1.3).

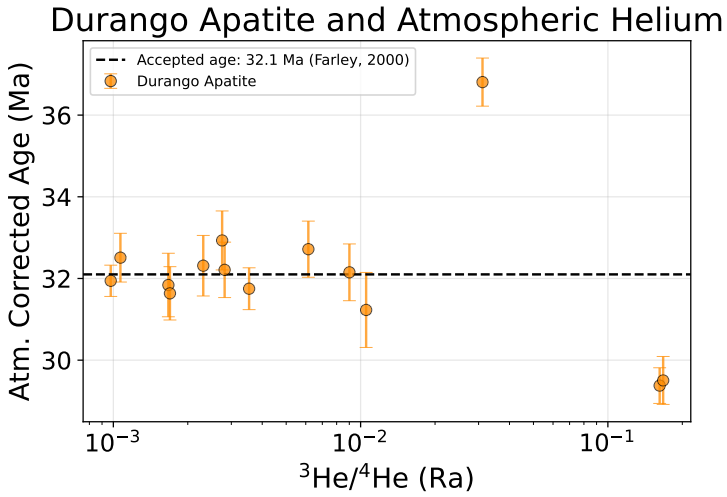

Figure 3.2: The atmosphere-corrrected age for encapsulated Durango apatite is plotted against the measured helium isotope ratio ($^3He/^4He$), normalized to atmosphere (Ra), illustrating the atmospheric-contamination problem motivating the experimental correction protocol described in 3.2. Aliquots with low measured Ra (i.e. a small atmospheric 4He contribution relative to radiogenic 4He) require insignificant mathmatical correction, preserving the accuracy of the ages and agreeing with the 32.1Ma accepted age (Farley, 2000). Aliquots measured before implementing the experimental atmosphere-degassing step contain more 3He and require stronger corrections. Because these corrections are unaware of the true isotopic composition of fractionated atmosphere in the ampoule, they yield inaccurate ages when they are large with respect to the sample's total measured helium. The weighted average age of the aliquots with mass >0.5 mg, 32.1 ± 0.2 Ma, agrees well with the accepted age, affirming the fundamental accuracy of the measurements.

After 12 hours, unheated furnace blank measurements were periodically taken until the measured value was similar to such blanks taken for batches of samples not encapsulated in silica glass, then samples were individually dropped into a resistance furnace and baked at 1000-1200°C for 15 minutes to degas their helium contents. Although pyrite and arsenopyrite are unusually retentive of helium at lower pressures, these temperatures were far above these phases' stability field, and so they were readily degassed. Helium abundances and isotope ratios were measured on a Thermo Fisher Helix SFT mass spectrometer. Gasses evolved during the bake were passed through a charcoal filter held at 77°K, then exposed to an SAES-10 getter and a heated titanium getter before being cryofocused at 14°K, then released at 34°K. Sensitivities were quantified using a calibrated 3He-enriched air standard several times throughout the analysis.

Trace Metal Analysis

The encapsulated, now degassed samples were removed from the furnace and were matched to their pre-bake photos and masses. Around 20% of samples would be typically lost during the helium analysis process, with ampoules either broken by the autosampler or by thermal stresses during the bakes. Samples bearing residual nickel-foil that had fused to the surface of the glass were scratched clean with a dental pick, and the remaining glass ampoule was sonicated in a sequence of ethanol then MilliQ water baths. Samples were then placed in an acid-washed Teflon beaker and were broken underwater by an agate pestle, to prevent loss of sample material. The pestle was rinsed with MilliQ water between samples to avoid cross-contamination. Excess water was pipetted away from each sample and disposed of, then the remaining water was evaporated on a hot plate. The vast majority of the samples were then spiked with a U-232 and Th-230, and vanadium enriched spike and digested in concentrated HNO_3 in closed Teflon beakers, leaving the glass intact. The uranium and thorium concentrations of the glass ampoules were measured to be $\approx$ 50 and 30 ppb by mass respectively, but this contaminant payload was only delievered when the glass was dissolved in HF; the uranium and thorium contents of concentrated nitric acid in extended contact with broken glass were below detection limits.

After this point, the Kidd Creek pyrite/arsenopyrite and Durango Apatite

solutions were pipetted and diluted to an ICPMS-ready solution. The gympie samples, however, required further processing to dissolve fine U-Th bearing rutile inclusions. The gympie and MS-43 hematite samples, now mostly digested in concentrated HNO_3, were dried down and put back up in small volumes of concentrated HCl, then were loaded into Parr vessels and baked at 200°C for 24 hours. The Gympie samples were then dried down again and put back up in concentrated HNO_3 before the concentrated acid was transferred and diluted to an ICPMS-ready solution. U and Th contents were quantified with isotope dilution using an Agilent 8800 ICP-MS. Other trace elements were measured as ratios with spike vanadium. The U-Th contents of the spikes and the sensitivities of the other quantified trace metals relative to vanadium were continuously montiored by analyzing a standard solution of known composition that was spiked with the same U-Th-V spike as the samples. The same empty tube procedural blank used for the helium analysis was also processed through the same acid digestion and ICPMS analysis, and the measured U, Th, and other trace metal contents were subtracted from the measured values of the samples. The U-Th contents of the procedural blank were below detection limits, and the other trace metals were typically 1-5% of the measured values of the samples.

The Gogebic Iron Range hematite standard (Farley and McKeon, 2015) was carried through this same Parr-vessel digestion step alongside the Gympie arsenopyrite, serving as an in-run check on how completely this digestion protocol recovers a sample's true U-Th-Fe budget. Fig. 3.3 plots each Gogebic aliquot's measured ^{4}He against its eU, normalized by the sample mass calculated from measured Fe content where available and by the pre-analysis balance mass otherwise. Except for one clearly erroneous outlier (with an age in excess of 1.5Ga), these samples agree with two independently published ages for Gogebic, Michigain hematite: 453 Ma (U-Th/^{21}Ne, aliquot MI-45; (Farley and McKeon, 2015)) and 579 Ma, (U-Th/He, (Hofmann, Treffkorn, et al., 2020)). Here the outlier is interpreted to have experienced U-loss, as its helium content is similar to the other aliquots, but it has low eU. Although incomplete digestion would also produce erroneous older ages, the apparent helium concentration would be higher and the eU would match the other samples if that was the case.

Encapsulated Sulfide Step Heat Experiments

Samples

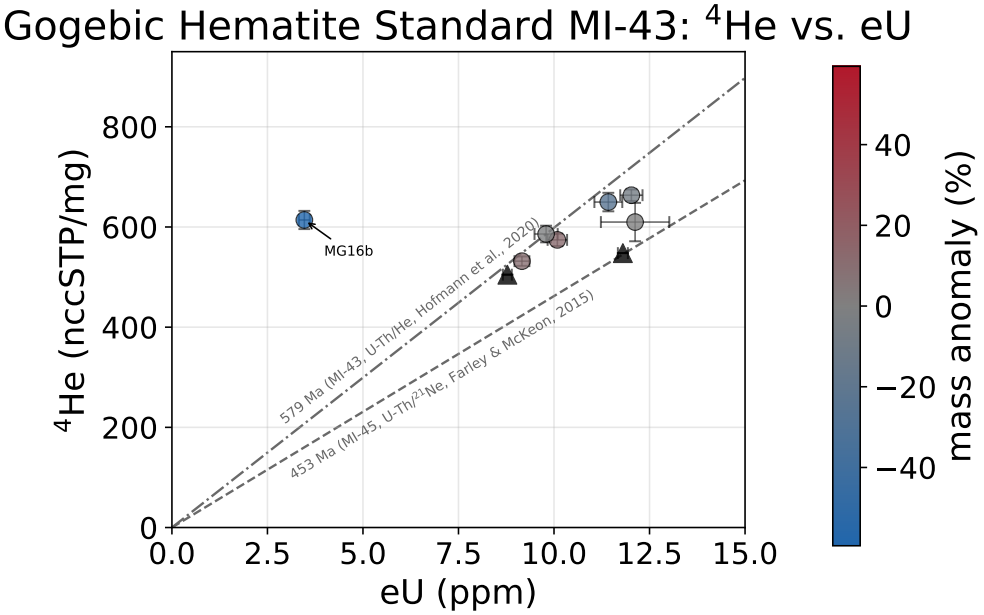

Figure 3.3: Atmosphere corrected ^{4}He (nccSTP/mg) vs. eU (ppm), with 2σ error bars, for the MI-43 Gogebic hematite standard aliquots analyzed alongside Gympie arsenopyrite (Table 3.6), normalized by mass calculated from Fe molar abundaces measured on ICPMS. Fe was not measured for two samples: they are plotted as triangles using their balance-measured mass. These two measures of mass are compared to assess internal consistency within samples. Points are colored by "mass anomaly" in percent: $\frac{M_{Fe}-M_B}{M_{Fe}} \times 100$. Anomalies are generally small: with absolute values 11% or smaller for all but 2 samples. The old outlier sample 041323MG16b with low eU, however, has a mass anomaly of 59%, meaning that the balanced-measured mass was significantly higher than the mass inferred from the iron abundance. Two isochrons of independently published ages for Gogebic hematite are plotted: 453 Ma (MI-45, U-Th/^{21}Ne; (Farley and McKeon, 2015)) and 579 Ma (Hofmann, Treffkorn, et al., 2020). Only the low eU outlier 041323MG16b falls outside of this age range. With anomalously low eU at an otherwise unremarkable ^{4}He content, is interpreted here as an example of U-loss. Its large mass anomaly may point to the presence of an insoluble phase such as quartz, however, it is difficult to explain why an insoluble phase would affect only the eU concentration and not the helium concentration of the hematite.

Nine helium step heat experiments of sulfides (pyrite, arsenopyrite, chalcopyrite, galena, chalcocite, and sphalerite) sealed in silica glass ampoules performed at the Noble Gas Laboratory at Caltech are discussed here. Table 3.1 shows the dimensions, geometric factors, and encapsulated materials of this work. All sulfides of this study were sourced from the Caltech Mineral Collection from visually pure, well-formed euhedral crystals with half prism widths greater than 200 microns, often larger than a centimeter. Samples were broken into smaller pieces under ethanol to avoid irreversible trapping of atmosphere (Cox et al., 2022). Except for the Rustenburg pyrite, all step heats measured 3He evolution from an initially homogeneous distribution of spallogenic 3He. To produce a homogeneous distribution of 3He, the samples were irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following standard procedures (Shuster and Farley, 2005).

Table 3.1: Silica Glass encapsulated sulfides that were the subject of helium step heat experiments in this work. The Rustenburg Pyrite experiment had evolved radiogenic 4He quantified by isotope dilution using a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer, whereas evolved spallogenic 3He was quantified using a Helix-MC-Plus^{10K} multi-collector mass spectrometer for the remainder of the samples, which were proton-irradiated. Experiments Spal230707 and CCT230707 contained no steps for which the uncertainty due to helium permeation through the silica glass capsule could be estimated, and therefore are discussed in a limited, qualitative capacity only in the Supplementary Information (§3.1s)

Experiment Name and Phase	Sample Locality	Tube Inner x Outer Diameters	Geometric factor G *(eqn 2.2, Ch.1)*	VR *(eqn 5.4, Ch.1)*	Encapsulated "Buffer" Materials	Mineral Half-prism Width (a)
LVCOPy230630 Pyrite	Leadville, Colorado	*Not Measured*	*3000 cm^{-2}*	-	~0.2 mg Fused Sulfur	~200 µm
BigLVCOPy230707 Pyrite	Leadville, Colorado	*1300 x 1650 µm*	1985 cm^{-2}	1.6	~0.2 mg Fused Sulfur	~400 µm
112123LVCOPyIrr Pyrite	Leadville, Colorado	830 x 1550 µm	1859 cm^{-2}	2.5	~ 0.2 mg Fused Sulfur and >5 mg silica glass shards	~300 µm
Rustenburg Pyrite Pyrite (4He on Quad)	Rustenburg, South Africa	520 x 810 µm	6675 cm^{-2}	1.4	~0.2 mg Fused Sulfur	106 µm

Experiment Name and Phase	Sample Locality	Tube Inner x Outer Diameters	Geometric factor G (*eqn 2.2, Ch.1)*	VR (*eqn 5.4, Ch.1)*	Encapsulated "Buffer" Materials	Mineral Half-prism Width (a)
112123SalAspIrr Arsenopyrite	Mitterberg, Austria	530 x 970 µm	4711 cm^{-2}	2.3	~0.5 mg Crystalline Realgar (AsS)	~175 µm
GalTS230630 Galena	Tri-State District, Missouri	620 x 920 µm	5273 cm^{-2}	1.2	~0.2 mg Fused Sulfur	~200 µm
CCpTS230630 Chalcopyrite	Baxter Springs, Kansas	540 x 1010 µm	4382 cm^{-2}	2.5	~0.2 mg Fused Sulfur	~200 µm
Spal230707* Sphalerite	Chitwood, Missouri	1110 x 1490 µm	2205 cm^{-2}	0.8	~0.2 mg Fused Sulfur	~200 µm
CCT230707* Chalcocite	Messina, South Africa	1100 x 1520 µm	2044 cm^{-2}	0.9	~0.2 mg Fused Sulfur	~200 µm

Encapsulation

Before encapsulating the sulfide samples, glass tubes were cleaned and pulled into pipettes. 10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thinner pipettes using a natural gas and oxygen welding torch and the thin ends were fused closed.

Aliquots of sulfides were weighed and loaded into room temperature glass tubes along with fused sulfur in the presence of an ionizing fan to mitigate static. The arsenopyrite ampoule was loaded with crystalline realgar (arsenic sulfide) instead of fused sulfur, and one pyrite ampoule was loaded with silica glass pieces above and below the pyrite aliquot to separate the pyrite from the sulfur liquid during the experiment. To seal, the top of each pipette was interfaced with a vacuum line and the tube was pumped to a pressure of <60 millitorr, which greatly aided the ease and quality of the final seal. After attaining rough vacuum, a fine-flamed welding torch was used to seal the top of each ampoule. To determine the geometric factors for permeation of the ampoules, the inner and outer diameters of most ampoules were measured using the measurement function of the Amscope stereo microscope software, which was calibrated and tested during early development of this method.

Two experiments (LVCOpy230630 and BigLVCOPy230707) were completed without these geometric measurements. For these experiments, a geometric factor of 3000 cm^{-2} was arbitrarily assigned to estimate uncertainties of the measurements due to the convolution of the process of helium permeation through silica glass. This assumption is unlikely to affect the interpretation of the results.

The sealed ampoules were placed in copper foil packets and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in detail by Farley, Reiners, et al. (1999). Before the experiment began, each ampoule was washed with ethanol, air-dried, loaded in the lamp cell, pumped to high vacuum, and was held at 100°C for at least 30 minutes under dynamic vacuum to desorb ethanol, water, and other atmospheric species on the ampoule, copper packet, and lamp cell.

Helium Mass Spectrometry

Heating steps and 3He measurements for all samples were automated using custom LabView routines on a Helix-MC-Plus^{10K} multi-collector mass spectrometer, or a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer for the Rustenburg Pyrite ampoule. For all ampoules, a Watlow temperature controller held the ampoule at the temperature set point for a pre-programmed duration, after which the ampoule was allowed to cool while the evolved gas was prepared for measurement for about 25 minutes.

For the irradiated samples, Helium was purified by exposing evolved gasses to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. the Helix-MC-Plus^{10K} mass spectrometer was operating in peak hopping mode and the majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech has shown this approach to be reliable, reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray. 4He signals were near-blank for all experiments of irradiated samples, and no attempt is

made here to interpret those data. 4He overspray likewise did not affect 3He signals. Measurements of high 3He MM gas standard were analyzed every 8 hours to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility. These standards show 3He was generally internally reproducible to within 1% (2σ) for rolling sets of 5 consecutive standards, except for the latest measurements on the 112123SalAspIrr Arsenopyrite, during which the ion source began to fail. Those measurements are excluded from further analysis, and any helium evolved during those steps is likely to be a very small proportion of the total helium contents of the sample (<0.2%) and thus would not significantly affect the calculation of the total helium content of the sample.

Radiogenic 4He evolved during step heats of the encapsulated Rustenburg Pyrite was quantified using isotope dilution with a high purity 3He spike on a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer. A standard gas with a known amount of 4He was measured before and after the experiment, to monitor drift in the relative sensitivities of 3He and 4He and to estimate reproducibility, which was about 6% (2σ). After each step, evolved gas from the ampoule in the lamp heating cell was allowed to mix with the 3He spike and the mixture was frozen on charcoal held at 14°K within a cryostat, then helium was selectively released to the mass spectrometer by raising the temperature to 36°K.

Scanning Electron Microscope Study

The methodology for the SEM study of the sulfides is described in detail in 1.6s.

Analytical Methods For Step Heat Experiments

Similarly to O. V. Yakubovich, Gedz, et al. (2019), these helium step heat experiments were designed according to the assumption that the silica glass ampoule was effectively "helium transparent", or in other words that helium permeation through the silica glass ampoule was nearly instantaneous with respect to the rate of helium evolution from the encapsulated sulfides. Thus, permeation steps (as discussed in chapter 1 and 2), were rarely used. This assumption was acceptable for most of the pyrite and arsonpyrite step heats, and much of the galena and chalcopyrite step heats. Unfortunately, the assumption was not valid for any of the sphalerite or chalcocite step heats, and for some of the lower temperature steps of the other sulfides, which can only be

discussed here qualitatively. The helium yields for steps at high temperatures are approximately the measured value, and the uncertainty of this estimation due to permeation lag constrained by upper bounds following the uncertainty-bounding approach presented in Chapter 1 (§1.4). This approach requires no permeation steps or knowledge of the helium content within ampoule from previous steps, instead calculating the maximum helium content remaining within the ampoule for a given measurement, provided the step was long and hot enough for the tube to have reached a steady state concentration and assuming that the rate of helium evolution from the encapsulated mineral is constant. The steady state assumption is taken to be effectively valid once the ampoule loses memory of the helium content remaining from the previous step. According to eqn. (1.57) in chapter 1, only 1.8% of the difference between the previous step's remaining helium content and the current step's steady state helium content remains when $\lambda t = 4$. In this work, the steady state assumption is only applied for steps long and hot enough for $\lambda t \geq 4$, where $\lambda = KGRT$, K is the temperature-dependent permeation coefficient for 4He or 3He through silica glass, G is the ampoule geometric factor, R is the gas constant, T is the temperature in Kelvin, and t is the time in seconds. The 4He permeation coefficient K is taken from Swets et al. (1961); and these coefficients are multiplied by the permeation coefficient ratio from Jones (1953) to yield the 3He permeation coefficient.

Many heating steps, especially near the beginning of the experiments where lower temperature steps were used, are not interpreted or plotted in this work because $\lambda t < 4$; however, some steps with $\lambda t < 4$ are used when eqn. (1.10) is used to place upper bound constraints on preceding steps, so that eqn. (1.57) can in turn be used to quantify the uncertainty on the yields step that has not reached steady state.

This method of calculating uncertainties produces an asymmetric upper and lower uncertainty for each step. These assymetric uncertainties are propagated through the diffusivity inversion of Fechtig and Kalbitzer (1966) with eqn. (2.1) in Chapter 2. In general, the 2σ analytical uncertainties as calculated using the reproducibility of standards is greatly exceeded by the uncertainties due to the permeation lag of the ampoule as constrained by the procedure outlined in §1.4.

3.3 Results

Sulfide Step Heat Experiments

The apparent diffusivities calculated from the helium step heat data are reported in Tables 3.2–3.5 (one per pyrite/arsenopyrite experiment; the analogous galena and chalcopyrite tables are given in the Supplementary Information, §3.1s). These experiments qualitatively show remarkably high helium retentivity at high temperatures for pyrite and arsenopyrite. For no pyrite or arsenopyrite helium step heat experiment did the final cumulative fractional helium yield exceed 6%, even after repeated heating cycles at $\geq$ 500 °C.

Table 3.2: ^{3}He step-heat dataset for Pyrite, Leadville, CO, 200 µm half-prism width (sample LVCOPy230630), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.4 (Arrhenius) and Fig. 3.5 (delta). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
31	475.0	100.0	0.0075	0.4382	0.0022	-24.682	-24.061	-23.805
32	465.0	120.0	0.0048	0.4430	0.0012	-25.283	-24.681	-24.457
33	455.0	140.0	0.0047	0.4476	0.0010	-25.149	-24.854	-24.650
34	510.0	60.0	0.0105	0.4581	0.0046	-23.282	-23.179	-22.810
35	520.0	60.0	0.0126	0.4708	0.0053	-23.412	-22.965	-22.612
36	530.0	60.0	0.0177	0.4885	0.0068	-22.943	-22.597	-22.263
37	539.9	60.0	0.0204	0.5089	0.0075	-22.815	-22.414	-22.096
38	555.0	60.0	0.0315	0.5404	0.0105	-22.193	-21.930	-21.632
39	545.0	60.0	0.0181	0.5585	0.0065	-23.302	-22.440	-22.127
40	535.0	80.0	0.0175	0.5760	0.0049	-23.190	-22.730	-22.480
41	525.0	100.0	0.0136	0.5895	0.0032	-23.622	-23.180	-22.962
42	515.0	120.0	0.0114	0.6009	0.0024	-23.848	-23.515	-23.319
43	505.0	140.0	0.0078	0.6087	0.0015	-24.400	-24.029	-23.849
44	560.0	60.0	0.0252	0.6339	0.0087	-22.046	-21.984	-21.682
45	570.0	60.0	0.0371	0.6710	0.0121	-21.810	-21.550	-21.258
46	580.0	60.0	0.0465	0.7176	0.0145	-21.553	-21.260	-20.979
47	590.0	60.0	0.0604	0.7780	0.0179	-21.188	-20.925	-20.654
48	600.0	60.0	0.0690	0.8470	0.0195	-20.999	-20.710	-20.450
49	604.7	60.0	0.0691	0.9161	0.0191	-20.945	-20.626	-20.372
50	595.0	60.0	0.0381	0.9542	0.0109	-21.846	-21.163	-20.904
51	585.0	60.0	0.0244	0.9786	0.0073	-22.167	-21.576	-21.310

Table 3.3: ^{3}He step-heat dataset for Pyrite, Leadville, CO, 400 µm half-prism width (sample BigLVCOPy230707), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.4 (Arrhenius) and Fig. 3.5 (delta). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
16	400.0	160.0	0.0201	0.7190	0.0056	-26.075	-26.029	-25.779
17	420.0	140.0	0.0386	0.7576	0.0107	-25.357	-25.204	-24.951
18	440.0	120.0	0.0461	0.8037	0.0132	-25.072	-24.815	-24.554
19	460.0	100.0	0.0521	0.8558	0.0161	-24.734	-24.449	-24.170
20	480.0	80.0	0.0490	0.9048	0.0175	-24.619	-24.229	-23.914
21	500.0	60.0	0.0538	0.9586	0.0245	-24.174	-23.791	-23.402
22	510.0	60.0	0.0559	1.0145	0.0236	-24.262	-23.696	-23.332
23	490.0	70.0	0.0237	1.0382	0.0094	–	-24.668	-24.330
24	470.0	90.0	0.0109	1.0491	0.0036	-27.670	-25.682	-25.396
25	450.0	110.0	0.0053	1.0544	0.0016	-27.693	-26.586	-26.327
26	430.0	130.0	0.0033	1.0577	0.0009	-27.872	-27.226	-26.978
27	410.0	150.0	0.0007	1.0584	0.0002	–	-28.887	-28.642
28	450.0	160.0	0.0055	1.0640	0.0010	-26.951	-26.914	-26.743
29	470.0	140.0	0.0100	1.0739	0.0019	-26.295	-26.187	-26.012
30	490.0	120.0	0.0219	1.0958	0.0043	-25.321	-25.232	-25.049
31	510.0	100.0	0.0378	1.1336	0.0082	-24.596	-24.476	-24.276

Table 3.4: ^{3}He step-heat dataset for Pyrite, Leadville, CO, sulfur-isolated (glass-shard-separated) (sample 112123LVCOPyIrr), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.4 (Arrhenius) and Fig. 3.5 (delta). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
46	500.0	120.0	0.0447	0.1998	0.0138	-22.605	-22.576	-22.268
48	510.1	120.0	0.0311	0.2327	0.0090	-23.201	-22.736	-22.461
49	520.1	120.0	0.0259	0.2586	0.0070	-23.204	-22.796	-22.543
51	530.0	120.0	0.0325	0.2921	0.0082	-22.643	-22.453	-22.213
53	540.0	120.0	0.0277	0.3203	0.0066	-22.823	-22.507	-22.283
55	550.0	120.0	0.0326	0.3531	0.0073	-22.460	-22.249	-22.036
57	560.1	384.0	0.1064	0.4600	0.0061	-22.099	-22.041	-21.978
58	570.1	60.0	0.0156	0.4755	0.0080	-22.460	-21.968	-21.543
59	580.0	60.0	0.0263	0.5018	0.0125	-21.756	-21.400	-20.997
60	590.0	60.0	0.0596	0.5614	0.0263	-20.721	-20.497	-20.107
61	600.0	60.0	0.0630	0.6244	0.0265	-20.852	-20.332	-19.959
62	595.0	60.0	0.0468	0.6712	0.0206	-21.358	-20.542	-20.162
63	590.1	60.0	0.0318	0.7030	0.0146	-21.891	-20.867	-20.480
64	585.1	60.0	0.0259	0.7290	0.0122	-21.846	-21.031	-20.637
65	580.1	60.0	0.0165	0.7455	0.0082	-22.786	-21.452	-21.045
66	575.1	60.0	0.0114	0.7569	0.0058	-23.073	-21.808	-21.390
67	570.1	60.0	0.0101	0.7669	0.0053	-22.775	-21.915	-21.489
68	565.2	60.0	0.0075	0.7744	0.0041	-23.431	-22.203	-21.761
69	560.1	60.0	0.0058	0.7802	0.0033	-23.671	-22.444	-21.988
70	555.2	60.0	0.0051	0.7853	0.0030	-23.638	-22.572	-22.104
71	600.1	60.0	0.0337	0.8190	0.0138	-20.747	-20.655	-20.303
72	551.7	79.4	0.0115	0.8305	0.0047	–	-21.986	-21.641
73	550.2	60.0	0.0044	0.8349	0.0028	–	-22.660	-22.161
74	550.2	60.0	0.0040	0.8388	0.0025	-23.985	-22.751	-22.266
75	500.0	420.0	0.0065	0.8453	0.0005	-24.680	-24.202	-24.131
76	500.1	180.0	0.0023	0.8476	0.0004	-24.621	-24.392	-24.220

Table 3.5: ^{3}He step-heat dataset for Arsenopyrite, Mitterberg, Austria (sample SalAspIrr), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.4 (Arrhenius) and Fig. 3.19 (delta, row 1). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
76	360.0	120.0	0.0052	0.3258	0.0017	–	-24.590	-24.302
78	370.0	120.0	0.0073	0.3342	0.0022	–	-24.224	-23.960
80	380.0	120.0	0.0059	0.3417	0.0016	–	-24.408	-24.166
82	390.0	120.0	0.0074	0.3504	0.0018	–	-24.159	-23.935
84	400.0	120.0	0.0111	0.3625	0.0025	–	-23.728	-23.520
86	395.0	120.0	0.0075	0.3704	0.0018	–	-24.086	-23.871
88	390.0	120.0	0.0049	0.3763	0.0012	–	-24.499	-24.277
90	385.0	130.0	0.0065	0.4182	0.0015	–	-24.194	-23.982
92	380.0	140.0	0.0044	0.4226	0.0010	–	-24.646	-24.443
93	410.0	120.0	0.0154	0.4379	0.0032	-23.277	-23.213	-23.020
95	420.0	120.0	0.0255	0.4693	0.0049	–	-22.647	-22.466
97	430.0	120.0	0.0547	0.5262	0.0097	–	-21.794	-21.621
99	440.0	120.0	0.1123	0.6419	0.0185	–	-20.915	-20.747
101	450.0	120.0	0.1449	0.7912	0.0222	–	-20.455	-20.297
102	445.0	120.0	0.0825	0.8737	0.0131	-21.171	-20.871	-20.716
103	440.0	120.0	0.0510	0.9247	0.0084	-21.566	-21.276	-21.119
104	435.0	130.0	0.0305	0.9552	0.0048	-22.142	-21.825	-21.678
105	430.0	140.0	0.0187	0.9739	0.0028	-22.651	-22.361	-22.221
106	425.0	150.0	0.0149	0.9888	0.0021	-22.849	-22.644	-22.510
107	420.0	160.0	0.0091	0.9979	0.0012	-23.453	-23.189	-23.060
108	415.0	170.0	0.0062	1.0040	0.0008	-23.848	-23.624	-23.499
109	410.0	180.0	0.0048	1.0089	0.0006	-24.113	-23.926	-23.804
110	405.0	190.0	0.0035	1.0124	0.0004	-24.501	-24.304	-24.184
111	400.0	206.0	0.0054	1.0178	0.0007	-24.021	-23.937	-23.823
112	460.0	120.0	0.0588	1.0765	0.0084	-20.992	-20.981	-20.844
113	470.0	100.0	0.0750	1.1515	0.0123	-20.608	-20.493	-20.336
114	480.0	80.0	0.0732	1.2247	0.0147	-20.409	-20.230	-20.041
115	500.0	60.0	0.1425	1.3672	0.0353	-19.292	-19.189	-18.954
116	495.0	60.0	0.0693	1.4366	0.0178	-20.532	-19.831	-19.596
117	485.0	60.0	0.0300	1.4666	0.0083	-21.527	-20.634	-20.387
118	475.0	60.0	0.0100	1.4766	0.0030	-23.470	-21.718	-21.458
119	465.0	60.0	0.0046	1.4812	0.0015	-23.503	-22.482	-22.204
120	455.0	60.0	0.0054	1.4866	0.0019	-22.649	-22.328	-22.025
121	445.0	60.0	0.0022	1.4888	0.0009	-25.234	-23.223	-22.887
122	435.0	60.0	0.0015	1.4903	0.0006	-24.523	-23.620	-23.261

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Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
123	425.0	80.0	0.0004	1.4907	0.0001	–	-25.195	-24.929
125	405.0	120.0	0.0015	1.4922	0.0003	-24.385	-24.298	-24.102
126	395.0	150.0	0.0018	1.4940	0.0003	-24.512	-24.317	-24.151
127	395.0	150.0	0.0009	1.4950	0.0002	-25.402	-24.974	-24.808
128	395.0	150.0	0.0022	1.4972	0.0004	-24.203	-24.122	-23.956
129	395.0	150.0	0.0017	1.4989	0.0003	-24.635	-24.372	-24.206
130	400.0	150.0	0.0018	1.5007	0.0003	-24.527	-24.336	-24.176
131	410.0	415.0	0.0041	1.5048	0.0002	-24.597	-24.519	-24.468
132	420.0	130.0	0.0010	1.5058	0.0002	-25.025	-24.780	-24.619
133	430.0	120.0	0.0022	1.5080	0.0004	-24.001	-23.918	-23.754
134	440.0	593.0	0.0103	1.5183	0.0003	-23.988	-23.950	-23.921

The irradiated Leadville pyrite was the most retentive sample of the sulfide suite. Three separate helium step heat experiments were conducted on this sample, two with varying grain size and a third that attempted to isolate the pyrite from the sulfur liquid by packing glass shards above and below the pyrite grains within the ampoule (3.1). The Arrhenius plots for all three experiments show similar diffusivities (Fig. 3.4). The three experiments shared a similar slope, and a regression was fit to the most linear region of 9 measurements of the LVCOpy230630 experiment with an R^2 of 0.998, yielding an apparent activation energy of 47.59 ± 0.85 Kcal/mol (199.1 ± 3.5 kJ/mol; Fig. 3.4), implying a closure temperature of 294.3 ± 6.7°C assuming a cooling rate of 10°C/Ma. The delta plot constructed with this activation energy showed a general agreement between the smaller grain-size experiments, seemingly unaffected by the direct contact with sulfur liquid or lack thereof. The more retentive, larger ampoule yielded lower delta values at the same cumulative fractional yield, revealing that a smaller proportion of its helium resided in these least retentive domains that produce the common diffusivity trend on the arrhenius plot for the three experiments (Fig. 3.5).

The octahedral Rustenburg pyrite is less retentive of helium than the pyritohedral Leadville pyrite. On the down-temperature retrogrades of heating cycles, the Rustenburg pyrite shows clearly developed reproducible linear arrays with the unusually high implied activation energy of 71.8 ± 1.7 Kcal/mol (300.6 ± 7.3 kJ/mol; Fig. 3.6: R^2 of 0.998 on 5 measurements). This feature is completely absent from the three experiments of proton-irradiated Leadville pyrite (Fig. 3.4). The delta plot constructed using this activation energy shows

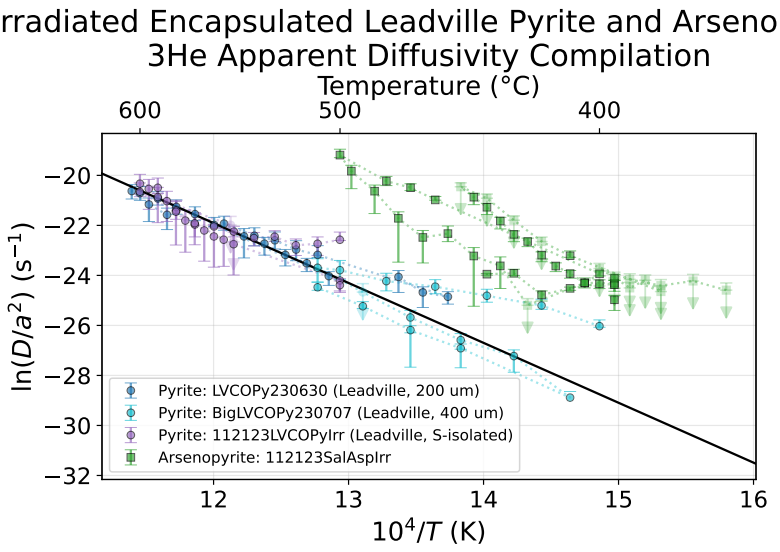

Figure 3.4: Compiled Arrhenius plot of apparent ^{3}He diffusivity for the three irradiated Leadville pyrite step-heat experiments (LVCOPy230630, BigLVCOPy230707, 112123LVCOPyIrr) and the proton-irradiated arsenopyrite experiment (112123SalAspIrr), each with permeation-lag uncertainty bars. Downward-pointing arrows mark steps with an unconstrained lower bound (see §3.2). The solid black line is the regression fit to the most linear region of the LVCOPy230630 experiment (9 points, $R^2 = 0.998$, Ea = 47.59 ± 0.85 kcal/mol), used throughout this section as the shared reference line for all three Leadville pyrite experiments and for arsenopyrite.

a monotonic decrease in delta values throughout the duration of the experiment (Fig. 3.7).

The encapsulated arsenopyrite helium step heat experiment produced an apparent diffusivity Arrhenius plot that showed generally high helium retentivity, but less so than the Leadville pyrites (Fig. 3.4). There were no representative linear regions produced by this experiment that were suitable for linear regression, although the general slope was similar to the Leadville pyrites' and so it was also used as a reference for this experiment. The delta plot using this reference showed complex behavior, where high delta values (far beyond the uncertainty related to the permeation lag of the ampoule itself) were observed for low temperature steps following high temperature steps (Fig. 3.19, row 1, Supplementary Information).

Step heat results for galena (GalTS230630) and chalcopyrite (CCpTS230630) are reported in the Supplementary Information (§3.5). Neither mineral seems to be a promising U-Th/He geochronometer or helium isotope target.

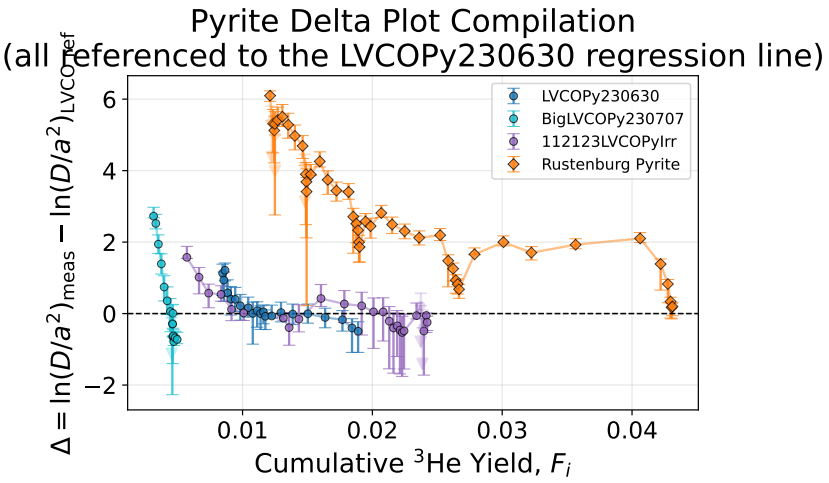

Figure 3.5: Delta plot ($\Delta = \ln(D/a^2)_{\text{meas}} - \ln(D/a^2)_{\text{LVCO ref}}$) vs. cumulative helium yield for all four pyrite step-heat experiments of this study, each referenced to the shared LVCOPy230630 regression line of Fig. 3.4. The three irradiated Leadville pyrite experiments (LVCOPy230630, BigLVCOPy230707, 112123LVCOPyIrr) initially decrease, then plateau near $\Delta = 0$, with the large-grained BigLVCOPy230707 aliquot recording lower diffusivities than they other two at a given cumulative fractional yield. The Rustenburg pyrite (§3.4) departs sharply from all three, reaching Δ values up to +6 at low cumulative yield before decreasing toward $\Delta = 0$ at higher yield.

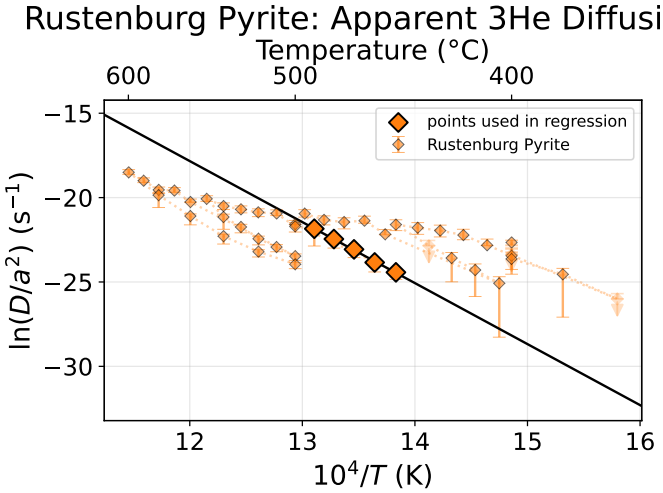

Figure 3.6: Apparent ^{4}He diffusivity Arrhenius plot for the Rustenburg pyrite step-heat experiment (isotope-dilution quantified on a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer; §3.2). The solid black line is the regression fit to the five down-temperature retrograde steps used to define the Rustenburg activation energy (highlighted points; Ea = 71.8 ± 1.7 kcal/mol, R^2 = 0.998, n = 5), markedly steeper than the shared Leadville/arsenopyrite reference slope of Fig. 3.4.

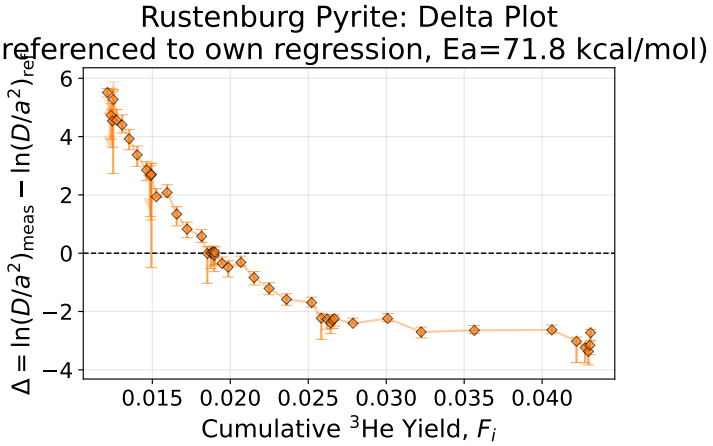

Figure 3.7: Delta plot for the Rustenburg pyrite step-heat experiment, referenced to its own regression (Fig. 3.6; Ea = 71.8 kcal/mol). Delta values decrease monotonically and from about +5.5 at low cumulative ^{4}He yield to about −3.5 at high yield, in contrast to the near-flat delta trends of the three Leadville pyrite experiments (Fig. 3.5).

Table 3.6: U-Th/He geochronology results for the four sample suites discussed in the main text: Kidd Creek pyrite, Gympie pyrite, and the Durango apatite and Gogebic (MI-43) hematite standards run alongside them, including U, Th, and Sm (the radiogenic-budget elements quantified by ICPMS, §3.2) plus Zr, also measured as a ratio to spike vanadium. Zr was only measured for some analyses. The remaining ICPMS trace elements measured the same way (Cu, As, Ag, Sb) are reported separately in the Supplementary Information (Table 3.9). Ages are the atmospheric-corrected Helium age. Excluded aliquots (§3.3) are noted in the last column, as are the first four Durango apatite aliquots (2211DApA, 2211DApC, 2211DApD, 2022StDApA), analyzed under the early "old method" protocol that predated the warm atmosphere-degassing step (§3.2, Fig. 3.2). The complete compilation, including localities not discussed in the main text, is given in the Supplementary Information (Table 3.10).

Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Zr (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)	Note
Kidd Creek Pyrite (Archean VMS, ∼2716 Ma host)									
2211KC_Py_ST05_C	4.77	0.918±0.054	1.641±0.040	0.099±0.006	–	1.312	470.80	694.9±29.1	
2211KC_Py_ST05_A	3.23	0.686±0.013	2.640±0.050	0.134±0.009	–	1.320	305.56	644.1±8.4	
2211KC_Py_ST05_B	1.74	0.879±0.052	1.593±0.033	0.203±0.012	–	1.261	159.83	621.3±25.6	
2211KC_Py_ST05_D	1.98	0.659±0.021	0.911±0.029	0.037±0.005	–	0.878	165.82	810.8±19.6	
011123KCPyST05C	1.28	0.510±0.019	0.732±0.051	0.384±0.070	9.13±2.64	0.686	158.07	1595.9±46.0	
011423KCPyST05A	2.41	1.706±0.024	4.047±0.115	1.176±0.071	28.91±2.25	2.678	389.20	591.0±7.7	
011423KCPyST05B	4.04	1.999±0.052	3.383±0.086	2.205±0.130	41.16±3.86	2.811	534.00	465.2±8.8	
011423KCPyST05C	3.93	0.834±0.038	1.969±0.062	0.492±0.047	15.46±1.10	1.307	281.60	539.9±16.3	
011423KCPyST05D	1.46	0.441±0.024	0.850±0.088	0.073±0.035	7.43±2.53	0.645	71.24	731.3±34.6	
013023KCPyST05As1	1.66	0.905±0.019	7.933±0.215	11.476±0.420	6.85±1.82	2.809	259.45	551.5±10.3	Silver Luster, Aspy Habit
013023KCPyST05Cu1	1.05	0.324±0.074	1.001±0.122	0.249±0.085	4.91±2.77	0.564	55.14	860.1±120.5	Brassy Luster, Cube Habit
013023KCPyST05Cu2	1.56	0.867±0.018	9.902±0.264	16.538±0.625	8.81±2.16	3.244	268.77	523.4±10.0	Brassy Luster, Cube Habit
Gympie Pyrite (Permo-Triassic orogenic Au)									
013023PVGyPy-1	3.81	0.065±0.004	0.064±0.011	0.128±0.022	1.22±0.75	0.080	34.71	928.3±50.9	excl.: early protocol

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Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Zr (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)	Note
013023PVGyPy-2	1.15	0.272±0.016	0.254±0.040	0.320±0.065	3.58±2.43	0.333	47.71	1008.0±51.7	excl.: early protocol
013023PVGyPy-3	1.81	0.128±0.006	0.159±0.024	0.173±0.047	–	0.167	31.64	763.3±34.8	excl.: early protocol
013023PVGyPy-4	1.82	0.280±0.041	0.358±0.031	0.647±0.049	7.15±3.71	0.366	52.23	638.1±69.5	excl.: early protocol
0413PyGy-1	3.54	0.462±0.006	0.587±0.027	0.589±0.069	56.95±3.43	0.603	53.03	233.1±3.4	
0413PyGy-2	1.67	0.232±0.014	0.362±0.032	0.375±0.058	40.37±3.17	0.319	15.12	257.4±12.6	
0413PyGy-3	1.32	0.747±0.018	0.721±0.047	0.775±0.124	72.56±3.28	0.920	30.29	235.2±5.3	
0413PyGy-4	1.85	0.862±0.016	0.791±0.037	0.489±0.071	77.41±5.18	1.052	126.53	622.9±10.2	excl.: outlier
062123GyPyHClb	2.15	0.636±0.022	0.583±0.050	0.498±0.028	89.48±7.17	0.776	32.69	191.1±6.1	
062123GyPyHClc	9.43	0.331±0.007	0.394±0.011	0.375±0.022	65.54±2.35	0.425	126.01	306.4±5.0	
230625GyPyHCl	2.38	0.477±0.007	0.463±0.041	0.971±0.053	64.39±2.99	0.588	35.29	227.5±4.5	
230625GyPyUnAbr	2.32	0.615±0.011	0.518±0.044	0.782±0.053	61.55±2.84	0.739	50.13	285.0±5.5	
230628GyPyHClab	6.17	0.627±0.007	0.716±0.034	0.785±0.051	96.43±3.82	0.799	133.56	264.7±3.6	
230628GyPyHClc	4.03	0.585±0.023	0.478±0.025	0.650±0.045	64.98±6.53	0.699	78.96	275.6±9.1	
230628GyPyUnAbrA	2.53	0.308±0.013	0.371±0.042	0.396±0.032	63.98±3.08	0.397	31.76	303.6±12.3	
230628GyPyUnAbrB	1.57	0.468±0.012	0.410±0.068	0.632±0.049	64.38±6.01	0.567	23.90	261.8±8.9	
Durango Apatite Standard (accepted age 32.1 Ma; Farley 2000)									
2211DApA	0.81	10.375±0.104	193.585±3.785	25.000±0.909	–	56.835	141.43	31.7±0.5	old method: pre-degassing
2211DApC	0.82	10.830±0.127	206.180±4.095	24.512±0.743	–	60.313	181.23	36.8±0.6	old method: pre-degassing
2211DApD	0.31	12.593±0.188	272.219±4.710	31.500±1.123	–	77.926	81.50	29.4±0.4	old method: pre-degassing
2022StDApA	0.30	12.615±0.216	230.680±5.529	71.500±1.955	–	67.979	69.61	29.5±0.6	old method: pre-degassing
011123DAp1	0.53	10.233±0.205	206.604±7.328	135.566±5.575	10.47±7.17	59.818	96.71	31.2±0.9	
011123DAp2	0.58	10.892±0.116	200.372±5.387	126.983±5.017	–	58.982	107.43	32.2±0.7	
011123DAp3	0.79	11.203±0.092	226.208±5.951	120.190±4.386	–	65.493	164.65	32.7±0.7	
011423DAp1	0.93	9.901±0.107	196.551±5.118	133.065±4.774	–	57.073	165.93	32.2±0.7	
011423DAp2	1.51	11.131±0.122	204.980±4.486	129.139±4.798	–	60.326	286.66	32.5±0.6	
011423DAp3	0.98	11.494±0.150	223.171±5.900	123.673±4.393	–	65.055	203.52	32.9±0.7	

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Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Zr (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)	Note
011423DAp4	0.81	11.773±0.142	235.763±6.994	125.741±5.204	8.92±4.03	68.356	170.67	31.8±0.8	
062123DAp	0.30	13.397±0.173	266.573±7.381	142.947±10.021	5.04±2.79	77.375	72.66	32.3±0.7	
062523DAp	0.85	11.074±0.097	211.843±5.315	123.774±8.205	2.71±1.71	61.916	161.32	31.6±0.7	
062823DAp	–	–	–	–	–	–	177.58	31.9±0.4	
Average for > 0.5 mg	–	–	–	–	–	–	–	32.1±0.2	
Gogebic Iron Range BIF Hematite Standard (MI-43; 571 Ma)									
2211HemMSA	1.37	11.793±0.131	0.039±0.012	0.344±0.029	–	11.802	750.58	450.1±4.8	
2211HemMSB	2.12	8.765±0.117	0.059±0.007	0.343±0.017	–	8.779	1070.17	554.8±7.0	
041323MG16a	0.46	9.199±0.134	0.000±0.000	1.559±0.192	–	9.199	240.72	549.4±7.7	
041323MG16b	0.49	2.153±0.057	0.089±0.083	0.866±0.154	19.36±5.03	2.174	188.74	1539.7±35.2	
041323MG16d	0.28	11.281±0.193	0.000±0.000	1.548±0.344	–	11.281	174.28	531.8±8.3	
041323MG16e	0.42	12.062±0.467	0.441±0.268	1.864±0.457	31.06±24.33	12.168	257.09	489.6±17.8	
062123MS	0.32	10.072±0.102	0.000±0.260	1.808±0.101	–	10.072	187.09	561.8±5.5	
230625MSHm	0.23	11.371±0.121	0.000±0.398	1.811±0.183	–	11.371	148.88	549.4±6.0	
230628MSHm	0.42	9.747±0.111	0.000±0.204	1.633±0.167	–	9.747	244.96	579.5±6.4	

Pyrite/Arsenopyrite U-Th/He Ages from Kidd Creek

Twelve aliquots of Kidd Creek pyrite/arsenopyrite were dated by U-Th/He step-degassing (Table 3.6) and give atmospherically corrected ages ranging from 465 to 860 Ma with the exception of one aliquot (011123KCPyST05C) that gives an older measured age of 1596 Ma Fig. 3.16. The mean age of the remaining eleven aliquots is 639 ± 36 Ma, 1σ. These dates are decisively older than the independent regional apatite cooling-age of 392 ± 32 Ma (Sturrock et al., 2024), yet far younger than the ∼2716 Ma age of the host rhyolite (Hannington et al., 1999). (Sec. 3.4).

Arsenopyrite U-Th/He Ages from Gympie

Sixteen aliquots of Gympie arsenopyrite were dated by the same method (Table 3.6; Fig. 3.8). The first four aliquots analyzed (013023PVGyPy-1 through -4) were prepared under an early iteration of the protocol that predated both the tube-blank atmospheric helium degassing step and the Parr-vessel HCl digestion step (Sec. 3.2) later adopted to fully dissolve fine U-Th-Sm-bearing rutile inclusions. The measured ages of these aliquots are unreasonable: ranging from 638 to 1008 Ma (mean 834 Ma, n=4), though it is unclear to what extent the bias is due to atmospheric contamination, incomplete digestion of rutile, or both. The ages of these four early aliquots are therefore considered unreliable and are excluded from further interpretation.

The remaining twelve Gympie aliquots, prepared after both protocol refinements were adopted, define a substantially younger and tighter age population. Eleven of these twelve give atmospherically corrected ages between 191 and 306 Ma (mean 258 ± 35 Ma, 1σ, n=11; these eleven are plotted in Fig. 3.8). The twelfth, 0413PyGy-4, is a clear outlier at 623 Ma and is also anomalous in its measured 4He concentration (68.4 ncc/mg), roughly three times that of the next most helium-rich aliquot in the accepted Gympie suite (230628GyPyHClab, 21.65 ncc/mg: supplementary Fig. 3.20). We interpret this combination as evidence for helium implantation from a U-Th-Sm-rich accessory phase concentrated at the sulfide grain margin and not fully digested during trace-element sample preparation. Aliquot 0413PyGy-4 is therefore also excluded from further interpretation.

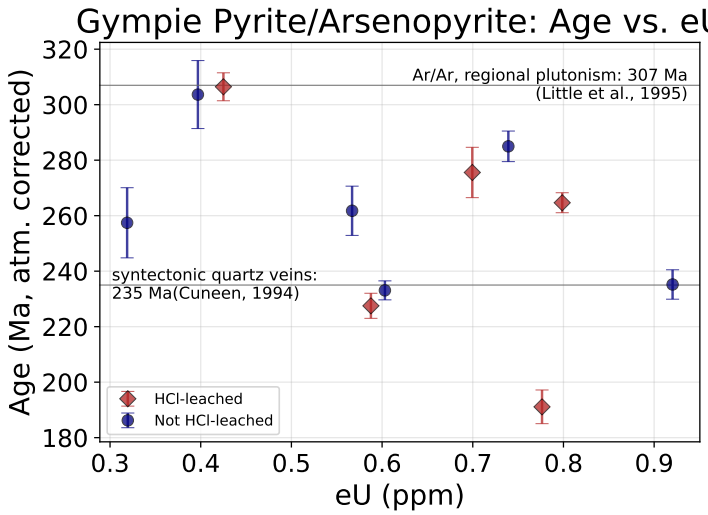

Figure 3.8: Atmospherically corrected age vs. eU for the eleven reliable, newer-protocol Gympie arsenopyrite aliquots (Table 3.6), split by HCl-leaching. Reference lines mark the 235 Ma K/Ar syntectonic quartz-vein age (Cuneen, 1994; Holcombe et al., 1997) and the 307 Ma Ar/Ar regional plutonism age (Little et al., 1995). The four early-protocol aliquots and the 0413PyGy-4 outlier (§3.3) are excluded here; see Fig. 3.20 (SI) for the outlier.

3.4 Discussion

Effectiveness of the Ampoule Method in Stabilizing Sulfides

The methodology for the heating treatments and SEM imagery of pyrite and arsenopyrite surfaces following thermal treatments in different conditions is discussed in 1.6s. The SEM imagery reveals deformed, variably raised, ~10 µm cracked surfaces on pyrite and marcasite that are heated in open air, and for arsenopyrite heated in a closed capsule without a realgar buffer (Fig. 1.1; Fig. 3.9). Pyrite and marcasite samples that are heated in ultra-high vacuum developed recessive ~1 µm rounded craters (Fig. 3.9). Contrarily, images of thermally treated, encapsulated pyrite and arsenopyrite with added fused sulfur or realgar show no obvious distinction from the control group: both of which showcase conchoidal fracture (Fig. 1.1). The buffered, encapsulated marcasite sample shows a raised layer of angular (presumably pyrite) crystals growing atop its surface (Fig. 3.9); evidence of phase transformation is present in all three heat-treated marcasite populations, demonstrating that the encapsulation method is not effective in preventing phase breakdown at elevated temperatures for marcasite. These experiments are not conclusive evidence that thermal phase transformation is prevented for pyrite or arsenopyrite (indeed, arsenopyrite's equilibrium stoichiometry varies with temperature, suggesting that some

reaction during step heats would be expected for this phase: Morimoto and Clark (1961) and Barton (1969)). However, the step heats presented here are consistent with the interpretation that reaction is suppressed, encouraging its application for helium step heat experiments.

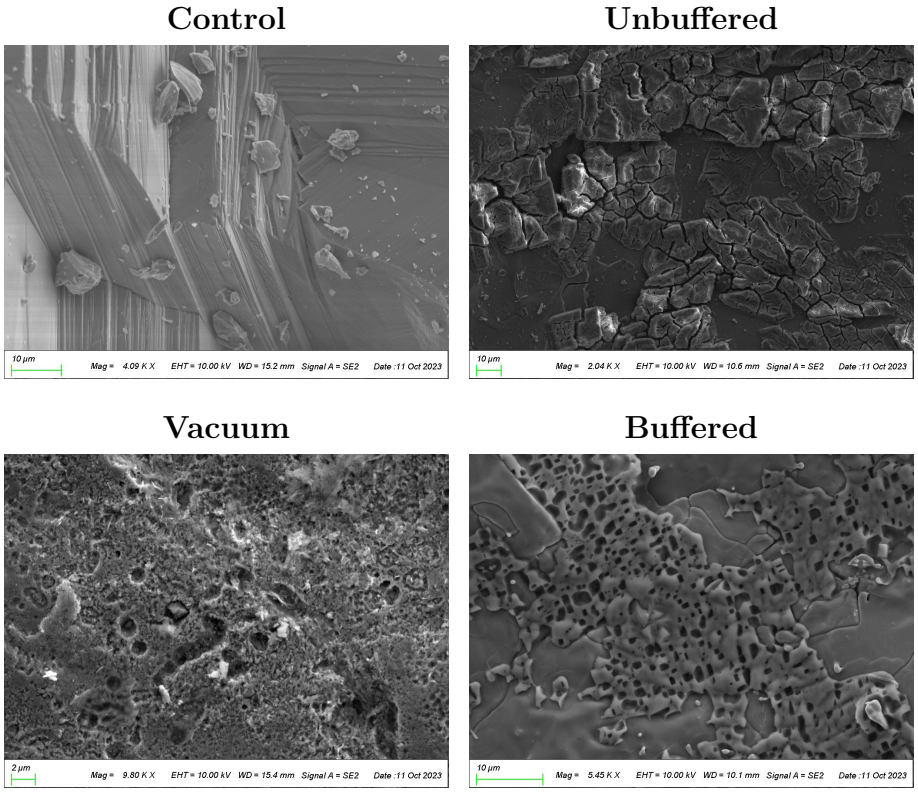

Figure 3.9: Secondary electron SEM images of marcasite surface textures before heat treatment (control) and after heat treatment unbuffered in open air (400°C, 5 hours), in ultra-high vacuum (400°C, 4.5 hours), and buffered (encapsulated in a silica glass ampoule with excess fused sulfur; 600°C, 8 hours). Unlike the buffered pyrite and arsenopyrite of Fig. 1.1, which retain their control-group surface texture under equivalent encapsulated treatment, the buffered marcasite instead develops a raised layer of angular, presumably pyrite, crystals, indicating that encapsulation does not suppress phase breakdown for marcasite as it does for the other two sulfides. See the supplement 1.6s for experimental methods and sample preparation.

Remarkable Helium Retentivity of Pyrite and Arsenopyrite

The central finding of this chapter's step heat experiments (§3.3) is that pyrite (and to a lesser extent, arsenopyrite) are extraordinarily retentive of helium aginst thermal loss: cumulative fractional yields never exceeded 7% for any pyrite or arsenopyrite experiment, and never exceeded 2% for the undeformed,

inclusion-poor irradiated Leadville pyrite specifically, even after repeated heating cycles at temperatures between 500-600 °C well in excess of any plausible natural burial or hydrothermal setting. These cumulative fractional yields are difficult to interpret without a reference diffusive timescale, so these step heat results are contextualized against the helium retentivity of other phases in Fig. 3.10, placing the sulfides of this study on the same diffusive-dose axis as step heat experiments of several minerals reported here or independently well-characterized helium retentivity: single-crystal and polycrystalline hematite, vacuum- and hydrothermally-degassed goethite, zircon, and monazite. To do this, the activation energy of hematite from Farley (2018) is used to convert the time and temperature of each step heat into a cumulative diffusive time, expressed as a Fourier number ($Fo = \frac{D}{a^2}_{\text{ref}} t$). The Fourier number can also be thought of as a dimensionless diffusive time, real time divided by the diffusive timescale of a reference domain, $\tau = \frac{a^2}{D}$. τ here is referenced to a 1 μm single-crystal hematite domain. The aim of such a plot is to flatten complex heating histories into a single value so that phases with complex helium migration behavior can be compared on a common scale. Incontrovertibly, this is not strictly correct because the acivation energy of different phases (and indeed diffusive domains within a single phase) are not identical, however this plot preserves the relative ordering of phases regardless of the choice of reasonable activation energy: we tested 149-171 kJ/mol. Therefore, this plot is a useful tool for comparing the relative retentivity of different phases, even though the absolute diffusive dose is not strictly correct.

On this comparison, pyrite and arsenopyrite sit at or beyond the retentive extreme of the entire compilation. Both plot at diffusive doses orders of magnitude beyond where the reference 1 μm single-crystal hematite domain (Farley, 2018) has gone fully to completion, and beyond essentially the full range spanned by zircon (Guenthner et al., 2013), comparable to or exceeding that of monazite (Farley, 2007). The polycrystalline hematite and vacuum-degassed goethite aggregates, by contrast, sit orders of magnitude further to the left. The hydrothermally degassed goethite (Chapter 2) plots closer to but less retentive than the sulfides here.

Taken together, this comparison indicates that sulfur-buffered, sulfide-encapsulated pyrite and arsenopyrite are not merely retentive relative to other sulfides (galena, chalcopyrite, chalcocite, sphalerite; §3.1s), but are among the most helium-

retentive phases yet characterized by step heating, on par with or exceeding zircon and monazite and substantially exceeding both single-crystal and polycrystalline hematite and vacuum-degassed goethite. This is the empirical basis for treating pyrite (and, by extension, arsenopyrite) as a credible U-Th/He geochronometer in this chapter's Kidd Creek and Gympie case studies (§3.3, §3.3), notwithstanding complexities introduced by the inclusion-hosted origin of the radiogenic budget (§3.4).

Inferring Bulk Helium Migration Behavior from Low Fractional Yields

Many of the step heats on encapsulated sulfides presented here show very high helium retentivity, and for those experiments only a very small fraction of the total yield evolved (<5% for the arsenopyrite and Rustenburg pyrite; <2% for the Leadville pyrite), similarly to previous work on highly helium retentive phases such as baddeleyite (Baughman and Flowers, 2018). For these sulfides, a mechanism responsible for bulk helium loss besides desulfurization cannot be identified. For some well-studied phases like zircon and titanite, Arrhenius plots of the earliest fractions of evolved helium often reflect processes or conditions that are accessory to the bulk mechanism for helium migration through the mineral, such as radiation damage annealing or surface roughness (Reiners and Farley, 1999; Guenthner et al., 2013), where the primary mechanism for helium migration through the pristine mineral is volume diffusion. These apparent departures from the bulk helium migration behavior can be caused by properties of natural samples, like surface roughness or fine-scale cracks and fractures (Reiners and Farley, 1999) that produce complex grain size distributions even in carefully sieved, (apparently) single grains. Contrastingly, some other complexities in the helium loss behavior at early loss fractions can be ephemeral and altered during experimental heating, such as the annealing of radiation damage (Shuster and Farley, 2009). For radiation damaged samples, understanding the migration behavior of helium before and during annealing is essential to understand the sample's retentively in nature and therefore the earliest fraction of helium evolved in step heats is the focus of many studies (Shuster and Farley, 2009; Flowers, Ketcham, et al., 2009; Flowers, Shuster, et al., 2007; Shuster, Flowers, et al., 2006).

The fraction of total helium evolved before the predominant migration mechanism's behavior can be characterized in helium step heats varies: Guenthner

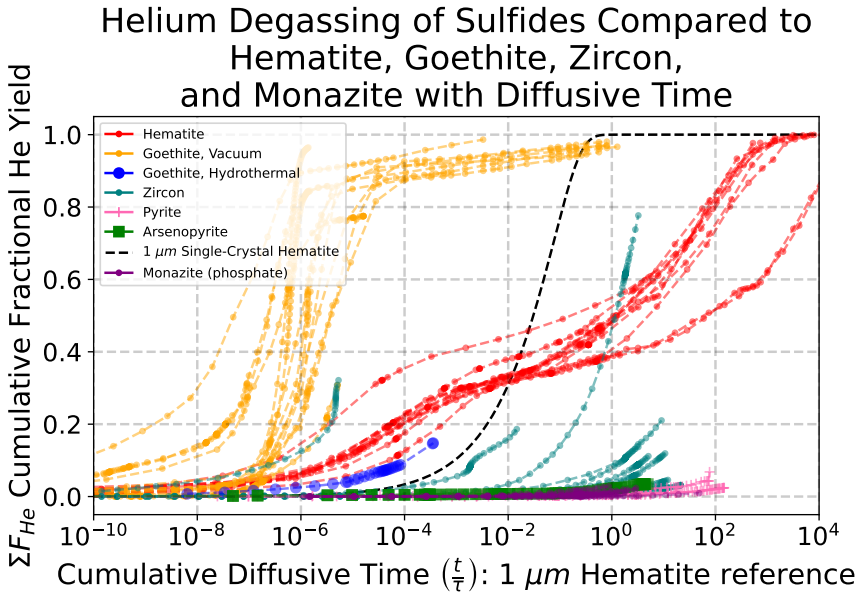

Figure 3.10: Cumulative fractional He yield vs. cumulative diffusive time (Fourier number, $Fo = D_{\text{ref}}t/a^2_{\text{ref}}$) for each heating step of every experiment plotted, referenced to a common 1 μm single-crystal hematite domain diffusivity (Farley, 2018) ($\ln(D_0/a^2) = 0.66$, $E_a = 171$ kJ/mol). Expressing step power this way, rather than as raw time or temperature, puts experiments run under very different heating schedules, temperatures, and durations onto one common, directly comparable dose axis: a phase reaching a given cumulative yield only at a much larger Fo is more helium-retentive than one reaching the same yield at a smaller Fo, regardless of how either was actually heated. Because cumulative fractional yield is dimensionless, isotope-specific normalization is not required to combine step heats of different helium isotopes on one axis: ^{3}He step heats of proton-irradiated pyrite, arsenopyrite, zircon, monazite, and vacuum-degassed hematite/goethite are plotted alongside ^{4}He step heats of the (non-irradiated, isotope-dilution-quantified) Rustenburg pyrite and the hydrothermally degassed goethite. Plotted data: this chapter's own sulfur-buffered pyrite and arsenopyrite step heats; 11 vacuum step heat aliquots of polycrystalline hematite from the Grand Canyon, Arizona (Farley and Flowers, 2012) and the Gogebic Range, Michigan (Farley and McKeon, 2015); 9 vacuum step heat aliquots of polycrystalline goethite (Win-06-01b and Capão, Minas Gerais, Brazil) (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024); the hydrothermal step heat of Win-06-01b goethite from Chapter 2 of this dissertation; and proton-irradiated zircon (Guenthner et al., 2013) and synthetic monazite (Farley, 2007) step heats. The dashed black reference curve is the forward-diffusion cumulative-yield curve for the reference 1 μm hematite domain itself (i.e. Fo vs. its own resulting cumulative yield, with no data plotted against it), included so every other curve's retentivity can be read directly relative to this single benchmark.

et al. (2013) show that significantly non-zero delta or ln(r/r0) values persist in zircon until about 2% of the total helium is degassed for typical natural samples; Reiners and Farley (1999) observe apparent diffusivities of helium in titanite converging to linearity in titanite after 5% of the total helium is evolved. Similar experiments on proton-irradiated synthetic samples seem not to be marred by the same complexities: helium step heat data for synthetic monazite converge nearly immediately to a linear array at very loss cumulative helium loss fractions (Farley, 2007).

The step heat experiments on pyrite and arsenopyrite in this study would have benefitted from longer steps at high temperature to degas more helium to better understand the bulk behavior of helium, but even at small loss fractions some key observations about helium loss in sulfides can be made.

Comparison of Helium Step Heat Experiments of Natural Pyrites

In this work, four encapsulated, sulfur-buffered aliquots of pyrite taken from two localities are subjected to helium step heat experiments. Step heats for the three irradiated pyrites from Leavldville, CO are broadly similar, but contrast markedly from step heats of non-irradiated pyrite from Rustenburg, South Africa, therefore differences between pyrites of different localities is discussed before the variation within the aliquots from Leadville, CO.

Radiogenic helium evolution during step heats of octahedral pyrite from Rustenburg, South Africa show that this sample is less retentive than the pyritohedral pyrite from Leadville, Colorado (Fig. 3.5). Apparent diffusivities of these samples also have fundamentally distinct geometries when plotted in an arrhenius plot. The Rustenburg pyrite shows a striking feature of very steep slopes (corresponding to an activation energy of 300.6 ± 7.3 kJ/mol) on retrograde-prograde heating cycles at temperatures lower than the previous maximum temperature step heat, whereas the maximum apparent diffusivity for a given temperature seemed to follow a much shallower slope, similar to the lower activation energy of the Leadville, CO pyrite: 199.1 ± 3.5 kJ/mol. The geometry of the step heats in an arrhenius plot for the irradiated Leadville, CO pyrite is generally simpler: generally slowly decreasing in a delta plot, or for isothermal steps (Fig. 3.5) as would be expected for helium release governed by MDD systematics (Farley, 2018).

Natural pyrite is known to develop fractures and crystalline dislocations in

metamorphic and experimental conditions (Gill, 1969; McClay and Ellis, 1983; Barrie et al., 2011). Abundant fractures of micron-scale widths were clearly visible in the Rustenburg pyrite of this study (Fig. 3.11). Those pyrites were surprisingly easy to crush, presumably breaking easily along fine reexisting fractures. Although pyrites from Leadville are known to be associated with faulted dolomites, especially along deformation planes which likely served as pathways for mineralizing hydrothermal fluids (Nash, 1979), the samples used here were exceptionally sturdy and visually homogeneous under SEM. Rare veins and coatings of dolomite within and around the pyrites were present but were avoided when preparing the sample for irradiation.

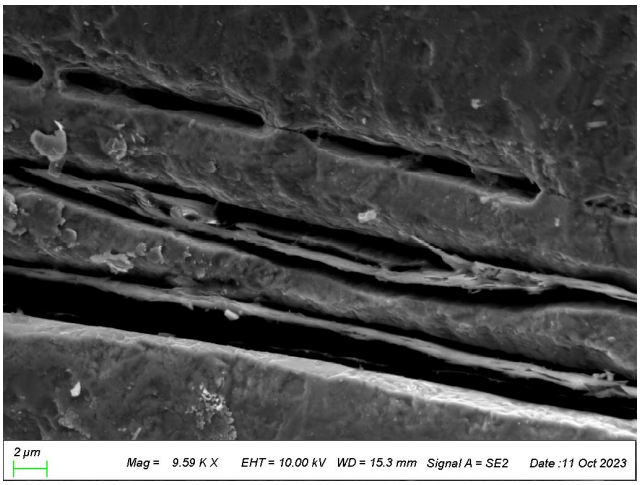

Figure 3.11: Secondary-electron SEM image of Rustenburg (Bushveld Complex) pyrite prior to heat treatment (control), showing the pervasive, micron-scale fractures characteristic of this sample (§3.4). These pre-existing fractures are interpreted as the pathways for the short-circuit helium diffusion mechanism invoked below to explain the Rustenburg pyrite's distinct retrograde Arrhenius behavior (Fig. 3.6).

Within this work, a pipe/short circuit diffusion mechanism (Lee and Aldama, 1992) is favored to explain the helium loss behavior of the Rustenburg pyrite, where helium populates the sample's fine fractures at increasing temperatures due to the primary helium migration mechanism, perhaps volume diffusion through crystalline pyrite. This primary mechanism is interpreted to be the same for all pyrite samples regardless of locality. On retrograde heating steps, short circuit diffusion of helium out of fine fractures in the Rustenburg pyrite is hypothesized to become the rate-limiting process.

Leadville, CO Pyrite: Evidence for a Grain Size Control?

Three helium step heat experiments for the irradiated Leadville, CO pyrite are discussed here. Grain size was varied by a factor of 2 between a half-prism width of 200 microns for LVCOPy230630 and 400 microns for LVCOPy230707 to investigate the dependence of grain size on helium migration in pyrite. The aim of the third experiment, 112123LVCOPy, was to test the effect of isolating the pyrite from the liquid sulfur during the experiment to investigate the role of dissolution and (re)crystallization of pyrite by the ampoule's sulfur fluid. Silica glass shards were added to this ampoule below and above the pyrite aliquot, forcing the pyrite grains to remain in the middle of the ampoule, and not at the edges/bottom, where sulfur liquid is presumed to accumulate. All three samples overlap in their Arrhenius plots (Fig. 3.4), and all three samples record declining diffusivities consistent with MDD systematics (Fig. 3.5). The larger grained aliquot recorded lower cumlative fractional yields at these common apparent diffusivities. The smaller grained aliquots: LVCOPy230630 and 112123LVCOPy have essentially identical delta plots (Fig. 3.5), providing no evidence for a sulfur-liquid control in driving helium loss for these experiments.

Very little helium was evolved from any of the exceptionally retentive Leadville pyrite samples: none evolved more than 3% of its cumulative fractional yield, in spite of the intense heating schedules (at temperatures up to 600°C) they were subjected to. These samples are only resolved by grain size. Because all arrhenius plots overlap, it is likely that the differences in the delta plots are due to the relative abundance of the smallest domain size: the smaller aliquots contained a greater proportion of the least retentive domains interogated by these experiments. One potential explanation is that only helium on extraordinarily small dusty material carried on the surfaces of the larger aliquots is degassed, and thus the primary control on the delta plot is the surface area to volume ratio. This is consistent with the observation that the smaller aliquots, with half the half prism width half that of the larger aliquot, record Δ=0 at twice the cumulative fractional yield of the larger aliquot (Fig. 3.5).

Relevance of Helium Step Heat Data of Encapsulated, Sulfur-Buffered Sulfides in Nature

Given the high helium retention of pyrite and arsenopyrite shown in this work, thermal helium loss in nature may be rare. A more likely mechanism for helium

loss in these common sulfides in nature may indeed be recrystallization. Pyrite in metamorphic conditions commonly recrystallizes and may either shrink or grow (Craig et al., 1998; McClay and Ellis, 1983). Until more work has characterized helium loss (or gain) for recrystallizing pyrite or arsenopyrite in metamorphic conditions, helium should cautiously be considered to have behaved as an open or partially open system in sulfides during metamorphic events, although O. Yakubovich, Podolskaya, et al. (2020) report that helium in the Uzelga pyrite was retained through metamorphism.

Arsenopyrite

Arsenopyrite (FeAsS) adopts a distorted, lower-symmetry variant of the marcasite structure in which arsenic substitutes irregularly for sulfur (Morimoto and Clark, 1961), in contrast to pyrite's simple cubic lattice. Because the degree of distortion scales with arsenic content (Morimoto and Clark, 1961), Helium diffusion is likewise expected to vary with composition. Unfortunately, equilibrium arsenopyrite composition is not itself fixed: arsenic content increases systematically with formation temperature (Barton, 1969; Morimoto and Clark, 1961), so the same crystal-chemical property expected to govern helium retentivity should also vary with an arsenopyrite sample's thermal history, complicating any attempt to generalize helium retentivity from a single experimental aliquot to arsenopyrite as a mineral species. Similarly, the composition of the arsenopyrite subjected to step heats herein may itself have varied composition during the experiment, further abstractifying these experimental results from helium loss from arsenopyrite in nature.

With that caveat, the step heat results obtained here (§3.3) show the encapsulated arsenopyrite experiment was less helium-retentive than the Leadville pyrite experiments of this study, and it lacking a well-defined linear Arrhenius region. Still, arsenopyrite remained markedly more retentive of helium than any other sulfide examined in this study besides pyrite, in agreement with the relative helium retentivity ranking of sulfides reported by O. V. Yakubovich, Gedz, et al. (2019).

Qualitatively, arsenopyrite appears to be more retentive of helium than most zircon (Fig. 3.10, Guenthner et al. (2013)).

Thermal Helium Migration Systematics for Other Sulfides

Discussion of the thermal helium migration systematics of the four remaining, non-pyrite/arsenopyrite sulfides of this study (chalcopyrite, chalcocite, sphalerite, and galena) is given in the Supplementary Information (§3.5); neither copper sulfide nor galena is considered a promising U-Th/He geochronometer or helium isotope target.

Origin of Radiogenic Helium in Sulfides

The pyrite and arsenopyrite of this study are not, themselves, appreciably radiogenic minerals. Pyrite in particular has very little affinity for U, Th, or Sm, which are large, highly charged cations poorly suited to substitution for Fe^{2+}. Consistent with this, only the aliquots (of any locality or mineral in the compilation, main text or SI) that host measurable U-Th-Sm-bearing accessory phases returned appreciable eU and, correspondingly, appreciable radiogenic 4He. For example, the Leadville pyrite samples are purest, most visually inclusion-free grains selected for the sulfur- buffered step heat experiments of §3.2, and they are essentially non-radiogenic. The U-Th/He ages for sulfides may not date growth of the sulfide lattice directly, even if it is perfectly retentive of helium, because some of the helium retentivity of the sample will be determined by the size, mode, and connectivity of inclusions. Sulfide ages are easiest to interpret for the case when the radiogenic helium-source inclusions are smaller that 10 microns across, when only the helium retentivity of the sulfide is significant. (Fig. 3.17).

Backscattered-electron imagery of the Kidd Creek arsenopyrite/pyrite shows a surprising abundance of apatite inclusions directly. Superficially, hand-picked crystals appear as optically clean, blocky euhedral crystals set within a finer silicate groundmass (Fig. 3.12), but in cross-section the same grains are pervasively spongy, riddled with abundant tiny inclusions rather than being the pure, homogeneous pyrite their exteriors suggest. These inclusions are dominantly apatite (Figs. 3.13, 3.14), ranging in size from sub-6 micron grains at the limit of SEM resolution up to ∼30 micron half-width. Because apatite preferentially incorporates U, Th, and especially Sm relative to coexisting sulfide and silicate phases, even a small volume fraction of included apatite can dominate a pyrite aliquot's bulk eU budget and, correspondingly, its radiogenic 4He production.

The Gympie arsenopyrite hosts a texturally distinct, but analogously inclusion-hosted, radiogenic budget. SEM imagery (Fig. 3.15) shows abundant fine, needle-like rutile inclusions. These rutile are the most likely U-Th-Sm-He source for the Gympie samples, and variable inheritance of older rutile engulfed by later sulfide may be responsible for its observed age dispersion (§3.3).

This inclusion-host origin of radiogenic helium in sulfides has a direct methodological consequence: sulfide U-Th/He geochronology cannot be treated as a single, generic protocol applied uniformly across localities. The identity, size distribution, and digestion behavior of the accessory phase or phases actually carrying the U-Th-Sm budget is a property of each deposit's own mineral paragenesis, and differs sharply even between the two localities studied here. Reliable sulfide U-Th/He geochronology of a new locality therefore requires SEM characterization of the dated material to identify its radiogenic inclusion population before analysis, and a dissolution protocol tailored to that population's specific mineralogy and grain size.

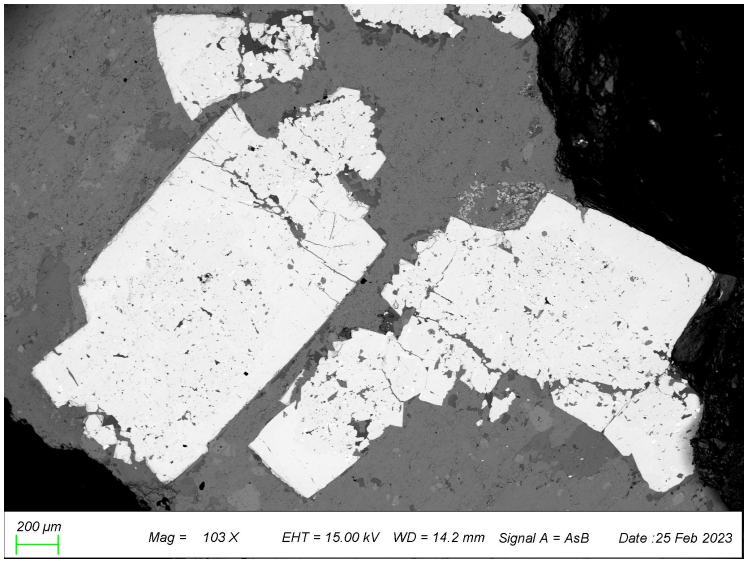

Figure 3.12: Backscattered-electron image of Kidd Creek pyrite (ST05) in cross-section: optically clean, blocky euhedral grains set within a finer-grained groundmass are, internally, spongy and riddled with abundant tiny (predominantly apatite) inclusions rather than homogeneous pyrite.

Interpretation of Kidd Creek Pyrite/Arsenopyrite Ages: Mixing Between a Retentive, He-Th-Poor Host and Apatite Inclusions

The Kidd Creek pyrite/arsenopyrite ages are not scattered randomly around some single "true" age. Eleven of the twelve dated aliquots define a coherent

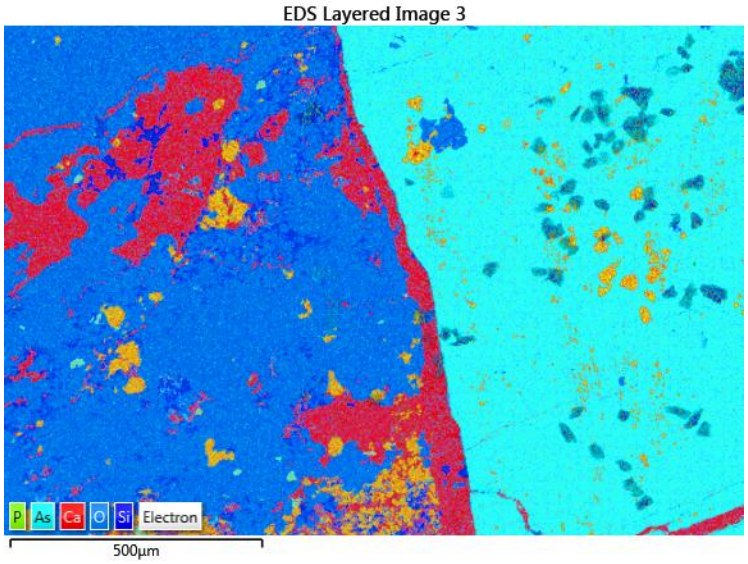

Figure 3.13: Backscattered-electron image of Kidd Creek pyrite showing apatite inclusions, the dominant host of the aliquots' U-Th-Sm budget.

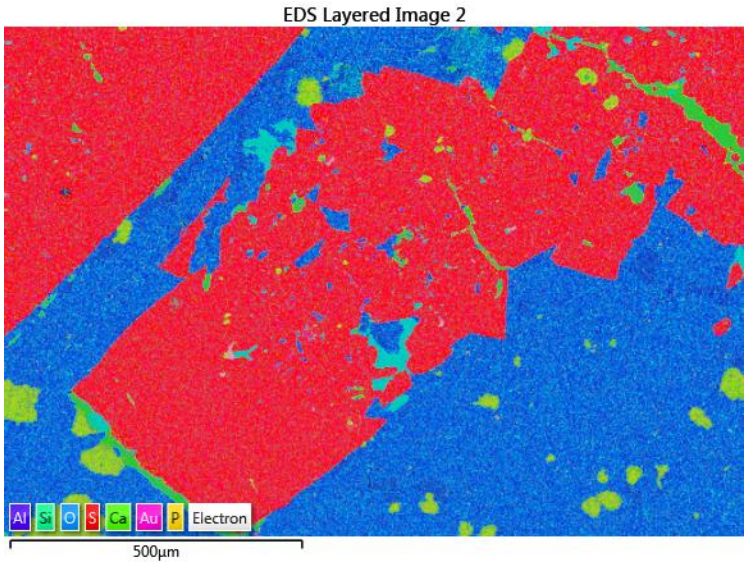

Figure 3.14: Backscattered-electron image of apatite inclusions in Kidd Creek pyrite, ranging from sub-6 micron grains up to ∼30 micron half-width.

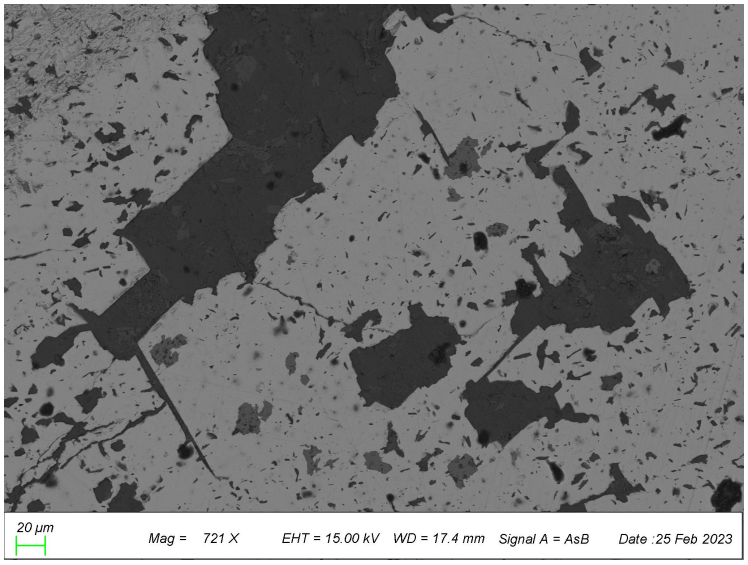

Figure 3.15: Backscattered-electron image of Gympie pyrite showing abundant fine, needle-like rutile inclusions and much larger, blocky feldspar inclusions. Ineffective rutile digestion is implicated in the anomalously old ages discussed in §3.3.

array in which the Th- and He-richest aliquots give the youngest ages and the Th- and He-poorest aliquots give the oldest, ranging from 465 to 860 Ma (Fig. 3.16). The youngest of the eleven approach, but are always higher than, the 392 ± 32 Ma regional apatite (U-Th)/He cooling age reported by Sturrock et al. (2024) for sample 01-OE-69, sampled only ∼90 km from Kidd Creek; the oldest approach ∼860 Ma, implying that this population is significantly more helium-retentive than the apatite it includes.

Given the inclusion-hosted origin of radiogenic helium established above (§3.4), the natural reading of this array is a two-component mixture: an old end-member that is essentially He-Th-poor but comparatively retentive, and a population of included apatite with a much younger (∼392 Ma) cooling age (Sturrock et al., 2024). A Kidd Creek aliquot's bulk age then falls between these two end-members depending on what fraction of its total measured Th and He budget was hosted in (and retained by) apatite inclusions since 392 Ma, versus alpha-ejected into the surrounding, retentive sulfide host and retained there since it was crystallized.

A simple model shows that lower age paired with higher Th and He can be explained with a single-age apatite population of variable inclusion size. Fig. 3.17 extends the alpha-ejection retention framework of §3.4 to compute

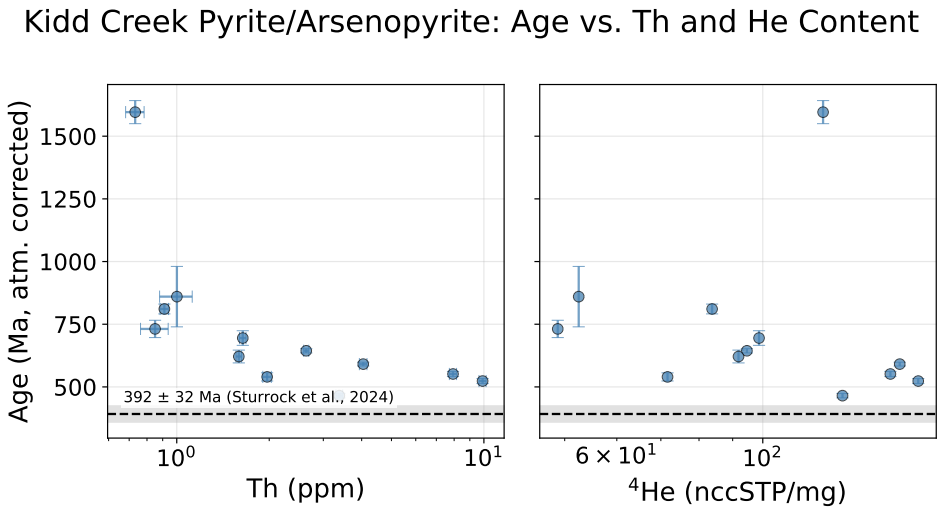

Figure 3.16: Atmospherically corrected Kidd Creek pyrite/arsenopyrite age vs. Th content (left) and vs. measured 4He content (right; nccSTP/mg), sharing a common age axis (§3.3). The shaded band is the independent regional apatite cooling-age estimate of 392 ± 32 Ma for sample 01-OE-69 (Sturrock et al., 2024). Both panels show the same sense of anti-correlation: Th- and He-rich aliquots give younger ages, Th- and He-poor aliquots give older ages.

the apparent bulk He age of a hypothetically perfectly helium-retentive sulfide aliquot hosting a single apatite inclusion of half-prism radius R, assuming the sulfide and apatite crystallized 2700 Ma ago, but the apatite itself only began retaining its own self-implanted helium after 380 Ma. At small R, nearly all the produced helium recoils out of the inclusion into the retentive sulfide, and the aliquot's bulk age approaches the old, 2700 Ma crystallization age, but at large R, an increasing fraction of the inclusion's own production is retained within the apatite inclusion itself, so the bulk age falls toward 380 Ma. Because a larger inclusion also simply contains more total U-Th-Sm mass, the same increase in R that drives the age down simultaneously drives up an aliquot's bulk Th and He content, producing exactly the anti-correlation seen in Fig. 3.16.

At the $\leq$30 μm half-width apatite inclusion size actually observed in the Kidd Creek material (§3.4, Fig. 3.14), Fig. 3.17 predicts an apparent age of $\sim$1.6 Ga or older for a sulfide host that is perfectly closed to helium loss over its entire 2.7 Gyr history. Eleven of the twelve measured Kidd Creek ages, plotted at this inclusion size in Fig. 3.17, fall far short of this prediction. This model cannot produce ages as young as 465-860 Ma, so most of these sulfides must

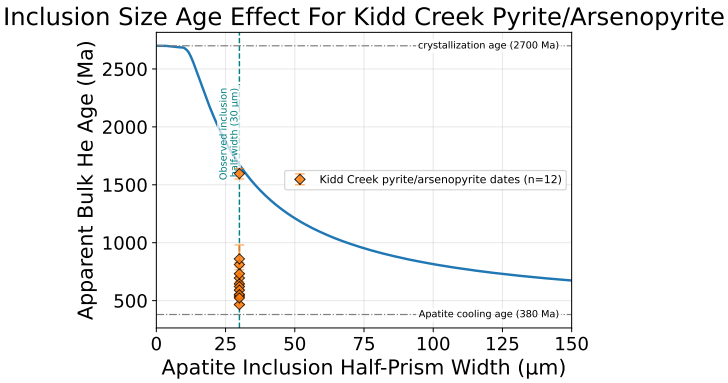

Figure 3.17: Modeled apparent bulk He age of a quantitatively He-retentive, 2.7 Ga Kidd Creek sulfide aliquot hosting a single fluorapatite inclusion serving as its U-Th-Sm source. The apparent age of such a system is a function of the inclusion's half-prism radius R. The alpha-ejection retention fraction $F_t(R)$ (fraction of produced helium self-implanted within the inclusion) is computed via a Monte Carlo simulation of 10^6 decays uniformly distributed by volume within a spherical inclusion with isotropic alpha-ejection directions, using per-isotope (^{238}U, ^{235}U, ^{232}Th, ^{147}Sm) stopping distances for fluorapatite and a fixed U:Th:Sm ratio of 1:2:2. This reproduces Farley, Wolf, et al. (1996)'s closed-form retention factor for $R \geq S$ (the stopping distance) and remains accurate for $R < S$, where the closed form is not valid. Helium is then accumulated separately for the inclusion-retained fraction $F_t(R)$, assuming retention only since 380 Ma, and the sulfide-retained fraction $1 - F_t(R)$, assuming retention since a 2700 Ma inclusion/host source age. The two helium contributions are summed to calculate a single apparent bulk age. Reference lines mark the 2700 Ma source age and 380 Ma cooling age end-members. A vertical line marks the maximum ~30 μm half-width apatite inclusion size actually observed at Kidd Creek (Fig. 3.14). Eleven of the twelve pyrite/arsenopyrite ages fall well below the model's closed-system prediction; but the oldest age (011123KCPyST05C, 1596 ± 46 Ma) agrees with the model prediction (§3.4).

have lost a substantial fraction of their radiogenic helium since crystallization, rather than behaving as a closed system simply mixing with variably sized apatite inclusions.

The twelfth, oldest aliquot, 011123KCPyST05C, is the one exception: its 1596 ± 46 Ma age sits close to the closed-system prediction above at the observed ~30 μm inclusion size, as though this one aliquot alone retained essentially all of its radiogenic and inclusion-sourced helium since crystallization while the rest of the suite lost helium over time. A single old date is weak evidence for anything on its own: an anomalously old U-Th/He age is also the expected signature of incomplete acid digestion, which would leave some

U-Th-Sm-bearing residue undissolved, underestimating the apparent age. We cannot distinguish between these two explanations with the data at hand.

This chapter's own step heat experiments show that sulfur-buffered pyrite is consistently among the most helium-retentive phases measured with step heat experiments while arsenopyrite, though still strongly retentive, is notably less so (Fig. 3.4, Fig. 3.10; §3.4). If 011123KCPyST05C were confirmed to be closer to pure pyrite than the rest of the Kidd Creek suite, this retentivity contrast would offer a ready explanation for why it alone approaches closed-system behavior. Indeed, the arsenic measurement (Table 3.9) supports this interpretation: 011123KCPyST05C has the lowest measured arsenic concentration of the Kidd Creek aliquots. Unfortunately, this study's arsenic measurements for both the Kidd Creek and Gympie sulfides are not reliable: they are inconsistent with observations of optical habit during sample picking, and they do not agree with estimates of arsenopyrite content implied by mass balance using sample mass and measured Fe.

Whether the rest of the suite's helium loss was thermal or accompanied recrystallization is similarly unclear, but recrystallization seems the less likely explanation, since the most recent regional metamorphic event that could have recrystallized these sulfide porphyroblasts predates 2.6 Ga (Smith et al., 1993; Hannington et al., 1999), well before the 465-860 Ma ages recorded here.

Arsenopyrite U-Th/He Ages for Gympie Gold Belt

There is no obvious difference in the geochronological sulfide U-Th-Sm/He data between the HCl leached and non-leached Gympie arsenopyrite populations (fig. 3.8), thus we interpret the leaching step to be a harmless but unnecessary step in the Gympie arsenopyrite U-Th/He geochronology protocol. This step was originally intended to remove any potential surface contamination, but these data don't support that the scatter in age is related to surface contamination. It is more likely that the scatter in age is related to the variable sizes and interconnectivity of inclusions, which could affect the sulfide's helium retentivity.

Lithostratigraphic age constraints that rely on foraminifera microfossils seem to suggest an Early Permian age for the Gympie Group rocks that host the gold-vein systems (§3.1, Holcombe et al. (1997)). However, the oldest ages reported here are indeed resolvably older than the inferred early permian upper bound,

instead suggesting that the mineralized volcanic pile may be of carboniferous age. Indeed, these biostratigraphic constraints are sparse and rest mainly on correlation higher in the sequence rather than direct radiometric dates on the volcanic pile itself, and Holcombe et al. (1997) note that the lower Gympie Block volcanics "may contain older volcanic units." The data here support the latter interpretation.

The oldest Gympie ages are interpreted as a record of hydrothermal alteration of the Gympie block volcanics contemporaneous with ∼300-310 Ma regional granitoid magmatism (§3.1, Little et al. (1995) and Holcombe et al. (1997)). Because the rutile inclusions identified as the U-Th-Sm source in the Gympie arsenopyrite (§3.4) are small (≤20 µm, Fig. 3.15), they would self-implant little of their own produced helium, and thus the measured dates cannot be explained by rutile inheritance before a hypothetical sulfide crystallization event at ∼235 Ma (Cuneen, 1994; Holcombe et al., 1997). Thus, the arsenopyrites likely crystallized at or before ∼306 Ma, well before the Au-quartz mineralization event at ∼235 Ma. It is possible that the rutile population predates ∼306 Ma, and only started retaining helium after being incorperated into the sulfide hosts, permitting the volcanic pile's emplacement age to be older than the oldest sulfide ages. We prefer the interpretation that the sulfide crystallized contemporaneously with metasomatism related to the ∼300-310 Ma regional granitoid magmatism because it is more feasible that such large sulfides would be metasomatic rather than magmatic in origin. Therefore, the volcanic pile was likely emplaced sometime before ∼306 Ma. Aside from a single young outlier, the youngest Gympie sulfide ages reported here are contemporaneous with the syntectonic Au-mineralization event at ∼235 Ma (Cuneen, 1994; Holcombe et al., 1997), which we interpret to represent the timing of a recrystallization event for some of the sulfides. At the very least, it is somewhat remarkable that the sulfide U-Th/He ages demonstrate that most samples retained helium through the ∼235 Ma hydrothermal event.

Under this interpretation, the eleven reliable Gympie ages (§3.3) record variable, partial resetting of the ∼300-310 population during the ∼235 Ma hydrothermal mineralizing event, rather than multiple discrete sulfide generations.

The one clear exception is 062123GyPyHClb, at 191 Ma, the youngest age in the suite, and the only one below the ∼235 Ma mineralization age itself. A partial-resetting model bounded between a ∼306 Ma inclusion age and a ∼235

Ma resetting event cannot, on its own, produce an age younger than 235 Ma, so this aliquot requires additional explanation. One speculative possibility is that this particular aliquot's contained U-Th-Sm rich inclusions near or on surface of the sulfide grains, and thus some of the radiogenic helium was implanted elsewhere and not measured in the aliquot. Some gympie samples contain coarser alkali feldspar inclusions identified alongside the rutile (Fig. 3.15). If there were rutile crystals at the interface of the sulfide and the feldspar, then some of the rutiles' radiogenic helium may have been implanted into the feldspar and lost over geologic time, since feldspar is not normally considered a particularly helium-retentive phase (Hurley and Goodman, 1941). In any case, we point out that this single age is only one of eleven, and therefore seems to be the exception rather than the rule.

3.5 Conclusions

Sulfide geochronology has so far relied almost entirely on the Re-Os system, which is compositionally restricted to a handful of phases and analytically restricted to a few specialized laboratories (§3.1). This chapter tested whether U-Th/He dating of pyrite and arsenopyrite can serve as an accessible complement to Re-Os geochronology through two lenses: direct application to two independently age-constrained deposits at Kidd Creek and Gympie (§3.1), and a set of sulfur-buffered helium step heat experiments designed to characterize the underlying diffusion and migration behavior of helium in pyrite, arsenopyrite, galena, and chalcopyrite.

The step heat experiments show that pyrite and arsenopyrite are remarkably helium-retentive against thermal loss: cumulative fractional yields never exceeded 5% for any pyrite or arsenopyrite experiment, and never exceeded 2% for the undeformed Leadville pyrite specifically, even after repeated heating cycles up to 600°C (§3.4). Placed on a common diffusive-dose axis against independently characterized phases, pyrite and arsenopyrite sit at or beyond the retentive extreme of the compilation, on par with or exceeding zircon and monazite and substantially exceeding hematite and goethite (Fig. 3.10). Arsenopyrite is less retentive than pyrite, consistent with its lower-symmetry, arsenic-bearing structure (§3.4), but is similarly retentive to zircon and markedly more retentive than galena or the copper sulfides, which are not considered promising U-Th/He targets (§3.1s).

The step heat data also point to two distinct helium-loss pathways within pyrite itself. Deformed, fractured Rustenburg pyrite shows steep, retrograde-dependent Arrhenius behavior best explained by a short-circuit or pipe-diffusion pathway along pre-existing fractures (Lee and Aldama, 1992), whereas the visually homogeneous, irradiated Leadville pyrite follows a simpler, shallower-sloped trend consistent with multiple domain volume diffusion (§3.4). Varying grain size and sulfur-liquid contact within the Leadville suite did not resolve a clear control on this behavior beyond a surface-area-driven nugget effect, in which smaller aliquots simply carry a larger proportion of fine pyrite dust (§3.4).

Applied to natural samples, the U-Th/He ages obtained here are not simple crystallization or cooling ages. Eleven of twelve Kidd Creek pyrite/arsenopyrite aliquots give ages of 465-860 Ma (mean 639 ± 36 Ma, 1σ, n=11), decisively younger than the ∼2716 Ma host rhyolite (Hannington et al., 1999) but older than the regional 392 ± 32 Ma apatite cooling age (Sturrock et al., 2024); the twelfth instead gives a much older, 1596 ± 46 Ma age. As measurements, mass-balance stoichiometry, and optical habit are mutually inconsistent for this suite, so individual aliquots cannot be reliably assigned to pyrite or arsenopyrite (§3.4). Gympie arsenopyrite gives ages of 191-306 Ma (mean 258 ± 35 Ma, 1σ, n=11; §3.3), bracketed by the regional ∼300-310 Ma magmatic episode and the ∼235 Ma gold-vein mineralizing event (§3.1). Neither pyrite nor arsenopyrite itself hosts appreciable U or Th, instead relying on U-Th-Sm-bearing accessory inclusions to supply the sulfide hosts' alpha-emitters and radiogenic helium: apatite at Kidd Creek, rutile at Gympie. The helium from small, dispersed inclusions is implanted entirely within the sulfide host (and is thus subject to the sulfide's own helium retentivity), but larger inclusions will partially self-implant, complicating the interpretation of the sulfide hosts' ages (§3.4). At Kidd Creek, a simple nugget-effect model explains the anti-correlation between age and Th/He content among the eleven-aliquot array, but the observed sizes of the apatite inclusions are too small to explain ages this young if this population behaved as a closed system: most of it must have experienced helium loss during its long history. The twelfth, oldest aliquot is instead consistent with the closed-system prediction at the observed inclusion size, though this rests on a single measurement that could equally reflect incomplete sample digestion rather than a genuinely more retentive phase (§3.4).

Together, these results support pyrite and arsenopyrite U-Th/He dating as a viable complement to Re-Os geochronology in multistage sulfide systems, but only once the identity and grain size of the actual radiogenic host phase are established for a given locality (§3.4): no single digestion protocol can be assumed to transfer between deposits. Extending this framework to the other sulfide-hosted geochronometers proposed in §3.1, and to a wider range of deposits than the two tested here, remains the natural next step.

Supplementary Information

for Chapter 3: *(U-Th-Sm)/He Sulfide Geochronology*

3.1s Chalcopyrite, Chalcocite, Galena, and Sphalerite Step Heat Results and Discussion

This section reports the step heat results and discussion for the two non-pyrite/arsenopyrite sulfides of this study – the copper sulfides chalcopyrite (CCpTS230630) and chalcocite (CCT230707), and galena (GalTS230630) – moved here from the main text because neither mineral is a promising U-Th/He geochronometer or helium isotope target, and neither is otherwise part of this chapter's main narrative (Table 1 in the main text still lists all nine step heat experiments, including these three, for completeness).

Results

Chalcocite and Chalcopyrite

Step heat data step heat experiments for chalcopyrite (CCpTS230630) and chalcocite (CCT230707) produced macroscopic evidence of phase transformation, including the development of an iridescent blue sulfide or oxide layer on both specimens and dramatic crystal growth on the ampoule walls for the chalcocite experiment Tube Photo Although permeation lag was not sufficiently constrained to calculate apparent diffusivities for the chalcocite experiment, the data for CCT230707 are consistent with a primary phase-breakdown control of helium migration at temperatures $\leq$300°C.

A linear regression is fit to 9 measurements defining a linear region of the Arrhenius plot of chalcopyrite sample CCpTS230630 for use as a reference for the delta plot (Fig. 3.19, row 3). This regression has an R^2 value of 0.9811, an implied activation energy of 41.8 $\pm$ 2.2 kcal (174.8 $\pm$ 9.2 kJ/mol), and an implied closure temperature of 128 $\pm$ 13°C assuming a cooling rate of 10 °C/Ma. The delta plot generated using this reference line for chalcopyrite clearly shows a dramatic mid-experiment increase in diffusivity following steps above 400°C, followed by a continuous decrease in delta values (Fig. 3.19, row 3), similarly to that observed for the dehydrating goethite sample Win-06-01b (Farley, H. B. Monteiro, et al., 2024).

Galena

The majority of the helium total yield was evolved during the step heat experiment of the encapsulated galena, in stark contrast to the comparatively retentive pyrites and arsenopyrite. The first 70% of helium released in the

galena experiment could not be reduced into diffusivities, because $\lambda t \leq 4$ and so steady state permeation out of the ampoule could not be assumed. There were no permeation steps for this experiment. These data are still included in Fig. 3.10, because the cumulative yields are less affected by the permeation lag, which mostly affects the short-term knowledge of helium degassing: since earlier degassed helium will have long since permeated out, allowing for generally accurate accounting of evolved helium throughout the experiment. The reference line from the chalcopyrite experiment showed a similar activation energy to the activation energy implied by curved retrogrades of the galena experiment (Fig. 3.19, row 2), so it was used as the reference for the delta plot of galena. This delta plot shows a monotonic decrease in delta values as the experiment proceeds (Fig. 3.19, row 2).

The full ^{3}He step-heat datasets for both experiments, in the same format as the main-text pyrite/arsenopyrite tables (§3.3), are given in Tables 3.7 and 3.8.

Sphalerite Although we conducted a step heat experiment on translucent sphalerite (SphTS230630), none of the sample's steps were above $\lambda t \leq 4$ and there were insufficient permeation steps to constrain the permeation lag, so no diffusivities could be calculated. Qualitatively, the sphalerite experiment showed that it is somewhat less helium retentive than apatite, and it was consistent with a single diffusion domain, although our data do not permit any confident interpretation. Future work on sphalerite should conduct disciplined permeation steps to track the helium inventory of the ampoule following the procedure outlined in §1.4.

Table 3.7: ^{3}He step-heat dataset for Galena, Tri-State District, Missouri (sample GalTS230630), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.19 (row 2). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
15	385.0	100.0	0.2968	11.3209	0.0835	-14.758	-13.974	-13.716
16	375.0	120.0	0.2004	11.5213	0.0491	-15.018	-14.489	-14.264
17	380.0	110.0	0.2022	11.7235	0.0528	-14.616	-14.344	-14.105
18	390.0	90.0	0.1906	11.9141	0.0586	-14.471	-14.153	-13.877
19	400.0	70.0	0.3558	12.2699	0.1413	-13.379	-13.207	-12.853
20	410.0	60.0	0.0947	12.3646	0.0473	–	-14.316	-13.904
21	300.0	60.0	0.0374	12.4020	0.1171	–	-15.228	-13.793
22	350.0	60.0	0.0241	12.4262	0.0366	–	-15.656	-14.729
23	375.0	60.0	0.0459	12.4720	0.0327	-16.600	-15.005	-14.462
24	400.0	60.0	0.1039	12.5760	0.0525	-14.539	-14.166	-13.750
25	415.0	60.0	0.1722	12.7481	0.0744	-13.977	-13.621	-13.251
26	460.0	60.0	0.6155	13.3637	0.1765	-12.343	-12.226	-11.943
27	470.0	60.0	0.6037	13.9673	0.1605	-12.349	-12.034	-11.765
28	480.0	60.0	0.4856	14.4529	0.1200	-12.389	-12.024	-11.771
29	489.9	60.0	0.3938	14.8467	0.0908	-12.325	-11.997	-11.787
30	499.7	60.0	0.3285	15.1752	0.0709	-12.280	-11.985	-11.764
31	504.6	60.0	0.2357	15.4109	0.0493	-12.437	-12.108	-11.896
32	495.0	60.0	0.1264	15.5374	0.0282	-13.040	-12.570	-12.355
33	485.0	80.0	0.0935	15.6309	0.0158	-13.390	-13.047	-12.882
34	475.0	100.0	0.0667	15.6976	0.0093	-13.777	-13.516	-13.380
35	465.0	120.0	0.0489	15.7465	0.0059	-14.144	-13.938	-13.820
36	455.0	140.0	0.0351	15.7816	0.0038	-14.549	-14.368	-14.262
37	510.0	60.0	0.0648	15.8464	0.0131	-12.894	-12.836	-12.643
38	519.9	60.0	0.0765	15.9229	0.0145	-12.740	-12.562	-12.376
39	529.9	60.0	0.0758	15.9987	0.0135	-12.638	-12.438	-12.261
40	539.1	60.0	0.0709	16.0696	0.0120	-12.557	-12.359	-12.190
41	535.8	60.0	0.0570	16.1266	0.0098	-12.650	-12.429	-12.258
42	538.6	60.0	0.0385	16.1651	0.0065	-12.977	-12.697	-12.531
43	532.9	80.0	0.0330	16.1980	0.0041	-13.243	-13.033	-12.908
44	525.0	100.0	0.0265	16.2245	0.0027	-13.542	-13.378	-13.276
45	515.0	120.0	0.0207	16.2452	0.0018	-13.857	-13.722	-13.633
46	503.7	140.0	0.0139	16.2590	0.0011	-14.348	-14.209	-14.129
49	480.0	60.0	0.0028	16.2618	0.0007	-14.929	-14.929	-14.706
50	489.9	60.0	0.0032	16.2651	0.0007	-15.010	-14.771	-14.562
51	500.0	60.0	0.0050	16.2701	0.0011	-14.468	-14.310	-14.113

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Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
52	505.0	60.0	0.0052	16.2753	0.0011	-14.495	-14.262	-14.070

Table 3.8: ^{3}He step-heat dataset for Chalcopyrite, Baxter Springs, Kansas (sample CCpTS230630), under the "treatment as uncertainty" method (§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.19 (row 3). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
16	374.9	120.0	0.0782	13.1758	0.0247	-18.321	-17.572	-17.296
17	380.0	110.0	0.0639	13.2396	0.0220	-18.167	-17.680	-17.383
18	390.0	90.0	0.0721	13.3118	0.0296	-17.713	-17.350	-17.005
19	400.1	70.0	0.0943	13.4061	0.0509	-17.198	-16.822	-16.388
20	410.0	60.0	0.0963	13.5024	0.0610	-17.385	-16.637	-16.143
22	291.8	60.0	0.0928	13.5961	3.0586	-17.704	-16.664	-12.980
23	345.2	60.0	0.2696	13.8657	1.0009	–	-15.578	-13.976
24	375.0	60.0	0.9931	14.8588	1.0408	–	-14.209	-13.439
25	400.0	60.0	2.0663	16.9251	1.4653	-13.975	-13.324	-12.714
26	415.0	60.0	2.2228	19.1480	1.3491	-14.056	-13.046	-12.507
27	429.9	60.0	2.8501	21.9981	1.4512	-13.143	-12.562	-12.080
28	420.0	60.0	2.4223	24.4204	1.3837	-13.328	-12.480	-11.959
29	410.0	60.0	1.4757	25.8962	0.9756	-15.497	-12.790	-12.232
30	400.1	60.0	0.7845	26.6806	0.6038	–	-13.309	-12.706
31	390.0	80.0	0.5091	27.1898	0.2693	–	-13.962	-13.523
32	380.0	100.0	0.3706	27.5604	0.1495	-15.738	-14.456	-14.108
33	370.0	120.0	0.2689	27.8293	0.0892	-15.728	-14.924	-14.632
34	360.0	140.0	0.1510	27.9803	0.0452	-16.521	-15.632	-15.367
35	355.0	150.0	0.1416	28.1219	0.0408	-16.131	-15.749	-15.493
36	365.0	130.0	0.1775	28.2994	0.0557	-15.621	-15.362	-15.085
37	375.0	110.0	0.2245	28.5239	0.0808	-15.219	-14.937	-14.625
38	385.0	90.0	0.3152	28.8391	0.1360	-14.657	-14.365	-13.999
39	395.0	80.0	0.4454	29.2845	0.2065	-14.213	-13.857	-13.463
40	405.0	70.0	0.4926	29.7771	0.2543	-14.095	-13.565	-13.132
41	415.0	60.0	0.4935	30.2707	0.2950	-14.053	-13.346	-12.857
42	460.0	60.0	0.4521	30.7228	0.1759	-14.407	-13.370	-13.029
43	469.7	60.0	0.1397	30.8624	0.0517	–	-14.504	-14.186
44	465.4	60.0	0.1654	31.0278	0.0601	-14.684	-14.313	-13.999

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Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
45	473.5	60.0	0.1554	31.1832	0.0531	-14.837	-14.352	-14.054
46	480.0	60.0	0.1691	31.3523	0.0538	-14.617	-14.244	-13.963
47	491.9	60.0	0.1549	31.5072	0.0452	-14.729	-14.307	-14.047
48	491.0	60.0	0.1219	31.6291	0.0357	-14.985	-14.525	-14.265

Discussion: Thermal Helium Migration Systematics

Chalcopyrite and Chalcocite

Neither copper-bearing sulfide mineral studied here was successfully stabilized by encapsulation with sulfur, so these results provide little insight into helium migration within copper sulfides in nature, though they suggest high potential for helium retention at surface temperatures, similar to apatite. Still, they are unlikely to be promising targets for U-Th/He geochronology or helium isotope studies as there exists many copper-bearing sulfide phases stable at temperatures between 200-500°C, for which phase stability and Cu/Fe ratios are sensitive functions of temperature, composition, and sulfur fugacity (Zhao et al., 2014; J. H. Chen and Harvey, 1975; Yund and Kullerud, 1966). These reactions are likely to result in helium loss during reheating, burial, diagenesis, or metasomatism after crystallization. Copper sulfides are unlikely to be promising targets for U-Th/He geochronology, but may be suitable for helium isotope studies in young or unaltered samples.

Galena

Because the majority of the cumulative yield was degassed in steps for which $\lambda t > 4$, the bulk helium behavior for galena is somewhat unclear. The data of the remaining 30% of the cumulative yield was consistent with a multiple diffusion domain helium loss mechanism. As expected for an MDD system, the delta plot for galena monotonically decreases as the fractional yield increases (Fig. 3.19, row 2; Farley (2018)). The majority of the 3He cumulative fractional yield was evolved at low temperatures where the uncertainties due to permeation lag could not be constrained, yet these early degassing domains were certainly less retentive than 150 µm half-prism width Durango apatite (Farley, 2000). Additionally, the total 3He concentration for the irradiated galena was less than half that of co-irradiated pyrite, implying poor helium retention even at

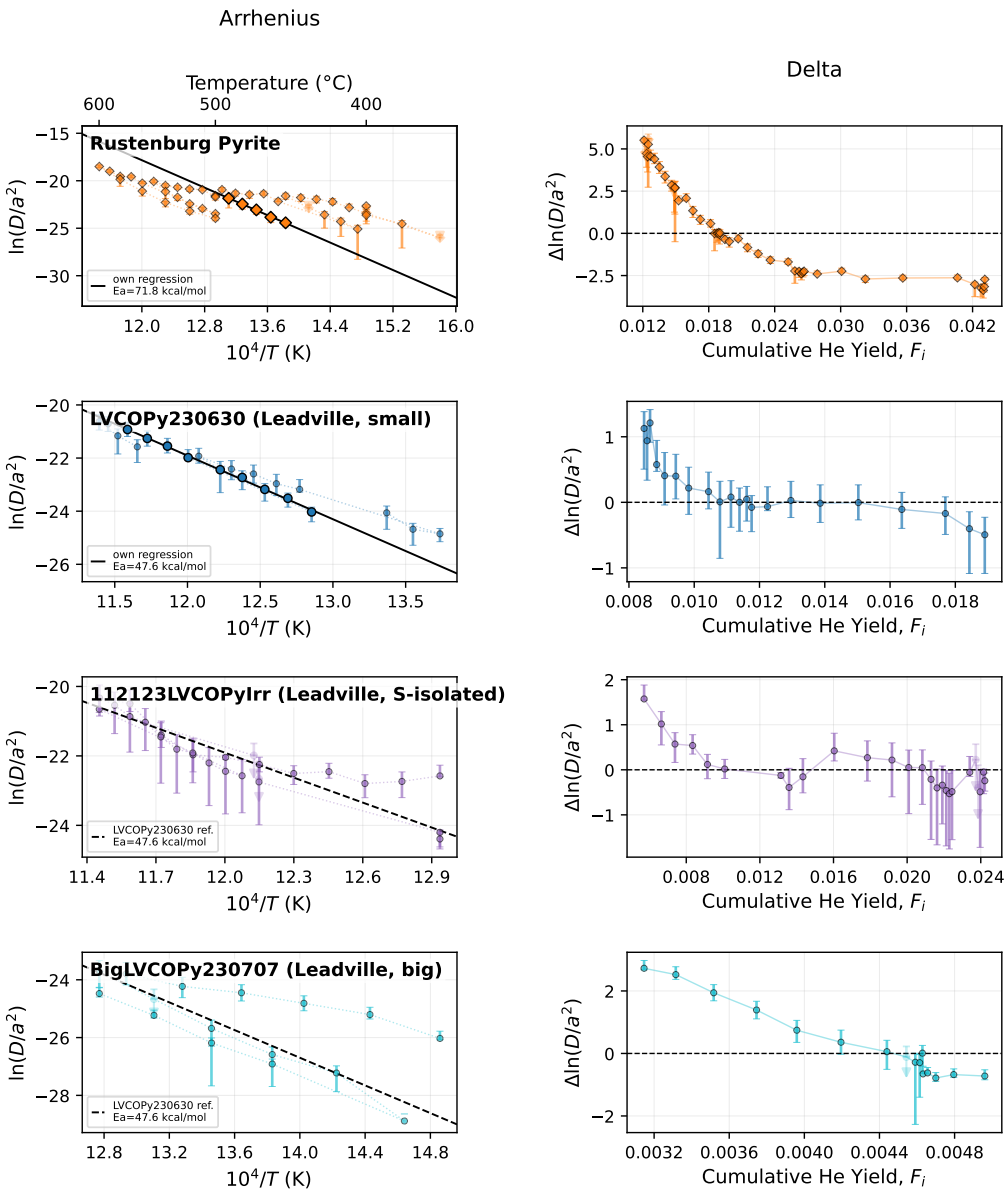

Figure 3.18: Compiled Arrhenius (left) and delta (right) plots for the four plotable Chapter 3 pyrite step-heat experiments: (1) Rustenburg pyrite, (2) LVCOPy230630 (Leadville, small), (3) 112123LVCOPyIrr (Leadville, sulfur-isolated), (4) BigLVCOPy230707 (Leadville, big). Each row's Arrhenius panel plots the reference regression used for that row's delta panel (solid where owned, dashed where borrowed; see §3.3). Each row has its own x-axis scale, labeled on every panel. The three non-pyrite sulfides are compiled separately in Fig. 3.19.

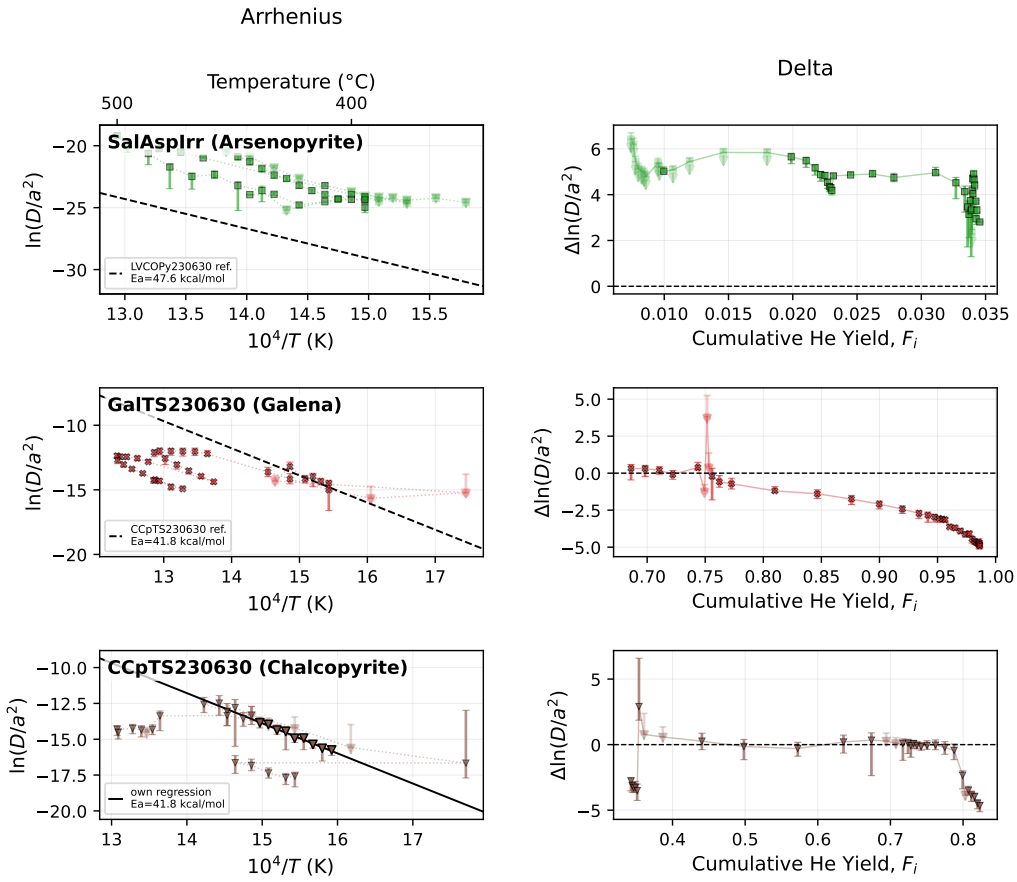

Figure 3.19: Compiled Arrhenius (left) and delta (right) plots for the three plotable Chapter 3 non-pyrite sulfide step-heat experiments: (1) SalAspIrr (arsenopyrite), (2) GalTS230630 (galena), (3) CCpTS230630 (chalcopyrite). Each row's Arrhenius panel plots the reference regression used for that row's delta panel (solid where owned, dashed where borrowed; see §3.3). Each row has its own x-axis scale, labeled on every panel. The four pyrite samples are compiled separately in Fig. 3.18.

ambient temperatures. Galena is unlikely to be a promising target for U-Th/He geochronology or helium isotope studies.

3.2s Complete U-Th/He Geochronology Compilation

Trace Elements for the Main-Text Sample Suites

The main-text geochronology table (Table 3.6) reports U, Th, Sm, and Zr for the Kidd Creek, Gympie, Durango, and Gogebic aliquots. The remaining ICPMS trace elements measured the same way – Cu, As, Ag, and Sb – are reported here instead (Table 3.9), in the same row order, to keep the main-text table within the page width.

Table 3.9: Cu, As, Ag, and Sb (measured as ratios to spike vanadium by the same ICPMS run as U, Th, Sm, and Zr, §3.2) for the same Kidd Creek, Gympie, Durango, and Gogebic aliquots reported in Table 3.6, in the same row order; '–' denotes that the value was not measured. Arsenic (As) in particular shows some unreasonable/impossible values for two gympie samples that were analyzed before HCl was added to its digestion protocol. These samples are already excluded from the analysis of this work, but it is unclear how the old protocol could have produced erroneously high As values, casting doubt on their reliability. Sample notes are repeated from Table 3.6 for reference.

Sample	Cu (ppm)	As (ppm)	Ag (ppm)	Sb (ppm)	Note
Kidd Creek Pyrite (Archean VMS, ∼2716 Ma host)					
2211KC_Py_ST05_C	–	–	–	–	
2211KC_Py_ST05_A	–	–	–	–	
2211KC_Py_ST05_B	–	–	–	–	
2211KC_Py_ST05_D	–	–	–	–	
011123KCPyST05C	248.36±27.86	49.38±12.69	3.88±0.98	15.91±0.94	
011423KCPyST05A	555.10±38.86	5922.20±397.89	7.82±1.12	35.69±0.95	
011423KCPyST05B	1365.56±58.41	1604.14±109.70	12.62±1.04	22.44±0.41	
011423KCPyST05C	780.58±52.49	8026.88±487.81	11.17±1.64	21.97±0.46	
011423KCPyST05D	146.71±19.63	11440.33±796.21	2.13±1.35	24.09±0.73	
013023KCPyST05As1	17.41±7.76	47121.03±2800.88	–	38.45±0.81	Silver Luster, Aspy Habit
013023KCPyST05Cu1	314.00±30.30	11923.72±782.77	3.52±2.05	23.93±1.33	Brassy Luster, Cube Habit
013023KCPyST05Cu2	102.88±20.20	410116.35±23437.59	–	1235.42±29.55	Brassy Luster, Cube Habit
Gympie Pyrite (Permo-Triassic orogenic Au)					
013023PVGyPy-1	42.50±7.50	202426.65±12489.88	0.73±0.41	138.00±3.18	excl.: early protocol
013023PVGyPy-2	196.52±14.56	1045127.96±63721.03	3.15±1.26	733.74±21.57	excl.: early protocol
013023PVGyPy-3	15.93±8.91	433911.06±27230.00	2.81±0.86	345.22±9.92	excl.: early protocol
013023PVGyPy-4	38.76±16.89	616561.59±36135.06	1.67±0.79	333.74±7.83	excl.: early protocol

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Sample	Cu (ppm)	As (ppm)	Ag (ppm)	Sb (ppm)	Note
0413PyGy-1	43.08±8.28	552.38±44.66	9.26±1.92	70.17±4.40	
0413PyGy-2	33.14±8.87	164.64±37.86	8.78±2.05	12.54±3.70	
0413PyGy-3	95.81±27.83	3445.18±214.54	7.91±4.05	61.53±5.65	
0413PyGy-4	30.30±10.79	3814.85±170.90	7.41±2.90	82.93±4.53	excl.: outlier
062123GyPyHClb	29.50±5.71	3594.31±91.81	4.76±1.64	186.80±5.22	
062123GyPyHClc	21.71±3.17	499.31±18.48	2.30±1.11	6.36±0.19	
230625GyPyHCl	30.60±5.78	13394.39±344.59	4.16±1.53	144.68±5.06	
230625GyPyUnAbr	11.48±2.49	101443.33±2285.09	2.85±1.18	166.23±5.89	
230628GyPyHClab	60.99±6.34	297125.64±8329.25	72.30±26.22	222.09±12.53	
230628GyPyHClc	23.00±4.19	33613.36±1092.63	7.65±3.09	183.41±11.95	
230628GyPyUnAbrA	30.71±4.04	1615.76±61.65	4.60±1.97	96.60±6.99	
230628GyPyUnAbrB	12.00±5.29	1498.79±70.73	7.27±3.08	45.68±3.40	
Durango Apatite Standard (accepted age 32.1 Ma; Farley 2000)					
2211DApA	–	–	–	–	old method: pre-degassing
2211DApC	–	–	–	–	old method: pre-degassing
2211DApD	–	–	–	–	old method: pre-degassing
2022StDApA	–	–	–	–	old method: pre-degassing
011123DAp1	–	924.97±72.80	118.08±10.69	89.07±4.68	
011123DAp2	–	910.07±100.20	–	26.32±2.87	
011123DAp3	–	803.92±55.21	–	49.44±1.01	
011423DAp1	–	787.88±50.76	1.79±1.42	28.71±0.83	
011423DAp2	–	846.83±57.62	–	17.66±0.58	
011423DAp3	–	820.23±71.82	–	23.65±0.54	
011423DAp4	–	810.88±61.16	–	42.87±2.29	
062123DAp	–	665.14±204.98	–	–	
062523DAp	24.94±8.99	799.92±156.54	–	–	

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Sample	Cu (ppm)	As (ppm)	Ag (ppm)	Sb (ppm)	Note
062823DAp	–	–	–	–	
Average for > 0.5 mg	–	–	–	–	
Gogebic Iron Range BIF Hematite Standard (MI-43; 571 Ma)					
2211HemMSA	–	–	–	–	
2211HemMSB	–	–	–	–	
041323MG16a	–	338.77±109.12	–	–	
041323MG16b	77.04±34.63	1310.34±209.47	10.81±9.03	–	
041323MG16d	–	3960.06±226.90	20.03±15.53	–	
041323MG16e	–	8490.93±1373.53	–	–	
062123MS	–	2717.17±112.31	3.63±1.42	18.83±0.96	
230625MSHm	–	1849.86±128.15	78.00±29.79	89.12±4.54	
230628MSHm	–	443.71±19.65	3.06±1.63	93.15±7.07	

Additional Sample Suites

This subsection is a data appendix, not a narrative: it reports the complete U-Th/He geochronology compilation (Chapter3/Tables/geochronology_ compilation_BJT.xlsx) for the sample suites and replicate aliquots not discussed in the main text – Kidd Creek replicate/mass-only aliquots, Franklin (NJ) pyrite and fluorite, Rustenberg/Pilanesberg pyrite, Magnet Cove carbonatite pyrite, French Creek magnetite/pyrite, and Salzburg arsenopyrite – so that the full underlying dataset is available even where it is not part of this chapter's argument. Later work extending this chapter's geochronology (e.g. to these additional localities) can build directly on this section.

Table 3.10: U-Th/He geochronology results for all remaining sample suites in the compilation, not discussed in the main text: Kidd Creek replicate/mass-only aliquots, Franklin (NJ) pyrite and fluorite, Rustenberg/Pilanesberg pyrite, Magnet Cove carbonatite pyrite, French Creek magnetite/pyrite, and Salzburg arsenopyrite. Reported here for completeness, with the full trace-element suite (Cu, As, Zr, Ag, Sb) retained since these aliquots are not duplicated in the main text; columns and conventions as in Table 3.6.

Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Cu (ppm)	As (ppm)	Zr (ppm)	Ag (ppm)	Sb (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)	Note
Kidd Creek Pyrite (replicate / mass-only aliquots)													
2211KC_Py_ST05_DU1	2.09	0.78±0.03	1.76±0.04	0.08±0.01	–	–	–	–	–	1.20	–	–	
2211KC_Py_ST05_DU2	2.98	0.80±0.01	2.19±0.06	0.02±0.00	–	–	–	–	–	1.33	–	–	
2211KC_Py_ST05_DU3	1.77	0.77±0.03	1.93±0.06	0.10±0.01	–	–	–	–	–	1.23	–	–	
12122022KCPyST05A	14.35	–	–	–	–	–	–	–	–	–	1422.90	–	
12122022KCPyST05B	10.52	–	–	–	–	–	–	–	–	–	960.74	–	
12122022KCPyST05C	12.19	–	–	–	–	–	–	–	–	–	1203.97	–	
12122022KCPyST05D	9.60	–	–	–	–	–	–	–	–	–	865.75	–	
Franklin, NJ Pyrite (marble/volcanics, ∼1.3–1.0 Ga)													
2211GFr_PyA	2.49	0.29±0.01	0.05±0.01	0.02±0.00	–	–	–	–	–	0.31	51.29	529.1±22.1	
2211GFr_PyB	3.00	1.78±0.02	0.01±0.01	0.01±0.00	–	–	–	–	–	1.78	66.87	106.9±1.3	
2211StFr_PyA	1.45	0.64±0.03	0.02±0.01	0.02±0.00	–	–	–	–	–	0.65	42.84	322.2±14.8	
2211StFr_PyB	2.53	0.12±0.01	0.00±0.00	0.91±0.04	–	–	–	–	–	0.12	57.76	1256.6±124.3	
2211AtmFr_PyC	1.93	1.00±0.02	0.01±0.01	0.02±0.00	–	–	–	–	–	1.00	55.14	223.2±4.6	
Franklin, NJ Fluorite													
011423FrFl2	1.40	3.19±0.03	0.24±0.07	0.21±0.04	32.17±10.03	–	1.82±1.13	10.23±1.72	35.81±0.74	3.25	76.81	168.7±1.9	
011423FrFl3	1.03	5.35±0.08	0.20±0.06	0.34±0.09	–	–	5.80±1.56	–	30.63±0.71	5.40	99.90	179.3±2.8	
Rustenberg/Pilanesberg(?) Pyrite													
013023BVPy H-3026-1	1.03	2.50±0.31	1.04±0.09	0.29±0.07	173.63±26.23	3328.64±233.02	10.86±5.67	–	152.60±3.84	2.75	226.12	747.3±79.0	
041323BVPy1	0.95	0.56±0.02	0.17±0.05	0.05±0.05	105.69±27.36	391.95±59.38	–	–	–	0.60	45.43	724.9±19.9	

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Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Cu (ppm)	As (ppm)	Zr (ppm)	Ag (ppm)	Sb (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)	Note
041323BVPy2	0.53	2.07±0.28	2.34±0.41	0.77±0.45	484.42±455.64	22900.08±4862.73	130.29±59.21	–	680.02±100.62	2.63	70.82	490.3±52.8	
062123BVPyHCla	1.22	3.13±0.12	0.81±0.07	0.25±0.01	95.87±10.25	582.38±13.15	12.17±2.15	2.43±0.97	10.07±0.42	3.33	241.19	571.5±19.2	
062123BVPyHClb	0.53	1.46±0.05	0.25±0.18	0.16±0.01	102.76±11.05	1499.31±44.51	7.89±4.05	3.30±1.37	10.48±0.72	1.52	1046.06	5109.8±58.3	
230628BVPyUnAbrB	0.75	2.15±0.07	0.42±0.13	0.14±0.02	64.34±12.08	996.13±55.99	–	3.09±1.39	3.54±0.32	2.25	233.66	1244.1±39.2	
230628BVPyUnAbrC	1.11	0.50±0.13	0.12±0.09	0.09±0.02	979.31±62.60	1028.86±34.43	–	7.63±3.34	4.60±0.29	0.53	723.21	5050.8±429.6	
230628BVPyUnAbrC	0.60	29.01±0.40	5.44±0.33	2.01±0.17	138.48±15.53	728.93±53.84	202.97±10.19	11.80±4.77	10.59±0.76	30.32	881.54	474.1±6.0	
Magnet Cove Carbonatite Pyrite, Arkansas (∼100 Ma)													
011123MCARPy1	3.38	0.06±0.00	0.41±0.03	0.31±0.03	15.72±4.66	3.86±1.22	1.63±0.65	1.10±0.48	18.58±0.78	0.16	9.67	177.1±7.5	
011123MCARPy2	3.90	0.01±0.00	0.20±0.01	0.02±0.01	56.98±6.51	62.10±8.60	1.58±0.65	–	11.49±0.88	0.06	3.25	52.2±4.7	
French Creek, PA Magnetite and Pyrite (∼130 Ma)													
062123FCHCl	4.36	0.01±0.00	0.00±0.02	0.01±0.00	8.97±2.49	29.58±2.10	–	21.25±6.89	7.65±1.10	0.01	1.55	218.0±52.4	
062123FCMagHCl	3.84	0.06±0.00	0.00±0.02	0.05±0.00	–	81.88±3.18	–	0.71±0.29	15.55±0.44	0.06	2.67	97.7±5.5	
230625FCHFa	3.41	0.63±0.05	0.71±0.03	0.34±0.02	209.62±15.43	1118.04±38.65	7.76±0.88	5.71±2.12	19.75±0.77	0.80	276.46	953.2±53.0	
230625FCMagHCl	4.07	0.05±0.00	0.00±0.02	0.02±0.00	–	85.58±8.67	–	0.74±0.33	4.22±0.19	0.05	3.60	137.1±7.7	
Salzburg, Austria Arsenopyrite													
230625SBAsPy	2.52	0.96±0.01	1.29±0.05	0.11±0.01	78.87±11.98	187365.67±4344.31	24.74±1.67	3.67±1.41	94.85±3.82	1.27	33.95	106.2±1.3	

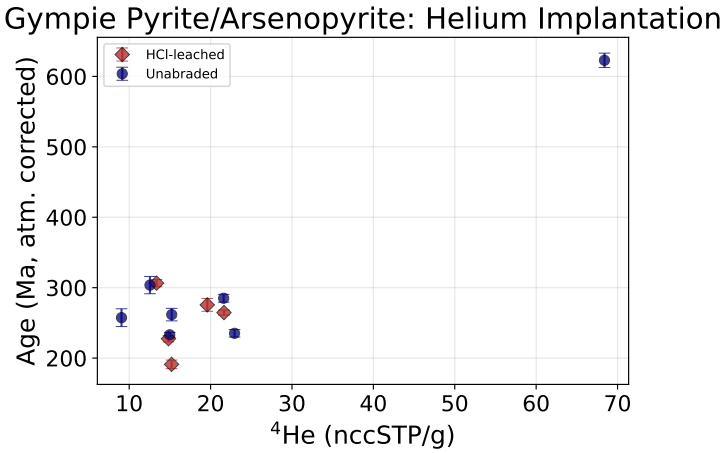

Figure 3.20: Companion to the main-text Gympie age-vs-eU figure (Fig. 3.8), plotting atmospherically corrected age against measured 4He content (ncc-STP/g) for the newer-protocol Gympie arsenopyrite suite with the 0413PyGy-4 outlier put back in (§3.3, main text). Its 4He content stands well off the trend defined by the other eleven aliquots, consistent with the helium-implantation interpretation offered there.

Chapter 4

NEW FRONTIERS IN THERMODYNAMICALLY INFORMED GEODYNAMIC MODELING: A GENERAL DEEP LEARNING ACCELERATED FREE ENERGY MINIMIZER FOR MULTICOMPONENT SILICATE SYSTEMS

[TODO: One-paragraph chapter summary.]

4.1 Introduction

[TODO: Motivate the need for fast thermodynamic forward models in geodynamics. Review current tools (Perple_X, THERMOCALC, MAGEMin) and their computational bottlenecks. State the deep learning acceleration approach.]

4.2 Background

Gibbs Free Energy Minimization in Petrology

[TODO:]

Machine Learning Surrogates for Physical Models

[TODO:]

4.3 Model Architecture

Training Data Generation

[TODO: Describe the strategy for generating phase equilibrium training data from classical minimizers across relevant P–T–X space.]

Neural Network Design

[TODO: Describe the architecture: input/output spaces, layers, activation functions, loss function, training procedure.]

Uncertainty Quantification

[TODO:]

4.4 Validation

[TODO: Benchmark speed and accuracy against classical minimizers on held-out P–T–X points and on known phase diagrams.]

4.5 Application to Geodynamic Modeling

[TODO: Demonstrate integration into a geodynamic simulation. Show a representative case study.]

4.6 Discussion

[TODO: Discuss accuracy/speed tradeoffs, generalizability to other bulk compositions, and extension to systems relevant to the Bushveld Complex (foreshadow Appendix 5.8 and the Conclusion).]

4.7 Conclusions

[TODO: Summarize the ML minimizer performance and implications for future thermodynamically coupled geodynamic models.]

Chapter 5

FIELD STUDIES REVEAL SYNMAGMATIC DEFORMATION AND EXTENSIVE CRUSTAL CONTAMINATION OF THE UPPER CRITICAL ZONE, RECESSIVE LOWER MAIN ZONE, AND FLOOR ROCKS AT THE MARGIN OF THE BUSHVELD COMPLEX AT DRAAIKRAAL, MPUMALANGA, SOUTH AFRICA

The paleoproterozoic Rustenburg Layered Suite (RLS) of the Bushveld Magmatic Province in northeast South Africa is the largest layered igneous intrusion in the world, and contains the world's largest source of platinum group elements as well as important reserves of chromium, vanadium, titanium, and iron. It is exceptionally well-preserved, and has been the subject of intense study and debate for generations. Studies focusing on the intrusion's margin have been especially fruitful in informing geologists' understanding of igneous petrogenesis and intrusion mechanisms, modes, and tempos of layered mafic intrusions and plutonic rocks more generally. The phaneritic RLS is stratigraphically divided into four zones based on geochemistry and modal mineralogy: the Lower, Critical, Main, and Upper Zones. These zones are often thought to have been constructed at differing times and from different magmatic sources and processes. Thus, the intersection of these stratigraphically defined zones at the intrusion's margin are particularly interesting locations for detailed field work. This field study focuses on several unusual but understudied structural and geomorphic features occurring at the economically significant intersection of the Main and Critical Zones with the intrusion's margin in the well-exposed Eastern Limb near the village of Draaikraal, Mpumalanga, South Africa: the Kliprivier synform, the Draaikraal Lineament, and the unexpected recessive topography of the Main Zone in the agricultural valley surrounding the village of Draaikraal.

The Kliprivier Synform represents a series of modestly rotated plutonic rocks with conspicuously absent modal layering, and country

rocks that include Quartzite, calc-silicate hornfels, and fine-grained Marginal Zone gabbros. The country rocks appear to be displaced ~ 1 km into the plutonic RLS near the mapped intersection of the Critical and Main Zones. The Draaikraal Lineament (named here) is a linear feature of recessive topography stretching from the country rocks beyond the intrusion's margin to the middle Main Zone. It overlaps with the Kliprivier Synform and separates typical, resistant Main Zone gabbronorites to the north from the recessive Main Zone gabbronorites to the south in the agricultural valley around Draaikraal. Using structural measurements and correlation of mottled anorthosite marker beds of the lower Main Zone, I find that the mottled anorthosite marker beds are concordant with the floor rocks at the intrusions margin and are not displaced across the Draaikraal Lineament. I argue that the Kliprivier Synform is a synmagmatic monocline that formed due to lopolithic deformation of the country rock roof by the intrusion of Critical Zone sills before the formation of the UG2 chromitite and overlying Merensky Reef and Main Zones. The orientation of the Draaikraal Lineament is very similar to the surface-projection of the axial plane of the Kliprivier Monocline, which likely continues at depth. The relative recessive topography of the Draaikraal Lineament and the Main Zone to the south is argued to be the result of higher degrees of contamination by hydrothermal fluids than the more resistant Main Zone gabbronorites of the surrounding area. The rocks at the Draaikraal Lineament are envisaged to be fluxed with upwardly migrating fluids that were trapped by the Critical Zone roof, escaping by exploiting the structural weakness of the Kliprivier Monocline. The agricultural valley to the south of the lineament is simply envisaged to be relatively contaminated by melting and devolatilizing sediments at the proximal floor, whereas the resistant Main Zone to the north was relatively buffered by interaction with such fluids by the thick Critical Zone rocks that underlie it there.

5.1 Introduction

[TODO: Expand into a standalone Introduction section (motivation, regional setting at a glance, and a roadmap of this chapter's sections)

distinct from the abstract above and the literature review immediately below – matching the Introduction/Background split used in Chapters 1–3. The abstract's second paragraph (the Kliprivier Synform / Draaikraal Lineament argument summary) can likely be adapted as a starting point.]

5.2 Field Methods

[TODO: Describe the field campaign: traverse strategy and coverage, GPS/geotagging protocol (Section 5.5 data were collected this way), sampling and photo-documentation conventions, and the point-count/mineral-percentage estimation approach used in the field (columns present but largely unused in Table 5.1). See also the analytical-methods description already written for the stratigraphic column (§5.4).]

5.3 Geological Background

Layered Mafic Intrusions

Layered mafic intrusions (LMIs) are compositionally zoned formations of mafic rocks characterized by distinct millimeter to meter scale layering of igneous minerals. Traditionally, the layering and chemical zonation of these geologic marvels have been attributed to processes such as fractional crystallization and gravitational settling within a magma chamber (Wager et al., 1960; Charlier et al., 2015). Notable examples of LMIs include the Stillwater (2.71 Ga) in Montana, USA; Great Dyke (2.57 Ga) in central Zimbabwe; the Bushveld's layered series (2.06 Ga) in South Africa's Limpopo, Mpumalanga, and Gauteng provinces; the Muskox intrusion (1.27 Ga) in Nunavut, Canada; the Duluth complex (1.01 Ga) in Minnesota, USA; the Sept Iles intrusion (565 Ma) in Quebec, Canada; Dufek (189 Ma) in Antarctica; the Rum intrusion (60.5 Ma) in the Inner Hebrides, Scotland; and the Skaergaard intrusion (55.4 Ma) in East Greenland. Despite their significance as supposedly simple natural laboratories for studying the chemistry evolution of magmas, numerous aspects of their formation remain subject to debate, even including the presence of significant liquid magma in their formation (Kruger and Latypov, 2020; Mungall et al., 2016; Latypov et al., 2022; Yao et al., 2021).

Experimental observations have revealed that plagioclase floatation occurs in mafic magmas at depths corresponding to the Earth's crust (Campbell

et al.„ 1978). This challenges the notion that gravitational settling alone is responsible for igneous layering. Instead, chemical means of phase separation are often suggested, referencing processes like periodic nucleation during in-situ crystallization (McBirney and Noyes, 1979), in-situ crystallization of a magma with compositional layering developed and sustained through double-diffusive convection (Huppert and Turner, 1981), or preferential dissolution and recrystallization of existing unstructured crystal cumulates or mushes to minimize crystal surface energy perpendicular to temperature gradients (Boudreau, 2011) or by preferential attack by chlorine-rich fluids (Sonnenthal, 1992). An alternative mechanical explanation proposes seismically aided hydrodynamic sorting within a tilting, crystal-rich cumulate pile has also enjoyed support (Maier et al., 2013).

The Bushveld Event and RLS

This study centers on the Rustenburg Layered Series (RLS): the layered, mafic component of the Bushveld Event. The RLS is renowned as the world's largest layered mafic intrusion, distinguished by its remarkably continuous layers, sometimes across hundreds of kilometers (Cawthorn, 2015; Latypov et al., 2022; Scoon and Tiegler, 1994). Within the RLS, pegmatoid "reefs" and chromitites host the world's largest reserves of platinum group elements or PGEs (Lee et al., 1996), In addition to the RLS, the Bushveld event encompasses several satellite intrusions, including the Uitkomst Intrusion, Phalaborwa Carbonatite, the Stavoren Granophyre at the top of the RLS, the massive felsic Rooiberg lava eruption covering 200,000 km3, and the world's largest A-type granitic pluton at the intrusion's roof: the Lebowa granite suite (Hill et al., 1996; Kleeman and Twist, 1989; Scoates et al., 2021; Walraven, 1985).

The mafic plutonic rocks of the RLS are subdivided into the Lower Zone, Critical Zone, Main Zone, and Upper Zone, with unlayered sills and chill sequences at the margin known as the Marginal Zone (Von Gruenewaldt et al., 1985). These zones are primarily distinguished by their mineralogy, which transitions from more primitive assemblages in the Lower Zone to more evolved ones in the Upper Zone. The Lower Zone uniquely features dunites and harzburgites, the Critical Zone features chromitites and norites, the Main Zone features Gabbronorites, and lastly the Upper Zone features diorites and magnetitites. The RLS exhibits significant compositional zoning, with increasing iron, silica, vanadium, and alkali content as well as chromium and

magnesium depletion toward the intrusion's roof (Cawthorn, 2015). While the RLS often maintains remarkable lateral continuity on the kilometer scale, variations in stratigraphy, PGE mineralization, and thickness sometimes present challenges to stratigraphic correlations (Ashwal et al., 2005; McDonald et al., 2005; Scoon and Tiegler, 1994).

Radiogenic Isotope Stratigraphy of the RLS

Kruger (1994) reported a comprehensive strontium isotope stratigraphy of the RLS, revealing radiogenic Sr isotopes indicative of varying degrees of crustal contamination throughout the intrusion. This stratigraphy differentiates the Lower, Critical, and lower Main zones from the upper Main zone and Upper zone. The lower portion of the intrusion records highly variable Sr isotopes, interpreted to arise due to repeated magma replenishments in an open system. Kruger (1994) refers to this zone as the "Integration Stage." Conversely, the middle Main Zone displays fairly consistent isotopic compositions, transitioning to more primitive values at the pyroxenite marker—a well-known bed in the upper Main Zone signifying a shift towards more primitive magmas (Cawthorn et al., 1991). These two stratigraphic regions with internally constant initial strontium isotope ratios are referred to as the “Differentiation Stage” and could possibly have evolved through closed-system fractional crystallization on the basis of strontium isotopes. Studies on neodymium isotopes within the RLS are less detailed than those available for strontium isotopes, but show a broadly similar pattern that requires crustal contamination throughout the entire complex, with the Main and Upper Zones requiring more radiogenic (crustal?) contamination than the Lower and Critical Zones (Maier et al., 2000).

Strontium isotope disequilibrium is well-known within the Merensky and Bastard reefs, where pyroxenes and plagioclase have different and variable initial strontium isotope values (Seabrook et al., 2005; Cawthorn, 2015; however, strontium isotope disequilibrium is not unique to the Merensky and Bastard reefs. Chutas et al. (2012) demonstrated with plagioclase throughout the Lower and Critical were isotopically zoned with primitive cores and evolved rims, and that coexisting pyroxenes and plagioclase were not in isotopic equilibrium. Roelofse and Ashwal (2012) similarly observed significant strontium isotopic disequilibrium in lower Main Zone rocks. This means that crystalline phases

from the entire Lower, Critical, and lower Main zone cumulates interacted with exotic magmas or hydrothermal fluids after substantial crystallization, either by later intrusion of an exotic magma into a preexisting crystal mush at the level of the RLS (Chutas et al., 2012) or by intrusion of xenocryst-bearing crystal rich magmas derived from a subcompartmentalized chamber at depth (Roelofse and Ashwal, 2012). Any rock in the Critical, Lower, and lower Main zones was thus formed by multiple magmatic ± hydrothermal events/environments, and any bulk isotopic or chemical measurement of these zones will inherently integrate over multiple crystallization episodes.

A comprehensive dataset of rhenium-osmium (Re-Os) isotopes provides valuable insights into the history of platinum group elements (PGEs) within the Bushveld Complex. Schoenberg et al. (1999) measured a 2.043 ± 0.011 Ga isochron age for the Bastard Reef, which contained little common osmium. Furthermore, they report a systematic trend in the initial $^{187}Os/^{188}Os$ ratios within chromitites spanning the Critical Zone. These ratios exhibit a distinct progression, starting from values closely resembling chondritic compositions and escalating towards highly radiogenic values as one moves upwards through the stratigraphic sequence reaching their most radiogenic values at the Merensky Reef and upper group (UG) chromitites (Schoenberg et al., 1999; McCandless et al., 1999).

The presence of heterogeneous initial $^{187}Os/^{188}Os$ values within chromitites is commonly invoked to support the argument for an orthomagmatic origin for PGE mineralization in the upper Critical Zone, arguing against a hydrothermal origin (Schoenberg et al., 1999; McCandless et al., 1999). However, the validity of the argument depends upon whether Re-Os isotopes faithfully record magmatic processes. Notably, experimental evidence has demonstrated that osmium diffuses rapidly within sulfides, displaying a closure temperature similar to the closure temperature of argon within biotite (Brenan et al., 2000). Ar-Ar thermochronology of biotite therefore can be used in this case as a proxy to determine the timing of the closure of the Re-Os system. Nomade et al. (2004) reported an average Ar-Ar age of 2.042 ± 0.003 Ga for biotite within the UG2 chromite, which agrees well with the Re-Os isochron age of the Bastard Reef. Furthermore, this congruence extends beyond the Bushveld Complex, as similar Ar-Ar ages are observed on a regional scale within syn-kinematic micas found within low-grade phyllitic metasediments of the Transvaal Supergroup (Alexandre et al., 2006). These Ar-Ar ages effectively date regional compression

within the Kaapvaal craton, including deformation into fold and thrust belts (Alexandre et al., 2006). Consequently, it can be inferred that the Re-Os isotope system underwent diffusive reequilibration within the chromitites approximately 10 million years after the cessation of magmatism within the RLS, which ended at 2.055 Ga (age derived from Scoates et al., 2021; Zeh et al., 2015).

The most enigmatic aspect of the Re-Os isotope geochemistry of the RLS centers around the coexistence of two unusual, isotopically distinct sulfide populations within the Merensky Reef. A substantial population of laurite (RuS_2) exhibits the most radiogenic initial $^{187}Os/^{188}Os$ values of the entire 8 kilometer stratigraphy of the RLS, while the same hand sample contains erlichmanite (OsS_2) with an osmium isotopic ratio mirroring a chondritic value, the least radiogenic $^{187}Os/^{188}Os$ within the entire RLS (Hart and Kinloch, 1989). It is difficult to imagine how sulfides with such different $^{187}Os/^{188}Os$ values were able to coexist within the same rock without isotopically equilibrating.

In a broader geological context, Richardson and Shirey (2008) conducted measurements of sulfide Re-Os isotopes, as well as silicate Rb-Sr and Sm-Nd isotopes, within kimberlitic diamond inclusions in close proximity to the Bushveld complex. Their aim was to constrain the isotopic reservoirs within the subcontinental lithospheric mantle (SCLM) beneath the Bushveld complex. Their findings reveal the enrichment of radiogenic isotopes across all three isotopic systems within the SCLM, which they use to argue that the radiogenic isotopic signatures observed within the RLS can be explained as a three-part mixture of mantle reservoirs that requires no contributions from the crust, instead requiring only the harzburgitic SCLM, eclogitic SCLM, and the depleted, convecting mantle.

Stable Isotopes of the RLS

Oxygen isotopic studies show that oxygen within the RLS was equilibrated at magmatic temperatures, remarkably unaffected by interactions with surface-derived fluids (Schiffries and Rye, 1989; Harris et al., 2005). Early studies indicated magmatic temperatures for the Upper Zone equilibrating at 1000°C, with the rest of the intrusion at 1200°C (Schiffries and Rye, 1989). However, later studies produced scattered equilibration temperatures as low as 550°C (Harris et al., 2005). Oxygen isotopes within the RLS exhibit remarkable consistency throughout the intrusion and are slightly heavier (~8‰ $\delta^{18}O$)

than mantle values (~6‰ δ^{18}O) suggesting substantial but well-mixed contamination from surface-derived material, possibly from Transvaal supergroup rocks exposed on the surface or from unexposed, isotopically heavy rocks at depth (Schiffries and Rye, 1989; Harris et al., 2005). Similarly, multiple-sulfur isotopes identified a well-mixed and uniform non-zero mass independent fractionation (MIF) of Δ^{33}S (Penniston-Dorland et al., 2012) across the RLS, requiring a component of surface derived sulfur that was very efficiently mixed throughout the entirety of the RLS. The remarkable homogeneity in volatile element isotopes throughout the RLS contrast with heterogeneous strontium and neodymium isotopes throughout the intrusion, alluding to a decoupling of magmatic and hydrothermal material sources within the RLS. The volatile element isotopic data require a single source for late-stage hydrothermal fluids across the complex.

Hydrogen isotopes indicate late-stage hydrothermal metasomatism at low water/rock ratios (Schiffries and Rye, 1990). Carbonated fluid within the RLS imparted a carbon isotope signature likely arising from the devolatilization of Transvaal carbonates (Schiffries and Rye, 1990). Sedimentary country rocks display oxygen and carbon isotopic depletions correlating with increasing metamorphic grade and proximity to the RLS margins (Buick et al., 2000; Schiffries and Rye, 1990). The lack of complete isotopic equilibration of metasedimentary country rocks and RLS fluids suggests that fluids flowed from devolatilized metasedimentary rocks into the RLS, and not vice versa (Buick et al., 2000).

Unusual Volatile Phases in the Lower RLS

Within the Lower and Critical zones of the RLS, volatile-bearing phases such as apatite, amphibole, and phlogopite are rare but often contain exceptionally high chlorine concentrations (Boudreau et al., 1986; Willmore et al., 2000). These phases are most concentrated in the lateral Merensky reef and the vertical metasomatic Driekop dunite pipe, often associated with magmatic graphite and platiniferous sulfides (Ballhaus and Stumpfl, 1985; Schiffries, 1982). Thermodynamic calculations suggest that the initial volatile phase exsolved from RLS magmas was predominantly hydrochloric acid (HCl) and chloromethane (CH3Cl), with minor amounts of water and sulfur (Mathez et al., 1989). This fluid likely exsolved sulfur during graphite crystallization.

Chlorine isotopic studies of the RLS show that these Cl-rich fluids were not derived from the local Transvaal Supergroup, but are consistent with an origin from SCLM charged with volatiles from dehydrating subducting altered oceanic crust (Willmore et al., 2002). The association of these C-Cl rich volatile phases with platiniferous horizons has led many authors to suggest that PGE mineralization is controlled by C-Cl rich fluids, perhaps by chromatographic extraction by reactive porous flow of C-Cl rich fluids within the existing Critical Zone cumulates (Boudreau and Meurer, 1999).

Petrological Debates Within the RLS
Open or Closed System

Another quintessential LMI, the Skaergaard intrusion, is thought to have formed primarily through closed-system fractional crystallization from the chamber's outer regions inward (McBirney and Noyes, 1979), albeit with strontium isotopic evidence of country rock assimilation in its early stages (Steward and DePaolo, 1990). In contrast, the RLS is interpreted as having crystallized mainly in an open system, featuring frequent magmatic recharge. This is supported by numerous chemical reversals in plagioclase and pyroxene compositions, as well as variations in initial strontium isotopes within the Lower, Critical, and lower Main zones (Kruger, 1994; Cawthorn, 2015). However, the upper Main and Upper Zones (sometimes lumped together as the UUMZ) of the RLS are often seen as products of closed-system fractional crystallization, as indicated by uniform strontium isotopes (Kruger, 1994), oxygen isotopes that seem to have evolved through closed-system fractional crystallization (Bucholz et al., 2017), and well-fitting crystallization sequences produced by MELTS simulations of closed-system fractional crystallization of plausible UUMZ parental compositions (Vantongeren et al., 2010). Nevertheless, this view has been challenged by the identification of significant reversals in plagioclase and noncomformable field relationships within the UUMZ (Yuan et al., 2017; Scoon and Mitchell, 2012).

Spatial and Temporal Scales of Magma Chambers in the Crust, Pluton Assembly Mechanisms

The RLS, along with other LMIs, has become a battleground for debates regarding the temporal and spatial scales of crustal magma chambers and pluton

assembly. Some argue that plutonic rocks represent fossilized magma chambers (Kruger and Latypov, 2020; Bachmann and Bergantz, 2004; Paterson, 2009), while others propose that these rocks are assembled incrementally through sill-like intrusions. Such arguments are applied to rocks within the RLS by Mitchell et al. (2019), Yao et al. (2021), Scoon and Mitchell (2023), and Bourdeau et al. (2023). The latter viewpoint first gained significant support when detailed, high-precision zircon U/Pb geochronology studies revealed wide ranges of crystallization ages within seemingly homogeneous, continuous plutons (Coleman et al., 2004; Glazner et al., 2004). These wide age ranges produced by zircon U/Pb geochronology in plutons greatly exceed the timeframes over which large magma chambers can be thermally sustained before cooling (Annen, 2009; Cawthorn and Walraven, 1998).

Zircons from the RLS and Stillwater layered mafic intrusions similarly yield some out-of-sequence U/Pb ages that challenge the idea that they represent large, fossilized magma chambers (Mungall et al., 2016; Wall et al., 2018; Scoates et al., 2021). However, resolvably out-of-sequence zircon are somewhat rare and were not detected in a slightly earlier study (Zeh et al., 2015). Additionally, some authors suggest these data are impossible to reconcile with some field relationships, challenging the reliability of U/Pb zircon geochronology for investigating layered mafic intrusions (Latypov and Christyakova, 2022). Nevertheless, U/Pb zircon geochronology successfully yields geologically expected narrow age ranges for other igneous rocks when geologically expected, such as for erupted ignimbrites (Wotzlaw et al., 2013).

Origin of Enigmatic Monomineralic Igneous Rocks

The RLS contains enigmatic monomineralic igneous rocks that continue to perplex igneous petrologists. For instance, the RLS contains dozens of centimeter to meter scale, near-pure chromitites which are challenging to explain. One common explanation involves the mixing of primitive and evolved magmas inducing chromite saturation (Irvine, 1977). However, an in-situ crystallization hypothesis is problematic because the Lower and Critical Zones contain significantly more chromium than could have been delivered by a plausible amount of chromium-saturated silicate melt (Eales, 2000), and the cumulate rocks above and below some chromitites are petrologically identical (Mondal and Mathez, 2007), seeming to preclude in-situ magma mixing unless over 300,000

km3 differentiated mafic lavas were lost from the chamber's top (Cawthorn and Walraven, 1998). The most popular alternative explanation posits that chromite crystallized within staging chambers and was injected into the RLS as chromite-phyric silicate magmas (Mondal and Mathez, 2007; Eales, 2000). Experiments have shown that chromitites can form through magmatic sedimentation from a chromite-bearing magma within realistic timescales (Manoochehri and Schmidt, 2014).

Stratiform anorthosites within the RLS pose another challenge to conventional explanations because plausible RLS magma compositions are rarely, if ever, saturated in plagioclase alone (Latypov et al., 2020; Maier et al., 2020). Various hypotheses propose that these anorthosites resulted from [1] density driven mechanical sorting a magma concomitantly saturated in multiple phases (Cawthorn and Ashwal, 2009;); [2] depressurization forced saturation of plagioclase alone, followed by crystal settling (Yuan et al., 2017) or in-situ growth on the magma chamber floor (Latypov et al., 2020); [3] tectonically-induced mechanical sorting of a preexisting cumulate pile (Maier et al., 2016); [4] reactive porous flow of acidic fluids (Maier et al., 2020; Sonnethal, 1992); or [5] restites from partial melting at the contact of new intrusions of magma (Mitchell et al., 2019).

The Origin of the Merensky Reef and PGE Mineralization

The origin of the Merensky Reef and PGE mineralization in the upper Critical Zone remains contentious. One theory suggests that immiscible sulfide fluids scavenged PGEs from a large magma chamber (Campbell et al., 1983). Alternatively, percolating fluids rich in chlorine, carbon, and sulfur may have redistributed PGE elements within an already-solidified critical zone (Boudreau et al., 1986; Ballhaus and Stumpfl, 1985), or magmatic differentiation controlled by chromite may have concentrated PGEs within this zone (Scoon and Tiegler, 1994). The platiniferous Merensky reef, a two-part pyroxenite and pegmatoidal horizon, is debated to be either a composite of multiple synmagmatic or out-of-sequence sills (Mitchell and Scoon, 2007; Mitchell et al., 2019) or a mixture of disequilibrated crystals precipitating synchronously from overlying density-stratified Main and Critical zone magmas (Seabrook et al., 2005). Most authors agree that the Merensky reef represents a mixture of Main Zone and Critical Zone magmas (Kruger, 1994; Sharpe, 1985; Cawthorn, 2015).

Structure and Relationship to Country Rocks

The layered series outcrops as distinct lobes on the present-day surface, with variations in stratigraphy, PGE mineralization, and mineralogy observed across limbs. These lobes are separated and influenced by geological features like the Thabazimbi-Murchison and Steelpoort lineaments (Du Plessis and Walraven, 1990; Ashwal et al., 2005; McDonald et al., 2005; Scoon and Tiegler, 1994). Despite these variations, there is remarkable continuity between layers. Chromitite layers with similar thickness and chemical signatures can be correlated across limbs separated by hundreds of kilometers of felsic igneous rocks and sedimentary cover. Additionally, crustal thickness models derived from seismic receiver functions and gravity anomalies support the possibility that the intrusion is connected at depth (Webb and Cawthorn, 2004; Cawthorn and Webb, 2001). Some researchers have further classified the intrusion into "distal" and "proximal" facies based on the hypothesis that the intrusion's feeders exploited the Thabazimbi-Murchison Lineament, supported by observations such as the best development of ultramafic zones closest to this lineament (Scoon and Tiegler, 1994; Clarke et al., 2000).

The RLS generally dips inward toward its center, with the most significant dips occurring at the margin (Button, 1976; Frick, 1967). While the RLS was historically considered a lopolith (Wilson, 1956), paleomagnetic studies favor an initially horizontal orientation that flexed isostatically after crystallization ceased (Hattingh, 1986). The floor of the RLS exhibits synmagmatic deformation, possibly resulting from interactions during the coalescence of offset interfingered sills (Pollard et al., 1975; Clarke et al., 2009). Abundant large sedimentary inliers within the complex can also be explained by the inhomogeneous propagation of finger sills along the complex's margin (Clarke et al., 2000; Clarke et al., 2009).

The Transformation Hypothesis

Van Biljon (1949) developed a controversial hypothesis for the origin of the RLS: that it represented an extreme case of in-situ metasomatism of the upper Pretoria Group sedimentary rocks of the Transvaal Supergroup on the basis of field relationships and geochemical arguments. Indeed, terminology and concepts from sedimentology and stratigraphy is often (though arguably erroneously) employed to describe features within the RLS and other LMIs.

Van Biljon's Transformation Hypothesis sought to explain the origin of the enormously laterally continuous markers within the RLS, the apparent problem of an enormous amount of displaced sedimentary rocks, and the transgressive nature of the RLS's contact with the Pretoria Group. Van Biljon (1949) argues that chert is transformed to anorthosites, carbonates are transformed to magnetitites/chromitites, variably impure quartzites are transformed to (gabbro)norties, and shales are transformed to pyroxenites. He presents geochemical evidence that seems to show a chemical continuum in different oxide spaces between unmetamorphosed sedimentary rocks, variably hornfelsized sedimentary rocks, and mafic rocks of the RLS.

Button (1976) emphasized Van Biljon's recognition of the "space problem" introduced by the transgressive RLS into relatively undeformed sedimentary rocks, instead arguing that the felsic Bushveld rocks represented the displaced (and transformed) sediments. However, Cawthorn (1998) showed through a simple geometric analysis that the apparently missing sedimentary rocks may have been preserved underneath the RLS given the nature of its transgressive and discordant contact with the Pretoria Group.

One simple counterpoint to Van Biljon's transformation hypothesis for the RLS is that similar LMI's exist that are emplaced entirely or nearly entirely within non-sedimentary granitic, basaltic, and gneissic country rock, such as the chromitite-bearing Great Dyke of Zimbabwe (Worst, 1958) and the finely-layered Skaergaard Intrusion (McBirney, 1996). Furthermore, the RLS itself onlaps granite in the North Limb (Van Der Merwe, 1976).

Generally, available stable and radiogenic isotopic data require whole-intrusion-scale magmatic and hydrothermal processes to be the dominant processes forming the RLS, and for local contamination with crustal rocks to play a subordinate role (Zirakparvar et al., 2014; Schiffries and Rye, 1989 and 1990; Willmore et al., 2002; Penniston-Dorland et al., 2012; Kruger, 1994; Richardson and Shirey, 2008). This general statement has notable exceptions: 1) The Marginal Zone contains abundant metapelitic migmatites and calc-silicates (Johnson et al., 2003; Van Biljon, 1949; Button, 1976; Frick, 1967; this work). 2) Interstitial phases such as anhydrite, almost certainly derived from sedimentary rocks, is documented within the ultramafic Lower Zone (Wilson et al., 2017; Yudovskaya et al., 2021), and 3) textural, geochemical, and isotopic data on some noritic sills within the Lower Zone strongly suggest an origin as (partially)

melted sedimentary Pretoria Group protoliths.

These observations together suggest that in-situ igneous sediment transformation is not an essential or dominant process in the formation of the RLS or other LMIs, but that it may still occur at small scales. Evidence for this process would become obscured and eventually destroyed by episodic melting and/or reaction with an abundance of incoming magmas and hydrothermal fluids, suggesting that this process could also have been locally important at higher levels within the RLS even when evidence was subtle or completely lost to igneous overprinting.

Previous work near Kliprivier and Draaikraal

Kliprivier was the focus of several early works focusing on the margin of the RLS to better understand the relationship between the RLS and the sedimentary country rock. Van Biljon (1949) studied the hornfelsized carbonates of the area and their relationship to the margin, observing an increase in metamorphic grade to wollastonite skarns near the margin of the RLS and reporting that the transgressive RLS mafic rocks are continuous along strike with the metasedimentary rocks, a finding that I also report here (see the Discussion below for elaboration). He used this and similar pieces of field and geochemical evidence to develop his provocative but unpopular Transformation Hypothesis: claiming that the RLS (and potentially other igneous rocks) were formed from in-situ metasomatic replacement of sediment (see critique and discussion within the comparatively well-cited work of Cousins 1959).

Kliprivier and the surrounding De Berg region was studied in detail by Frick (1967), who distinguished different facies of marginal sills, documented variable assimilation of country rock by marginal sills and the presence of Critical Zone and two chromitite seams in the area, and characterized contact metamorphism within the sedimentary rocks. Frick (1967) emphasized the discordance of the contact between the sediments and the RLS and reports that chromite seams are disaggregated at the contact and not present in the Marginal Zone. Button (1976) reports that the extent of the metamorphic thermal aureole decreases southward in the area as the RLS package thins.

In their influential paper geochemically categorizing the Marginal Zone sills, Harmer and Sharpe (1985) included a broad map of the Marginal Zone along the Bushveld's Eastern limb in addition to a detailed map of the marginal

rocks at the boundary of the Main and Critical Zones at the Kliprivier synform, described more in the "Structural Observations" section. These maps show that the field area at Draaikraal includes two distinct classes of Marginal Zone sills. The large map shows "B2" gabbroic sills to the North abutting the Critical Zone and "B3" gabbroic sills to the South abutting the Main Zone (although later works argue that the upper Critical Zone continues south within and beyond the field area: Van Der Merwe, 2007; Latypov et al., 2022). Furthermore, fine gabbros of the "B2" and "B3" lineages are also subdivided into "B2N" and "B3N". Because both "B2N" and "B3N" sills are magnetic and visually identical in outcrop they are both described here with the lithologic code "Rz", for fine-grained Marginal Zone. Their coarse-grained "B2" gabbros may (in part) correspond to the Disturbed Gabbronorite (Dcg) lithology mapped in this work.

Van De Merwe (2007) mapped and described the geology on Kliprivier and several nearby farms along the eastern Lobe of the RLS to the end of assessing the area's PGE resource potential. He first described the Kliprivier "Syncline", further detailed in the structural observations, unit descriptions, and discussion in this work. Within, he references several private studies which were contracted by mining companies that are not publically available. Van De Merwe reported a core log from the area that is helpful to interpret field data.

Latypov et al. (2022) did a systematic but coarse field study of seven separate traverses across ~60 kilometers of the lower Main Zone in order to correlate the mottled anorthosites of the lower Main Zone, arguing that these correlated units, supposedly laterally continuous across a 4 km change in depth of the RLS floor, imply the existence of a kilometer-scale melt column. They argue their correlation requires a voluminous ~10^5 km^3 magma chamber during the emplacement of the Main Zone.

Setera and VanTongeren (2018) conducted a systematic study of the Main Zone in the Draaikraal area, reporting major and trace elements of major cumulus phases. They focused on characterizing magma recharge at the level of the Pyroxenite Marker in the upper Main Zone, but they also used chromium concentrations in pyroxene to identify a less significant magma influx at the level above the mottled anorthosites. This geochemical cross-section was compared with a similar cross-section to the north performed in an earlier work by Vantongeren and Mathez (2013), which notably featured more primitive

Main Zone rocks than those observed near Draaikraal. Setera and VanTongeren (2018) argue that this southerly evolving compositional gradient within the Main Zone arose from evolution of migrating melts away from the feeder of the Bushveld, interpreted to be at the Thabazimbi-Murchison Lineament.

Topographic 'Inversion' and the Draaikraal Lineament

The agricultural village of Draaikraal in the Mpumalanga Province of South Africa sits in a local sedimentary basin. Geologically, it is situated within the lower Main Zone of the RLS. The Main Zone is known to be highly resistant to erosion and dominates surface exposure of the RLS in the comparatively poorly exposed North and West Limbs of the RLS. This is thought to be because of the common observation of inversion of small pigeonite grains into larger, continuous OPX grains, serving as a sort of resistant "cement", contributing to the effectiveness of Main Zone rocks as dimension stone commonly used as a building material (Cawthorn, 2015). Figure 1a shows that this considerable change in topography at Draaikraal from the northern mountains to the southern valley occurs despite the continuity of the Main Zone through the lineament.

Figure 1a: Geology and Topography of Draaikraal. Yellow represents the Main Zone, Oranges are Quartzites and Hornfels, beige is Marginal Zone, and Green is Critical Zone. The Critical Zone is readily eroded and represents the lowest elevations within the mapped area. The Main Zone forms high mountains to the North and West, but is topographically recessive near Draaikraal. The Northern mountains are separated from the recessive valley to the South by a linear feature (the Draaikraal Lineament) marked by a dashed line that does not appear to offset high ridges defined by resistant strata highlighted by the gray box. The geology was compiled by Molyneux et al. (2008), who did not consider more recent mapping that revealed the presence of the Upper Critical Zone to the South, terminating beyond the map area (Van de Merwe, 2007).

5.4 Field Observations

Lithologic Descriptions

Eastern Margin

Much of this study focused on the eastern margin of the Eastern limb of the Rustenburg Layered Series (RLS) east of Draaikraal. The margin displays the

most unambiguous bedding planes of the study area, generally dipping 25° to the northwest towards the center of the Bushveld Complex, except for the Kliprivier Synform, which is detailed below in the "Structural Observations" section. The marginal rocks within the field area are composed of gabbroic intrusive igneous sills concordantly intercalated with variably metamorphosed, melted, or otherwise devolatilized Transvaal sediments of the Steenkampsberg and Houtenbek formations. Occasionally, melting and/or reactive porous flow of melts through sedimentary layers remineralizes sedimentary layers with igneous mineralogy and texture (photos 35, 51, and 52). Concordant sills of marginal zone magmas and packages of sediment range in thickness from 10 to 100s of meters thick, and can be found mutually interbedded at least a kilometer from the margin of the RLS. Here, the marginal rocks are divided into three units: "Rz", the fine-grained Marginal Zone gabbroic sills/magmas; "Qz", relatively undisturbed Houtenbek quartzite; and "ScP", hornfelsed metapelites, marbles, and quartzites with related igneous/hydrothermal pegmatites.

Rz – Fine grained gabbro The Marginal Zone gabbroic sills are commonly fine grained (<1 mm), but locally are coarser with some grains up to 2 mm in the groundmass. Unlike much of the plutonic RLS, the Marginal Zone is not visibly modally layered. Marginal Zone magmas within a meter of Steenkampsberg Quartzite often contain conspicuous evidence of silicic contamination: although they retain an igneous texture, the color significantly lightens and the rocks become much harder and less magnetic approaching the contacts with Steenkampsberg Quartzite. Aside from these immediate contacts, the Marginal Zone gabbro is always magnetic. The Marginal Zone contains more pyroxene (distinguishing clinopyroxene and orthopyroxene is difficult with such small crystals) than plagioclase with variable magnetite, sometimes up to 10%, and minor phlogopite. The heavy, fine fraction of mineral separates from one sample was briefly studied under optical microscope to identify trace phases, which include apatite, zircon, sulfides, and euhedral chromite.

The Marginal Zone magmas contain igneous xenoliths and are rich in paleo-vertical, across-strike features that likely record rising, buoyant fluid/magma. Perfect planar resistance banding (photo 95) is the most common structure in the fine grained Marginal Zone and is nearly ubiquitous within the unit. Though the resistance bands have seemingly random strikes, they most often are steeply dipping and may represent fluid conduits along brittle fractures

within the Marginal Zone.

In previous works, isolated meta-sedimentary xenoliths (Buick et al., 2000) were emphasized, but in this study, meta-sedimentary rocks were never found as minor xenoliths within magnetic marginal sills, but rather as continuous, concordant strata (photos 42, 70, 71) or as locally abundant disaggregated broken strata cemented within a pink, evolved diatexite (photo 100), both designated as a distinct unit: ScP.

The magnetic sills of the Marginal Zone (Rz) did include enclaves of coarse-grained leucogabbronorite, mottled anorthosite or pyroxenite (photos 2,4,6,96). The anorthosite-rich enclaves are rounded, 6-10 cm along their longest dimension, and are imbricated to be preferentially lying flat paleohorizontally within the Marginal Zone, suggesting they were weathered/eroded by reaction and/or mechanical transport through Marginal Zone magma. Pyroxenitic enclaves are also often spherical (photos 4, 96) and may be related to the formation of the pyroxenite "Tennis Ball Marker" (TBM) of the Main and Critical Zones, described further in the discussion. These plagioclase-rich igneous xenoliths contain large (<3 cm) optically continuous oikocrystic augite that enclose blocky plagioclase (<5 mm).

More commonly observed within the magnetic Marginal Zone sills east of Draaikraal are paleovertical structures that suggest buoyant upwelling of fluids and melts sourced from the hornfelsized sedimentary rocks. These structures included sheared, visco-elastically deformed lenses of thin pegmatites, including anorthosite (photo 6), magnetite gabbros (photo 60, sheared gabbro ~ 10 x 3 cm), and evolved quartz-biotite pegmatites (photo 67: 1x3 cm). In addition to homogeneous pegmatites, the Marginal Zone gabbros host paleovertical invasions of anastomosing anorthosite derived from anatexis of underlying metasedimentary rocks (photos 6, 97). These leucosome intrusions occasionally carry blocks of meta-sedimentary paleosome or brecciate and enclose the Marginal Zone gabbronoritic host rock itself (photo 98).

Near the contact with the migmatite, differential weathering sometimes reveals proud, thin, pinched ovate paleovertical structures (5x1 cm scale) on weathered faces of Marginal Zone norite that are otherwise invisible on a fresh rock face [photo 40].

The contact between the layered series (Critical and Main Zones) and the

Marginal Zone is likely a fault contact at the Kliprivier Synform, but elsewhere appears gradational, requiring a somewhat arbitrary categorization. Since the Lower Main and Upper Critical Zones are non-magnetic and marginal zone rocks are magnetic, I chose field-detectable magnetism as the simple discriminant to distinguish marginal rocks from the massive main and critical zones of the RLS, though these main and critical zone rocks near the margin may inherit some features typical of the marginal zone, such a resistance banding. Fine-grained Marginal Zone gabbros are mapped up to 120 meters below the Merensky Reef stratigraphically.

Qz – Houtenbek Quartzite The Houtenbek quartzite dips concordantly with marginal zone sills and migmatized sedimentary rocks towards the center of the Bushveld, here dipping about 25° to the northwest. Even within meters of a contact with Marginal Zone sills (Rz), the quartzite often retains a clastic sedimentary texture and possibly could be considered a proper clast-dominant quartzose sandstone with well-sorted and well-rounded coarse quartz grains ~1 mm in diameter with no other mineral visible. While there is little structure visible internally because of the purity of the quartzite, the bedding planes occasionally reveal conspicuous asymmetric ripple structures [photos 2, 68] <1 cm in height. One in-place ripple structure suggests a paleoflow direction towards the South-Southwest. Some quartzites are hornfelsized: recrystallized into an interlocking crystalline texture with large, irregular, angular, coarse grains <2 cm across. Many quartzites with preserved clastic textures near the margin of the RLS are stained bright red within rocks' unweathered interior. This staining is hypothesized to be derived from mobilized Fe-rich fluids percolating along grain boundaries, probably caused by heat and/or fluid circulation related to the emplacement of the RLS. Steenkampberg Quartzite is mapped up to 140 meters below the Merensky Reef stratigraphically.

ScP – Houtenbek Hornfels, Migmatites, Pegmatites In addition to quartzite, the Houtenbek formation contains carbonate and pelitic horizons (Button, 1976). At the margin East of Draaikraal, there is a 60 m thick package of melted and metamorphosed sediment containing stratiform pegmatites with variable mineralogy. Hornfelsized sediments come in two main varieties near Draaikraal: pink, diatexite-supported breccias [photo 100] or dark, high-grade schists [photo 29] with thin but conspicuous navy, gray, white, tan, and pink

bands of relatively mafic and felsic minerals have been recrystallized and no longer bear clastic textures except for thin seams of clast-dominated quartzite in a low-fraction glassy biotite-rich matrix [photo 35, large light seam is ~4 cm thick]. This unit is rarely magnetic and most often is visible as continuous, concordant strata [photo 42]. Even when highly brecciated, this unit can be followed along the strike of the quartzites and Marginal Zone, implying that the sediments were not incorporated into Marginal Magmas, but that they rather were melted and devolatilized in-situ due to heating from the emplacement of adjacent marginal sills. Fluids and melts produced in this process are visible as a wide variety of stratiform pegmatites and skarns with undulating erosive contacts within comparatively planar meta-sediments [photo 30: feldspar/wollastonite(?) + Amphibole skarn; photo 69: Magnesite, quartz pegmatite with minor siderite] and as pockets following fluid flow pipes and void spaces that are accommodated by the host strata through ductile deformation [photo 37, 38: Orthopyroxene and quartz pegmatite]. Hornfels in the field area are mapped up to 190 meters below the Merensky Reef stratigraphically.

The Rustenburg Layered Series (RLS)

The Rustenburg Layered Series is the World's largest individual igneous intrusion (Cawthorn, 2015; Latypov, 2022) and is subdivided into the Lower Zone, Critical Zone, Main Zone, and Upper Zone. In this study, RLS rocks of the lower Main zone, Transitional Zone, and upper Critical zone are mapped. The rocks of the Marginal Zone (Rz) are not layered, nor considered here to be a part of the RLS on a textural basis. Marginal zone rocks weather distinctly from the plutonic RLS phase with abundant, sharp, ~3 cm-scale potholes and generally are more textured and rough [photo 36] on a weathered surface than RLS rocks, which weather spheroidally and proudly [photos 8, 11: "porphyritic" gabbronorite]. Xenolith blocks of Marginal Zone norites bearing their distinctive resistance banding are found in both the Critical and the Main zones [Critical Zone: photo 76; Main Zone: 91], though each xenolith sighting within the Main Zone is proximal to a mottled or spotted anorthosite layer. These xenoliths are variably annealed, recrystallized, and even ductily deformed [Critical Zone: photo 77; Main Zone: 84].

The RLS is famous for its distinctive and numerous laterally continuous stratiform marker horizons, sometimes stretching on a 100-km scale (Cawthorn,

2015), which include the Critical Zone chromitite cycles, Merensky and Bastard pegmatoid reefs, three mottled anorthosite packages in the lower Main Zone, the Pyroxenite Marker of the upper Main Zone, and the magnetitite cycles of the Upper Zone. In this study, I observe and describe the three lower Main Zone magnetic mottled anorthosites [photos 49, 53, 86, 93, 105, 106, 108, 111], the magnetic chromitite UG2 [photos 65, 66], and a non-magnetic upper Critical Zone mottled anorthosite that is probably a part of the Merensky Reef footwall [photo 81], and several packages of xenolith-bearing modally layered gabbronoritites and spotted anorthosites within the Main Zone (photos). These laterally continuous reefs are mapped and described separately from the bulk cumulate rocks of the RLS.

Cz: Critical Zone The Critical Zone was previously mapped to pinch out due East of Draaikraal (shown in the compilation of Molyneux et al., 2008), however many now recognize that a thin layer of the upper Critical Zone persists much further south towards Tontledoos (Van De Merwe, 2007). In this study, diagnostic Critical Zone lithologies were only observed north of the Kliprivier syncline, where typical finely modally layered norites and anorthosites are well-exposed (photos 64 and 80), as well as the UG2 chromitite. The Critical Zone above the Kliprivier Syncline dips ~15° to the northwest similarly to the majority of the Main and Marginal Zone rocks in the field area, but gradually rotates to a southwestern dip as it approaches the contact with the rotated block of metasediments and gabbronorite sills of the Kliprivier Synform. It is essential to note that some rotated Critical Zone rocks are structurally beneath the large, rotated package of intruded metasediments (Rz + Qz + ScP) in the area of the Kliprivier Synform, meaning that this package of country rock is not only serving as the floor for RLS rocks to the west, but is also (at least locally) a roof for Critical Zone rocks below the level of the UG2.

Dcg: Disturbed Gabbronorite This lithology is restricted to the immediate area of the Kliprivier synform. Disturbed Gabbronorites are coarse-grained, even pegmatoidal (photo 62), with feldspars and pyroxenes up to 3 cms long. This unit was named in response to Van Der Merwe (2007), who said that the "structural change [of the Klipriver Synform] is reflected in the Bushveld rocks, which become highly disturbed and fragmented". Disturbed Gabbronorites are most recognizable for their abundance of blobby autoliths

of anothosite (photo 61) and pyroxenite (photos 70 and 72). While it exhibits abundant examples of modal separation, they are never remotely planar, unlike the majority of the RLS. I interpret this unit to be formed by synmagmatic deformation of nascent layered upper Critical Zone mushes during the formation of the Kliprivier Synform, effectively breaking up and redistributing the nascent layering into many blobs.

This lithology is most often non-magnetic, but a few specimens were magnetic. Besides the mottled anorthosites of the Main Zone and the anothosite-hosted chromitite leader of the UG2, no other rocks of the RLS within the field area are magnetic. Many but not all Disturbed Gabbronorites are rotated within the Kliprivier synform. Generally, the Disturbed Gabbronorites are CPX-poor and sulfide-rich with abundant, euhedral <1 cm pyroxenes set in an anorthosite matrix that effectively isolates the large, euhedral pyroxenes from one another [photo 62]. This unit generally becomes more feldspathic as one moves away from the intruded metasediments of the Kliprivier Syncline into the Critical Zone. This lithologic unit contains recrystallized and annealed xenolith blocks [photos 74, 75, and 77] of fine-grained Marginal Zone gabbros (Rz) as well as coarse-grained pockets containing very large (< 5cm) oikocrystic pyroxene [photo 73] and large clusters of pyroxene often rimmed by thick anorthosite halos [photo 72]. Disturbed Gabbronorite is mapped between 120-145 meters below the Merensky Reef stratigraphically, around the level of the UG1 in core logs (Van Der Merwe, 2007). Van Der Merwe (2007) and Latypov et al. (2022) report isolated patches of massive chromite here, but I did not see any.

UG2: Upper Group 2 Chromitite Near the top of the upper Critical Zone in the extreme northern portion of Kliprivier farm, there is a small but high-quality exposure of chromitite (>30 cm thick). The bulk of the exposure is a melanocratic, non-magnetic, anorthositic chormite, which is modally ~50% <1mm euhedral chromite set within a fine matrix of <1 mm plagioclase (photo 66). Structurally above this massive layer are some thin (<1 cm thick) chromitite leaders hosted within bright white mottled anorthosite (photo 65). Both layers contain amoeboid oikocrystic pyroxene-olivine mottles (<2cm). The mafic silicate mottles and anorthosite matrix seem to have crystalized within the interstices of a preexisting framework of a continuous chromite crystal slurry [photos 65, 66], (see Voordouw et al., 2009 and Mukherjee et al., 2017 for similar interpretations). The massive melanocratic layer includes

pure (relatively chromite-free) white anorthosite blocks (<4x15 cm) that are lying flat and dipping west towards the center of the Bushveld complex. These pure anorthosite blocks contain thin chromitite stringers that together have been ductilely deformed, and a thin (~1 mm) white seam of pure, apparently undeformed anorthosite forms a plane bounding the top of some of these blocks, also dipping to the west.

The anastamosing chromite leaders within the overlying mottled anorthosite are sometimes highly magnetic and are a much deeper, vitreous black: appearing under hand lens to be >80% modally chromite/magnetite [photo 65].

This chromitite package is interpreted as UG2 for a few reasons. First, although the anastomosing chromite leader resembles the UG1 stringers, the leader is structurally above the massive chromitite, whereas the UG1 stringers are recognized below the massive chromitite (Voordouw et al., 2009 and Mukherjee et al., 2017). The UG2 is known to have several small "leaders" a small distance above the main seam (Maier and Barners, 2024). Second, the magnetic chromitite within the leader bears explaining. One heterogeneous magnetic chromitite at the top of the UG2 with discrete magnetic and non-magnetic bands was described by J.J. Frankel (1942), 20-30 km to the north of Kliprivier at Tweefontein. Third, a rock at this location was also mapped as the UG2 by Harmer and Shaper (1985: Fig. 4) and Van De Merwe (2007: Fig. 3), who also identified the UG2 and closely overlying mottled anorthosites in drill cores of the area using PGE grades. Van Der Merwe (2007) also reports that the UG2 at Kliprivier is unusually enriched in iron, which may explain the magnetism of the leader. The UG2 is 86 meters below the Merensky Reef in the direct vicinity of the Kliprivier Synform (Borehole 3: Van Der Merwe, 2007).

Ln: Layered Norite: Spectacular outcrops of modally layered norites within the upper Critical Zone are observed within the field area at Kliprivier around and above the stratigraphic level of the UG2 chromitite (photos 64 and 80), at stratigraphic heights from 100 to 50 meters below the Merensky Reef. Modal layering is pronounced, with pyroxenite-rich layers being separated by near-pure centimetric anorthosite bands. This layering is undulose and irregular.

Px: Pyroxenite This refers to rocks predominantly composed of pyroxenes. Pyroxenites are rare in the field area. There was one coarse-grained

sulfide-bearing feldspathic pyroxenite horizon in the Critical Zone just below the UG2 (photo 78) as well as a pyroxenite layer just below a mottled anorthosite layer in the UMA in the lower Main Zone (photos 53-54).

Tz: Transitional Zone There is no perfect division for the Main and Critical zones. Moving upsection from the Critical Zone into the Main Zone, Cr2O3 concentration within OPX abundance decreases, Strontium isotopes become more radiogenic, and cumulus clinopyroxene becomes important— but these changes occur at different stratigraphic heights within the transition zone (Cawthorn, 2015) between the Merensky Reef and Giant Mottled Anorthosite (or above). Within the literature, the bottom of the Main Zone is considered to be above the Giant Mottled Anorthosite at the top of the Bastard unit, and the Merensky reef is taken to be the top of the Upper Critical Zone. (Cawthorn, 2015; Mitchell and Manthree, 2002; Van de Merwe, 2007). The Bastard and Merensky pyroxenites and Giant Mottled Anorthosite were never observed in the field area, but a ~50 cm thick mottled anorthosite [photo 81] was observed ~50 meters upsection from the UG2 with only finely layered norite [photo 80] and porphyritic gabbronorite outcropping between the UG2 and the mottled anorthosite. Since Van De Merwe (2007) reports in core logs that the Merensky Reef is 86 meters above the UG2 on Kliprivier and Latypov et al. (2022) observe unusual pyroxenites just above this mottled anorthosite, this anorthosite layer is interpreted to be a part of the footwall to the Merensky Reef, and the pyroxenites described in Latypov et al., 2022 are interpreted to be the Merensky Pyroxenite.

Pg: Porphyritic Gabbronorite The "Pophyritic Norite" is a well-recognized marker of the lower Main Zone, with large, rounded pyroxene phenocrysts set in a white, feldspar-rich matrix, used often in mining to locate the Merensky Reef (SACS, 1980; Von Gruenewaldt et al., 1985). I observe a general textural similarity in lower Main Zone Gabbronorites with the unlayered upper Critical Zone gabbronorites below the Merensky Reef (photos 10, 79). These gabbronorites are distinct from the dark and homogeneous gabbronorites that compose the majority of the Main Zone (Hg). The porphyritic gabbronoite contains large (~< 1cm), equant pyroxenes set in a leuconoritic matrix that is predominantly plagioclase with small spots of interstitial orthopyroxene. I refer here to them collectively as the "porphyritic gabbronorite" because I see

cumulus clinopyroxene present throughout, becoming more modally abundant upsection until about half of the cumulus pyroxenes are CPX near the Lower Mottled Anorthosite (LMA). Generally, cumulus CPX is present in the Main Zone and absent from the Critical Zone (Cawthorn, 2015). However, some cumulus CPX has been reported at the top of the Critical Zone (Cameron, 1982; SACS, 1980).

The rocks of this unit have well developed cleavage [photo 63]; however, they are not visibly layered. I located porphyritic gabbronorite south of the Kliprivier Synform at the beginning of the second field day and assumed it was Main Zone given the cumulus CPX I saw, but now I believe that the porphyritic gabbronorite occurs in the upper Critical Zone through the Transition Zone and into the lower Main Zone in the vicinity of Draaikraal. The Porphyritic Gabbronorite in this field area hosts spherical pyroxenite tennisballs (photos 10, 79), in the same locations as mapped by Harmer and Sharp (1985) and Mitchell and Scoon (2023). Porphyritic Gabbronorite is mapped in the Critical and Main Zones 480 meters below the Merensky Reef to 380 meters above, stratigraphically.

Mz: Main Zone The Main Zone is the most volumetrically significant portion of the RLS: as thick as 5 km (Latypov et al., 2022). The Main Zone stratigraphy is (in general) relatively featureless aside from three prominent mottled anorthosite packages in the lower Main Zone and the Pyroxenite Marker at the upper Main Zone, which was not studied in this work. Mottled anorthosites and associated strongly modally layered sequences are mapped and described here in order to study the effect of the Kliprivier Synform and Draaikraal lineament on the stratigraphy, as these unique layers can be traced great distances through the Lower Main Zone (Latypov et al., 2022).

Hg: Homogeneous Gabbronorite: The massive Homogeneous Gabbronorite of the Main Zone varies slightly in color due to variations in the modal abundance of pyroxenes and plagioclase, though these variations are subtle and continuous on a scale of 10s of meters. There is no obvious fabric visible in outcrops or hand samples of these massive rocks, although I was shown a polished core of upper Main Zone gabbronorite from a dimension stone mine east of Roossenekal that obviously displayed a fabric where crystals preferentially were aligned with the gravitational plane. Stonemasons favor this rock for building,

but are careful to allow only faces cut against this fabric to outwardly show for aesthetics and durability. Homogeneous Gabbronorite (Hg) begins about 450 meters above the Merensky Reef and continues upsection stratigraphically through the Pyroxenite Marker (SACS, 1980; Von Gruenewaldt et al., 1985).

(L/M/U)MA: (Lower/Middle/Upper) Mottled Anorthosites Mottled Anorthosites are enigmatic igneous rocks composed of a nearly pure, white anorthosite matrix containing (often quite large: <5 cm) mossy green "mottles" of oikocrystic pyroxene +/- olivine that enclose abundant plagioclase. In the RLS they are found in large stratiform horizons interpreted to be continuous across the majority of the Main Zone in the Bushveld (>100 of kilometers: Latypov, 2022). The lower Main Zone near Draaikraal contains 3 large mottled anorthosite packages. These mottled anorthosites are magnetic and notably contain rare large (< 2mm) grains of magnetite within the mottles. They are found in association with larger packages of distinct but non-magnetic anomalous lithologies. The associated unusual rocks include: 1) fine green gabbronorites with blebby, anhedral phenocrystic (<2 cm) clinopyroxene (lithologic code Fgog: Fine-grained oikocrystic gabbronorite) associated with the UMA; 2) feature-rich, coarse pyroxenites (lithologic code: Px, [photos 53, 54]) and 3) blobby pyroxenitic gabbronorites (lithologic code Bap: Blobby anorthosite-pyroxenite) that include deformed, irregular, decimeter scale blobby pyroxene agglomerations within an anorthositic matrix that are interpreted here to be recrystallized blocks [photos 104, 108]. Sometimes these blobby anorthosite-pyroxenites exhibit modal layering of coarse (<2 cm) pyroxene oikocrysts [photo 105] within anorthosite. The mottled anorthosites and their associated pyroxenites and blobby gabbronorites have undulating, nonplanar contacts with one another [photo 49] and with the underlying common gabbronorites [photos 107, 103]. Sometimes these contacts may be drastically discordant from the local strike and dip. In photo 106, the mottled anorthosite - gabbronorite contact is dipping ~70° to the East towards the margin of the chamber: a much steeper dip and opposite strike than observed in the leucogabbronorites of the immediate vicinity. Occasional 3D exposure of the anorthosite-gabbronorite contact allowed me to measure the contact rotating 90+ degrees in this outcrop within meters (photo 107). It is unclear if these undulating contacts are evidence for discordance of the mottled anorthosites, since in other places the contact appears concordant [photo 110], and local

apparent discordance could be the result of thermal erosion of the cumulate pile by overlying magmas (Roberts et al., 2007).

It is important to note that one outcrop of mottled anorthosite clearly showed disagreeing cleavage and planar modal layering, with modal layers dipping 50° more eastwardly than the rock cleavage [photo 115, annotated below in figure 1]. This suggests that cleavage and modal layering are formed by distinct processes that may have occurred at different times, with modal layering likely predating the formation of rock-cleavage (see Structural Observations section for more details).

In this study, all three lower Main Zone anorthosites described by Latypov et al. (2022) and mapped by Van Der Merwe (2007) were located within the low-lying valley south of the Draaikraal Lineament at stratigraphic heights between 200 and 600 meters above the level of the Merensky Reef. The Upper and Middle "mottled" anorthosites in the mountains to the north reported by Latypov et al., 2022 were located on this study, but are visually quite distinct and occupy much higher stratigaphic heights (>1000 meters) argued here to be distinct units that stratigraphically overly the thick, massive mottled anorthosites in the valley to the south (see the Discussion of this work). Von Gruenewaldt et al. (1985) report multiple subordinate anorthositic layers near and at the boundary of the Main Zone subzone A and B much higher above the level of the Main Mottled Anorthosite. Those higher anorthosites are referred to here as spotted anorthosites (SA) to reflect their distinct texture. Though this particular contested correlation by Latypov et al. (2022) is of minor significance to the overall argument put forward in their work, it is essential evidence to understand the origin and significance of the Draaikraal Lineament as the surface expression of a synmagmatic monocline formed by laccolithic deformation during the intrusion of much of the Critical Zone.

SA: Spotted Anorthosites North of the Draaikraal Lineament where exposure is better, two packages of anorthosites were found in the lower Main Zone. These anorthosites are somewhat distinct from the common, unlayered gabbronorites of the Main Zone, but anorthositic beds are significantly smaller (<1 m thick) and altogether less visually striking than the Mottled Anorthosites (MA) unit described above. The spotted anorthosites are non-magnetic and bear <1 cm spots of equant pyroxenes composing <20% of the rock by volume

[photos 86, 87] that sometimes contain recrystallized pyroxenite-gabbronorite blocks [photo 111]. The Spotted Anorthosites are strongly associated with much thicker (<10 m) packages of planar, finely layered gabbronorites that are unusual within the Main Zone and are more reminiscent of the Critical Zone [photos 85, 112]. These modally layered sections also include xenoliths [photos 91, 84], which were only observed within the Main Zone in these modally layered packages. These Spotted Anorthosite packages are argued here to be distinct and stratigraphically above the remarkable and prominent Mottled Anorthosites described above. Two or three packages of Spotted Anorthosites (often including thick layers of Layered Gabbronorite) are interpreted here at stratigraphic heights 700-1050 meters above the Merensky Reef. More packages may exist, although they are unlikely to be exposed to the south of the Draaikraal Lineament.

Lg: Layered Gabbronorite Modal layering is generally less common in the Main Zone than the underlying Critical Zone, and in outcrop it is comparatively understated. Whereas the Ln (Layered Norite) unit of the Critical Zone contains vivid, centimeter-scale, near-monomineralic anorthositic bands, the modal layering in the Main Zone gabbronorites is millimeter-scale, represented by subtle differences in cumulus grain size and modal proportions, and the contacts between layers are planar (photos 85 and 112). The Layered Gabbronorites are associated with Mottled Anorthosites (MA) or Spotted Anorthosites (SA). When Layered Gabbronorites (Lg) are observed in the same outcrops as Mottled or Spotted Anorthosites, the locality is assigned to be Mottled or Spotted Anorthosite. Layered Gabbronorites are mapped here at stratigraphic heights between 400 and 1050 feet in association with Mottled and Spotted Anorthosites.

Post Bushveld Rocks: In the valley surrounding Draaikraal, alluvium primarily is composed of well-rounded quartzite cobbles (~10 cm) and ironstone laterites. The ironstone laterites are commonly found as both float in alluvium and as in-place outcrops. Bushveld mafic rocks are rare in alluvium, probably due to their relative susceptibility to chemical weathering on Earth's surface. Quartzite was substantially dominant over RLS mafic rocks in alluvium even 8 km away from the nearest known quartzite source within the RLS, leading Dr. James Roberts and I originally to suspect that there were closer quartzite

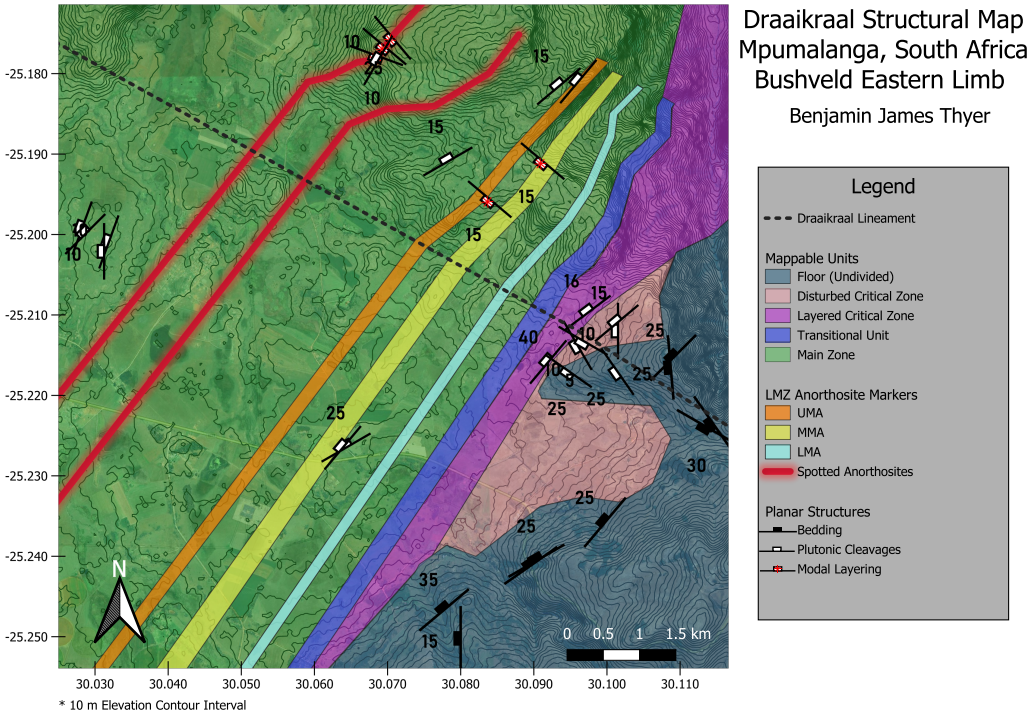

Figure 5.1: Topographic and geological map of the Draaikraal field area, compiled from Molyneux et al. (2008) with the Upper Critical Zone contact to the south added following Van der Merwe (2007). The Draaikraal Lineament (dashed line) separates topographically resistant Main Zone gabbronorites to the north from the recessive Main Zone in the agricultural valley to the south, without offsetting the high ridges of resistant strata (gray box).

[TODO: Insert annotated field Photo 115 here (cleavage and modal layering in a Mottled Anorthosite outcrop) – source image not yet organized into Chapter5/Figures/source/; original reference was to a pandoc-extracted media/image1.jpg that was never carried over with the raw report text.]

Figure 5.2: Cleavage and modal layering forming distinct planes in a Mottled Anorthosite outcrop (annotated Photo 115).

sources; however we were unable to identify quartzite sources west of Draaikraal. It is likely instead that the creeks which source their headwaters from the Houtenbek and Steenkampsberg quartzite highlands to the East and flow past Draaikraal to the West, transported and rounded these cobbles, which are much more stable on the surface than the RLS rocks, eventually coming to dominate the regional alluvium.

IS: Ironstone Laterites Ironstone laterites are common throughout the relatively low and flat valley surrounding Draaikraal. They are somewhat weakly-consolidated and highly porous non-magnetic amalgamations of <1cm iron concretions composed of 1 mm thick deep black vitreous goethite rims surrounding pockets of fine-grained yellow limonite matrix. Interspersed between these concretions are variably rounded medium-coarse Steenkampsberg quartzite gravels. The goethite nodules and quartz gravels are cemented by fine grained, red and yellow hematite-limonite matrix with mm-scale angular, coarse, quartz sand. The existence and abundance of these ironstone laterites implies that the valley surrounding Draaikraal is a longstanding, persistent topographical feature that has existed on geological timescales. They were likely formed by in-place weathering of RLS rocks, explaining the relative dearth of RLS material within the valley alluvium.

Dl: Dolerite Sill: Post-Bushveld Dolerite sills are mapped throughout the field area (Merwe, 2007; Molyneux, 2008), and one such discordant sill was observed at the levels of the UMA and MMA north of the Draaikraal Lineament within the field area (photo 89). Despite being younger than the surrounding RLS, the sill is significantly more weathered, likely because of extensive blocky fracturing hypothesized to result from rapid contraction during cooling. The dolerite is moderately magnetic.

Stratigraphic Column Analytical Methods

The field data of this study north of the Kliprivier Monocline is projected onto a stratigraphic column following the assumption that all rocks in the field area north of the Kliprivier Monocline follow Stenno's Law of Original Horizontality and were only uniformly deformed by tilting due to isostatic flexure towards the center of the RLS after emplacement, so that all regional bedding and structures are currently dipping 15 degrees to the northeast relative

to their original orientations. Geolocated field data is projected onto a line above the Kliprivier Monocline designed to be orthogonal to the regional strike (trending 301.79 degrees: W-NW) using the 'Snap Geometries to Layer' in the 'Vector Geometry' Processing Toolbox in QGIS, with a tolerance of 1 degree set to "prefer closest point". Distances from an arbitrary point along this line stratigraphically below all points were calculated using the "Distance Matrix" function in the "Vector Analysis" Processing Toolbox. A correction was added to these lateral distances to account for elevation, calculated as the product of the elevation differential and the tangent of the regional dip, assumed to be 15°. The column heights are normalized to the height of the UG2, which in turn is normalized to the height of the Merensky Reef, determined to be 86 meters above the UG2 stratigraphically within the Kliprivier Synform (Merwe, 2007). Note that no changes in strike and dip are considered, although the dip of the Main Zone likely shallows and rotates to be roughly due east further from the margin of the intrusion. The stratigraphy calculated here was optimized for discriminating units within the upper Critical and lower Main zones. Therefore, the stratigraphic heights presented here for the Homogeneous Gabbronite lithology are likely overestimated. The interpreted schematic stratigraphic column for the region is graphically presented in figure 5.

Structural Observations

Kliprivier Synform

The most interesting large-scale structure in the mapping area is a rotated package of disturbed, pegmatoidal upper Critical Zone pyroxenites and gabbronorites [photos 61,62, 72-78], well-cleaved leuconorite layers [photo 63], quartzites [photos 28 and 124], migmatized/hornfelsed argillites and calc-silicate layers [photos 29, 30, and 125], and marginal sills [photo 57] in the northern portion of Kliprivier Farm. The structure is remarkable because it apparently displaces a marginal package of metasediments and fine-grained sills over 1.25 kilometers into the plutonic RLS. This structure is described accurately but qualitatively by Van Der Merwe (2007), who investigated the area to assess its platinum resource potential. Van Der Merwe (2007) refers to this feature the "Kliprivier Syncline", though it is not clear if he is naming the structure or it was named elsewhere. A displaced quartzite layer within the structure was also observed during the field mapping of the Main Zone mottled anorthosites by Latypov et al. (2022) and is described in their supplementary

information as a "Large Quartzite Xenolith?".

The structure is referred to here instead as the "Kliprivier Synform" to recognize that it is not yet well understood. For instance, no previous work to date has correlated marginal magmas and country rocks from the two limbs of the putative syncline to demonstrate that they are the same and that the structure is indeed a fold and not another structure, such as a laccolithically deformed roof remnant of an RLS subchamber or a package of floor rocks entrained in a rising magma. The map presented here offers additional details that elucidate the nature of this structure. Below, a series of observations of the Kliprivier Synform is presented:

1. To the north of the structure, there is a continuous, 90° gradual change in strike moving from finely-banded noritic Critical Zone, which strikes southwest and dips northwest like the majority of the mapped lithologies east of Draaikraal, to a southeastern strike dipping southwest. This gradual change in structure is accompanied by lithologic changes in the Critical Zone: Near and at the deformed areas, grain-size increases (photo 61) and stratiform modal layering ceases and is replaced by a heterogeneous rock with abundant fine-grained Marginal Zone xenoliths, anorthosite autoliths (photo 62) and irregular clumps of pyroxene surrounded by anorthosite (photos 72-77). I have mapped these lithologies as "Disturbed Gabbronorite" (Lithologic code: Dcg).

2. To the south, there is little exposure and no structural data could be gathered to pinpoint the reversal of the structure back to the common northwestern dip. However, outcrops constrain the contact between the Marginal Zone magmas and the porphyritic gabbronorite of the upper Critical Zone to occur within 10 meters. The "disturbed" lithologies of the Marginal/Critical zones present on the north side of the synform are not present on the south side. Van De Merwe (2007) interprets faulting with downthrows to the south at this contact; however, Van De Merwe (2007) does not map any vertical displacement of marker beds such as the UG2 chromitite at this fault.

3. Stratigraphically higher plutonic rocks are undeformed, suggesting that they were intruded and crystallized after the deformation forming the Kliprivier Synform. Interestingly, the western termination of the synform

as mapped here occurs at or slightly below the level of the Merensky Reef, where a significant magma recharge event corresponding to the emplacement of the Main Zone is thought to have occurred (Seabrook et al, 2005; Cameron, 1982).

4. Displaced plutonic rocks to the north are stratigraphically below the quartzites and Marginal Zone sills, meaning that the section of sills and sediments serve as a roof to the RLS to north.

5. Within this study, there is one strike and dip measurement immediately to the West of the structure, that is striking SW like the majority of the field area but is dipping more steeply than other rocks in the region. Latypov et al. (2022) also report an unusually steeply dipping plutonic rock along strike.

6. To the east, the structure is attached to the floor of the RLS. Unfortunately, that area was not mapped in detail and the nature of the rotation from the synform's exotic dip to the common dip of the marginal series was not observed.

Strike and dip measurements taken from the Kliprivier Synform were plotted on a stereonet and analyzed using Stereonet 11 (Almendinger et al., 2012, Cardozo and Almendinger, 2013). Poles to bedding planes of quartzites, hornfels, mafic sills, and plutonic rocks were calculated and plotted together in stereonet projection in Figure 3. In Figure 4, the fold axis is calculated using the poles to the bedding planes. This calculated fold axis trends roughly W5°N, plunging 15°, similarly to the Draaikraal Lineament. The axial plane also was calculated, assuming it bisected the obtuse angle formed by the limbs. The obtuse fold angle bisector for the axial plane was selected because the rotation within the fold observed is quite gentle. An acute bisector for the axial plane would've been better supported if field evidence showed dramatic changes in dip along the fold, requiring a complete structural overturn that locally annihilated the original horizontality of the deformed rocks in addition to steeply dipping rocks intermediate to the two limbs. The calculated obtuse bisector axial plane strikes W°5N and dips 85°.

The axial plane is similar in orientation to the WNW-ESE trending Draaikraal Lineament that bounds the topographically high and low Main Zone rocks.

Considering that the Kliprivier Synform also rests upon this lineament, it is likely that the formation of the Kliprivier Synform is related to the Draaikraal Lineament.

[TODO: Insert Figure 3 (Kliprivier Synform structural map/cross-section), Figure 4, and Figure 5 (interpreted schematic stratigraphic column referenced in §5.4) here. Source figures are not yet organized into Chapter5/Figures/source/.]

5.5 Sample and Station Data

The field data underlying §5.4 – GPS-located stations, stratigraphic heights, interpreted units, and lithologic descriptions – are compiled in Table 5.1. Structural measurements (strike, dip, and structure type) recorded at a subset of these and additional dedicated structural stations are compiled separately in Table 5.2. Both tables are generated from `Chapter5/Tables/Draaikraal 2023 BJT.xlsx` by `Chapter5/Figures/write_chapter5_data_table.py`; regenerate after any change to the underlying spreadsheet.

Table 5.1: Field stations from the 'Geotagged Stratigraphy' sheet of the Draaikraal field GPS/GIS compilation (Chapter5/Tables/Draaikraal 2023 BJT.xlsx): station name, stratigraphic height relative to the Merensky Reef (§5.4), latitude/longitude (WGS84), elevation, the final interpreted map unit (lithology codes defined in §5.4), and a merged lithology-description/field-notes column; '–' denotes no description or notes recorded for that station. Structural (strike/dip) readings are tabulated separately in Table 5.2.

Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
RZ22	-932.7	-25.24760	30.11812	2146.3	Rz	–
QZ27	-724.3	-25.24040	30.10576	2076.1	Rz	–
QZ7	-705.6	-25.23020	30.11221	1957.7	Rz	–
RZ54	-699.5	-25.23980	30.10496	2043.7	Rz	–
CONT MARG E Q W?	-697.1	-25.22880	30.11288	1959.5	Rz	Quartzite; Quartzite follows topo line locally here
D3NB	-683.4	-25.22540	30.11481	1959.9	Rz	–
QZ1	-676.4	-25.22530	30.11455	1956.0	Rz	–
D3NA	-674.0	-25.22590	30.11390	1929.4	Rz	compass trouble, resolved
RZ43	-669.3	-25.23920	30.10392	2015.1	Rz	–
D7NC	-661.4	-25.23900	30.10369	2009.6	Rz	Alternating quartzite-marginal zone photo, not very good one
RZ8	-653.7	-25.22500	30.11369	1945.4	Rz	–
D3NC	-648.8	-25.22510	30.11337	1940.4	Rz	No faulting; the quartz and marginal zone is interbedded
RZ21	-644.3	-25.22500	30.11326	1947.1	Rz	–
RZ24	-642.9	-25.23900	30.10279	1996.9	Rz	–
RZ36	-641.9	-25.23860	30.10304	1995.1	Rz	–
QZ112	-638.6	-25.23860	30.10289	1995.3	Rz	–

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
D3SF	-637.8	-25.22610	30.11207	1929.2	Rz	Qz-Px Pegmatite; Hosted in migmatitmatic quartzite. Schist banding deformed around pegmatite
SCHIS	-636.0	-25.22580	30.11224	1937.3	Rz	–
QZ2	-631.5	-25.22510	30.11260	1950.0	Rz	–
QZ3	-631.1	-25.22460	30.11297	1953.6	Rz	–
RZ11	-629.4	-25.22490	30.11266	1952.0	Rz	–
QZ6	-628.6	-25.22520	30.11238	1946.3	Rz	–
QZ5	-625.5	-25.22480	30.11255	1950.0	Rz	–
RZ31	-624.7	-25.22600	30.11153	1925.8	Rz	–
RZ33	-621.8	-25.22460	30.11252	1946.4	Rz	–
D3SE	-621.4	-25.22460	30.11251	1949.9	Rz	Fine grained, light rock. Dacite?; Silicified. Pockmarks weathering proud in paleovertical pipes
D3SD	-619.1	-25.22450	30.11247	1946.4	Rz	Migmatized quartzite; Banded. Light bands of quartz preserve original sedimentary texture. Grains rounded and 2 mm. Some bands pink (feldspar) spme green (chlorite, epidote?)
RZ71	-616.9	-25.22690	30.11044	1910.5	Rz	–
SCHIST	-615.3	-25.22440	30.11236	1943.5	Rz	–
QZ4	-614.7	-25.22430	30.11241	1945.2	Rz	–
RZ41	-614.1	-25.22410	30.11254	1947.5	Rz	–
RZ110	-613.8	-25.23840	30.10183	1976.0	Rz	–
RZ51	-613.7	-25.22290	30.11342	1944.9	Rz	–
RZ61	-599.5	-25.22550	30.11068	1907.3	Rz	–
RZ32	-588.0	-25.22590	30.10982	1899.9	Rz	–
QZ26	-574.9	-25.23760	30.10054	1947.8	Rz	–

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
D7SE	-569.4	-25.23690	30.10081	1946.2	Rz	Silicate Pegmatite; within migmatized qzite very close to magnesite skarn. Mymerkitic quartz-K-spar, huge OPX, small vibrant rare green and blue crystals: apatite, CPX, beryl?
SHIST	-568.7	-25.23690	30.10078	1946.7	Rz	–
RZUP	-565.0	-25.23740	30.10019	1934.8	RzRz	–
D7SD	-558.8	-25.23740	30.09991	1936.3	Rz	Magnesite Skarn; within migmatized qzite. In ground, hard to tell if dike or concordant with schist host
D2NC	-555.1	-25.22110	30.11222	1993.1	Rz	Contact; On this ridge, there are multiple bands of quartzites with Marginal Zone between them. Is the marginal zone a sill between layers of quartzite, or is there a fault? (Update, I think it's a strike-slip(?) fault striking E-W dipping to the South with a component of left-lateral shear along the ridge)
D7NA	-551.9	-25.23700	30.09985	1923.7	Rz	Fine marginal gabbronorite with shearing pegmatite veins; sheared pegmatite veins in pockets/pipes orthogonal to bedding, paleoup. This layer is just above migmatized quartzite and just below fine magnetic marginal zone. Represents fluid transport from migmatized sedimentary rocks, origin of resistance banding? Strike and Dip are for migmatized Qz
SCH1	-551.2	-25.23700	30.09984	1930.2	Rz	–

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
RZMAG	-549.3	-25.23690	30.09979	1919.2	Rz	–
RZ	-532.3	-25.22390	30.10878	1908.6	Rz	–
D1ND	-506.5	-25.23730	30.09753	1926.2	Rz	Bladed iridescent white crystals (Plagioclase) in 2 cm thick planar features intersect orthogonally in outcrop. The plagioclase planes are surrounded by 10 cm zone of oikocrystic augite. The southwestern limb pinches in and out and occasionally overlaps itself in parallel, a la boudinage.
D1NC	-500.0	-25.23720	30.09730	1924.3	Rz	Note that all marginal zone rocks are characterized by dark resistance bands, perhaps cataclastic, reflecting warm-brittle deformation with finely recrystallized zones. They stand proud. They frequently strike E-W, secondarily, NE-SW, tertiarily. NW-SE, but can vary and appear pretty random. This feature is diagnostic of the marginal zone.
RZ3	-485.0	-25.22250	30.10765	1910.3	Rz	–
RLS	-481.1	-25.22350	30.10665	1889.9	Cz	–
RZ1	-480.6	-25.22030	30.10931	1973.2	Rz	–
D1Sb	-478.9	-25.23690	30.09644	1890.9	Rz	Common Marginal zone Medium-fine grained gabbronorite

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
D1A	-478.9	-25.23690	30.09644	1890.9	Rz	Fine-grained Magnetite Gabbro with gabbronoritic blocks; Fine-grained magnetite gabbro includes rounded, elongate blocks of oikocrystic gabbronorite preferentially horizontally imbricated, ~6-10 cm along axis. Also a bunch of rounded gabbronorite inclusions that may represent the spherical aggregates of the TBM
MZ3	-477.8	-25.22290	30.10699	1902.2	Cz	–
D2SB	-476.7	-25.21950	30.10983	2004.2	Rz	Sampled because quartzite here all have iron stained interiors. The weathered exteriors are clear. Quartzites served as a conduit for mobilized iron-rich fluids?
RZ2	-470.3	-25.22270	30.10679	1900.7	Rz	–
D2SH	-468.9	-25.22070	30.10839	1949.5	Rz	Schist with Skarn; Swirled and deformed schist with marginal zone, with thin clear banding. Non-magnetic. Xenolith? Associated with skarn pegmatite veins: 60% Feldspar, 40% amphibole? Big very dark crystals that I think I see 120 degree cleavage on... Likely devolatilizing xenolith.
RQCONT	-465.1	-25.21920	30.10955	2013.8	Rz	–
RZ20	-455.8	-25.23540	30.09654	1902.8	Rz	–
RZ5	-440.7	-25.23760	30.09423	1917.8	Rz	–
D2ND	-440.4	-25.21820	30.10918	2019.6	Rz	Measureable bedding planes at top of ridge: Strike: 265, Dip: 25. Also, One bedding plane preserved lovely sedimentary ripples striking E-W

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
D11SA	-439.8	-25.23770	30.09405	1898.6	Rz	Coarse grained gabbronorite; Is it critical-margin zone? Are resistance bands chromite bearing?
QZ FOOLLOW TOPO	-433.0	-25.21900	30.10814	1989.6	Rz	–
RZ4	-430.9	-25.23700	30.09419	1905.5	Rz	–
RZ6	-429.7	-25.23860	30.09298	1923.1	Rz	–
D2NE	-426.0	-25.21670	30.10968	2030.8	Rz	Different strike, same dip. Strike: 315, dip: 25. Quartzites are rotated around the top of the ridge. NW-SE vertical fault or E-W South dipping fault can neatly explain rotation.
D4NB	-422.3	-25.24150	30.09048	1932.3	Rz	Contact between floor quartzite and marginal zone; Bed plane of quartzite measured
QZ51	-416.6	-25.21880	30.10754	1993.1	Rz	–
D10NK	-415.4	-25.24210	30.08966	1917.5	Rz	–
RZ7	-413.5	-25.23870	30.09213	1915.4	Rz	–
D4SA	-412.5	-25.24090	30.09036	1895.3	Rz	Contact between marginal zone and migmatized quartzite; Migmatitic quartzite ~10m thick. At contact is non-magnetic leucocratic gneiss, then fine magnetic marginal zone: 10cm, then coarse grained marginal zone: 30 cm, then common fine marginal zone
SCHI	-402.9	-25.24040	30.09025	1882.1	Rz	–
QZ61	-402.3	-25.21890	30.10676	1980.1	Rz	–
QZ41	-401.8	-25.21870	30.10691	1986.8	Rz	–
QZ	-395.1	-25.21680	30.10808	2000.8	Rz	–

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
QZ29	-393.5	-25.24210	30.08861	1906.9	Rz	–
QZ71	-390.6	-25.21900	30.10613	1975.3	Rz	–
RZ28	-388.4	-25.24200	30.08845	1906.1	Rz	–
SCH	-387.4	-25.24010	30.08972	1869.7	Rz	–
RZ12	-380.7	-25.23990	30.08955	1866.2	Rz	–
D2SA	-379.9	-25.23000	30.09696	1860.0	Cz	Common Main Zone Coarse Grained Gabbronorite (Office interp: Leucogabbronorite; porphyritic gabbronorite); On Dirt road to Klipriver Farms, Boulder piles outcropping of main zone. These rocks are the building materials for much of the structures in the area. Weathering is quite distinct from Marginal zone. Both have exfoliation/onion skin weathering. Main Zone is prouder, forming impressive rounded boulders like a granite. Weathered faces have distinct dalmatian pattern. A large boulder (float) at this outcrop contains a characteristic pyroxene "tennisball", a well-known marker at the base of the Main Zone.
RZ35	-379.6	-25.21890	30.10566	1964.7	Rz	–
D2SF	-378.9	-25.21550	30.10829	1994.2	Rz	Coarse Grained Magnetite Gabbro; Actually I am now thinking this sample is similar to the magnetite gabbro D1Sa, but coarser grained. That sample too had zones of coarse grains, and this one also had blocks of oikocrystic augite.
RZ42	-378.6	-25.21940	30.10519	1949.6	Rz	–

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
QZ8	-377.2	-25.21920	30.10529	1954.4	Rz	–
RZ9	-369.8	-25.23920	30.08952	1849.8	Rz	–
RAT	-368.2	-25.23630	30.09174	1882.9	Rz	–
RZ511	-367.9	-25.21970	30.10442	1934.5	Rz	–
RZLQZR	-365.6	-25.21960	30.10440	1938.2	RzRz	–
RZ53	-362.0	-25.21930	30.10448	1945.2	Rz	–
RZ113	-359.1	-25.24170	30.08722	1876.0	Rz	–
IS1	-358.5	-25.23030	30.09577	1868.1	IS	–
RZ62	-355.1	-25.21960	30.10390	1934.4	Rz	–
D2SG	-351.6	-25.21390	30.10811	1955.9	Cz	Norite; Weathering more closely resembles main zone than nearby marginal zone, but this rock does not look like main zone to me. Non-magnetic.
QZ9	-351.6	-25.22000	30.10342	1929.3	Rz	–
SCI	-350.5	-25.23900	30.08890	1888.3	Rz	–
D9NE	-348.4	-25.25130	30.07965	1928.7	Rz	–
QZ10	-346.9	-25.22000	30.10320	1927.8	Rz	–
SCH11	-343.1	-25.24070	30.08725	1880.7	Rz	–
RZ27	-341.4	-25.24060	30.08725	1881.7	Rz	–
D9SD	-340.3	-25.25150	30.07907	1912.0	Rz	Fine Grained Marginal Norite; Sampled for prominent resistance band for later analyses
QZ11	-338.8	-25.21970	30.10308	1937.7	Rz	–
SCH2	-338.1	-25.23940	30.08802	1887.5	Rz	–
RZ82	-337.6	-25.23960	30.08784	1885.7	Rz	–
D9NC	-334.9	-25.25180	30.07858	1907.8	Rz	Chloritized Norite; Plag, OPX, abundant phlogopite, alteration grade lessens up section
QZ12	-333.7	-25.21970	30.10285	1938.8	Rz	–

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
D9NF	-333.0	-25.25050	30.07951	1918.4	Rz	Fine Grained Marginal norite; Droplet forms of coarse pyroxenite xenoliths, pegmatite inclusions?
RZ23	-330.8	-25.21720	30.10466	1957.1	Rz	–
RZ72	-330.0	-25.21940	30.10291	1940.5	Rz	–
RZ81	-317.5	-25.21920	30.10248	1939.6	Rz	–
RZ19	-316.1	-25.21700	30.10410	1948.2	Rz	–
RZ25	-312.4	-25.24970	30.07913	1910.1	Rz	–
D9NG	-312.1	-25.24980	30.07904	1909.6	Rz	Paleovertical Pegmatite Dike Photograph, Marginal zone coarsens upsection towards quartzite
D5NC	-299.1	-25.21880	30.10195	1946.4	Rz	Cleavage
RZ44	-289.9	-25.24030	30.08497	1844.2	Rz	–
RZ18	-285.9	-25.21730	30.10245	1939.7	Rz	–
RZ91	-280.1	-25.21900	30.10086	1928.3	Rz	–
QZ31	-264.0	-25.21640	30.10209	1933.1	Rz	–
QZ28	-252.4	-25.24750	30.07796	1893.5	Rz	–
RZ10	-252.2	-25.21880	30.09972	1928.7	Rz	–
QZ13	-247.6	-25.21950	30.09891	1907.0	Rz	–
D5SB	-247.6	-25.21430	30.10287	1919.0	Cz	Amphibole Gabbro; Weakly magnetic, amphibole bearing? Rock clearly rotten
D9NH	-243.7	-25.24740	30.07764	1895.2	Rz	–
QZ23	-241.6	-25.21610	30.10128	1932.0	Rz	–
QZ14	-240.8	-25.21960	30.09852	1906.7	Rz	–
QZ15	-232.8	-25.21970	30.09807	1906.5	Rz	–
RPEG1	-229.4	-25.21390	30.10229	1906.6	Cz	–
RZ111	-223.4	-25.21950	30.09777	1901.7	Rz	–

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
D7NF	-218.0	-25.21350	30.10206	1905.5	Cz	pegmatitic gabbronorite; large (5 cm) oikocrystic pyroxene
QZ110	-215.1	-25.21610	30.10002	1922.7	Rz	–
QZ113	-214.3	-25.24740	30.07621	1874.5	Rz	–
RZ121	-213.4	-25.21890	30.09779	1910.6	Rz	–
QZ16	-213.4	-25.21910	30.09763	1908.3	Rz	–
QZ17	-211.3	-25.21880	30.09778	1914.8	Rz	–
RZ13	-208.8	-25.21860	30.09780	1909.1	Rz	–
QZ18	-205.1	-25.21850	30.09772	1914.0	Rz	–
D9NI	-204.2	-25.24670	30.07624	1865.2	Rz	Schist (Pebble marker?); well mixed clasts of schists, zones of pink felsic melt
QZ19	-200.2	-25.21860	30.09741	1912.4	Rz	–
RZ112	-198.6	-25.24650	30.07613	1864.7	Rz	–
D7NG	-197.3	-25.21220	30.10208	1904.9	Cz	pegmatitic gabbronorite; Here magnetic, slightly upsection (~2 m) rocks non-magnetic. Large (30 cm) blobs of pyroxenite rimmed by anorthosite. Pyroxene/Anorthosite ranging from 1/4 to 4. Included blocks with preserve reistance banding
SCHI1	-193.0	-25.24620	30.07609	1863.1	Rz	–
RZQZLEFT	-193.0	-25.21920	30.09654	1886.8	Rz	–
QZ20	-187.4	-25.21880	30.09663	1901.1	Rz	–
RZ14	-184.5	-25.21870	30.09657	1899.8	Rz	–
RZ15	-175.4	-25.21840	30.09636	1895.5	Rz	–
RZ26	-172.5	-25.24550	30.07565	1856.0	Rz	–
MZ11	-166.4	-25.22800	30.08851	1836.7	Cz	–

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
D5SA	-165.8	-25.21540	30.09820	1902.6	Cz	thought it looked like the "dark facies" main zone in the field,; Compare to D25g (porhpyitic gabbronorite), Quartzite in float (alluvial)
D6SL	-165.1	-25.21490	30.09851	1891.9	Cz	porphyritic gabbronorite; In notes, described as main zone porphyritic gabbronorite. However these rocks are structurally beneath chrome layers. Closer inspection reveals that pyroxenes are often oikocrystic and large... Incompletely formed Mottled Anorthosite?
RZ16	-161.9	-25.21870	30.09548	1886.7	Rz	–
QZ22	-160.6	-25.21810	30.09591	1897.9	Rz	–
D5NE	-159.8	-25.21870	30.09537	1883.1	Rz	–
RZ17	-153.5	-25.21950	30.09445	1875.0	Rz	–
QZ21	-153.3	-25.21950	30.09444	1875.0	Rz	–
QZ25	-147.4	-25.21770	30.09553	1876.6	Rz	–
D6SG	-145.7	-25.21840	30.09491	1872.2	Rz	magnetite pegmatite; Host rock magnetic. Pegmatite weathers proud, cutting the rock paleo-up, "sandy" texture in host rock
D6NK	-144.9	-25.21500	30.09749	1888.0	Cz	porphyritic gabbronorite
D7SH	-143.8	-25.21020	30.10109	1898.3	Cz	Sulfide-pyroxenite
D6SH	-143.7	-25.21770	30.09534	1871.5	Cz	Coarse-grained Feldspatic Pyroxenite; Non-magnetic, Floor of critical zone? "Flatreef"?
D6SI	-143.7	-25.21770	30.09534	1871.5	Cz	Coarse-grained Feldspatic Pyroxenite; Coarser grained rocks here are more rotten
QZ111	-140.5	-25.21760	30.09526	1869.3	Rz	–
QZ24	-138.6	-25.21800	30.09487	1869.2	Rz	–

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
RPEQ	-135.4	-25.21480	30.09718	1882.8	Cz	–
PEG	-133.2	-25.21140	30.09968	1892.7	Cz	–
D6NJ	-131.7	-25.21540	30.09655	1880.5	Cz	Pegmatite gabbronorite
RPEG	-129.6	-25.21540	30.09644	1876.7	Cz	–
RZ34	-124.2	-25.21910	30.09334	1857.8	Rz	–
RZ37	-120.5	-25.24380	30.07447	1839.6	Rz	–
RZ52	-117.9	-25.22040	30.09203	1846.3	Rz	–
D7NI	-100.2	-25.20893	30.09996	1875.8	Cz	Pyroxene tennisballs in Gabbronorite; Structurally below chromite?
D7SJ	-99.5	-25.20890	30.09995	1875.8	Cz	gabbronorite; small amount of interstitial CPX. Transitional zone?
MZPOR	-96.0	-25.20910	30.09963	1873.1	Cz	–
CZ3	-94.1	-25.20880	30.09978	1877.1	Cz	–
D6NM	-93.3	-25.21080	30.09820	1867.6	Cz	Fine Layered Anorthosite-norite
MZ5	-86.9	-25.24180	30.07439	1829.2	Cz	No field notes. Assigned Cz because it is stratigraphically below my modeled TU
D6SN	-86.0	-25.21040	30.09815	1862.6	UG2	magnetic chromite seam w/ oikocrystic pyroxenite and anorthosite; abundant blocks. chromite hosted with pure anorthosite and in massive layer with oikocrystic pyroxene. Top layers are magnetic (magnetite-bearing?)
CZ2	-83.4	-25.21040	30.09804	1865.3	Cz	–
CZ1	-72.4	-25.20930	30.09835	1862.1	Cz	–
D4ND	-71.8	-25.21690	30.09254	1847.4	Cz	Coarse Gabbronorite; Indirect measurement, bedding planes in a marsh
CZ	-57.9	-25.20720	30.09933	1884.0	Cz	–
IS	-51.8	-25.24350	30.07152	1840.5	IS	–

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
MZ2	-50.6	-25.21740	30.09118	1846.9	Cz	Assigned Mz because it is undeformed / overlying Kliprivier synform
CZ5	-47.9	-25.20630	30.09959	1897.6	Cz	–
D7SK	-33.0	-25.20600	30.09920	1919.5	TU	Mottled Anorthosite; Non-magnetic. Interpreted as proximal Merensky Footwall based distance from UG2
D11SB	161.8	-25.22810	30.07313	1803.4	Mz	Porphyritic Gabbronorite; Rotten. Initially confused for mottled anorthosite
D11SC	206.9	-25.22700	30.07188	1806.2	LMA	Mottled Anorthosite; Lower Mottled anorthosite; outcrop source of James' favorite boulder
D11NH	333.1	-25.19250	30.09242	1916.5	MMA	Large cliffside mottled anorthosite outcrop. ; Modal Layering
D5NF	361.8	-25.22690	30.06477	1803.5	Mz	–
D10SJ	373.0	-25.22700	30.06419	1807.6	MMA	Mottled anorthosite with blebby/blocky recrystalized pyroxenite; Big Outcrop, undulating contacts of MA, often dipping east. Sometimes appear discordant, though others appear concordant: Can be explained by thermal erosion, deposition of surface with meter-scale hummocks and swales
MZ1	376.5	-25.22870	30.06271	1796.5	Mz	–
MA	377.6	-25.22920	30.06219	1767.3	MMA	–
DS	378.7	-25.19100	30.09140	1902.6	Dl	–
D8SI	397.6	-25.18210	30.09749	1981.1	UMA	Mottled Anorthosite; Interpreted as LMA. Magnetic mottles. Includes fine-grained gabbronorite with large anhedral CPX phenocrysts (<5 cm)

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
D8SH	413.2	-25.18210	30.09678	1984.4	Dl	Dolerite Sill; Strongly magnetic, with well-developed fine blocky cleavage from stress field. Inherited phenocrystic blocky 1 cm plagioclase with strong ocillitory zoning. Likely later intrusive, not necesarily concordant
D11NF	415.5	-25.19710	30.08507	1896.0	Mz	Layered/porhpyritic gabbronorite; Fine modal layering, near "peace" rock
MA1	422.6	-25.22600	30.06264	1805.3	MMA	–
D10SI	433.5	-25.22550	30.06250	1801.1	MMA	Mottled anorthosite with blebby/blocky recrystalized pyroxenite; MMA
D8NJ	446.1	-25.18260	30.09486	1979.0	Mz	–
D11SG	451.4	-25.19410	30.08572	1911.4	UMA	Mottled Anorthosite; Very large magnetic mottles (<10 cm) with large (< 0.5 cm) magnetite
RESB1ND	453.6	-25.18230	30.09475	1982.6	Mz	–
D5SG	508.8	-25.21960	30.06343	1789.5	UMA	Mottled Anorthosite; Float, outcrop in sight. Didn't want to trespass, Also found green fine grained rock with large anhedral CPX
D3SG (D4NC)	564.4	-25.22740	30.05494	1780.5	UMA	Mottled Anorthosite over strongly graded pyroxenite-anorthosite over fine grained Gabbronorite with large (<5 cm) anhedral CPX; Gave a ride to an african man walking home from work at klipriver farms who spoke no english. Stopped by an outcrop and sampled in the dark. Vantongeren maps as Upper Mottled anorthosite

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
D11NI	622.4	-25.19160	30.07954	1865.4	Mz	Finely Layered Gabbronorite; Good cleavage, Mottled Anorthosite Nearby?
MODALLAYER	686.5	-25.18070	30.08540	2053.8	Mz	–
D8SG	708.1	-25.18270	30.08279	2021.6	Mz	Sheared Spotted Anorthosite; Strongy foliated, suggesting co-crystalization shearing. Biggest mottles: ~3cm x 0.5 cm. Interpreted as MMA by RL et al., 2022
MZD	739.0	-25.23480	30.04130	1789.6	Mz	–
MODAL LAYER	742.5	-25.19140	30.07401	1828.4	Mz	–
IS2	865.5	-25.19950	30.06205	1780.6	IS	–
D11SE	866.6	-25.18590	30.07260	1886.4	Mz	finely layered gabbronorite; Mottled anorthosite itself is in float here, but graded rocks with blocks are in place.. Mottles are small (<3mm) and non-magnetic
D11ND	892.7	-25.18530	30.07186	1890.6	Mz	~20 cm Graded spotted anorthosite with irregular contacts and variably recrystalized pyroxenite blocks/blebs. Proper mottled anorthosite in float; non-magnetic; In field interpreted as Upper/main mottled anorthosite (and related package).
D8SC	1030.4	-25.17670	30.07222	1967.9	Mz	Spotted Anorthosite; ~8cm thick, with prominent igneouss layering and including small amounts of pyroxenite below. Interpreted as UMA. I would not call this a mottled anorthosite, the pyroxenes are do not immediately appear oikocrystic
D8NE	1034.9	-25.17780	30.07114	1954.5	Mz	–
XENOLITH	1034.9	-25.17860	30.07051	1946.4	Mz	A second xenolith near photo 91

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
UMA2	1035.2	-25.17640	30.07222	1967.4	Mz	–
D8ND	1035.7	-25.17900	30.07014	1936.5	Mz	Layered Gabbronorite; Centimeter-scale igneous layering with variable color and grain size
D8ND	1035.7	-25.17900	30.07014	1936.5	Mz	Homogeneous gabbronorite; Different bedding in nearby unlayered rocks...
D8NA	1244.1	-25.17050	30.06651	1814.1	Mz	Porphyritic gabbronorite; Blebby Xenolith noted and photographed
D6SE	1416.8	-25.21910	30.02170	1770.8	Mz	green gabbronorite with amphibole "mottles"; Amphibole mottles (<5 cm) are strongly imbricated, undulose shine, not perfectly optically continuous
D6SF	1416.8	-25.21910	30.02170	1770.8	IS	Ironstone concretion; <2 cm Goethite nodules enclosing limonite cores with Hematite-Limonite matrix. Rounded clastic quarts <2 mm, in float
D10NC	1467.8	-25.20220	30.03204	1754.9	Mz	–
MZ31	1472.9	-25.20990	30.02600	1756.4	Mz	–
D10NB	1475.5	-25.20090	30.03275	1781.4	Mz	Included a range of measured strikes from 250-315, on fault?
D6NB	1550.5	-25.21840	30.01618	1815.6	Mz	Measured another dip (slump?) indirectly from road: 050:30
D10ND	1561.2	-25.19970	30.02966	1772.4	Mz	–
MZD1	1561.7	-25.21950	30.01472	1779.9	Mz	–
D10NA	1581.8	-25.19910	30.02914	1767.3	Mz	–
MZD2	1623.2	-25.22540	30.00751	1808.8	Mz	–
MZ4	1777.8	-25.19310	30.02448	1732.7	Mz	–
D10NH1	2049.9	-25.18090	30.02113	1745.9	IS	Ironstone Concretion in place

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Station	Strat. Ht. (m)	Lat	Lon	Elev. (m)	Unit	Description / Notes
D10NE	2056.3	-25.17710	30.02374	1757.4	Mz	–
MZ12	2057.2	-25.17750	30.02334	1739.5	Mz	–
D10NH	2135.9	-25.17770	30.01951	1728.7	IS	Ironstone Concretion in float
MZ21	2141.1	-25.17190	30.02370	1746.3	Mz	–
D10NG	2175.3	-25.17710	30.01813	1726.0	Mz	Amphibole 'mottled' gabbronorite in float. Fault?
D10NF	2182.5	-25.17820	30.01698	1729.8	Mz	–
D6ND	2682.5	-25.21330	29.96669	1526.2	Mz	Dark and Light Facies
D6NC	3266.3	-25.19900	29.95029	1475.6	Mz	Dark Facies

Table 5.2: Structural measurements from the 'Geotagged Structure Only' sheet of the Draaikraal field GPS/GIS compilation. Strike is recorded with west = 0° (right-hand-rule dip direction follows); dip in degrees from horizontal. '–' denotes not recorded for that station.

Station	Lat	Lon	Elev. (m)	Unit	Structure Type	Strike	Dip	Notes
D3NB	-25.22540	30.11481	1959.9	Rz	Bedding	235	30	–
D3NA	-25.22590	30.11390	1929.4	Rz	Bedding	210	30	compass trouble, resolved
D7NA	-25.23700	30.09985	1923.7	Rz	Beddin	310	25	sheared pegmatite veins in pockets/pipes orthogonal to bedding, paleoup. This layer is just above migmatized quartzite and just below fine magnetic marginal zone. Represents fluid transport from migmatized sedimentary rocks, origin of resistance banding? Strike and Dip are for migmatized Qz
D2ND	-25.21820	30.10918	2019.6	Rz	Bedding	265	25	Measureable bedding planes at top of ridge: Strike: 265, Dip: 25. Also, One bedding plane preserved lovely sedimentary ripples striking E-W

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Station	Lat	Lon	Elev. (m)	Unit	Structure Type	Strike	Dip	Notes
D2NE	-25.21670	30.10968	2030.8	Rz	Bedding	315	25	Different strike, same dip. Strike: 315, dip: 25. Quartzites are rotated around the top of the ridge. NW-SE vertical fault or E-W South dipping fault can neatly explain rotation.
D4NB	-25.24150	30.09048	1932.3	Rz	Bedding	330	25	Bed plane of quartzite measured
D10NK	-25.24210	30.08966	1917.5	Rz	Bedding	325	25	–
D9NE	-25.25130	30.07965	1928.7	Rz	Bedding	270	15	–
D5NC	-25.21880	30.10195	1946.4	Rz	Cleavage	235	25	Cleavage
D9NH	-25.24740	30.07764	1895.2	Rz	Bedding	320	35	–
D7NF	-25.21350	30.10206	1905.5	Cz	Bedding	270	10	large (5 cm) oikocrystic pyroxene
D7NG	-25.21220	30.10208	1904.9	Cz	Cleavage	320	15	Here magnetic, slightly upsection (~2 m) rocks non-magnetic. Large (30 cm) blobs of pyroxenite rimmed by anorthosite. Pyroxene/Anorthosite ranging from 1/4 to 4. Included blocks with preserve reistance banding
D5NE	-25.21870	30.09537	1883.1	Rz	Cleavage	215	25	–
D6NK	-25.21500	30.09749	1888.0	Cz	Cleavage	210	5	–
D6NJ	-25.21540	30.09655	1880.5	Cz	Cleavage	240	10	–
D6NM	-25.21080	30.09820	1867.6	Cz	Cleavage	323	16	–

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Station	Lat	Lon	Elev. (m)	Unit	Structure Type	Strike	Dip	Notes
D4ND	-25.21690	30.09254	1847.4	Cz	Cleavage	310	40	Indirect measurement, bedding planes in a marsh
D11NH	-25.19250	30.09242	1916.5	MMA	Modal Layering	220	15	Modal Layering
D5NF	-25.22690	30.06477	1803.5	Mz	Cleavage	330	25	–
D10SJ	-25.22700	30.06419	1807.6	MMA	Cleavage	310	15	Big Outcrop, undulating contacts of MA, often dipping east. Sometimes appear discordant, though others appear concordant: Can be explained by thermal erosion, deposition of surface with meter-scale hummocks and swales
D8SI	-25.18210	30.09749	1981.1	UMA	Cleavage	310	20	Interpreted as LMA. Magnetic mottles. Includes fine-grained gabbronorite with large anhedral CPX phenocrysts (<5 cm)
D8SH	-25.18210	30.09678	1984.4	Dl	Contact	275	25	Strongly magnetic, with well-developed fine blocky cleavage from stress field. Inherited phenocrystic blocky 1 cm plagioclase with strong ocillitory zoning. Likely later intrusive, not necesarily concordant
D11NF	-25.19710	30.08507	1896.0	Mz	Modal Layering	220	15	Fine modal layering, near "peace" rock

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Station	Lat	Lon	Elev. (m)	Unit	Structure Type	Strike	Dip	Notes
D8NJ	-25.18260	30.09486	1979.0	Mz	Cleavage	320	15	–
D11NI	-25.19160	30.07954	1865.4	Mz	Cleavage	330	15	Good cleavage, Mottled Anorthosite Nearby?
D8SC	-25.17670	30.07222	1967.9	Mz	Modal Layering	230	25	~8cm thick, with prominent igneouss layering and including small amounts of pyroxenite below. Interpreted as UMA. I would not call this a mottled anorthosite, the pyroxenes are do not immediately appear oikocrystic
D8NE	-25.17780	30.07114	1954.5	Mz	Modal Layering	220	10	–
D8ND	-25.17900	30.07014	1936.5	Mz	Modal Layering	200	10	Centimeter-scale igneous layering with variable color and grain size
D8ND	-25.17900	30.07014	1936.5	Mz	Cleavage	300	10	Different bedding in nearby unlayered rocks...
D10NC	-25.20220	30.03204	1754.9	Mz	Cleavage	270	10	–
D10NB	-25.20090	30.03275	1781.4	Mz	Cleavage	290	10	Included a range of measured strikes from 250-315, on fault?
D6NB	-25.21840	30.01618	1815.6	Mz	Cleavage	16	6	Measured another dip (slump?) indirectly from road: 050:30
D10ND	-25.19970	30.02966	1772.4	Mz	Cleavage	315	15	–
D10NA	-25.19910	30.02914	1767.3	Mz	Cleavage	290	10	–
D10NE	-25.17710	30.02374	1757.4	Mz	Cleavage	270	10	–
D10NF	-25.17820	30.01698	1729.8	Mz	Cleavage	315	15	–

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Station	Lat	Lon	Elev. (m)	Unit	Structure Type	Strike	Dip	Notes
D6ND	-25.21330	29.96669	1526.2	Mz	Cleavage	250	13	Dark and Light Facies
D6NC	-25.19900	29.95029	1475.6	Mz	Cleavage	292	12	Dark Facies

5.6 Discussion

Origin of the Kliprivier Monocline and Draaikraal Lineament

Van Der Merwe (2007) described a "synformal structure" on Kliprivier farm and referred to it as the "Kliprivier Syncline", but I identified only a single, exotically-dipping limb within the structure with a gradual southerly change in strike, therefore I interpret it here as a monocline.

Van Der Merwe (2007) brings attention to the PGE grade of such structures along the Eastern margin of the RLS, which often are rich in sulfides and contain unusually high grade Pt ores aside from the traditional targets: the UG2 chromitite layer and the Merensky Reef. Van Der Merwe (2007) believes that the Kliprivier structure and these other sulfide traps predate Bushveld magmatism, however I identified multiple deformed coarse-grained plutonic rocks of the upper Critical Zone included within the structure. No Main Zone or Transition Zone rocks exposed on the surface to the west of the Kliprivier are deformed within the Kliprivier Synform, suggesting that deformation preceded the crystallization of the Main Zone. This relative ordering requires that the structure is synmagmatic.

Structural analysis of the synform is consistent with gentle folding along a fold axis plunging shallowly to the west with a subvertical axial plane trending roughly E(SE)-W(NW) bisecting the limbs' obtuse intersection angle (Figure 4). This deformation is subtle and would not reverse the original ordering of the layered series and the marginal series. A syncline alone cannot explain how the Marginal Zone of the deformed, SW dipping limb came to be structurally higher than some of the deformed plutonic rocks. It is however, consistent with the idea that much of the stratigraphically lower Critical Zone was intruded as stacked sills from the North and terminated here, lopolithically deforming the overlying country rocks. Within this paradigm, the intruded Transvaal sediments "displaced" deeper within the RLS at the Kliprivier Synform are actually roof rocks to the Critical Zone below the level of the UG2. Given the intense melting and devolatilization of the floor rocks in the area, it would not be surprising to find that the thin strand of country rocks dividing the Critical Zone here beneath the level of the UG2 would have been completely reacted to igneous minerals and textures by reactive porous flow of magmas and hydrothermal fluids, as has been argued for some Lower Zone and Marginal Zone rocks (Yudovskaya et al., 2021). Indeed the ghosts of such transformed

strands may be expected at many locations throughout the complex and may have separated sills of distinct composition or served as traps for percolating melts or hydrothermal fluids. They may therefore be expected especially at geochemical discontinuities within the stratigraphic column, such as the Merensky Reef, Giant Mottled Anorthosite, Pyroxenite Marker, Upper Mottled Anorthosite, etc.

Sharpe and Chadwick, 1982 also reported that Critical Zone rocks but not Main Zone rocks were deformed in the area, but attributed similar small-scale folding on the margin to isostatic flexure. Isostatic flexure has affected the modern expression of the Bushveld Complex immensely, but a paleomagnetic study shows that isostatic flexure only commenced within the Bushveld after cooling beneath the curie temperature of magnetite: 575°C (Hattingh, 1986). This means that flexure had to occur after termination of Bushveld magmatism, and that related deformation would also have affected the Main Zone, ruling out a postmagmatic origin.

If the southwesterly dipping limb of the Kliprivier Syncline is rotated ~20° about a level rotational axis trending ~45° NE to "undo" the local effects of postmagmatic isostatic flexure that affected the RLS and its margin, the exotic limb associated with the Kliprivier Synform would be dipping ~20° due south. This observation is consistent with the hypothesis that an intrusion originating from the Thabazimbi-Murchison Lineament to the North terminated in this area, producing a southerly dipping monocline in its roof rocks.

Origin of Spheroidal Pyroxenite Aggregates (Tennis balls)

Spheroidal aggregates (SAs) of pyroxenite up to ~5 cm are known to occur as inclusions in the (gabbro)norites near the margin of the RLS, and larger pyroxenite "boulders" within the Merensky Reef are well-documented in the western limb of the RLS (Lee and Sharpe, 1979; Smith et al., 2023; Khoza, 2021). The spherical agglomerates (also called "tennis balls" due to their similar size) are richer in pyroxenes, higher in An# and pyroxene Mg#, and richer in quartz and sulfides than their (gabbro)noritic hosts (Lee and Sharpe, 1979; Khoza, 2021). Originally, dense groupings of these spherical aggregates were thought to represent laterally continuous marker beds, but they cannot be traced along strike of RLS layering for long distances relative to the named reefs or mottled anorthosites– instead they are most commonly found <200m stratigraphically

above the margin with the floor rocks irregardless of the stratigraphic position within the layered series [Lee and Sharpe, 1979]. This author located isolated pyroxenite tennisballs in porphyritic leuconorite in two places above and below the Kliprivier Synform [photos 8, 79], on regions previously mapped to contain tennisballs (Lee and Sharpe, 1979). More interestingly, this author located spheroidal aggregates of pyroxenite in the fine-grained Marginal Zone [photos 2, 4, 96]. Only very recently have authors argued the marginal zone hosted spheroidal aggregates at all. Mitchell and Scoon (2023) used geochemical evidence to claim the coarsely crystalline plutonic gabbronorites hosting the Tennis Ball Marker (TBM), famously known on the Middelkraal farm, ought to be considered as Marginal Zone and not as Main Zone as previously mapped (Lee and Sharpe, 1979; Molyneux et al. 2008; Khoza, 2021). Before this work, no fine-grained marginal zone rocks had reported to contain pyroxenite tennis balls or spheroidal aggregates, although those observed here in Draaikraal are much more lumpy and irregular than the *spheroidal* aggregates (SA) found in the Middelkraal TBM.

In photo 96, the largest pyroxenite SA (~4 cm) resembles a diapir: It has a long (~15 cm) and thin (~<1 - 2 mm) pyroxenite tail developed behind it, suggesting that it was migrating through a comparatively rigid, stationary crystal-rich mush. The formation of a diapir suggests that the SA was moving relative to the host rock and that the host rock was more rigid than the pyroxenite. This requires explanation, as the spheroidal aggregates contain more primitive pyroxenes and plagioclase than the host matrix and are equilibrated at higher temperatures, meaning that the pyroxenitic SAs would be more rigid than the host rocks if one assumes the two are in thermal equilibrium.

There are three main hypotheses for formation of the spheroidal pyroxenite aggregates in the RLS. The model of Lee and Sharpe, 1979 posits that pyroxenite aggregates are formed when a second, immiscible fluid called the “bridging” liquid intrudes a magma body with suspended pyroxene crystals. The bridging liquid preferentially “wets” the suspended pyroxene crystals, and entrains them as it coagulates with other smaller, individual bubbles of the immiscible fluid entraining the suspended pyroxenes of the host magma.

The second model of Khoza, 2021 posits that a magma recharge event introduces magma into a pyroxenite cumulus pile, breaking the pile into pieces that are then subsequently entrained into the secondary magma. This magma does not

equilibrate with the pyroxenite, instead it evenly melts the rough edges of the pyroxenite aggregates, fashioning them into spheres.

The third model posits that a pyroxenitic liquid or mush intrudes a cooler crystal mush and is disaggregated into pyroxenite aggregates, analogously to the formation of mafic enclaves in felsic plutons from the disaggregation of intruded basalts (Mitchell and Scoon, 2023; Smith et al., 2023). This is the model most compatible with the observation of diapirism of spherical Aggregates in the fine-grained marginal sill phase, since it allows for the SAs to be formed by disaggregation of a comparatively hot ultramafic magma intruded into comparatively cool (and therefore rigid) mafic sills.

The diapiric movement of the pyroxenite spheroid in photo 96 serves as good evidence against the second model of Khoza (2021), as Khoza's requires the pyroxenite spheroids to be more rigid than the host magma, so that they remain consolidated against the flowing host magma. If the spheroidal aggregate was more rigid than the host magma, then it would not have left a pyroxenitic tail behind it. Finally, if magma recharge introduced into a pyroxenite cumulate pile forms SAs, then one would expect them to be a ubiquitous feature throughout the Bushveld complex regardless of the distance from the margin, as the conventional and most accepted model for the assembly of the Bushveld complex evokes many instances of magma recharge atop a preexisting cumulate pile (Holness et al., 2017; Cawthorn, 2015).

The model of Lee and Sharpe, 1979 is interesting because secondary fluid input both can explain the relative softness of the primitive SA migrating through more evolved host rocks, and because it explains the abundant quartz (>10%: Lee and Sharpe, 1979) in the pyroxenite SA, which would not be expected for an ultramafic magma with high Mg# and An#. Photos 97 and 98 exhibit the migration of a distinct secondary comagmatic silica-rich fluid through the marginal zone magmas. However, Mitchell and Scoon (2023) point out that multiple distinct SA-rich horizons of the TBM exist but those that they report were over 100 meters from the alleged source of the fluids: the Dullstroom lavas at floor of the magma chamber. Their model simply requires injection of hot pyroxenite lavas into cooler mafic mushes. This author, however, suggests that the secondary bridging liquids of Lee and Sharpe's model could have percolated through homogeneous (gabbro)norites until ultimately meeting a coarse, pyroxenititc horizon susceptible to differential wetting.

The model of Lee and Sharpe (1979) offers a satisfactory explanation to the SA's ties to the intrusion's margin. The secondary fluids that circulated throughout the Bushveld chamber were predominantly sourced from devolatilized and variably melted country rocks (Schiffries and Rye, 1990; Buick et al., 2000), so Bushveld magmas near the margin would be the most likely to encounter poorly mixed, newly-introduced immiscible (or poorly miscible) fluids than magmas further from the margin. The model of Mitchell and Scoon (2023) also offers a plausible explanation for the SAs ties to the intrusion's margin, as the margin would cool faster to create a larger temperature (and viscosity) contrasts between new pulses of pyroxenitic magmas and their host rocks.

Correlation of the Mottled Anorthosites in the Lower Main Zone

Although there are three main named mottled anorthosite packages within the lower Main Zone, there are also many other minor anorthosites throughout the Main Zone (SACS, 1980; Von Gruenewaldt et al., 1985). There is agreement between Van Der Merwe (2007), Setera and VanTongeren (2018), and Latypov et al. (2022) about the location of the UMA on the Roossenekal-Lydenburg road. I allow this agreement to calibrate the naming of the area's mottled anorthosites. Additionally, Van Der Merwe (2007) and Latypov et al. (2022) also mapped the MMA and LMA along the Roossenekal-Lydenburg road at the same location as I have.

Van Der Merwe (2007) maps the surface exposure of the UMA, MMA, and LMA throughout the southern sector of the eastern limb of the RLS, and later works with Latypov et al. (2022), who undertook 6 stratigraphic traverses throughout the southeast RLS to prove that these 3 marker beds are laterally extensive throughout the entire Main Zone and do not abut the country rocks, arguing that these continuous mottled anorthosite packages were emplaced at the floor of an entirely molten 5 km thick magma chamber. Three of these traverses overlapped with the field area of this work. Within the field area, the UMA and MMA mapped by Van Der Merwe (2007) and Latypov et al. (2022) are discordant; they do not follow the regional dip to the northwest. I did locate anorthosites at the coordinates provided in the SI by Latypov et al. (2022); however I conclude that the UMA and MMA that they have mapped in their traverse I, north of the Draaikraal Lineament, are not the same anorthosites as their mapped UMA and MMA south of the Draaikraal Lineament on the Roossenekal-Lydenburg road in their traverse III. I further

argue that what they have mapped as UMA and MMA north of the Draaikraal lineament are actually spotted anorthosites and some lack mottles entirely.

The UMA outcrop on the Roossenekal-Lydenburg road (photos 53, 54) contains a distinct package of a striking magnetic mottled anorthosite, ~0.5m thick underlain by a ~0.5m coarse-grained feldspathic pyroxenite with some reacted anorthositic blocks and pure anorthosite bands. This pyroxenite is in turn underlain by a unique rock interpreted to be a fine-grained gabbronorite containing blebby anhedral bottle-green clinopyroxene phenocrysts. Fresh faces of this rock are a uniquely vivid green color. I was able to locate the vivid green fine grained gabbronorite with blebby anhedral pyroxene phenocrysts (Fgog) above the Draaikraal Lineament near a cliff with a view of the Booysendal PGE mine to the east (Fgog not pictured, photos 88 and 90). This outcrop is located along strike to the UMA outcrop south of the Draaikraal lineament, undisplaced and at the same stratigraphic height: ~ 500 meters above the Merensky Reef.

The anorthosite mapped by Van Der Merwe (2007) and Latypov et al. (2022) as the UMA north of the Draaikraal Lineament (photo 86) is very small (~10 cm), non-magnetic, and and lacks the associated coarse pyroxenite and fine-green gabbronorite lithologies; it does not resemble the UMA outcrop on the Roossenekal-Lydenburg road. Furthermore, it is significantly stratigraphically higher: approximately 1000 meters above the Merensky Reef. Van Der Merwe (2007) does not report displacement of the UG2 over the Draaikraal Lineament, so it is interesting that their UMA is given 500 meters of stratigraphic displacement.

Similarly, the discordant MMA correlation they offer does not stand up. The MMA is supposed to be a striking and very large marker in the lower Main Zone (photos 103-109), and yet the outcrop they correlate (photo 87) is a small exposure with subtle, non-magnetic pyroxene spots/mottles, and no associated blebby or coarsely layered pyroxenites. I instead find the concordant correlation of a very large package of magnetic mottled anorthosites with coarsely layered pyroxenites at photo 115 much more plausible.

Because there are essentially no detailed studies of the lower Main Zone anorthosites, (similar to Mitchell and Manthree, 2002 for the GMA of the transitional unit), it is easy to fall prey to one's confirmation bias: one can find an anorthosite in the lower Main Zone without too much trouble. Detailed

and careful petrography of these marker beds is necessary to make correct correlations. The strategy of Latypov et al. (2022) to use these marker beds to understand the structure and petrogenesis of the RLS is excellent, however, I would argue instead that one should first focus on one particularly remarkable anorthosite package (perhaps the MMA or UMA), and should follow it along strike. For this field area, I would want to do a detailed characterization of the Transitional unit, LMA, MMA, and UMA stratigraphy to the north of the Draaikraal Lineament to get complete sections of the remarkable, but incomplete, outcrops exposed along the Roossenekal-Lydenburg road. Then I would follow the anorthosites up north to examine how their strike might change, as their regionally concordant NW dip would quickly become discordant to the north. I consider the following open questions to remain:

1) Are the LMA, MMA, and UMA generally planar features, irregular contacts aside?

2) For how long are they laterally continuous?

3) Where and why are they discordant to the regional layering, if ever?

4) Do they abut the floor?

5) How do the distances between them vary at different locations within the intrusion?

Origin of the Mottled Anorthosites in the Lower Main Zone

Surprisingly little attention has been given to the spectacular anorthositic sequences of the lower Main Zone. More often, anorthosites from the Critical (or Transitional) Zone are studied (Mitchell and Manthree, 2002; Maier et al., 2016; Mitchell et al., 2019; Maier et al., 2020; Barnes et al., 2021), or at the top of the Main Zone at the Pyroxenite Marker (Bourdeau et al., 2021 and 2023). The Upper Mottled Anorthosite (UMA) and Pyroxenite Marker of the Main Zone are known to demarcate significant geochemical and isotopic changes in the RLS stratigraphy (Klemm et al., 1985; Kruger, 1994; VanTongeren and Mathez, 2013), most often ascribed to magma recharge.

In the East and West Limbs, the mottles of the lower Main Zone mottled anorthosites (LMA, MMA, UMA) are surprisingly iron rich– the pyroxenes have much lower Mg# than the surrounding gabbronorites (Scoon and Mitchell, 1994; Mitchell et al., 1998). Though I was surprised to find the mottles of

these three units were magnetic at first, it had already been known that they produce subtle but distinct magnetic anomalies observable in aerial surveys, at least within the West limb (Viljoen and Scoon, 1985; Cole et al., 2013). Viljoen and Scoon (1985) point out that the stratigraphic magnetic anomalies of the mottled anorthosites of the lower Main Zone are not present near structural weaknesses, like faults.

Scoon and Mtichell, (1994) make the provocative argument that the silicate iron-rich ultramafic pegmatites (IRUPs) of the upper Critical and lower Main Zones are sourced from accumulated evolved fluids interstitial to these mottled anorthosites of the lower Main Zone. These putative residual melts are argued to flow laterally, eventually exploiting structural weaknesses to percolate downward to replace plagioclase-rich, magnesian cumulates with olivine-pyroxene Iron-Rich Ultramafic Pegmatites (IRUP), a lithology were plagioclase is conspicuously absent. This bold argument seems unnecessary. It is easy to explain, as Scoon and Mitchell (1994) also do, that IRUPs likely originated from downward-percolating immiscible Fe-oxide melts from the Upper Zone. Strontium isotopes within IRUPs are uniform regardless of stratigraphic height and are identical to the Upper Zone (Scoon and Mitchell, 1994). IRUPs are known to occupy any location along the stratigraphic column between the Upper Zone and their lowest occurrences in the upper Critical Zone, and form a geochemical continuum that records them becoming more magnesian and silica-rich with depth, transitioning from Fe-Ti oxide pegmatites to Fe-rich silicate pegmatites, telling a clear story of evolution/hybridization of downward-percolating, dense Fe-rich liquids.

Still, the strontium isotopes of these mottled anorthosites and some stratigraphically proximal gabbronorites are often discontinuous with the otherwise homogeneous, isotopically evolved Main Zone immediately surrounding them, and instead are, remarkably, identical to the Upper Zone and IRUPs (Karykowski et al., 2017; Kruger, 1994). It may very well be true that the IRUPs are related to the mottled anorthosites of the lower Main Zone, though I would instead propose that these anorthosites might have served as porous “aquifers” that trapped IRUP melts originating from the Upper Zone, rather than pose the mottled anorthosites as the source of these iron-rich melts.

Additionally, compelling geochemical arguments have been presented on texturally similar mottled anorthosites in the Critical Zone that argue for formation

by reactive porous flow of a hydrothermal fluid including the presence of evolved but Ni–rich olivine, hydrous phases and sulfides, and extreme reverse zoning of plagioclase surrounding mottles (Maier et al., 2020). The presence of water is known to favor olivine crystallization at the expense of pyroxenes, as well as favoring more anorthitic plagioclase over albite (Baker and Boudreau, 2019).

I envisage mottled, stratiform anorthosites to form where low-permeability structural traps collect upward-percolating reactive hydrothermal fluid that locally favors plagioclase crystallization and pyroxene dissolution. EPMA work on the mottles of the LMA, MMA, and UMA would bring clarity to the relative importance of evolved interstitial fluids, potential IRUP contaminants, and hydrothermal fluids on the formation of mottled anorthosites in the lower Main Zone and to stratiform anorthosites in general.

Rethinking the Marginal Zone: Arguments for Incremental Assembly

The Marginal Zone has been traditionally understood to be the chilled margin of the RLS (Harmer and Sharpe, 1985), However, detailed work reveals that the lowest stratigraphy of the RLS is complex. The Marginal Zone is not always present at the floor of the RLS, especially to the southeast (Molyneux, 2008; Van Der Merwe, 2007; Laypov et al., 2020). Marginal Zone norites may locally intrude the Lower Zone (WIlson et al., 2017; Yudovskaya et al., 2021), and the Marginal Zone overlies an additional Bushveld-related ultramafic sequence referred to as the Apiesdooringdrai peridotite body or Basal Ultramafic Sequence (BUS) at Burgersfort that is more likely to contain chilled components of the earliest RLS-related intrusions (WIlson, 2012; Wilson et al., 2017). The BUS, Marginal Zone, and Lower Zone all contain reacted sedimentary "rafts" or xenoliths with igneous mineralogy (Yudovskaya et al., 2021). Thinner ultramafic and mafic sills likely related to Bushveld magmatism intrude the floor rocks kilometers below the mapped Marginal Zone (Sharpe and Hulbert, 1985). In my field work, I observed that Marginal Zone sills seemed to concordantly intrude the Transvaal sediments below the RLS for many kilometers.

The lack of a cohesive igneous stratigraphy and the geochemical and isotopic variability of the Marginal and Lower Zones lead Mitchell and Scoon (2023) and Scoon and Mitchell (2023) to propose that the Marginal and Lower Zones was assembled incrementally by magmas of different compositions. I would

extend their arguments to the entire RLS. I posit that a series of stacked peridotites, pyroxenites, norites, gabbronorites, and gabbros were emplaced, somewhat self-sorted according to density and the compositional control on stalling depth (more primitive magmas will crystallize at higher temperatures) at the location of the BUS, RLS and Marginal Zones presently. I suggest that postcumulus modification by reactive porous flow of late stage magmas and/or hydrothermal fluids allowed for dissolution and recrystallization to produce phaneritic textures within the RLS, and that these fluids interacted with the Marginal Zone to a far lesser extent than the greater RLS. These fluids reequilibrated and reacted with the RLS within a depth-controlled thermal gradient which produced the RLS-scale cohesive geochemical trends that are mostly uncorrelated with isotopic and lithological boundaries (see Eales et al., 1986). Suubconcordant Intrusion of magmas as sills into the country rock results in the inclusion of abundant but comparatively volumetrically small strands of sedimentary rock in the Lower, Marginal, Critical, and Main Zones of the RLS. These strands react to igneous mineralogy by reactive porous flow along grain boundaries and develop planar modal layering dues to lateral transit of fluids and evolved magmas sourced from the Thabazimbi-Murchison Lineament (TML) and they serve as traps for hydrothermal fluids that locally supported the reaction, dissolution, and crystallization of mafic lithologies into mottled anorthosites below. The Merensky Reef represents a structural trap for a reduced supercritical carbon-based fluid (Ballhaus and Stumpfl, 1985). The isotopic homogeneity of the Upper Zone allows for a more orthomagmatic interpretation: perhaps it represents a magma chamber where evolved tholeiitic magmas and hydrothermal fluids collected, and magnetite/anorthosite layers were crystallized by cycles of depressurization, cooling, and perhaps magma recharge.

Significance of Rock Cleavage and Modal Layering Orientation in the RLS

Within many of the phaneritic plutonic rocks, there was measurable, near-planar cleavage visible (e.g. photos 58 and 63). All measurements of plutonic rocks reported here are orientations of these cleavage structures. Plutonic rock cleavages without modal layering yielded measurements that were similar to the regional dip of bedding in the floor metasedimentary rocks to the east, but in areas with prominent modal layering these cleavages had strikes rotated 90

degrees from the regional strike, interestingly dipping similarly to the exotic limb of the Kliprivier Monocline. These 90 degree offset structures could be measured very near to one another, sometimes within the same outcrop. Photo 115 shows rock cleavage that is discordant with modal layering in a striking outcrop of the MMA.

There may have been multiple processes generating superimposed fabrics in the plutonic RLS. For instance, rock fabrics may have been developed in two phases, perhaps a primary fabric (modal layering) developed by gravity and crystal settling, then a second by lateral flow of hydrothermal fluids, or by differential contractions during cooling. Care must be taken when interpreting planar measurements in plutonic rocks of the RLS. The bimodal planes of the RLS in this field area may be due to an initial development of south-dipping igneous fabric above the Kliprivier Monocline as magmas filled a local basin towards the south before another generation of paleohorizontal fabric developed perpendicular to thermal gradients due to differential contraction during cooling.

5.7 Conclusions

1) The field data of this study supports the interpretation that the Kliprivier Synform is a Monocline synmagmatic with the Critical Zone below UG2, predating the UCZ at UG2 and up. There is no displacement of marker beds above UG2, including all mottled anorthosites. The country rocks and Marginal Zone sills of the Kliprivier Synform are roof rocks to the Critical Zone below UG2 and are the floor to undeformed Transitional and Main Zone.

2) The Draaikraal Lineament is best explained as the surface projection of the subvertical axial plane of the Kliprivier Monocline. It is hypothesized that trapped fluids from the emplacement of the underlying Critical Zone may have exploited the deformed Kliprivier Monocline, developing a vertical, planar fluid conduit at the Draaikraal Lineament where upwardly migrating fluids may have escaped, developing recessive lithologies. The regional recessive topography of the Main Zone south of the Draaikraal Lineament is proposed to originate from relatively high degrees of contamination from devolatilized/melted underlying sediments relative to the Main Zone lithologies at the same stratigraphic height to the north

of the Draaikraal Lineament.

3) The discordant mottled anorthosite correlation above and below the Draaikraal Lineament by Van Der Merwe (2007) and Latypov et al. (2022) is locally incorrect. The field work presented here instead suggests that the mottled anorthosite marker beds of the lower Main Zone are concordant and undisplaced by the Draaikraal Lineament. More careful, detailed field work will be necessary to ensure correct correlation of the many understudied anorthosites of the lower Main Zone.

4) Pyroxenite aggregate tennisballs are found in fine-grained Marginal Zone Gabbros. This supports the interpretation that the coarse-grained porphyritic gabbronorite host of the TBM at Middelkraal is a member of Marginal Zone after Mitchell and Scoon (2023). It is difficult to explain the development of a diapiric tail on the nascent pyroxene tennis balls in the fine-grained Marginal Zone with the model of Khoza, 2021, although this observation allows the models of Mitchell and Scoon (2023), Smith et al. (2023), and Lee and Sharpe (1979).

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CONCLUSION

[TODO: Write short dissertation conclusion. No data, few or no figures. Synthesize contributions. Allude to the Bushveld Complex fieldwork (Appendix 5.8) as motivating nGibbs work and sulfide/goethite geochronology. Mention the laterite geomorphology at Draaikraal as an unexpected and compelling target for goethite geochronology.]

REFERENCES

Ackerman, L. (2025). "Re-Os Cookbook: From Sample Decomposition to Elemental and Isotopic Analyses". In: *Analytical Isotope Geochemistry: Techniques and Data Interpretation*. Ed. by G. R. Tripathy. Cham: Springer Nature Switzerland, pp. 267–289. DOI: 10.1007/978-3-031-88388-0_12.

Allard, T., C. Gautheron, S. Bressan Riffel, E. Balan, B. F. Soares, R. Pinna-Jamme, A. Derycke, G. Morin, G. T. Bueno, and N. do Nascimento (2018). "Combined dating of goethites and kaolinites from ferruginous duricrusts: Deciphering the Late Neogene erosion history of Central Amazonia". In: *Chemical Geology* 479, pp. 136–150. DOI: 10.1016/j.chemgeo.2018.01.004.

Altemose, V. O. (1961). "Helium Diffusion through Glass". In: *Journal of Applied Physics* 32, pp. 1309–1316.

Ansart, C., C. Quantin, D. Calmels, T. Allard, J. Y. Roig, R. Coueffe, B. Heller, R. Pinna-Jamme, J. Nouet, S. Reguer, D. Vantelon, and C. Gautheron (2022). "(U-Th)/He Geochronology Constraints on Lateritic Duricrust Formation on the Guiana Shield". In: *Frontiers in Earth Science* 10. ISSN: 2296-6463. DOI: 10.3389/feart.2022.888993.

Barrie, C. D., M. A. Pearce, and A. P. Boyle (2011). "Reconstructing the Pyrite Deformation Mechanism Map". In: *Ore Geology Reviews* 39.4, pp. 265–276. DOI: 10.1016/j.oregeorev.2011.03.006.

Barton, P. B. (1969). "Thermochemical Study of the System Fe-As-S". In: *Geochimica et Cosmochimica Acta* 33.7, pp. 841–857. DOI: 10.1016/0016-7037(69)90031-3.

Bassal, F., B. Heller, J. Roques, H. Balout, L. Tassan-Got, T. Allard, and C. Gautheron (2022). "Revealing the radiation damage and Al-content impacts on He diffusion in goethite". In: *Chemical Geology* 611, p. 121118. DOI: 10.1016/j.chemgeo.2022.121118.

Baughman, J. S. and R. M. Flowers (2018). "Deciphering a 2 Gyr-Long Thermal History From a Multichronometer (U-Th)/He Study of the Phalaborwa Carbonatite, Kaapvaal Craton, South Africa". In: *Geochemistry, Geophysics, Geosystems* 19.5, pp. 1581–1594. DOI: 10.1029/2017GC007198.

Carman, P. C. (1956). *Flow of Gases through Porous Media*. New York: Academic Press.

Carslaw, H. S. and J. C. Jaeger (1959). *Conduction of Heat in Solids*. 2nd. Oxford: Oxford University Press.

Chen, J. H. and W. W. Harvey (1975). "Cation Self-Diffusion in Chalcopyrite and Pyrite". In: *Metallurgical Transactions B* 6.2, pp. 331–339. DOI: 10.1007/BF02913577.

Chen, M., J. Mao, C. Li, Z. Zhang, and Y. Dang (2015). "Re–Os Isochron Ages for Arsenopyrite from Carlin-like Gold Deposits in the Yunnan–Guizhou–Guangxi "Golden Triangle", Southwestern China". In: *Ore Geology Reviews* 64, pp. 316–327. DOI: 10.1016/j.oregeorev.2014.07.019.

Cherniak, D. J. (2019). "Diffusion of helium in radiation-damaged zircon". In: *Chemical Geology* 529, p. 119308.

Cost, J. R. and R. G. Hickman (1975). "Helium Release from Various Metals". In: *Journal of Vacuum Science and Technology* 12.1, pp. 516–519. DOI: 10.1116/1.568578.

Cox, S. E., H. B. D. Miller, F. Hofmann, and K. A. Farley (2022). "Short Communication: Mechanism and Prevention of Irreversible Trapping of Atmospheric He during Mineral Crushing". In: *Geochronology* 4.2, pp. 551–560. DOI: 10.5194/gchron-4-551-2022.

Craig, J. R., F. M. Vokes, and T. N. Solberg (1998). "Pyrite: Physical and Chemical Textures". In: *Mineralium Deposita* 34.1, pp. 82–101. DOI: 10.1007/s001260050187.

Crank, J. (1979). *The Mathematics of Diffusion*. Clarendon Press.

Cuneen, R. (1994). "New Insights into the Gympie Goldfield". In: *Queensland Exploration Potential 1994, Symposium Abstracts*, p. 26.

Danišík, M., N. J. Evans, E. R. Ramanaidou, B. J. McDonald, C. Mayers, and B. I. A. McInnes (2013). "(U-Th)/He chronology of the Robe River channel iron deposits, Hamersley Province, Western Australia". In: *Chemical Geology* 354, pp. 150–162. DOI: 10.1016/j.chemgeo.2013.06.012.

Davis, K. M. and M. Tomozawa (1995). "Water diffusion into silica glass: Structural changes in silica glass and their effect on water solubility and diffusivity". In: *Journal of Non-Crystalline Solids* 185, pp. 203–220.

Deng, X.-D., J.-W. Li, and D. L. Shuster (2017). "Late Mio-Pliocene chemical weathering of the Yulong porphyry Cu deposit in the eastern Tibetan Plateau constrained by goethite (U-Th)/He dating: Implication for Asian summer monsoon". In: *Earth and Planetary Science Letters* 472, pp. 289–298.

Dodson, M. H. (1973). "Closure temperature in cooling geochronological and petrological systems". In: *Contributions to Mineralogy and Petrology* 40, pp. 259–274.

Doremus, R. H. (1966). "Physical Solubility of Gases in Fused Silica". In: *Journal of the American Ceramic Society* 49, pp. 461–462.

Farley, K. A. (2000). "Helium diffusion from apatite: General behavior as illustrated by Durango fluorapatite". In: *Journal of Geophysical Research: Solid Earth* 105, pp. 2903–2914.

Farley, K. A. (2007). "He diffusion systematics in minerals: Evidence from synthetic monazite and zircon structure phosphates". In: *Geochimica et Cosmochimica Acta* 71, pp. 4015–4024.

– (2018). "Helium diffusion parameters of hematite from a single-diffusion-domain crystal". In: *Geochimica et Cosmochimica Acta* 231, pp. 117–129.

Farley, K. A. and R. M. Flowers (2012). "(U–Th)/Ne and multidomain (U–Th)/He systematics of a hydrothermal hematite from eastern Grand Canyon". In: *Earth and Planetary Science Letters* 359-360, pp. 131–140.

Farley, K. A. and R. McKeon (2015). "Radiometric dating and temperature history of banded iron formation-associated hematite, Gogebic iron range, Michigan, USA". In: *Geology* 43, pp. 1083–1086.

Farley, K. A., H. B. Monteiro, P. M. Vasconcelos, and K. Waltenberg (2024). "Dehydration of goethite during vacuum step-heating and implications for he retentivity characterization". In: *Chemical Geology* 663, p. 122254.

Farley, K. A., P. W. Reiners, and V. Nenow (1999). "An Apparatus for High-Precision Helium Diffusion Measurements from Minerals". In: *Analytical Chemistry* 71, pp. 2059–2061.

Farley, K. A., R. A. Wolf, and L. T. Silver (1996). "The effects of long alpha-stopping distances on (U-Th)/He ages". In: *Geochimica et Cosmochimica Acta* 60, pp. 4223–4229.

Fechtig, H. and S. Kalbitzer (1966). "The Diffusion of Argon in Potassium-Bearing Solids". In: *Potassium Argon Dating*. Ed. by O. A. Schaeffer and J. Zähringer. Berlin, Heidelberg: Springer, pp. 68–107.

Flowers, R. M., R. A. Ketcham, D. L. Shuster, and K. A. Farley (2009). "Apatite (U-Th)/He thermochronometry using a radiation damage accumulation and annealing model". In: *Geochimica et Cosmochimica Acta* 73, pp. 2347–2365.

Flowers, R. M., D. L. Shuster, B. P. Wernicke, and K. A. Farley (2007). "Radiation damage control on apatite (U-Th)/He dates from the Grand Canyon region, Colorado Plateau". In: *Geology* 35, pp. 447–450.

Fortier, S. M. and B. J. Giletti (1989). "An Empirical Model for Predicting Diffusion Coefficients in Silicate Minerals". In: *Science* 245.4925, pp. 1481–1484. DOI: 10.1126/science.245.4925.1481.

Gill, J. E. (1969). "Experimental Deformation and Annealing of Sulfides and Interpretation of Ore Textures". In: *Economic Geology* 64.5, pp. 500–508. DOI: 10.2113/gsecongeo.64.5.500.

Goldsmith, A. S., R. A. Ketcham, and D. F. Stockli (2020). "Simulating effects of heterogeneous 4He concentration profiles and radiation damage annealing on whole-grain zircon diffusivity analyses". In: *Geochimica et Cosmochimica Acta* 284, pp. 239–253.

Goss, C. J. (1987). "The kinetics and reaction mechanism of the goethite to hematite transformation". In: *Mineralogical Magazine* 51, pp. 437–451.

Guenthner, W. R., P. W. Reiners, R. A. Ketcham, L. Nasdala, and G. Giester (2013). "Helium diffusion in natural zircon: Radiation damage, anisotropy, and the interpretation of zircon (U-Th)/He thermochronology". In: *American Journal of Science* 313, pp. 145–198.

Guo, H. and A. Barnard (2011). "Thermodynamic modelling of nanomorphologies of hematite and goethite". In: *Journal of Materials Chemistry* 21, pp. 11566–11577.

Haar, L., J. S. Gallagher, and G. S. Kell (1984). *NBS/NRC Steam Tables: Thermodynamic and Transport Properties and Computer Programs for Vapor and Liquid States of Water in SI Units*.

Hannington, M. D., W. Bleeker, and I. Kjarsgaard (1999). "Sulfide Mineralogy, Geochemistry, and Ore Genesis of the Kidd Creek Deposit: Part I. North, Central, and South Orebodies". In: *Economic Geology Monograph* 10, pp. 163–224.

Hanson, G. N., K. R. Simmons, and A. E. Bence (1975). "40Ar/39Ar spectrum ages for biotite, hornblende and muscovite in a contact metamorphic zone". In: *Geochimica et Cosmochimica Acta* 39, pp. 1269–1278.

Harrison, M. T. (1982). "Diffusion of 40Ar in hornblende". In: *Contributions to Mineralogy and Petrology* 78, pp. 324–331.

Harrison, M. T., I. Duncan, and I. McDougall (1985). "Diffusion of 40Ar in biotite: Temperature, pressure and compositional effects". In: *Geochimica et Cosmochimica Acta* 49, pp. 2461–2468.

Heim, J. A., P. M. Vasconcelos, D. L. Shuster, K. A. Farley, and G. Broadbent (2006). "Dating paleochannel iron ore by (U-Th)/He analysis of supergene goethite, Hamersley province, Australia". In: *Geology* 34, pp. 173–176.

Hnatyshin, D., D. van Acken, S. Paradis, R. A. Creaser, and M. Brodbeck (2026). "Pyrite Re-Os Geochronology of Carbonate-Hosted Zn-Pb Deposits of the Canadian Cordillera". In: *Economic Geology* 121.1, pp. 75–92. DOI: 10.5382/econgeo.5215.

Hofmann, F., B. Reichenbacher, and K. A. Farley (2017). "Evidence for >5 Ma paleo-exposure of an Eocene–Miocene paleosol of the Bohnerz Formation, Switzerland". In: *Earth and Planetary Science Letters* 465, pp. 168–175.

Hofmann, F., J. Treffkorn, and K. A. Farley (2020). "U-loss associated with laser-heating of hematite and goethite in vacuum during (U–Th)/He dating and prevention using high O2 partial pressure". In: *Chemical Geology* 532, p. 119350.

Holcombe, R. J., C. J. Stephens, C. R. Fielding, D. Gust, T. A. Little, R. Sliwa, J. Kassan, J. McPhie, and A. Ewart (1997). "Tectonic evolution of the northern New England Fold Belt: the Permian-Triassic Hunter-Bowen event". In: *Tectonics and Metallogenesis of the New England Orogen: Alan H. Voisey Memorial Volume*. Ed. by P. M. Ashley and P. G. Flood. Special Publication 19. Geological Society of Australia, pp. 52–65.

Hurley, P. M. and C. Goodman (1941). "Helium retention in common rock minerals". In: *GSA Bulletin* 52. ISSN: 0016-7606. DOI: 10.1130/GSAB-52-545.

Jones, W. M. (1953). "Permeability and Solubility of He^3 and He^4 in Vitreous Silica". In: *Journal of the American Chemical Society* 75, pp. 3093–3096.

Ketcham, R. A., C. Gautheron, and L. Tassan-Got (2011). "Accounting for long alpha-particle stopping distances in (U-Th-Sm)/He geochronology: Refinement of the baseline case". In: *Geochimica et Cosmochimica Acta* 75, pp. 7779–7791.

King, E. M., C. T. Barrie, and J. W. Valley (1997). "Hydrothermal alteration of oxygen isotope ratios in quartz phenocrysts, Kidd Creek mine, Ontario: Magmatic values are preserved in zircon". In: *Geology* 25.12, pp. 1079–1082.

Lee, J. K. W. (1993). "The argon release mechanisms of hornblende in vacuo". In: *Chemical Geology* 106, pp. 133–170.

Lee, J. K. W. and A. A. Aldama (1992). "Multipath diffusion: A general numerical model". In: *Computers & Geosciences* 18. ISSN: 0098-3004. DOI: 10.1016/0098-3004(92)90093-7.

Lee, J. K. W., I. S. Williams, and D. J. Ellis (1997). "Pb, U and Th Diffusion in Natural Zircon". In: *Nature* 390.6656, pp. 159–162. DOI: 10.1038/36554.

Li, Y., S. Glorie, and D. Selby (2025). "Re–Os Geochronology for Sulfides and Organic-Rich Sediments". In: *National Science Review* 12.9, nwaf300. DOI: 10.1093/nsr/nwaf300.

Little, T. A., M. O. McWilliams, and R. J. Holcombe (1995). "40Ar/39Ar thermochronology of epidote blueschists from the North D'Aguilar block, Queensland, Australia: Timing and kinematics of subduction complex unroofing". In: *GSA Bulletin* 107.5, pp. 520–535. DOI: 10.1130/0016-7606(1995)107<0520:AATOEB>2.3.CO;2.

Lo, C.-H., J. K. W. Lee, and T. C. Onstott (2000). "Argon release mechanisms of biotite in vacuo and the role of short-circuit diffusion and recoil". In: *Chemical Geology* 165, pp. 135–166.

Lollar, G. S., O. Warr, J. Telling, M. R. Osburn, and B. Sherwood Lollar (2019). "'Follow the Water': Hydrogeochemical Constraints on Microbial Investigations 2.4 km Below Surface at the Kidd Creek Deep Fluid and Deep

Life Observatory". In: *Geomicrobiology Journal* 36.10, pp. 859–872. DOI: 10.1080/01490451.2019.1641770.

Lovera, O. M., F. M. Richter, and T. M. Harrison (1991). "Diffusion domains determined by 39Ar released during step heating". In: *Journal of Geophysical Research: Solid Earth* 96, pp. 2057–2069.

Lyu, C., J.-F. Gao, L. Qi, and X.-W. Huang (2020). "Re-Os Isotope System of Sulfide from the Fule Carbonate-Hosted Pb-Zn Deposit, SW China: Implications for Re-Os Dating of Pb-Zn Mineralization". In: *Ore Geology Reviews* 121, p. 103558. DOI: 10.1016/j.oregeorev.2020.103558.

Majzlan, J., K.-D. Grevel, and A. Navrotsky (2003). "Thermodynamics of Fe oxides: Part II. Enthalpies of formation and relative stability of goethite (α-FeOOH), lepidocrocite (γ-FeOOH), and maghemite (γ-Fe2O3)". In: *American Mineralogist* 88, pp. 855–859.

Masaryk, J. S. (1968). *Permeation in Fused Silica*. As cited in the fused-silica helium permeability literature (e.g., Shelby, 1972); original venue not confirmed.

McClay, K. R. and P. G. Ellis (1983). "Deformation and Recrystallization of Pyrite". In: *Mineralogical Magazine* 47.345, pp. 527–538. DOI: 10.1180/minmag.1983.047.345.14.

McInnes, B. I. A., N. J. Evans, F. Q. Fu, and S. Garwin (2005). "Application of Thermochronology to Hydrothermal Ore Deposits". In: *Reviews in Mineralogy and Geochemistry* 58, pp. 467–498.

McMurtry, G. M., J. R. DeLuze, D. R. Hilton, and J. E. Blessing (2019). "Differential Diffusion of Helium Isotopes in Glass, Quantum-tunneling 3He Enrichment, and Portable 3He/4He Monitoring of Mantle Processes". In: *Scientific Reports* 9, p. 5213.

Monteiro, H. (2017). "Paleoenvironmental Evolution of Continental Landscapes through Combined High-Resolution Geochronology and δ^{18}O Ion Microprobe Analysis of Goethite". PhD thesis. The University of Queensland. DOI: 10.14264/uql.2018.6.

Monteiro, H. S., P. M. Vasconcelos, K. A. Farley, C. A. Spier, and C. L. Mello (2014). "(U-Th)/He geochronology of goethite and the origin and evolution of cangas". In: *Geochimica et Cosmochimica Acta* 131, pp. 267–289.

Monteiro, H. S., P. M. P. Vasconcelos, K. A. Farley, and C. A. M. Lopes (2018). "Age and evolution of diachronous erosion surfaces in the Amazon: Combining (U-Th)/He and cosmogenic 3He records". In: *Geochimica et Cosmochimica Acta* 229, pp. 162–183.

Morimoto, N. and L. A. Clark (1961). "Arsenopyrite Crystal-Chemical Relations". In: *American Mineralogist* 46.11–12, pp. 1448–1469.

Naono, H., K. Nakai, T. Sueyoshi, and H. Yagi (1987). "Porous texture in hematite derived from goethite: Mechanism of thermal decomposition of goethite". In: *Journal of Colloid and Interface Science* 120, pp. 439–450.

Nash, J. T. (1979). *Geology, Petrology, and Chemistry of the Leadville Dolomite: Host for Uranium at the Pitch Mine, Saguache County, Colorado*. Tech. rep. 79-1566. US Geological Survey.

Navrotsky, A., L. Mazeina, and J. Majzlan (2008). "Size-Driven Structural and Thermodynamic Complexity in Iron Oxides". In: *Science* 319, pp. 1635–1638.

Nomade, S., P. R. Renne, and R. K. W. Merkle (2004). "40Ar/39Ar age constraints on ore deposition and cooling of the Bushveld Complex, South Africa". In: *Journal of the Geological Society* 161, pp. 411–420.

Norton, F. J. (1953). "Helium Diffusion Through Glass". In: *Journal of the American Ceramic Society* 36, pp. 90–96.

Oehler, A. and M. Tomozawa (2004). "Water diffusion into silica glass at a low temperature under high water vapor pressure". In: *Journal of Non-Crystalline Solids* 347, pp. 211–219.

Ostendorf, J., F. Henjes-Kunst, T. Seifert, and J. Gutzmer (2019). "Age and Genesis of Polymetallic Veins in the Freiberg District, Erzgebirge, Germany: Constraints from Radiogenic Isotopes". In: *Mineralium Deposita* 54.2, pp. 217–236. DOI: 10.1007/s00126-018-0841-1.

Ostrogorsky, A. G. and M. E. Glicksman (2015). "Segregation and Component Distribution". In: *Handbook of Crystal Growth*. 2nd. Vol. IIA. Amsterdam: Elsevier. Chap. 25.

Park, Y.-R., E. M. Ripley, J. D. Miller, C. Li, J. Mariga, and P. Shafer (2004). "Stable Isotopic Constraints on Fluid-Rock Interaction and Cu-PGE-S Redistribution in the Sonju Lake Intrusion, Minnesota". In: *Economic Geology* 99.2, pp. 325–338. DOI: 10.2113/gsecongeo.99.2.325.

Protin, M., P.-H. Blard, Y. Marrocchi, and F. Mathon (2016). "Irreversible adsorption of atmospheric helium on olivine: A lobster pot analogy". In: *Geochimica et Cosmochimica Acta* 179, pp. 76–88.

Qi, L., M.-F. Zhou, J. Gao, and Z. Zhao (2010). "An Improved Carius Tube Technique for Determination of Low Concentrations of Re and Os in Pyrites". In: *Journal of Analytical Atomic Spectrometry* 25.4, pp. 585–589. DOI: 10.1039/b919016c.

Rätzke, K., P. W. Hüppe, and F. Faupel (1992). "Transition from Single-Jump Type to Highly Cooperative Diffusion during Structural Relaxation of a Metallic Glass". In: *Physical Review Letters* 68.15, pp. 2347–2349. DOI: 10.1103/PhysRevLett.68.2347.

Reiners, P. W. and K. A. Farley (1999). "Helium diffusion and (U-Th)/He thermochronometry of titanite". In: *Geochimica et Cosmochimica Acta* 63, pp. 3845–3859.

Rogers, W. A., R. S. Buritz, and D. Alpert (1954). "Diffusion Coefficient, Solubility, and Permeability for Helium in Glass". In: *Journal of Applied Physics* 25, pp. 868–875.

Sander, R. (2015). "Compilation of Henry's law constants (version 4.0) for water as solvent". In: *Atmospheric Chemistry and Physics* 15, pp. 4399–4981.

Schoenberg, R., F. J. Kruger, T. F. Nagler, T. Meisel, and J. D. Kramers (1999). "PGE Enrichment in Chromitite Layers and the Merensky Reef of the Western Bushveld Complex; a Re–Os and Rb–Sr Isotope Study". In: *Earth and Planetary Science Letters* 172.1, pp. 49–64. DOI: 10.1016/S0012-821X(99)00198-3.

Schwertmann, U., P. Cambier, and E. Murad (1985). "Properties of Goethites of Varying Crystallinity". In: *Clays and Clay Minerals* 33, pp. 369–378.

Scoates, J. S., C. J. Wall, R. M. Friedman, D. Weis, E. A. Mathez, and J. A. VanTongeren (2021). "Dating the Bushveld Complex: Timing of Crystallization, Duration of Magmatism, and Cooling of the World's Largest Layered Intrusion and Related Rocks". In: *Journal of Petrology* 62.2, egaa107. DOI: 10.1093/petrology/egaa107.

Selby, D., R. A. Creaser, C. J. R. Hart, C. S. Rombach, J. F. H. Thompson, M. T. Smith, A. A. Bakke, and R. J. Goldfarb (2002). "Absolute Timing of Sulfide and Gold Mineralization: A Comparison of Re-Os Molybdenite and Ar-Ar Mica Methods from the Tintina Gold Belt, Alaska". In: *Geology* 30.9, pp. 791–794. DOI: 10.1130/0091-7613(2002)030<0791:ATOSAG>2.0.CO;2.

Shackelford, J. F. (1999). "Gas solubility in glasses – principles and structural implications". In: *Journal of Non-Crystalline Solids* 253, pp. 231–241.

Shackelford, J. F., P. L. Studt, and R. M. Fulrath (1972). "Solubility of Gases in Glass. II. He, Ne, and H2 in Fused Silica". In: *Journal of Applied Physics* 43, pp. 1619–1626.

Shelby, J. E. (1971). "Diffusion of Helium Isotopes in Vitreous Silica". In: *Physical Review B* 4, pp. 2681–2686.

– (1972). "Helium Migration in Natural and Synthetic Vitreous Silica". In: *Journal of the American Ceramic Society* 55, pp. 61–64.

– (1976). "Pressure dependence of helium and neon solubility in vitreous silica". In: *Journal of Applied Physics* 47, pp. 135–139.

Shukolyukov, Y., O. Yakubovich, A. Mochalov, A. Kotov, E. Sal'nikova, S. Yakovleva, S. Korneev, and B. Gorokhovskii (2012). "New Geochronometer for the Direct Isotopic Dating of Native Platinum Minerals (Pt-He Method)". In: *Petrology* 20.6, pp. 491–505. DOI: 10.1134/S0869591112060033.

Shukolyukov, Y. A., M. M. Fugzan, I. P. Paderin, S. A. Sergeev, and D. P. Krylov (2009). "Geothermochronology Based on Noble Gases: I. Stability of the U-Xe Isotopic System in Nonmetamict Zircons". In: *Petrology* 17.1, pp. 1–24. DOI: 10.1134/S0869591109010019.

Shuster, D. L. and K. A. Farley (2005). "4He/3He Thermochronometry: Theory, Practice, and Potential Complications". In: *Reviews in Mineralogy and Geochemistry* 58, pp. 181–203.

– (2009). "The Influence of Artificial Radiation Damage and Thermal Annealing on Helium Diffusion Kinetics in Apatite". In: *Geochimica et Cosmochimica Acta* 73.1, pp. 183–196. DOI: 10.1016/j.gca.2008.10.013.

Shuster, D. L., K. A. Farley, J. M. Sisterson, and D. S. Burnett (2004). "Quantifying the diffusion kinetics and spatial distributions of radiogenic 4He in minerals containing proton-induced 3He". In: *Earth and Planetary Science Letters* 217, pp. 19–32.

Shuster, D. L., R. M. Flowers, and K. A. Farley (2006). "The influence of natural radiation damage on helium diffusion kinetics in apatite". In: *Earth and Planetary Science Letters* 249, pp. 148–161.

Shuster, D. L., P. M. Vasconcelos, J. A. Heim, and K. A. Farley (2005). "Weathering geochronology by (U-Th)/He dating of goethite". In: *Geochimica et Cosmochimica Acta* 69, pp. 659–673.

Sivell, W. J. and M. T. McCulloch (2001). "Geochemical and Nd-isotopic systematics of the Permo-Triassic Gympie Group, southeast Queensland". In: *Australian Journal of Earth Sciences* 48, pp. 377–393.

Sivell, W. J. and J. B. Waterhouse (1988). "Petrogenesis of Gympie Group volcanics: evidence for remnants of an early Permian volcanic arc in eastern Australia". In: *Lithos* 21, pp. 81–95.

Smith, P. E., E. S. Schandl, and D. York (1993). "Timing of metasomatic alteration of the Archean Kidd Creek massive sulfide deposit, Ontario, using 40 Ar- 39 Ar laser dating of single crystals of fuchsite". In: *Economic Geology* 88, pp. 1636–1642. ISSN: 0361-0128. DOI: 10.2113/gsecongeo.88.6.1636.

Stein, H., A. Schersten, J. Hannah, and R. Markey (2003). "Subgrain-Scale Decoupling of Re and ^{187}Os and Assessment of Laser Ablation ICP-MS Spot Dating in Molybdenite". In: *Geochimica et Cosmochimica Acta* 67.19, pp. 3673–3686. DOI: 10.1016/S0016-7037(03)00269-2.

Studt, P. L. (1962). "Mechanism of Gaseous Permeation through Glass, Single-Crystal Silicon, and Germanium; and Stress-Enhanced Gaseous Permeation through Alumina Bodies". PhD thesis. University of California. URL: https://escholarship.org/uc/item/9pj1h2x1.

Sturrock, C. P., R. M. Flowers, B. P. Kohn, and J. R. Metcalf (2024). "Phanerozoic Burial and Erosion History of the Southern Canadian Shield from Apatite (U-Th)/He Thermochronology". In: *Minerals* 14.1, p. 57. DOI: 10.3390/min14010057.

Suzuki, K., M. Feely, and C. O'Reilly (2001). "Disturbance of the Re-Os Chronometer of Molybdenites from the Late-Caledonian Galway Granite, Ireland, by Hydrothermal Fluid Circulation". In: *Geochemical Journal* 35.1, pp. 29–35. DOI: 10.2343/geochemj.35.29.

Swets, D. E., R. W. Lee, and R. C. Frank (1961). "Diffusion Coefficients of Helium in Fused Quartz". In: *The Journal of Chemical Physics* 34, pp. 17–22.

Toulmin, P. and P. B. Barton (1964). "A thermodynamic study of pyrite and pyrrhotite". In: *Geochimica et Cosmochimica Acta* 28, pp. 641–671. ISSN: 0016-7037. DOI: 10.1016/0016-7037(64)90083-3.

Tsuru, T., R. Igi, M. Kanezashi, T. Yoshioka, S. Fujisaki, and Y. Iwamoto (2011). "Permeation properties of hydrogen and water vapor through porous silica membranes at high temperatures". In: *AIChE Journal* 57.3, pp. 618–629. DOI: 10.1002/aic.12298.

Vasconcelos, P. M., J. A. Heim, K. A. Farley, H. Monteiro, and K. Waltenberg (2013). "40Ar/39Ar and (U-Th)/He–4He/3He geochronology of landscape evolution and channel iron deposit genesis at Lynn Peak, Western Australia". In: *Geochimica et Cosmochimica Acta* 117, pp. 283–312.

Verhallen, P. T. H. M., L. J. P. Oomen, A. J. J. M. v. d. Elsen, J. Kruger, and J. M. H. Fortuin (1984). "The diffusion coefficients of helium, hydrogen, oxygen and nitrogen in water determined from the permeability of a stagnant liquid layer in the quasi-steady state". In: *Chemical Engineering Science* 39, pp. 1535–1541.

Virtanen, P. et al. (2020). "SciPy 1.0: Fundamental algorithms for scientific computing in Python". In: *Nature Methods* 17, pp. 261–272.

Waltenberg, K. M. (2012). *Mineral Physics and Crystal Chemistry of Minerals Suitable for Weathering Geochronology: Implications to 40Ar/39Ar and (U-Th)/He Geochronology*. PhD thesis; institution not confirmed in the source draft.

Wilson, W. D. and C. L. Bisson (1973). "Atomistics of helium diffusion in copper and tungsten". In: *Radiation Effects* 19, pp. 53–58.

Wolf, R. A., K. A. Farley, and D. M. Kass (1998). "Modeling of the temperature sensitivity of the apatite (U-Th)/He thermochronometer". In: *Chemical Geology* 148, pp. 105–114.

Wolf, R. A., K. A. Farley, and L. T. Silver (1996). "Helium diffusion and low-temperature thermochronometry of apatite". In: *Geochimica et Cosmochimica Acta* 60, pp. 4231–4240.

Yakubovich, O., M. Podolskaya, I. Vikentyev, E. Fokina, and A. Kotov (2020). "U-Th-He Geochronology of Pyrite from the Uzelga VMS Deposit (South Urals) – New Perspectives for Direct Dating of the Ore-Forming Processes". In: *Minerals* 10, p. 629.

Yakubovich, O., I. Vikentyev, E. Ivanova, M. Podolskaya, I. Sobolev, E. Tyukova, and A. Kotov (2021). "U-Th-He Geochronology of Pyrite from Alteration of the Au-Fe-Skarn Novogodnee-Monto Deposit (Polar Urals, Russia) – The Next Step in the Development of a New Approach for Direct Dating of Ore-Forming Processes". In: *Geosciences* 11.10, p. 408. DOI: 10.3390/geosciences11100408.

Yakubovich, O. V., A. M. Gedz, I. V. Vikentyev, A. B. Kotov, and B. M. Gorokhovskii (2019). "Migration of Radiogenic Helium in the Crystal Structure of Sulfides and Prospects of Their Isotopic Dating". In: *Petrology* 27, pp. 59–78.

Yakubovich, O. V., A. G. Mochalov, and S. F. Sluzhenikin (2015). "Sperrylite ($PtAs_2$) as a ^{190}Pt-^{4}He Geochronometer". In: *Doklady Earth Sciences* 462.1, pp. 472–474. DOI: 10.1134/S1028334X15050098.

Yakubovich, O. V., Y. A. Shukolyukov, A. B. Kotov, S. Z. Yakovleva, and E. B. Sal'nikova (2010). "Geothermochronology based on noble gases: II. Stability of the (U-Th)/He isotope system in zircon". In: *Petrology* 18, pp. 555–570.

Yao, Z. and J. E. Mungall (2021). "Linking the Siberian Flood Basalts and Giant Ni-Cu-PGE Sulfide Deposits at Norilsk". In: *Journal of Geophysical Research: Solid Earth* 126.3, e2020JB020823. DOI: 10.1029/2020JB020823.

Yapp, C. J. and H. Poths (1995). "Stable hydrogen isotopes in iron oxides: III. Nonstoichiometric hydrogen in goethite". In: *Geochimica et Cosmochimica Acta* 59, pp. 3405–3412.

Yapp, C. J. and D. L. Shuster (2011). "Environmental memory and a possible seasonal bias in the stable isotope composition of (U-Th)/He-dated goethite from the Canadian Arctic". In: *Geochimica et Cosmochimica Acta* 75, pp. 4194–4215.

Yund, R. A. and G. Kullerud (1966). "Thermal Stability of Assemblages in the Cu–Fe–S System". In: *Journal of Petrology* 7.3, pp. 454–488.

Zeitler, P. K., A. L. Herczeg, I. McDougall, and M. Honda (1987). "U-Th-He dating of apatite: A potential thermochronometer". In: *Geochimica et Cosmochimica Acta* 51, pp. 2865–2868.

Zhao, J., J. Brugger, Y. Ngothai, and A. Pring (2014). "The Replacement of Chalcopyrite by Bornite under Hydrothermal Conditions". In: *American Mineralogist* 99.11–12, pp. 2389–2397. DOI: 10.2138/am-2014-4825.

FIELD STUDIES REVEAL SYNMAGMATIC DEFORMATION AND EXTENSIVE CRUSTAL CONTAMINATION OF THE UPPER CRITICAL ZONE, RECESSIVE LOWER MAIN ZONE, AND FLOOR ROCKS AT THE MARGIN OF THE BUSHVELD COMPLEX AT DRAAIKRAAL, MPUMALANGA, SOUTH AFRICA

[TODO: One-paragraph summary of the field campaign and key findings.]

Introduction

[TODO: Introduce the Bushveld Complex as one of the world's largest and best-preserved layered intrusions. State the scientific motivation for fieldwork at Draaikraal: tectonic setting, unexplained recessive topography, and the potential for goethite laterite geochronology.]

Literature Review

[TODO: Review Bushveld stratigraphy, magmatic layering, and prior work on the Critical Zone, Main Zone, and floor rock relationships. Summarize existing interpretations of crustal contamination and syn-magmatic deformation. Review goethite laterite formation in the context of Bushveld-derived lithologies.]

Geology of the Bushveld Complex

[TODO:]

The Upper Critical Zone and Merensky Reef

[TODO:]

The Lower Main Zone and Its Recessive Expression

[TODO:]

Floor Rocks and Crustal Contamination

[TODO:]

Laterite Formation and Goethite Geochronology Potential

[TODO:]

Field Methods

[TODO: Describe traverses, mapping strategy, sampling, and photographic documentation at Draaikraal.]

Field Observations

[TODO: Describe structural and lithological observations: deformation fabrics, contamination textures, lateral facies changes.]

Sample and Station Data

[TODO: Insert data table of field stations, GPS coordinates, lithologic descriptions, and samples collected.]

Field Map

[TODO: Insert field map figure.]

Annotated Field Photographs

[TODO: Insert annotated field photographs with captions. Use `subcaption` for multi-panel figures.]

Discussion

[TODO: Interpret field evidence for synmagmatic deformation and crustal contamination. Discuss the recessive topography at Draaikraal as a geomorphological signature of the Lower Main Zone. Connect to the motivating synthesis described in the Conclusion.]

Conclusions

[TODO: Summarize key field findings. Point to the application of ML-accelerated thermodynamic models and goethite/sulfide geochronology (Chapters 4, 2, 3) as the next research steps.]